ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

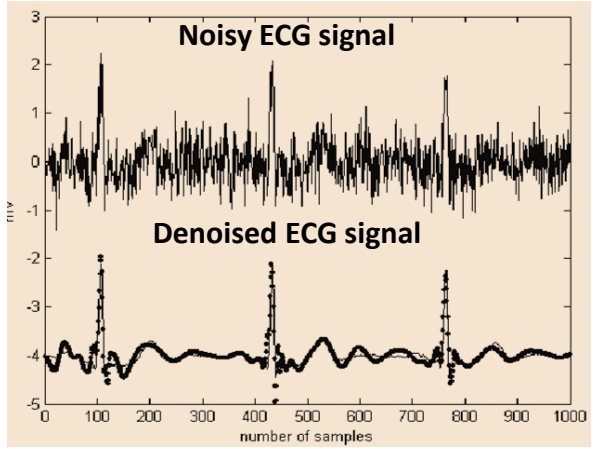
- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Digital Signal Processing (DSP) Fundamentals

- In the 1990s the digital signal processing revolution started, both in terms of the consumer boom in digital audio, digital telecommunications and the wide used of technology in industry.
- The professional music industry uses spectrum analysers, digital filtering, sampling conversion filters etc and is one of the biggest users and exploiters of DSP technology. This technology is found inside headphone, smartphones, smart speakers and is actually a cornerstone of modern audio products..
- In the medical field, DSP systems are widely utilized for recording data analysis and the interpretation of ECG signals.
- Audiologists and speech therapists are exposed to DSP systems for both testing a person's level of hearing and subsequently DSP hearing aid filtering.
- DSP is applied in the area of control and power systems, biomedical engineering, instrumentation, automotive engineering, telecommunications, mobile communication, speech analysis and synthesis, audio and video processing, seismic, radar and sonar processing and neural computing.





Advantages using DSP

- High reliability and reproducibility
- High accuracy
- Flexibility and programmability
- Absence of component drift problem
- Compressed storage facility
- Cheaper

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Discrete-time signals & Difference equations

✓ There are **two** types of signals:

Continuous - time signals

A continuous-time signal, $x(t)$, the independent variable t is continuous and thus $x(t)$ is defined for all t $(-\infty < t < \infty)$ – *see Figure 1*

Discrete - time signals

- Discrete-time signals are defined only at discrete times and consequently the independent variable n takes on only a discrete set of values .
- A discrete- time signal is thus a sequence of numbers where n is a discrete time - independent variable ($n = \ldots$ -2, -1, 0, 1, 2,…) - *see Figure 1*

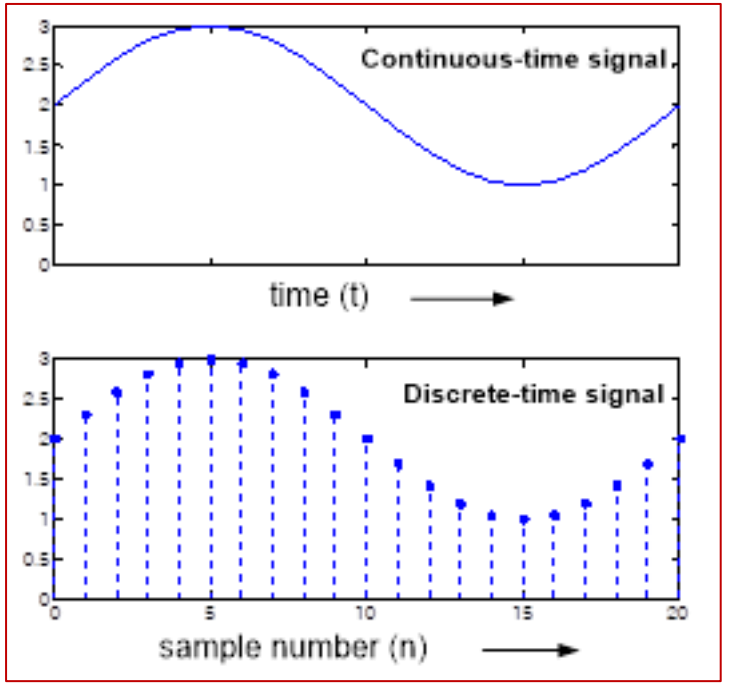


Figure 1: An example of continuous-time and discrete-time signals

Examples

- A person's body temperature is a continuous-time signal.
- The prices of stocks printed in the daily newspapers are discrete-time signals.
- Voltages and currents are usually represented by continuous-time signals. They are also represented by discrete-time signals if they are specified only at a discrete set of values of t.

Discrete-time signals: Sampling

Sampling

A discrete-time signal is often formed by sampling a continuous -time signal *x(t)*. If the samples are equidistant then;

$$x[n] = x(t)\big|_{t=nT} = x(nT)$$

Square brackets [ ] ⇒ Discrete time signals
Round Brackets ( ) ⇒ Continuous signals

The constant *T* is the sampling interval or period and $f_s=1/T$ is the sampling frequency.

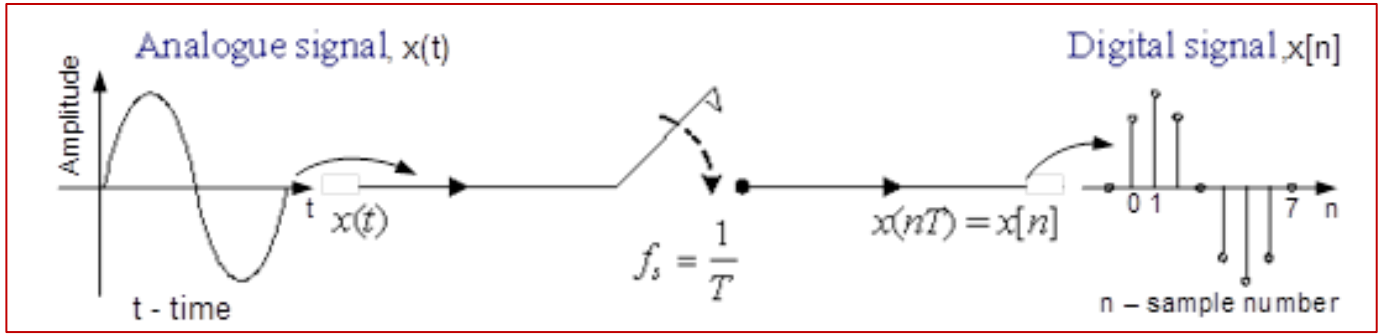
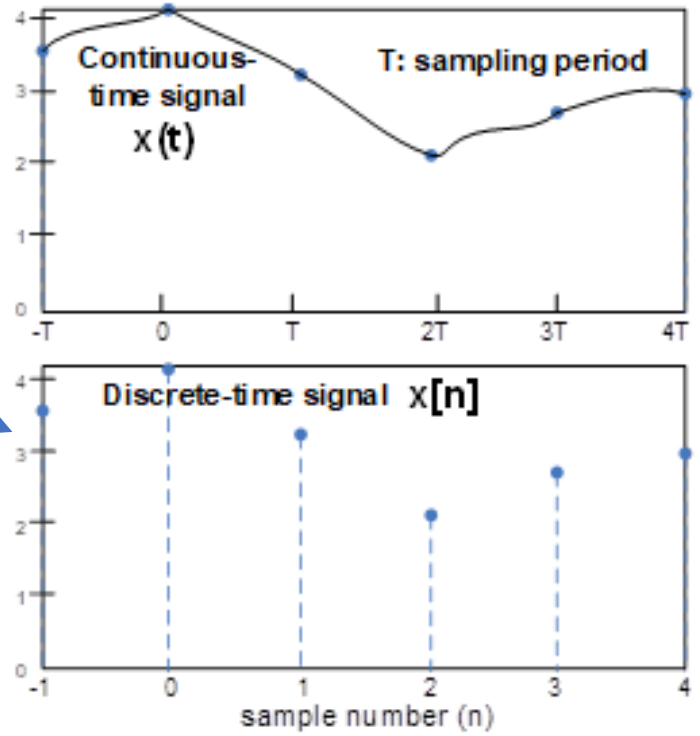


Figure 2: An example of acquiring discrete-time signal by sampling continuous-time signal

$$x[n] = \{\ 3.5,\ \ 4,\ \ 3.25,\ \ 2,\ \ 2.5,\ \ 3.0\ \}$$

n=-1 n=0 n=2 n=4

It is important to recognize that *x[n]* is only defined for integer values of *n*. It is not correct to think of *x[n]* as being zero for *n* not an integer, say *n* =1.5. *x[n]* is simply undefined for non-integer values of *n*.



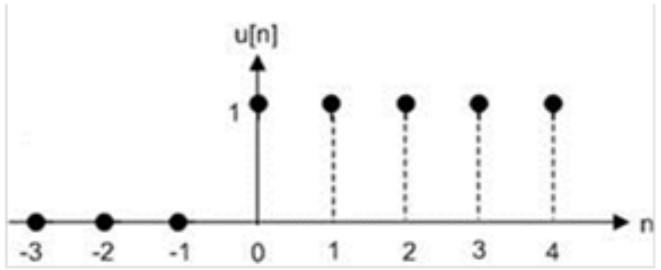
Sampling frequency $(f_s) = \frac{1}{T}$

Figure 3: An example of acquiring discrete-time signals by sampling continuous-time signals.

Discrete-time signals: Unit step, Unit impulse & Exponential

Unit step function

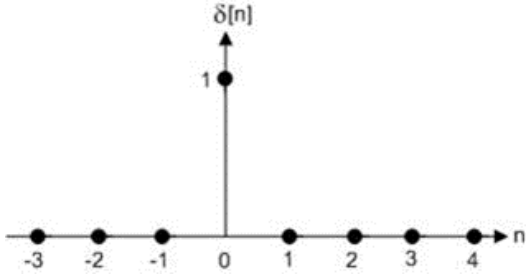
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



The Unit Impulse

- An important concept in the theory of linear systems is the continuous time unit impulse function.
- The discrete-time unit impulse function $\delta[n]$ is defined in a manner similar to its continuous time counterpart. We also refer $\delta[n]$ as the unit sample

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Discreate –time exponential signals

- In discrete time, it is common practice to write a real exponential signal as: $\boldsymbol{x[n] = c\,\alpha^{n}}$
- If c and α are real and if $|\alpha| > 1$ the magnitude of the signal grows exponentially with n, while if $|\alpha| < 1$, we have decaying exponential (see Figure 4)

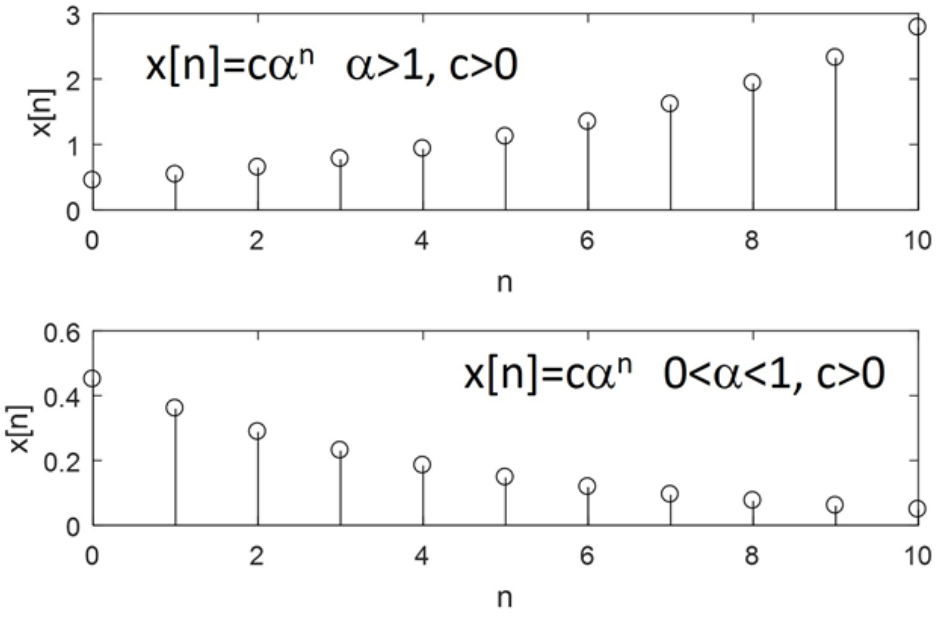


Figure 4: Examples of discrete-time exponential signals

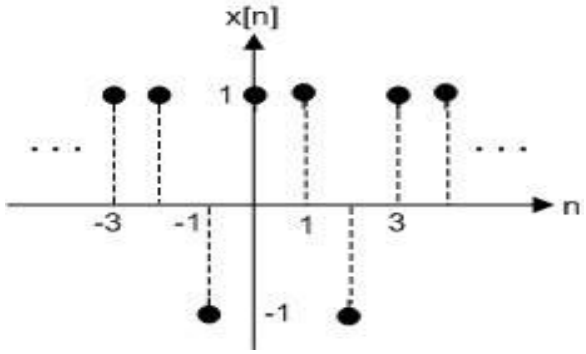
Discrete-time signals : Periodic signals

Periodic Signals

- A discrete-time signal *x[n]* is periodic with period *N*, where *N* is a positive integer, if for all values of *n*.

$$x[n] = x[n+N]$$

x[n] with Period, *N* = 3 samples



Discrete-time sinusoidal signals

- A continuous-time sinusoidal signal is given by:

$x(t) = A\sin(\omega_a t) = A\sin(2\pi f_a t)$ f_a: Analogue frequency

- A discrete - time sinusoidal signal may be expressed as:

$$x[n] = x(t)|_{t = nT} = x(nT)$$

$$x[n] = A\sin(n\boldsymbol{\omega_a}\mathrm{T}) = A\sin\left(2\pi\frac{f_a}{f_s}\mathrm{n}\right) = Asin(n\theta)$$

$$\boldsymbol{Digital\ frequency}\ (\theta) = \boldsymbol{2\pi\frac{f_a}{f_s}} = \boldsymbol{\omega_a T}\ \text{radians}$$

$$\boldsymbol{-\pi \le \theta \le \pi}$$

$$\boldsymbol{Sampling\ frequency}\ (f_s) = \frac{1}{T}\ \text{Hz}$$

- A discrete-time signal is said to be periodic with a period length *N*, if *N* is the smallest integer for which

$$x[n+N] = x[n]$$

$$\therefore A\sin((n+N)\theta) = A\sin(n\theta)$$

which can only be satisfied for all *n* if, $N\theta = 2\pi k$ (where *k* is an arbitrary integer); $N = \frac{2\pi k}{\theta} = \frac{2\pi k}{2\pi\frac{f_a}{f_s}}$ $\boxed{N = \frac{f_s}{f_a}k}$

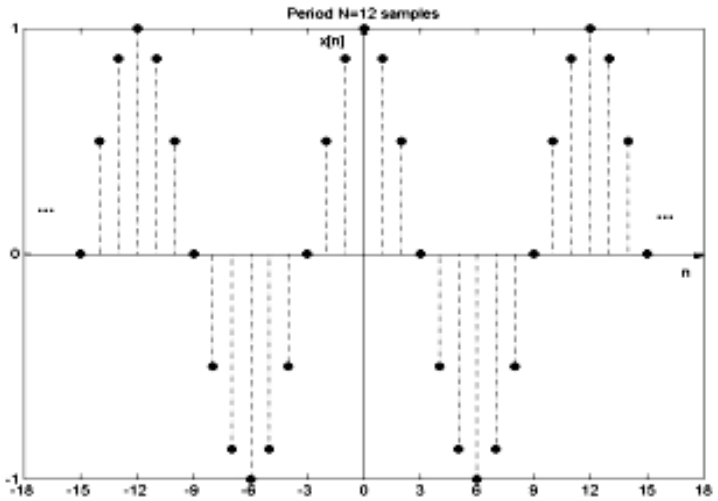
- For example, f_a = 1000 Hz and f_s = 8000 Hz then $N = f_s / f_a$ = 8 samples (assume *k* is the smallest positive integer, i.e. *k* = 1). Therefore the period length N = 8 samples.

Discrete-time signals: Examples

Example 1:

- The following diagram depicts the cosine wave *with* period *N* = 12 samples.

$$x[n] = \cos\left(\frac{2\pi n}{12}\right)$$



Example 2:

- Determine the fundamental period of $x[n] = 10\cos\left(\frac{2\pi}{15}n + \frac{\pi}{3}\right)$

$$Digital\ frequency\ (\theta) = 2\pi/15$$

Recall the equation: $N = \frac{2\pi k}{\theta}$; k =1 is the smallest positive integer.

$$\therefore N = \frac{2\pi.1}{2\pi/15} = 15 \text{ samples}$$

Example 3:

- The sinusoidal signal *x[n]* has fundamental period *N=10* samples. Determine the smallest θ for which *x[n]* is periodic:

$$\theta = {}^{2\pi k}/_{N} = {}^{2\pi k}/_{10};$$

Smallest value of θ is obtained when *k* =1;

$$\therefore \theta = \frac{2\pi}{10} = \frac{\pi}{5} \text{ radians}$$

Example 4:

- Determine if or not the discrete-time signal, *x[n]*, below is periodic and if it is periodic, determine the fundamental period. $x[n] = \sin[\frac{2}{10}\,n + \pi]$.

Ans: $x[n] = \sin[0.2\,n + \pi] = \sin[0.2\,(n + N) + \pi]$ where N is an integer period.

∴ 0.2 N = 2 π k (k is an arbitrary integer and sine function is periodic with a period of 2 π.

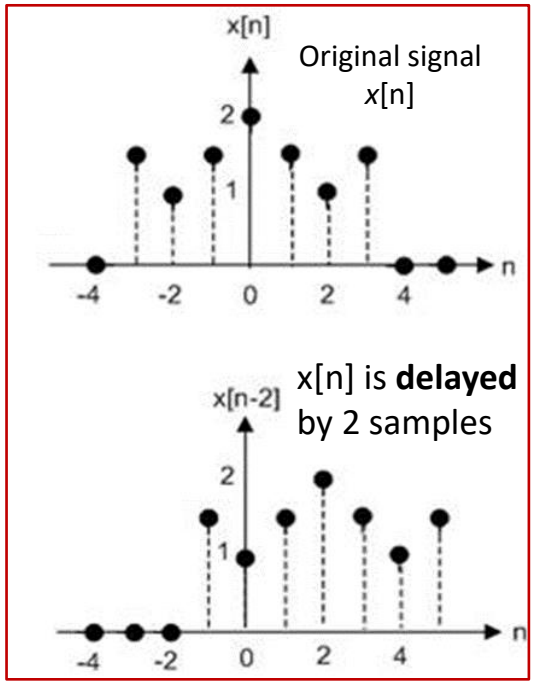
∴ N = 5 (2 π) k [non integer value of N exists]

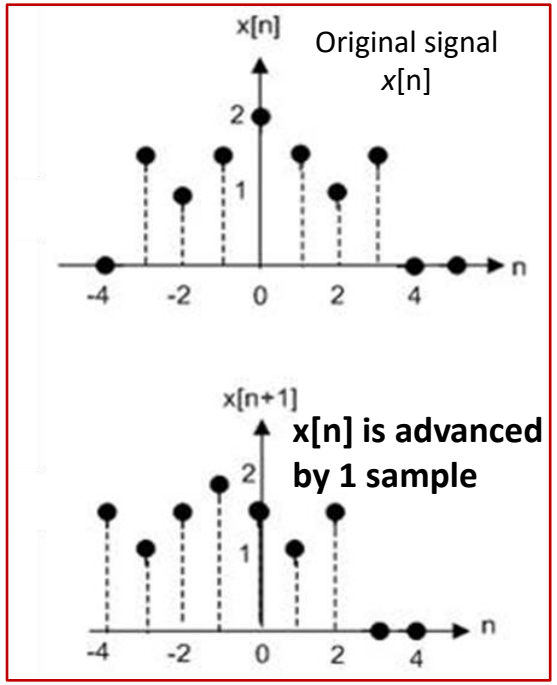
Since π is a irrational number (cannot be expressed as a ratio of two integers) no integer value of N exists. Therefore, the signal is aperiodic.

Discrete-time signals : shifted signals

Simple manipulation of discrete-time signals

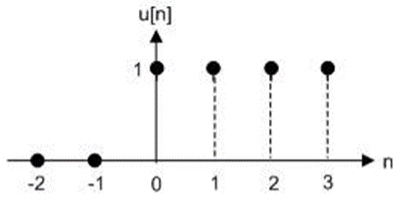
- A signal $x[n]$ may be shifted in time by replacing the independent variable n by $n-k$ where k is an integer.
- If $k > 0 \Rightarrow$ the time shift results in a **delay** of the signal by k samples [i.e. shifting a signal to the right]
- If $k < 0 \Rightarrow$ the time shift results in an **advance** of the signal by k samples. [i.e. shifting a signal to the left]



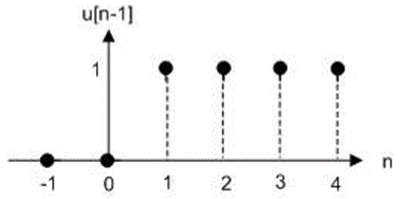


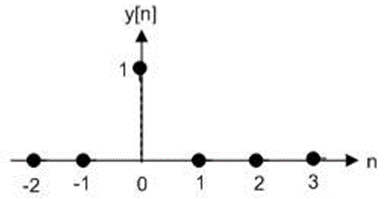
Example 5:

- Sketch the waveform : $y[n] = u[n] - u[n-1]$



$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$





Example 6:

- A continuous time signal is given by the following equation: $x(t) = 10e^{-t} - 5e^{-0.5t}$

  Write an equation for the discrete-time signal x[n] and find $x[n-2]$.

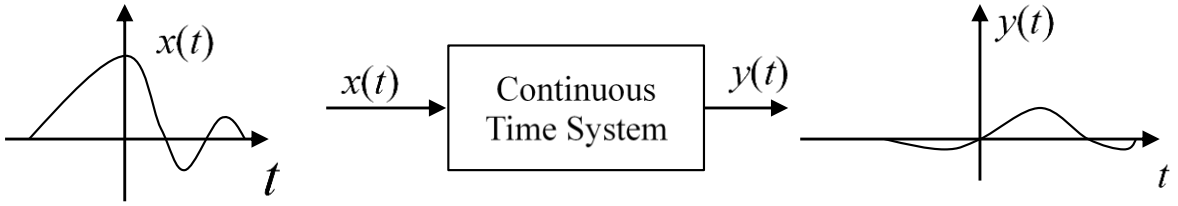
Ans: Substitute $t = nT$ in $x(t)$; [n - sample number & T – sampling period] $x[n] = 10^{-nT} - 5e^{-0.5nT}$

$$x[n-2] = 10^{-(n-2)T} - 5\,e^{-0.5(n-2)T}$$

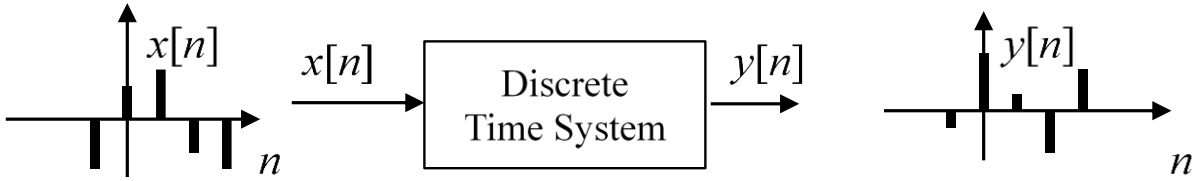
Difference equations

Systems

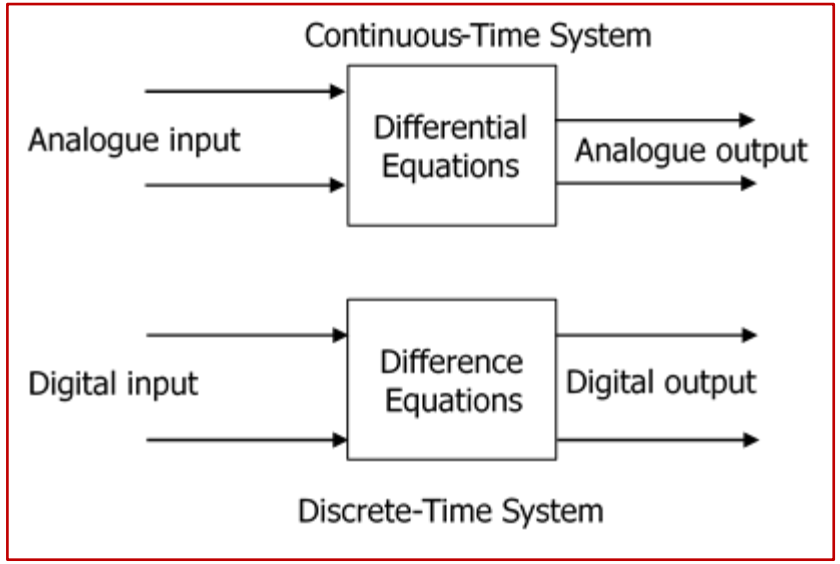
- A continuous-time system is one whose input *x*(t) and output *y*(t) are continuous time functions related by a rule as shown below:



- A discrete system is one whose input *x*[n] and output *y*[n] are discrete time function related by a rule as shown below:



- An important mathematical distinction between continuous-time and discrete-time systems is the fact that the former are characterized by differential equations whereas the latter are characterized by differen**ce** equations.



Difference equations

Example 7:

- The RC circuit (known as a low-pass filter or an integrator) shown below is a continuous-time system.
- If $e(t)$ is the input signal and $v_c(t)$ is the corresponding output signal, then we obtain using simple circuit analysis

$$\frac{dv_C(t)}{dt}+\frac{1}{RC}v_C(t)=\frac{1}{RC}e(t) \qquad (1)$$

Differential equation

- From equation (1), a discrete-time system can be developed. If the sampling period T is sufficiently small, by applying Backward Euler approximation , we obtain

$$\left.\frac{dv_C(t)}{dt}\right|_{t=nT}=\frac{v_C(nT)-v_C(nT-T)}{T} \qquad (2)$$

- By substituting equation (2) into (1) and replacing t by nT, we obtain

$$\frac{v_C(nT)-v_C(nT-T)}{T}+\frac{1}{RC}v_C(nT)=\frac{1}{RC}e(nT)$$

- The difference equation is:

$$\frac{v_C[n]-v_C[n-1]}{T}+\frac{1}{RC}v_C[n]=\frac{1}{RC}e[n]$$

Output

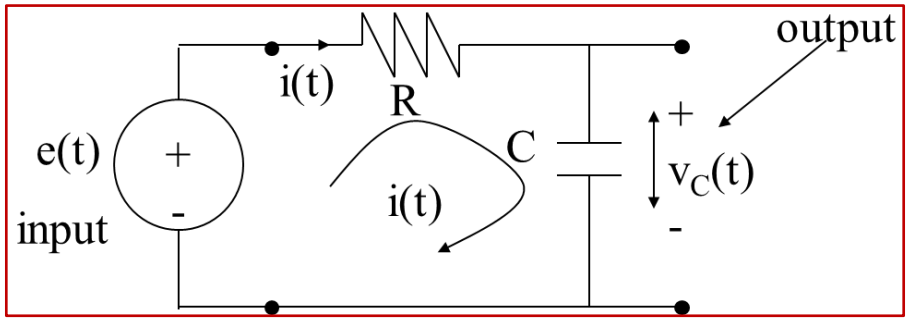
input

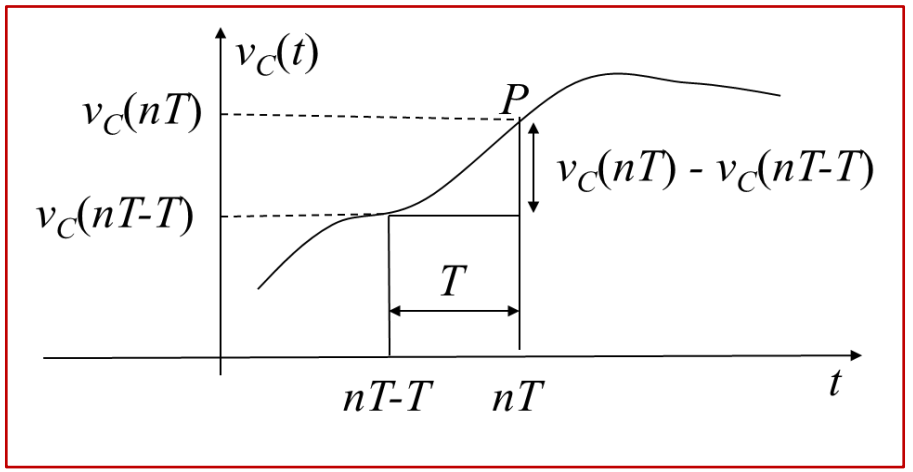
$$v_C[n]=\frac{RC}{RC+T}v_C[n-1]+\frac{T}{RC+T}e[n]$$

Previous output

Difference equation

RC circuit





Backward Euler approximation
[Assuming T is sufficiently small]

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Time and frequency domain analysis

Continuous-time Fourier Transform (FT)

- The Fourier transform for non-periodic continuous-time signals is a mathematical transformation to transform signals between time domain and frequency domain, which has many applications in engineering. Most signals of practical importance are non-periodic. The FT pairs is given by:

  Fourier Transform (FT): $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

  Inverse Fourier Transform (IFT): $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$

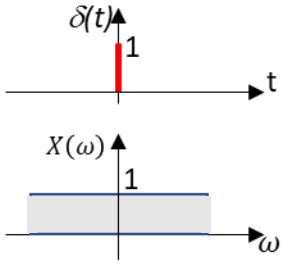
- The FT converts the time domain signal, *x*(t), into its frequency domain representation, $X(\omega)$, and the IFT converts the frequency domain representation, $X(\omega)$, back into the time domain *x*(t).

Example 8:

- Find the Fourier Transform of $x(t) = \delta(t)$.

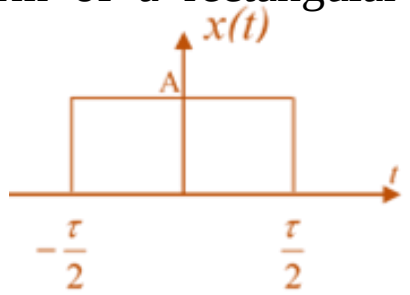
$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}dt = 1$$

- The unit impulse $\delta(t)$ contains all frequencies $(-\infty \ to \ +\infty)$ of unit amplitude.



Example 9:

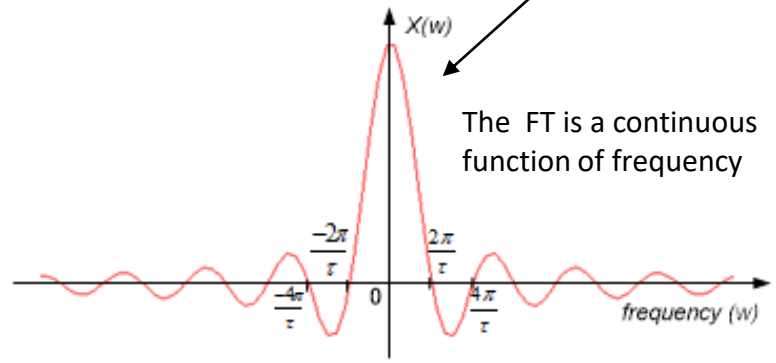
Evaluate the Fourier transform of a rectangular pulse shown below:



$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$= \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} x(t)e^{-j\omega t}\,dt \quad = -\frac{A}{j\omega}\left[e^{-\frac{j\omega\tau}{2}} - e^{\frac{j\omega\tau}{2}}\right]$$

$$X(\omega) = A\tau \frac{sin\frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}$$ ← Sinc function



The FT is a continuous function of frequency

Time and frequency domain analysis

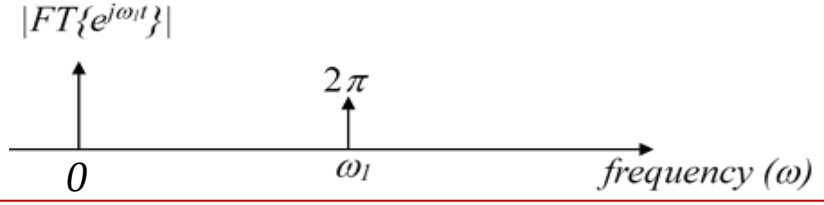
Example 10:

- Find the Fourier transform of $x(t) = e^{j\omega_1 t}$.

$$FT\{e^{j\omega_1 t}\} = \int_{-\infty}^{\infty} e^{j\omega_1 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j(\omega_1 - \omega)t} dt = 2\pi\delta(\omega_1 - \omega)$$

$$FT\{e^{j\omega_1 t}\} = 2\pi\delta(\omega - \omega_1)$$

Hence the magnitude spectrum of $e^{j\omega_1 t}$ is shown below:



Note: Use the following properties:

$\int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$

and

$\delta(\omega - \omega_1) = \delta(\omega_1 - \omega)$

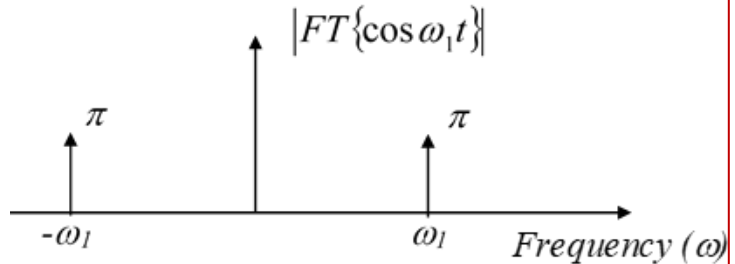
Example 11:

- Find the Fourier transform of $x(t) = \cos(\omega_1 t)$.

$$FT\{\cos\omega_1 t\} = FT\left\{\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\right\} = \frac{1}{2}FT\{e^{j\omega_1 t}\} + \frac{1}{2}FT\{e^{-j\omega_1 t}\}$$

$$= \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$ Use results from Example 10.

The magnitude spectrum of $\cos(\omega_1 t)$ is shown below:



- The spectrum of cosine signal has two impulses at positive and negative frequencies.

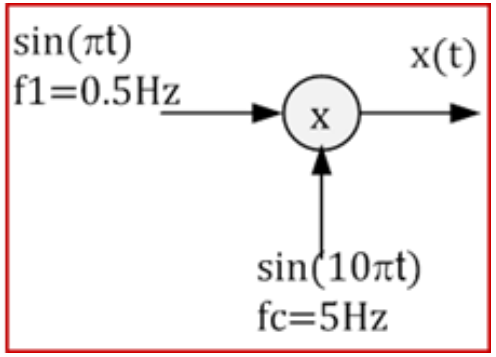
Time and frequency domain analysis

Multiplication of sinusoids

- The product of two real sinusoids (frequencies f_1kHz and f_2kHz) results in the sum and difference frequencies $[(f_1+f_2)$kHz and (f_1-f_2)kHz$]$

- Consider a signal $x(t)$ which is a product of two sinusoids, as given by:

$$x(t) = \sin(\pi t) \text{ x } \sin(10\pi t)$$



$x(t)$ can be re-written as

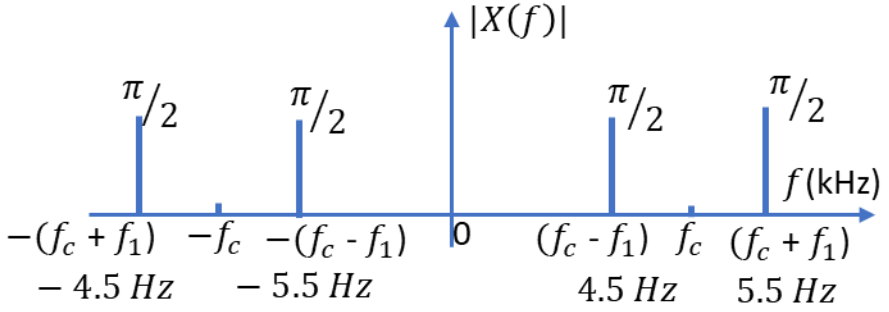
$$x(t) = \left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j}\right)\left(\frac{e^{j10\pi t} - e^{-j10\pi t}}{2j}\right) = \frac{-1}{4}\left(e^{j\pi t} - e^{-j\pi t}\right)\left(e^{j10\pi t} - e^{-j10\pi t}\right)$$

$$x(t) = \frac{-1}{4}[e^{j11\pi t} - e^{-j9\pi t} - e^{j9\pi t} + e^{-j11\pi t}] = \frac{1}{2}\cos(9\pi t) - \frac{1}{2}\cos(11\pi t) = \frac{1}{2}\cos(2\pi 4.5t) - \frac{1}{2}\cos(2\pi 5.5\,t)$$

- Recall example 11: $FT\{\cos\omega_1 t\} = \pi\delta(\omega-\omega_1)+\pi\delta(\omega+\omega_1)$

- From $x(t)$ it can be seen that there are four spectral components at frequencies f_c+f_1= 5.5Hz, f_c-f_1 = 4.5Hz, $-(f_c+f_1)$ = 4.5Hz, $-(f_c-f_1)$ =-5.5Hz, as shown below

Magnitude Spectrum of $x(t)$



ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

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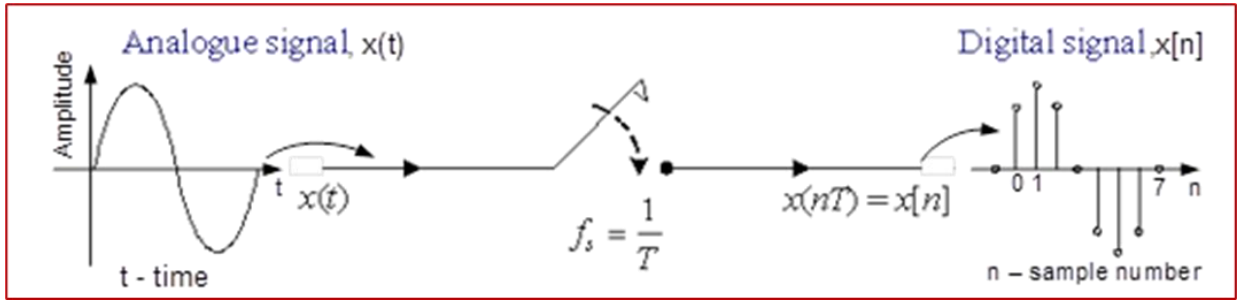
Sampling Theorem & Aliasing

Nyquist –Shannon Sampling theorem

- If the highest frequency contained in an analogue signal $x(t)$ is f_{max} and the signal is sampled at a rate $f_s \geq 2\, f_{max}$ then $x(t)$ can be exactly recovered from its sample values using an interpolation function.
- The maximum frequency (f_{max}) that is generally accepted as upper limit of human hearing and perception of music sounds is 20kHz.
- Audio CDs use a sampling rate $f_s = 44.1kHz$ for storage of the digital audio signal. This sampling frequency is slightly more than $2 \times f_{max}$ [f_{max} = 20kHz] to avoid aliasing.

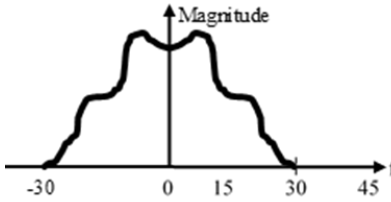
- For example, a sinewave of 50Hz (shown on the right) need to be sampled at higher than 100 Hz (i.e. 100 samples per second) in order NOT to lose any data. That is, to be able to reconstruct the 50Hz sinewave exactly.

Acquiring discrete-time signal, x[n] by sampling continuous-time signal x(t).

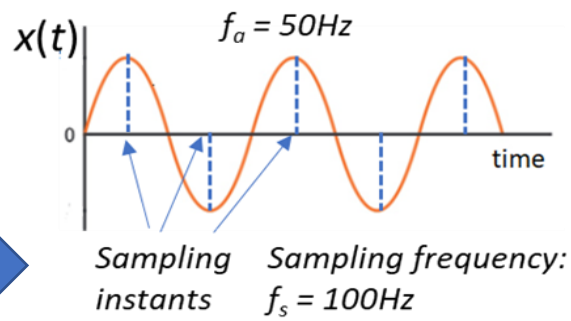


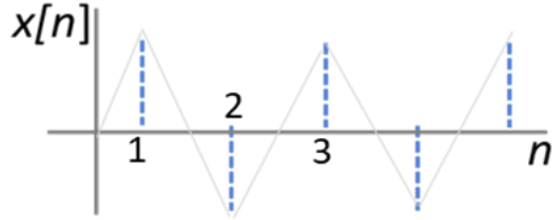
Example 12:

- What is the minimum sampling frequency, f_s, required to avoid aliasing when sampling a signal with the following spectrum?



a) 30 kHz
b) 60 kHz
c) 45 kHz
d) None of the above





Sampling Theorem & Aliasing

Overview of a DSP system

- An analogue signal processing system is shown Figure 5, in which both the input signal and output signal are in analogue form.
- A digital signal processing system in Figure 6 provides an alternative method for processing the analogue signal.

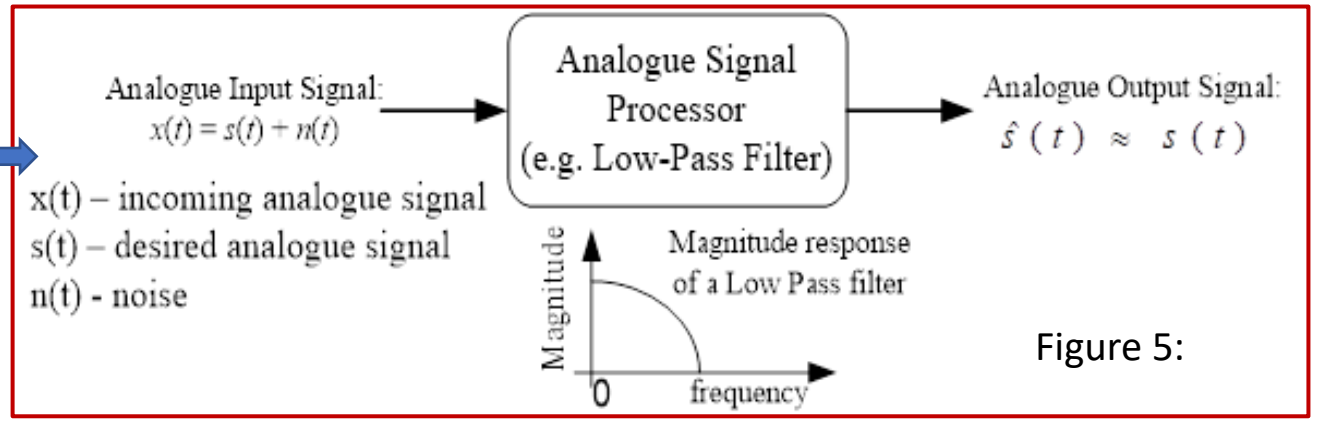


Figure 5:

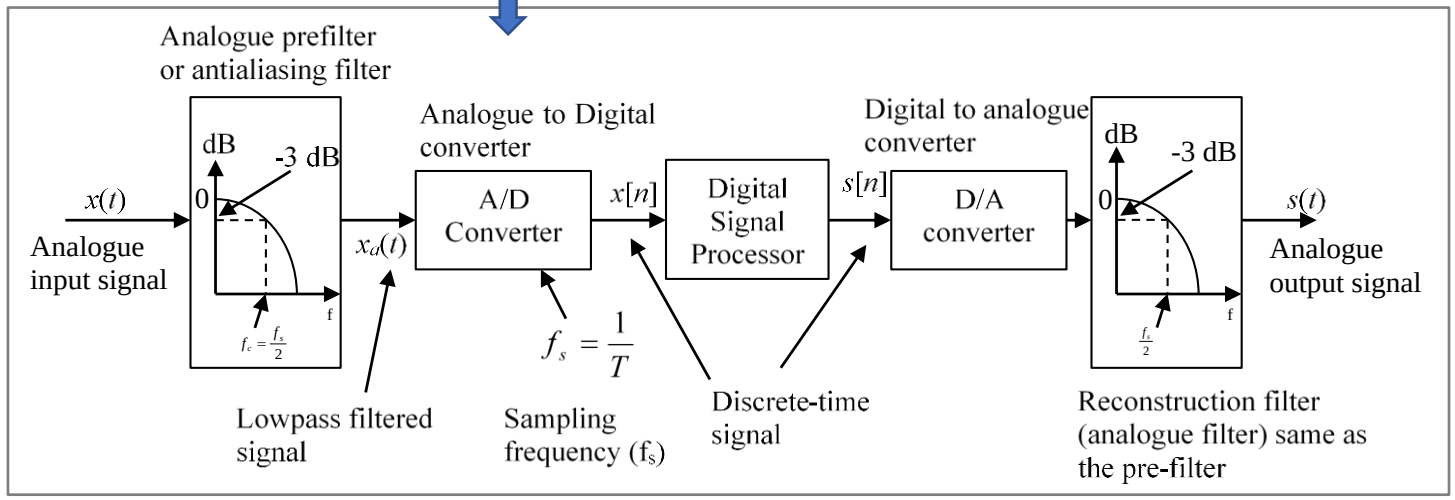


Figure 6: A general process of converting analogue signals into digital signals and back to analogue form.

The **digital signal processor** may implement one of the several DSP algorithms, for example digital filtering (low-pass filter) mapping the digital input signal $x[n]$ into digital output signal $s[n]$.

Name of the Block	Function
Anti-aliasing filter ($f_c = \frac{f_s}{2}$)	To band-limit the analogue input signal prior to digitisation to reduce aliasing
Analogue-to-digital converter (A/D)	To convert analogue input signal into digital output signal by sampling ($f_s = 1/T$)
Digital Signal Processor (heart of the system)	To process the digital signal according to the pre-defined rules.
Digital-to-analogue converter (D/A)	To convert the digital input signal into analogue output signal by interpolating.
Reconstruction filter ($f_c = \frac{f_s}{2}$)	To smooth out the output of D/A converter and to remove unwanted high frequency components.

Table 1: Description of blocks contained in a general DSP process

Sampling Theorem & Aliasing

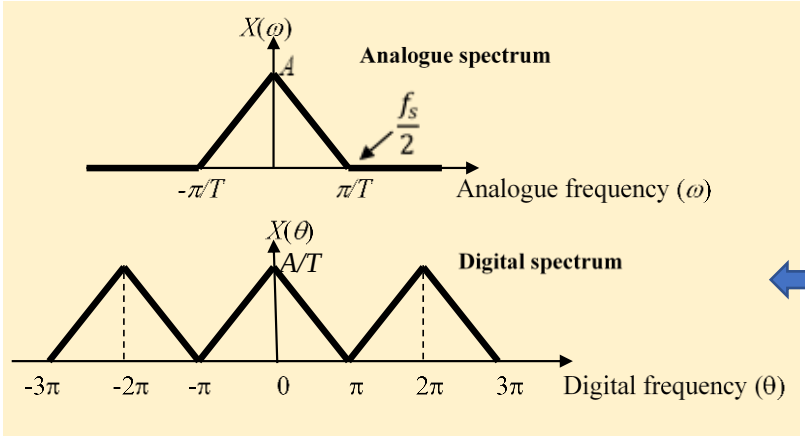
Sampling of Continuous –time signal

- If a continuous-time signal *x*(t) is sampled every *T* seconds, then at the output of the Analogue to Digital Converter, as shown below we obtain a discrete-time signal x[n]



- where t = time; n = sample number; ω = analogue frequency; θ = digital frequency $(-\pi \leq \theta \leq \pi)$; $\theta = \omega T$, $\theta = 2\pi\frac{f_a}{f_s}$, $X(\theta)$ = digital spectrum; $X(\omega)$ = analogue spectrum.

Note:
$\omega = \frac{\pi}{T}$ (or $f = \frac{f_s}{2}$) corresponds to $\theta = \pi$



Discrete-Time Fourier Transform Pair

- The Discrete-Time Fourier Transform (DTFT) pair is given by:

Discrete Time Fourier Transform (DTFT):
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}$$

Inverse Discrete Time Fourier Transform (IDTFT):
$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{jn\theta} d\theta$$

$$where \ -\pi \leq \theta \leq \pi, \quad \theta = \omega T, \quad \theta = 2\pi\frac{f_a}{f_s}$$

- Using Inverse Fourier Transforms in Continuous-time and discrete-time domains, we can show that [see Topic 1: Tutorial question 3 (SOLO – Level 3)].

$$X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T}) \qquad (3)$$

- It is seen that the digital spectrum $X(\theta)$ is periodic with period 2π. The digital spectrum is a repetition of the bandlimited analogue spectrum.

Sampling Theorem & Aliasing

Aliasing

- Aliasing arises when a continuous-time signal is sampled at a rate that is insufficient to capture the changes in the signal. If aliasing occurs, the original continuous time signal cannot be recovered.
- The Nyquist sampling theorem states that the sampling frequency, f_s, should be at least twice the highest frequency, f_c, contained in the signal ($f_s \geq 2f_c$) to avoid aliasing.
- Figure 7 illustrates the relationship between the digital spectrum $X(\theta)$ and the analogue spectrum $X(\omega)$ for the case $X(\omega) = 0$ where $|\omega| \geq \frac{\pi}{T}$ (i.e. $|f| \geq \frac{f_s}{2}$)

Case 1: $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$; (Sampling theorem holds)

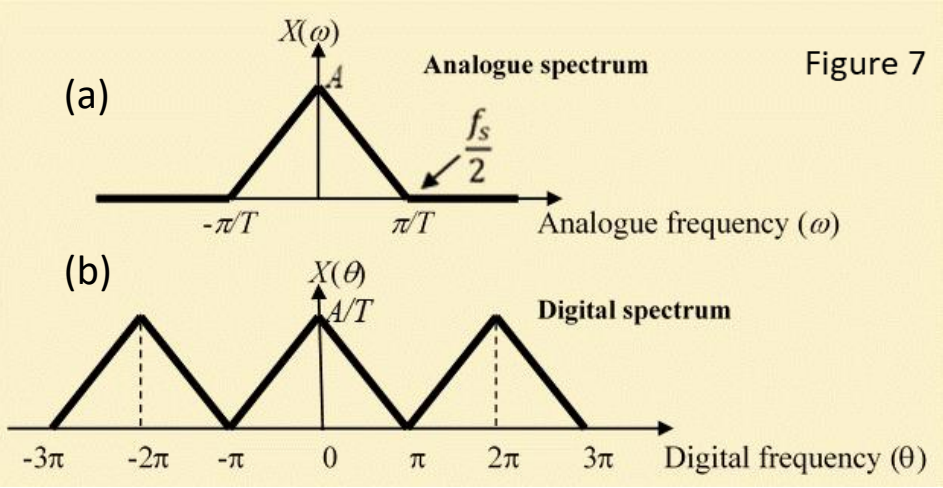


Figure 7

The digital spectrum is the same as the original analogue spectrum and repeats at multiples of the sampling frequency f_s

Case 2: $X(\omega) = 0$, $|\omega| \geq \frac{3\pi}{2T}$; (Sampling theorem does not hold)

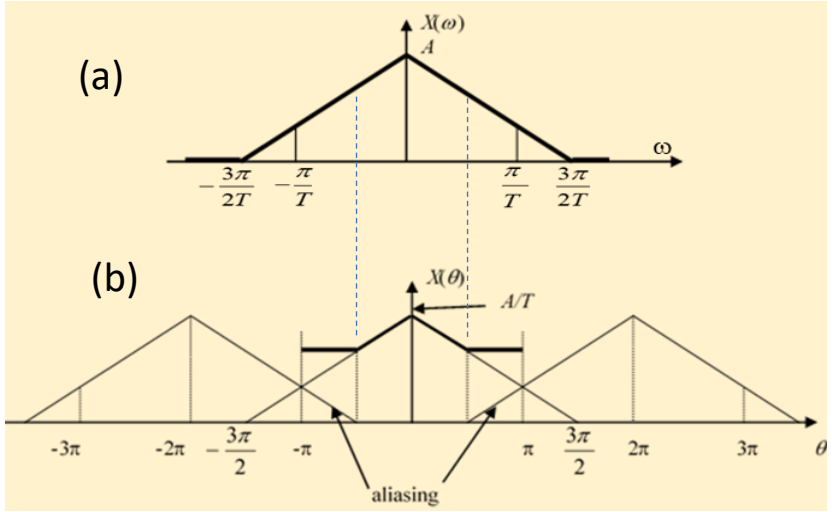


Figure 8: (a) Magnitude response of an analogue signal whose highest frequency component is greater than the sampling frequency. (b) Magnitude response of the sampled analogue signal. The overlapped region represents aliasing.

- If the sampling frequency, f_s is not sufficiently high, the spectrum centered on f_s will fold over or alias into the base band frequencies (Figure 8).
- Aliasing can only be avoided if the analogue signal is band limited such that $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$; [i.e. $|2\pi f| \geq \frac{\pi}{T} \Rightarrow \boldsymbol{f \geq \frac{f_s}{2}}$]

Sampling Theorem & Aliasing

Aliasing

- The digital spectrum consists of original band limited analogue spectrum and repeats at multiples of the sampling frequency f_s (refer to Figure 9) as given by:

$$f_k = f_0 + k f_s \qquad (4)$$

- where k is an integer ($k = \pm1, \pm2, \pm3, \ldots$) and f_0 is the frequency present in the fundamental region of the original analogue spectrum.
- It is clear that the frequency f_k is outside the fundamental frequency range

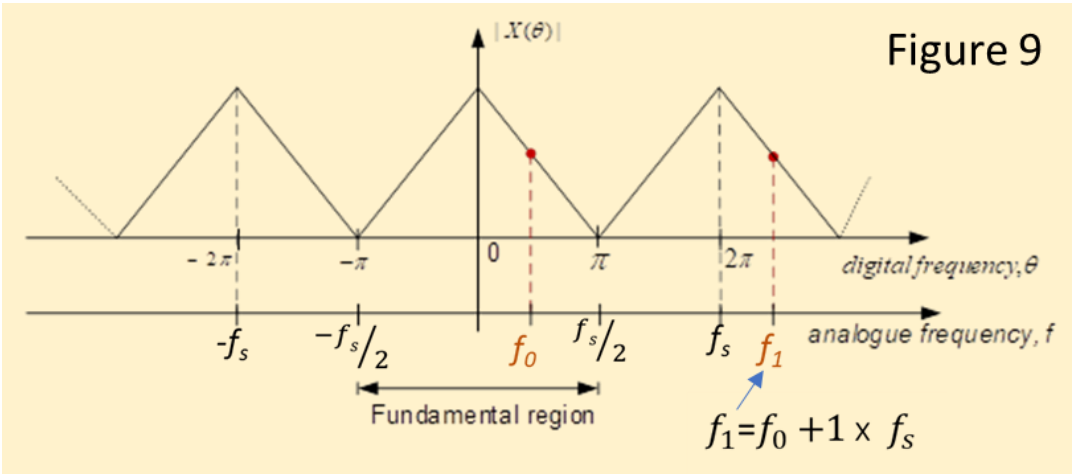
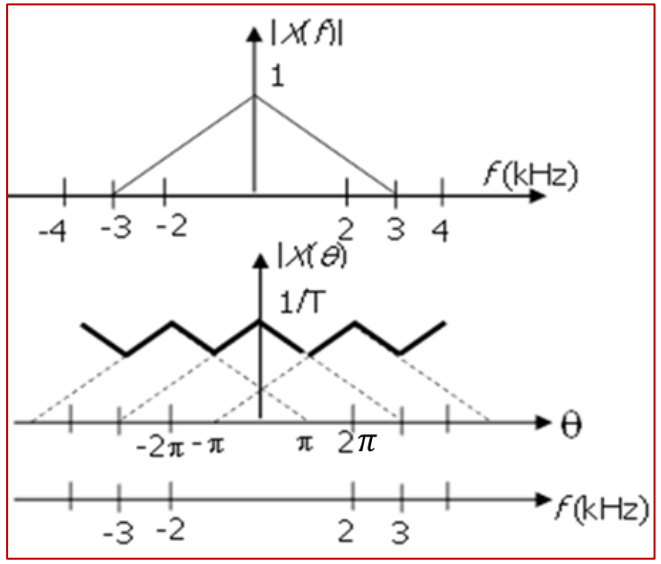


Figure 9

Example 13:

- Suppose $x(t)$ has the magnitude spectrum $|X(f)|$ as shown as shown on the diagram. Sketch the digital spectrum $|X(\theta)|$ if the sampling frequency f_s = 2 kHz.



Example 14:

- Consider the following analogue signal:

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)\ ;$$

What is the Nyquist rate for this signal?

Ans: The frequencies present in the signal above are:

$f_1 = 25$Hz; $f_2 = 150$ Hz; $f_3 = 50$ Hz; Hence, $f_{max} = 150$ Hz

$f_{sampling} > 2 f_{max} = 300$ Hz;

The Nyquist rate is $f_N = 2 f_{max} = 300$ Hz.

Sampling Theorem & Aliasing

Example 15:

- Consider the following analogue signal

$$x(t) = 3\,cos(2000\pi t) + 5\,sin(6000\pi t) + 10\,cos(12000\pi t)$$

(a) What is the Nyquist rate for this signal?

The frequencies existing in the analogue signal are:

f_1 = 1 kHz; f_2 = 3 kHz; f_3 = 6 kHz

Thus f_{max} = 6 kHz and according to the sampling theorem,

$f_s > 2\,f_{max}$ = 12 kHz

The Nyquist rate is = 12 kHz.

(b) Assume now that we sample this signal $x(t)$ using a sampling rate f_s = 5 kHz (samples/sec). What is the discrete-time signal obtained after sampling?

$$f_s = 5kHz \Rightarrow \frac{f_s}{2} = 2.5kHz$$

Recall equation (4): $\boldsymbol{f_k = f_0 + kf_s}$ [f_0 is the frequency present in the fundamental region of the original analogue spectrum] . Using equation (4), we can calculate the new frequency ($\boldsymbol{f'_n}$)caused by aliasing effect as follows:

$$\boldsymbol{f'_n = f_n - kf_s} \qquad -\frac{f_s}{2} \le \boldsymbol{f'_n} \le \frac{f_s}{2}; \qquad (k = \pm1, \pm2, \ldots)$$

$\boldsymbol{f'_n}$ - new frequency due to aliasing effect; $\boldsymbol{f_n}$ – frequency causing the aliasing (which is above the folding frequency)

- The frequency f_1 = 1000 Hz is $< \frac{f_s}{2}$ (= 2500 Hz) and thus it is not affected by aliasing.
- However, the other two frequencies f_2 & f_3 are above the folding frequency and they will be changed by the aliasing effect.

$$f_2' = f_2 - 1\,f_s = 3000 - 5000 = -2 \text{ kHz}$$

$$f_3' = f_3 - 1\,f_s = 6000 - 5000 = 1 \text{ kHz}$$

Therefore,

$$x[n] = 3\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right) + 5\,sin\left(2\pi\left(-\frac{2000}{5000}\right)n\right) + 10\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right)$$

(C) What is the analogue signal $y(t)$ we can reconstruct from the samples if we use ideal interpolation?

Since only frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analogue signal we can recover is,

$$y(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$$

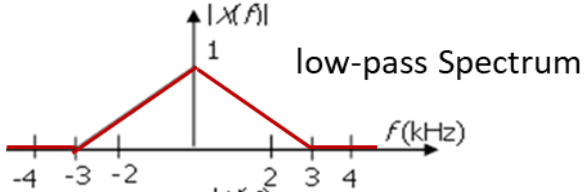
which is obviously different from the original signal x(t).

The distortion of the original analogue signal was caused by the aliasing effect, due to the low sampling rate used.

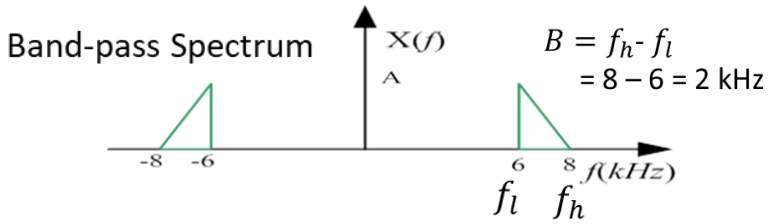
Sampling Theorem & Aliasing

Sampling Bandpass Signal

- The sampling of low-pass signals (see diagram below), is mainly the sampling scheme used in practice.



- We can use a technique known as **bandpass sampling** to sample a continuous bandpass signal that is centered about some frequency other than zero Hz (see diagram below).

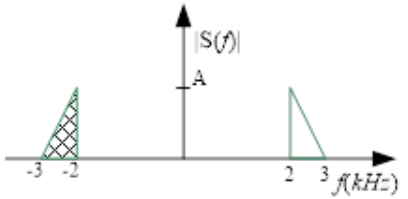


- The minimum sampling frequency (f_s) for a bandpass signal (see above) with a bandwidth (B) of 2 kHz is only about 4kHz ($f_s = 2\,B$), as opposed to twice the highest frequency of 16 kHz.
- $f_s = 2\,B$ holds (no aliasing will occur) only if f_h is an **integer** multiple of the Bandwidth *B*.

Example 16:

- Consider a bandpass signal, *s(t)*, with magnitude spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2\text{kHz} < |f| < 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$



f – frequency in kHz. The signal *s(t)* is sampled uniformly with a sampling frequency of 2kHz. Sketch the digital spectrum $|S(\theta)|$, if it is sampled at a sampling frequency of 2kHz.

Ans: We have $f_s = 2$ kHz, and $f_k = f_o + kf_s$

[where *k* is an integer (*k* = ±1, ±2, ±3, …) and f_0 is the frequency present in the fundamental region of the original analogue spectrum]

The new frequency due to aliasing is: f_o'

$f_o' = f_o - kf_s, \qquad k = 0, \pm1, \pm2, \ldots$

For $k = 1$:

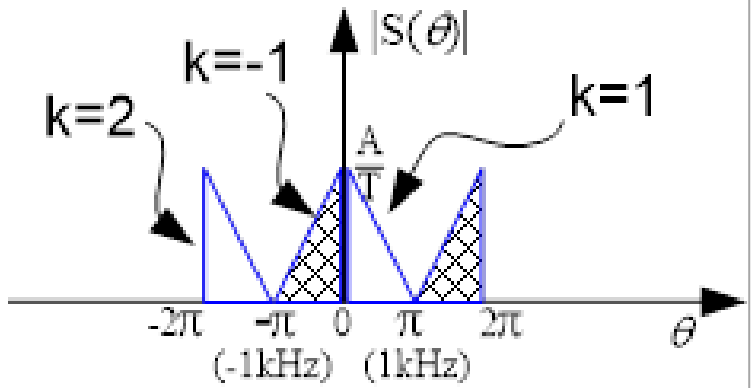
$f_o' = 2 - 1(2) = 0;$

$f_o' = 3 - 1(2) = 1$

For $k = -1$:

$f_o' = -2 + 1(2) = 0;$

$f_o' = -3 + 1(2) = -1$



ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

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Term 3, 2021

A/D Conversion & quantisation errors

Analogue to Digital (A/D) Conversion Process

- Before any DSP algorithm can be performed, the signal must be in a digital form [$x_a(t)$ to x[n] transformation – see Figure 10.].
- The A/D conversion process involves the following steps:
  - The signal (Band-limited) is first sampled, converting the analogue signal into a discrete-time signal
  - The amplitude of each sample is quantised into one of 2^B levels (where B is the number of bits used to represent a sample in the A/D converter)
  - The discrete amplitude levels are represented or encoded into distinct binary words each of length B bits.
- A practical representation of the A/D conversion process is shown in Figure 11.
- Sample and hold (S/H) takes a snapshot of the analogue signal every T sec and then holds that value constant for T secs until the next snapshot is obtained.

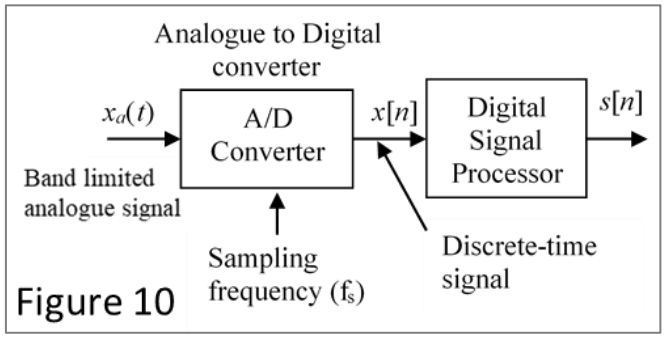


Figure 10

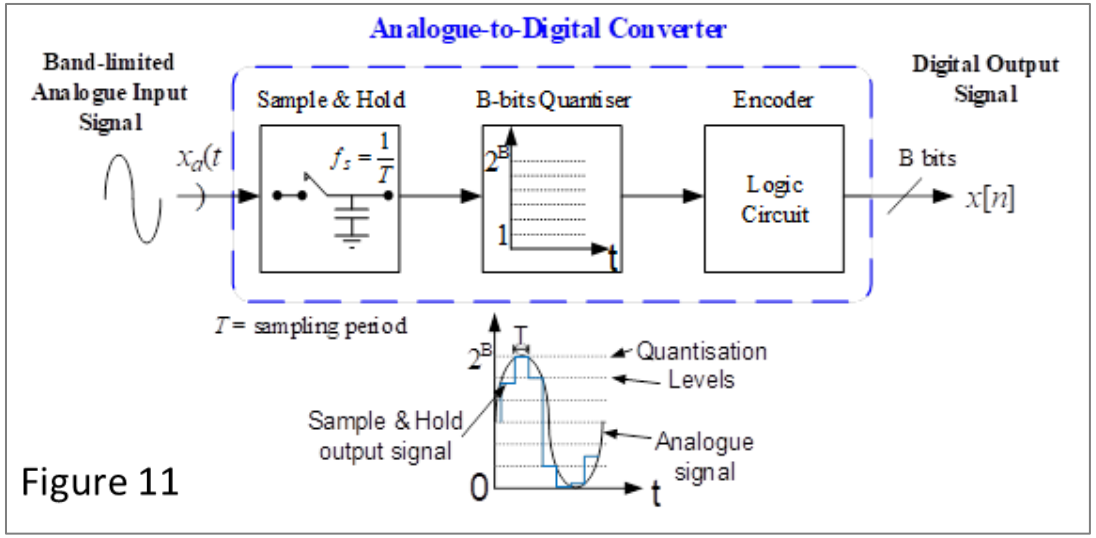


Figure 11

A/D Conversion & quantisation errors

Quantisation and Encoding

- Quantizing/encoding is the process of mapping the sampled analog voltage values to discrete voltage levels, which are then represented by binary numbers (B bits).
- Before conversion to digital, the analogue sample is assigned one of 2^B values (see Figure 12: **unipolar** 3 bit quantiser – 8 discrete levels). This process, termed quantization, introduces an error, which cannot be removed.
- A/D systems are characterized by the number of bits (e.g. 8-bit A/D; 12-bit A/D, 16-bit A/D) they have available to perform quantization. The number of bits determines the number of quantization levels.
- Input-output characteristics of a **bipolar** 4-bit A/D converter, with linear quantisation (two's complement notation) is shown in Figure 13.

An 8-level unipolar A/D converter coding scheme

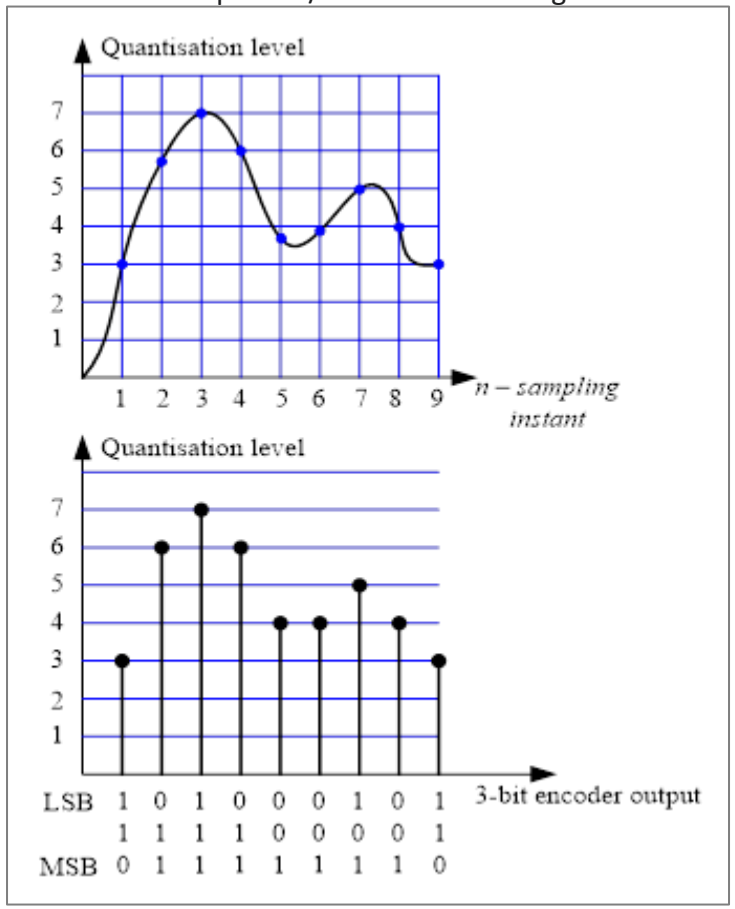


Figure 12: Quantisation of discrete-time signals

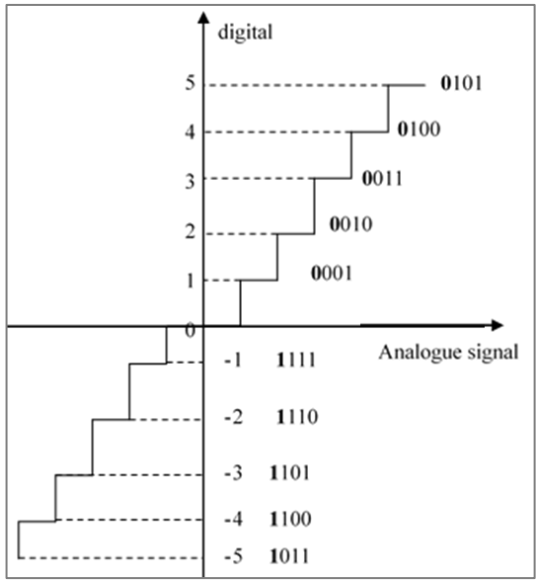


Figure 13: A 4 bit bipolar A/D converter with 2's complement coding scheme

Quantisation errors

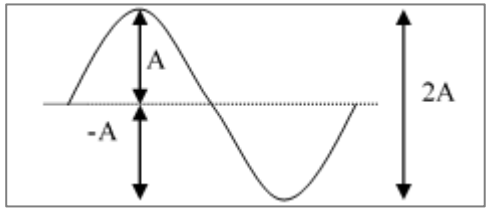
Quantisation step size

- For a B-bit A/D converter, the number of quantisation level is 2^B, and the interval between levels, that is known as the quantisation step size or resolution (ΔV) is given by:

  $\Delta V = \frac{V}{2^B - 1} \approx \frac{V}{2^B}$ [The approximation holds when B is Large (B >10 bits)

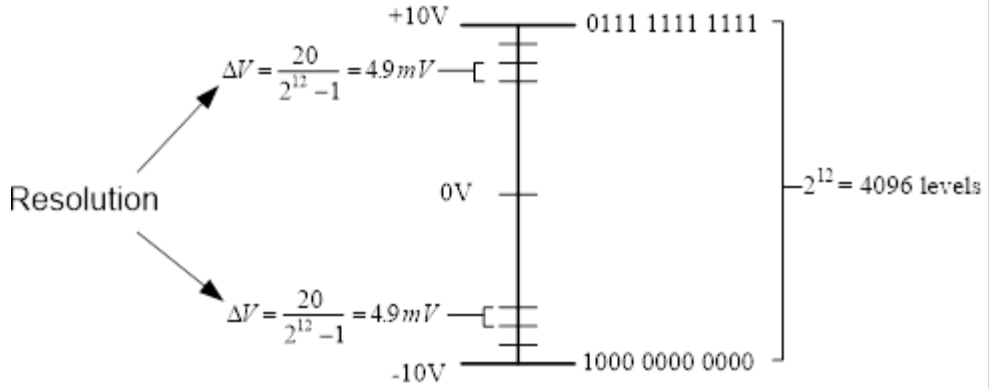
- where ΔV = resolution, B = number of bits and V = peak-to-peak amplitude.

- For a sine wave input of amplitude, A, the quantisation step size becomes: $\Delta V \approx \frac{2A}{2^B}$



Example 17:

- A 12 bit A/D converter (bipolar) with an input voltage range of ±10V will have a least significant bit (LSB) of $\frac{20V}{2^{12}-1} mV = 4.9mV$ (resolution or quantisation step size, ΔV)



Quantisation errors

Quantisation error (e)

- There is always error introduced with the A/D conversion process.
- The quantisation error, e[n], is the difference between the actual amplitude of the input signal and the quantised amplitude: $e[n] = A - A_q$

Actual amplitude — Quantized amplitude

- Figure 14 shows a portion of a signal that has been quantized with N bits. The green plot is the quantization error e[n]. The quantization error manifests as noise (it is also called quantization noise) in the reconstructed analogue signal.

- Quantization error can be reduced by increasing the number of bits N for each sample. This will make the quantization intervals smaller, thus reducing the difference between the input analogue signal and the quantization levels.

- For an A/D converter with bipolar signal inputs, the quantisation error (e[n]) is rounded up or down $\pm\ \Delta V/2$ (see Figure 15). The quantisation error ($e[n]$) for each sample, is normally assumed to be random and uniformly distributed in the interval $\pm\ \Delta V/2$ with zero mean.

- We can't use an infinite number of bits for the A/D conversion, so some quantization noise will be always there. For human brain (e.g. for audio signals) – anything beyond 16 bits for each sample, the associated quantization noise becomes imperceptible.

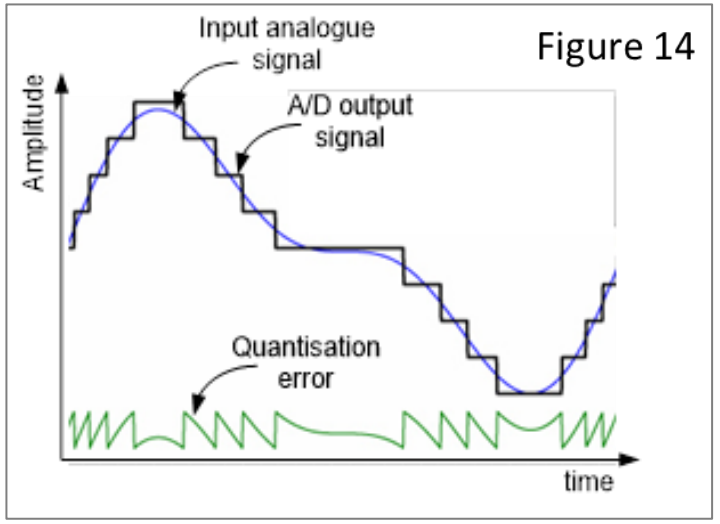


Figure 14

Quantisation (error) noise is random and uniformly distributed with zero mean.

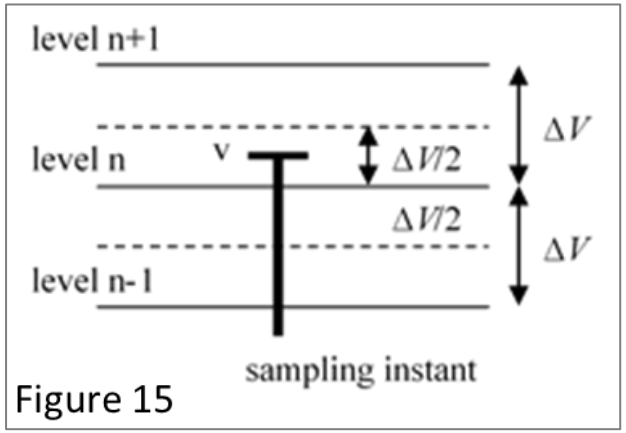


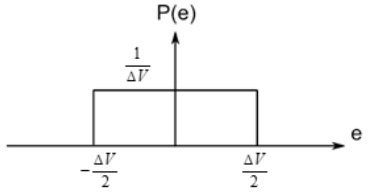
Figure 15

Maximum Quantisation error (e[n]) = $\frac{\Delta V}{2}$

Quantisation errors

Quantisation error (e)

- The probability density function of the error $P(e)$ has the form as shown below:



- The quantisation noise power or variance $\sigma_e{}^2$ is hence given by:

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e)de$$

$$= \frac{1}{\Delta V}\int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de = \frac{1}{\Delta V}\left[\frac{e^3}{3}\right]_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} = \frac{\Delta V^2}{12}$$

Note: Probability of quantisation error is constant:

$P(e)=1/\Delta V$

- For uniform quantisation [all steps (ΔV) are of equal size], the quantisation noise power:

$\sigma_e^2 = \frac{\Delta V^2}{12}$ {**rms** quantisation error $\boldsymbol{\sigma_e}= \Delta/\sqrt{12}$ }

Signal-to-Quantisation ratio (SQNR)

- Signal-to-quantisation noise power ratio (SQNR) is defined as:

$$SQNR = \frac{P_s\ (signal\ power)}{P_n\ (noise\ power)}$$

$$P_s = \frac{1}{N}\sum_{n=1}^{N}\{x[n]\}^2; \quad P_n = \frac{1}{N}\sum_{n=1}^{N}\{e[n]\}^2; \quad P_n = \sigma_e^2 = \frac{\Delta V^2}{12}$$

$$\therefore SQNR(dB) = 10\,log\frac{P_s}{P_n} = 10\,log\frac{\sum_{n=1}^{N} x^2[n]}{\sum_{n=1}^{N} e^2[n]}$$

- The dynamic range, R, of the signal is defined as

$$R = \{x[n]\}_{\max} - \{x[n]\}_{\min}$$

- The quantisation step size of resolution is defined as $\Delta V = \frac{R}{L}$ (L- number of levels in the quantiser).

$$\therefore SQNR(dB) = 10\,log\frac{P_s}{\frac{(R/L)^2}{12}} = 10\,log\frac{12P_sL^2}{R^2}$$

$$SQNR(dB) = 10\,log\,P_s + 20\,log\,L + 10\,log\,12 - 20\,\log R$$

Quantisation errors

SQNR

- For the sine wave input, the average signal power is $A^2/2$, i.e. $(A/\sqrt{2})^2$ rms value. The signal-to-quantisation noise power ratio (SQNR) in decibels is:

$$SQNR = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{\Delta V^2}{12}}\right) = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{(2A/2^B)^2}{12}}\right) = 10\,log\left(\frac{3\times 2^{2B}}{2}\right)$$

$$\boldsymbol{SQNR = 6.02B + 1.76 \quad dB}$$

- The SQNR increases with the number of bits, *B*. In many DSP applications, an A/D converter resolution between 12 and 16 bits is adequate.
- It can be seen from the table below that the signal-to-quantisation noise ratio increases approximately 6dB for each bit.

Number of bits	Number of Levels	SQNR
3	8	19.7 dB
4	16	25.3 dB
5	32	31.6 dB
6	64	37.7 dB
7	128	43.8 dB

Example 18:

- A sampled signal that varies between -2V and 2V is quantised using B bits. What value of B will ensure that the quantisation noise power is less than 25×10^{-6}?

$$\text{Resolution } (\Delta V) = \frac{R \text{ (Dynamic range)}}{L \text{ (No of levels in the quantiser)}}$$

$\sigma_e^2 = \frac{\Delta V^2}{12} = 25 \times 10^{-6}$; $\therefore\ \Delta V^2 = 12 \times 25 \times 10^{-6}$

R = 2 – (-2) =4; $\therefore \Delta V^2 = \left(\frac{R}{L}\right)^2 = \frac{16}{L^2}$

$\frac{16}{L^2} = 12 \times 25 \times 10^{-6}$; L = 230.94

$L = 2^B - 1$; $\therefore 2^B = 231.94$

B = 7.86; Number of bits = 8;

Digital to Analogue Conversion (D/A) – Signal recovery

D/A converter

- The D/A conversion process is employed to convert the digital signal into analogue form, after it has been digitally processed.
- The reason for this conversion may be, for example, to generate an audio signal to drive a loudspeaker, or to sound an alarm. The D/A process is shown in Figure 16.
- A register is used to buffer the D/A's input to ensure that its output remains the same until the D/A is fed with the next digital input.
- The inputs to the D/A are a series of impulses, while the output of the DAC has a staircase shape as each impulse is held for a duration of *T* sec.
- The D/A shown in Figure 16 is referred to as a zero-order hold (i.e. holding each sample value for one sample interval)
- By comparing its output and its input y[n], it is evident that for each digital code fed into the D/A, its output is held for a duration of *T*. The result is the characteristic staircase shape at the D/A output.
- The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse.

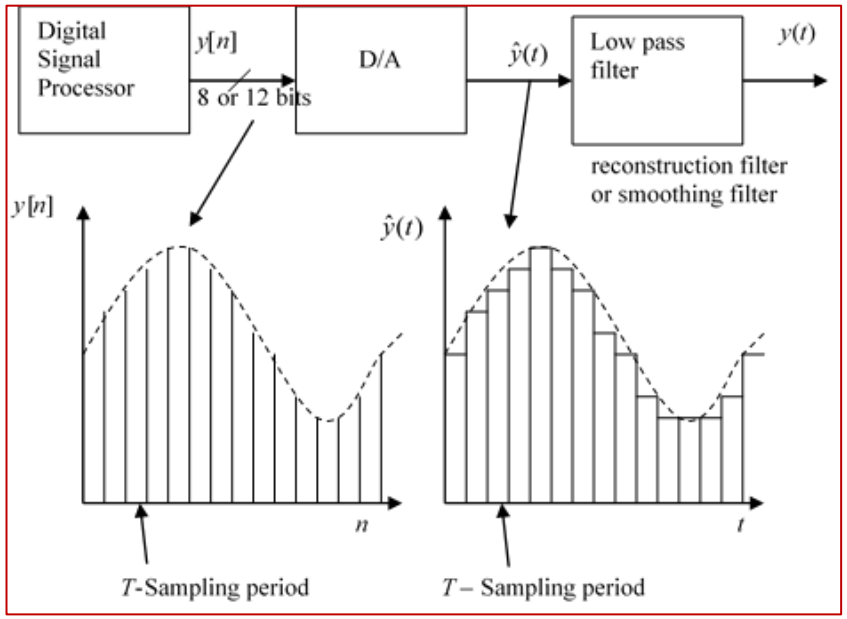


Figure 16: Conversion process from digital signals to analogue signals.

Reconstruction filter

The output of the D/A converter contains unwanted high frequency at multiples of the sampling frequency as well as the desired frequency components. The role of the output filter is to smooth out the steps in the D/A output thereby removing the unwanted high frequency components. In general, the requirements of the smoothing filter are similar to those of the anti-aliasing filter.

ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

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Term 3, 2021

Topic 1: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. Examine what will happen to the frequency f_1 = 5 kHz? (Tick the correct answer)
   a) 5kHz frequency will remain unchanged
   b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

2. Which of the following is a not periodic signal ? (Tick the correct answer)
   a) x[n] = sin[πn/9]
   b) x[n] = sin[nπ]
   c) x[n] = cos[πn^2/15]
   d) x[n] = sin[πn/5 + π] + cos[πn/10 – π]

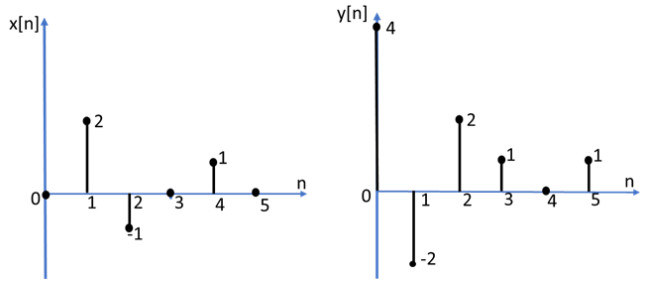
3. Given a real-valued bandpass signal x_a(t) with a bandpass spectrum of X_a(f) nonzero only for $9\ kHz\ < |f| < 12kHz$, determine the minimum sampling frequency (f_s) for which no aliasing occurs? (Tick the correct answer)
   a) f_S = 18 kHz
   b) f_S = 6 kHz
   c) f_S = 24 kHz
   d) None of the above

4. A discrete-time signal x(n) = 6.35 cos(nπ/10) is quantised with a resolution Δv=0.2 $volts$. How many bits are required in the Analogue to Digital Converter? (Tick the correct answer)
   a) 6 bits
   b) 8 bits
   c) 12 bits

SOLO – Level 1

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine its spectral frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) $\pm$1.5kHz and $\pm$0.5kHz
   b) $\pm$3kHz and $\pm$1kHz
   c) $\pm$2kHz
   d) $\pm$2kHz and $\pm$1kHz

6. Given the discrete-time sequences x[n] and y[n] shown, which difference formula best describes their relationship ? (Tick the correct answer)



   a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)

   (a) 1 sample (b) 5 samples
   (c) π samples (d) Non of the above

Topic 1: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. A sinusoidal signal with peak-to-peak amplitude of 5V is sampled at 50kHz with uniform quantisation. Find the minimum number of bits for the analogue to digital converter to achieve a SQNR of at least 92dB. State any assumptions made.
2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

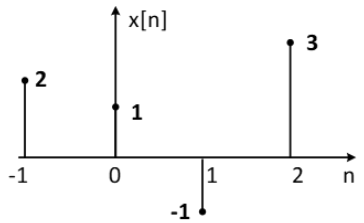
$$SQNR = 6.02B + 4.77 - 20\log\left({}^{A}/_{\sigma_{sig}}\right)$$

where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if *B* is 16 bits and the input is a signal with a rms value of A/5.

3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s=4kHz. Determine the resulting analogue and digital magnitude spectra.
4. An analogue signal $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per sec:
   (a) Determine the Nyquist sampling rate for x(t).
   (b) Determine the folding frequency (or half the sampling frequency).
   (c) What are the frequencies, in radians, in the resulting discrete time signal x[n]?
   (d) If x[n] is passed through an ideal D/A converter what is the reconstructed signal y(t)?
5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of $\pm$5 Volts is used to convert sampled voltages to integer codes.
   (a) Calculate the quantization interval of the A/D converter.
   (b) Determine the maximum quantization error.

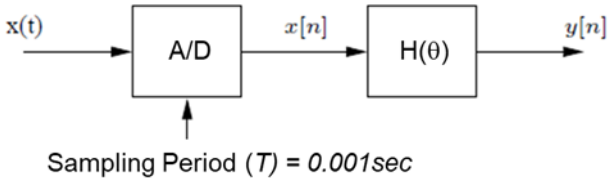
SOLO – Level 2

6. Consider the discrete-time sequence x[n] = cos[(nπ/8) + (π/4)] .
   (i) Determine the fundamental period of x[n].
   (ii) If the sampling frequency f_s = 10 kHz, write the equation of the analogue signal x(t).
7. Sketch the following:
   (c) x[n] = δ[n+2] + 2 δ[n-1] - 0.5 δ[n-3]
   (d) x[n] = u[-n-2]
   (e) x[n] = 0.5^n u[n-2]
8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses



9. For the system shown below , the input x(t) = cos(400πt) + cos(600πt). Find the output y[n] where

$$H(\theta) = \begin{cases} 1 & |\theta \le 0.5\pi| \\ 0 & otherwise \end{cases}$$



Topic 1: Tutorial questions (SOLO - Levels 3 & 4)

SOLO – Level 3

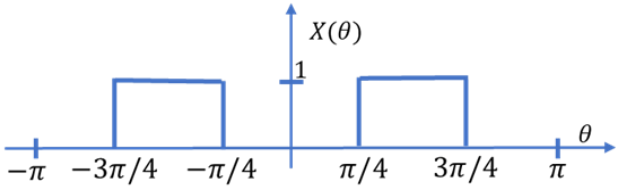
1. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ± 5 Volts is used to convert sampled voltages to integer codes. At what time (t>0) does the quantization error first reach its maximum value?
2. A system samples a signal x(t) = $\cos(\omega_o t)\cos(3\omega_o t)$ at f_s = 7 kHz and reconstructs the signal using an ideal LPF with a cutoff frequency of 3.5 kHz. The value of ω_o is $2\pi \times 10^3$ rad.
   (a) What is the highest frequency component present in the signal?
   Hint : $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$
   (b) Sketch $|X(\omega)|$ and $|X(\theta)|$ – the magnitude spectrum of the signal before and after sampling for (a) above.
   (c) Can aliasing be prevented if the sampling frequency is increased to 8kHz? Explain.
3. If an analogue signal *x*(*t*) is sampled every *T* seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below. Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$



SOLO – Level 3

4. Find the Inverse DTFT of $X(\theta)$ for the figure shown below and show that $x[n] = \frac{2\sin(n\pi/4)\cos(n\pi/2)}{n\pi}$



$$X(\theta) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\theta}$$

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}d\theta$$

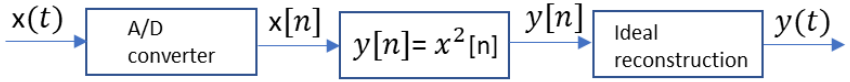
DTFT pair

θ = digital frequency; $-\pi \le \theta \le \pi$

5. Find the Fourier Transform of x(t) = δ(t-a). Hence, show that Hence, show that $\int_{-\infty}^{\infty} e^{j\omega t}dt = 2\pi\delta(\omega)$.

SOLO – Level 4

1. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s, and the sampled signal $x[n]$, is processed by squaring operations $y[n] = x^2$[n], whose output is ideally reconstructed to obtain y(t) as follows.



   What minimum sampling rate f_s, will ensure $y(t) = x^2(t)$?
2. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12 bit quantiser. What is the rms quantisation error and the SQNR?

Topic 1: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

1. Given a real-valued bandpass signal x_a(t) and $X_a(f) = 0$ for $|f| < f_1$ and $|f| > f_2$, the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2f_2$. However , in some cases , the signal may be sampled at a lower rate.

   (a) Suppose that f_1 = 8 kHz and f_2 = 10 kHz. Draw a sketch of $X_a(\theta)$, if the sampling frequency f_s = 4 kHz.

   (b) If the bandwidth of the bandpass signal is defined as B = f_2 - f_1 and the centre frequency to be $f_c = \frac{(f_2+f_1)}{2}$, Show that if $f_c > {}^{B}/_{2}$ and f_2 is an integer multiple of the bandwidth B, no aliasing will occur if x_a(t) is sampled at a sampling frequency $f_s = 2B$.

   (c) Repeat part (b) for the case in which f_2 is not an integer multiple of the bandwidth B.

2. The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse as shown in Figure A.

   (a) Consider one pulse, $\hat{y}\ (t)$, with a height k and show that the corresponding frequency response $\hat{Y}\ (\omega)$ is a *sinc* function given by the following equation:

$$\hat{Y}(\omega) = k\,\frac{T}{2}\,e^{-\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$

   (b) Explore the effect of $\hat{Y}(\omega)$ on the output signal spectrum $Y(\omega)$ and illustrate your understanding of this using a spectral plot. If compensation is required for the $sinx/x$, roll-off inherent in the D/A process, what function would you use to compensate? Briefly explain why.

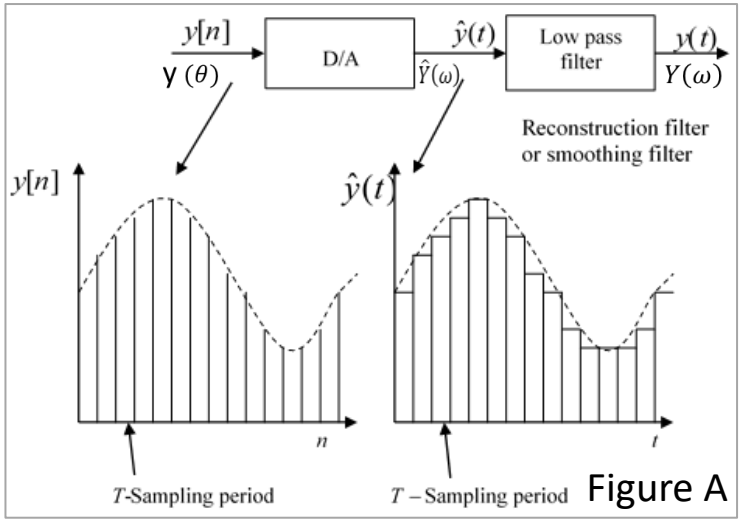


Figure A

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
- Tutorial questions (pp. 43-47)

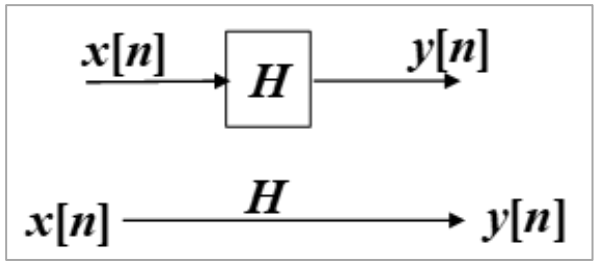
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Term 3, 2021

Discrete-time systems:

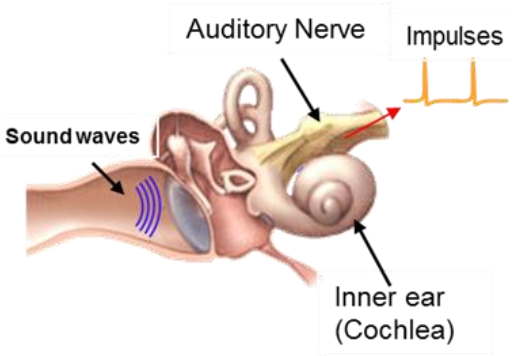
Introduction

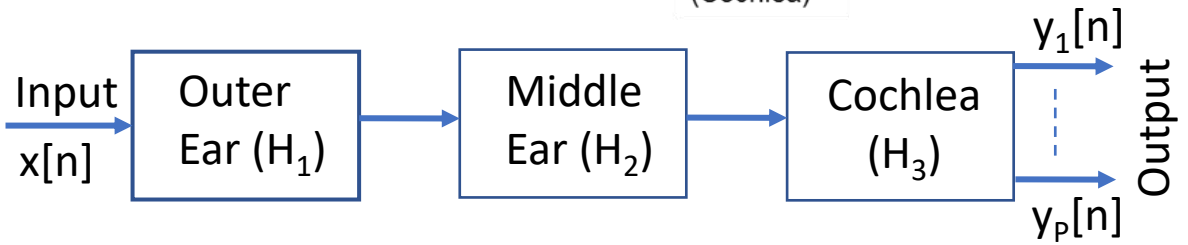
- A discrete-time system, *H*, is a device (or an algorithm) that operates on a discrete-time signal called the input *x*[*n*] (or excitation) according to some well defined rule, to produce another discrete-time signal, *y*[*n*], called the output (or response) – see diagram below



- The input signal *x*[*n*] is transformed by the system *H* to produce a signal *y*[*n*], is expressed by:
$y[n] = H\{x[n]\}$

Peripheral auditory system





For example, it is possible to implement a discrete time model of the peripheral auditory system comprising of a model of the outer ear (H_1) and the middle ear (H_2) and the model of the cochlea (H_3).

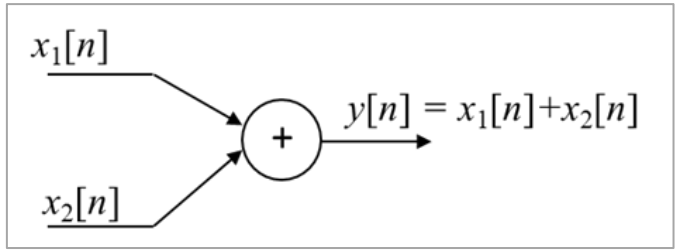
Discrete-time systems:

Block Diagram Representation

- In order to introduce a block diagram representation of discrete-time systems, we need to define some basic blocks that can be interconnected to form complex systems.

An adder

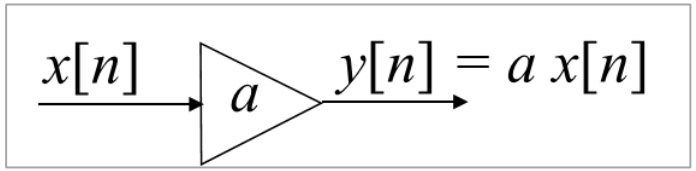
- A system that performs the addition of two signal sequences (x_1[n] & x_2[n]) to form another sequence $y[n]$.



- Note that it is not necessary to store either one of the sequences in order to perform the addition. In other words, the addition operation is **memoryless**.

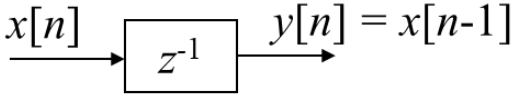
A constant multiplier

- This operation simply represents applying a scale factor 'a' on the input $x[n]$. Note that this operation is also **memoryless**.



A unit delay element

- The unit delay is a special system that simply delays the signal passing through it by one sample (or sample period).
- If the input signal is $x[n]$, the output is $x[n-1]$. In fact, the sample $x[n-1]$ is stored in memory at time $n-1$ and it is recalled from memory at time n to form $y[n] = x[n-1]$.
- Thus this basic building block **requires memory**. We use the symbol T or z^{-1} to denote the unit of delay.



n	-1	0	1	2	3
x[n]	x[-1]=0	x[0]=1	x[1]=0	x[2]=5	x[3]=7
y[n]	Y[-1]=?	Y[0]=0	Y[1]=1	Y[2]=0	Y[3]=5

Discrete-time systems:

Difference Equations

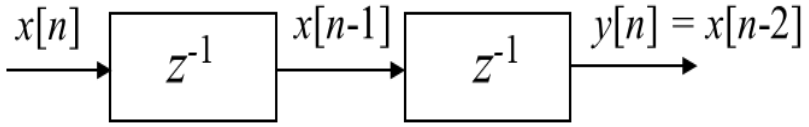
- A discrete-time system consisting of combinations of adders, multipliers and unit delays can always be described by a set of **difference equations**.
- The equations would be ordinary algebraic equations if no delays were present.
- For example, the system described by:

$$y[n] = a_0 x[n] + b_1 y[n-1]$$

produces the present output $y[n]$ as the weighted sum of the previous output $y[n-1]$ and the present input $x[n]$. The quantities a_0 and b_1 are called *coefficients*.

Example 1:

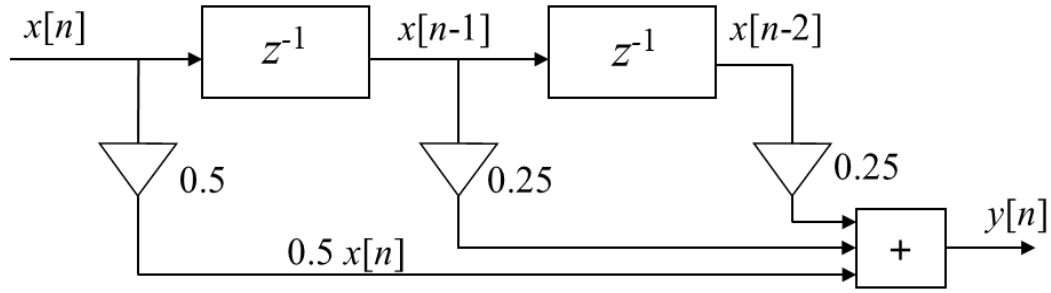
- Draw a system implementation for the difference equation: $y[n] = x[n-2]$



Example 2:

- Draw a system implementation for the difference equation:

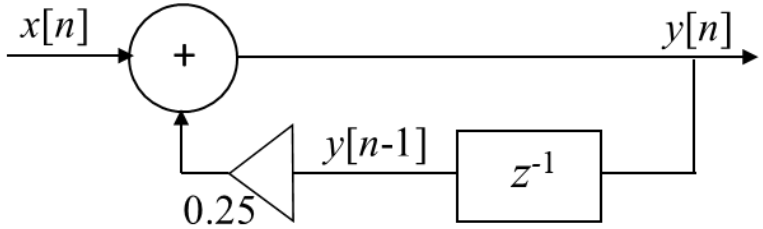
$$y[n] = \frac{1}{2}x[n] + \frac{1}{4}x[n-1] + \frac{1}{4}x[n-2]$$



Example 3:

- Draw a system implementation for the difference equation:

$$y[n] = x[n] + 0.25y[n-1]$$



Discrete-time systems

Example 4:

- Draw a system implementation for the difference equation: $y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$

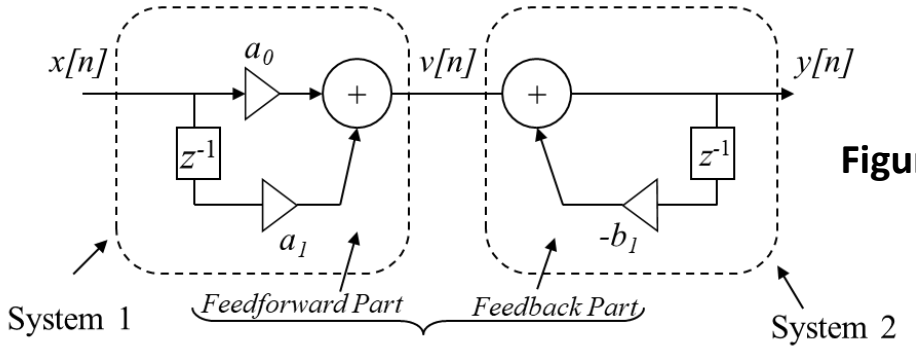


Figure 1

- Without changing the input-output relationship, we can reverse the ordering of the two systems in the cascade representation.

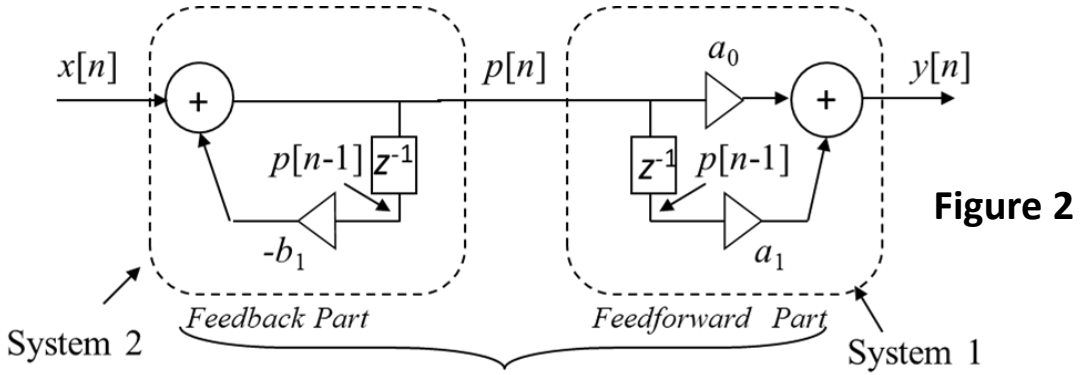
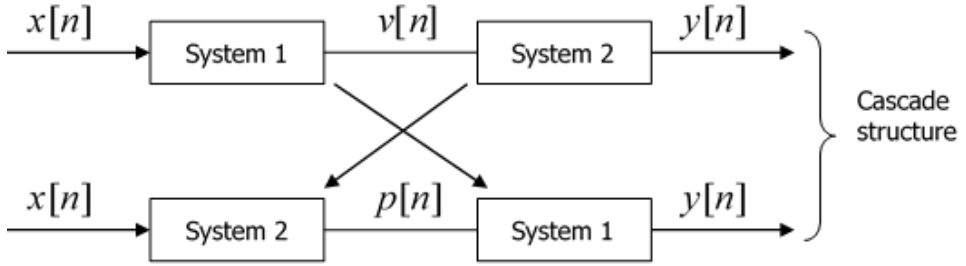


Figure 2

Note: Two systems forming a cascade structure can be interchanged without affecting the final output.



- There is no need for two delay operations; they can be combined into a single delay as shown below:

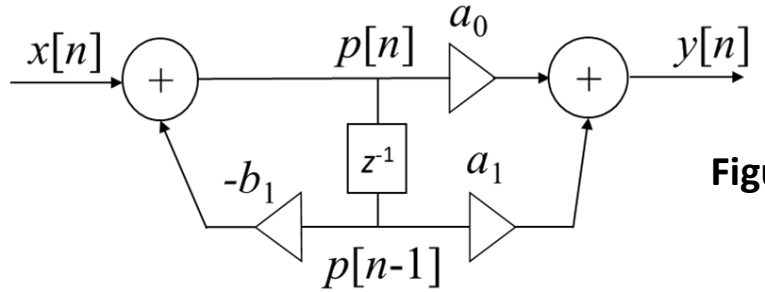


Figure 3

- It can be proven that Figures 1, 2 & 3 all represent the same difference equation:

$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$$

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
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Classification of discrete-time systems

Classification of Systems

- In the analysis as well as in the design of systems, it is desirable to classify the systems according to the general properties they satisfy. General Categories are:
  1. Static systems
  2. Time - invariant systems
  3. Linear systems
  4. Causal systems
  5. Stable systems

1. Static systems

- A discrete-time system is called static or memoryless if its output at any instant '*n*' depends at most on the input sample at the same time, but not on past or future samples of the input.

  For example, $y[n] = ax[n]$; $y[n] = nx[n] + bx^3[n]$ are both static or memoryless
- On the other hand, the systems described by the following input-output relations are dynamic systems or system with memory:

  $y[n] = x[n] + 3x[n-1]$; $y[n] = \sum_{k=0}^{N} x[n-k]$

2. Time-invariant systems

- A time-invariant system is defined as follows:

  $x[n-n_0] \xrightarrow{H} y[n-n_0]$ where y[n] = H{x[n]}.
- Specifically, a system is time invariant if a time shift in the input signal results in an identical time shift in the output signal.

3. Linear systems

- An input $x_1[n]$ to a system (H) results in an output $y_1[n]$; Given a different input $x_2[n]$, the system output is y_2[n]. $x_1[n] \xrightarrow{H} y_1[n]$; $x_2[n] \xrightarrow{H} y_2[n]$

  For a system to be **linear**, when its input is sum $x_1(n)$ + $x_2(n)$, then the output must be the sum of the individual outputs $x_1[n] + x_2[n] \xrightarrow{H} y_1[n] + y_2[n]$
- A linear system is defined as follows:

  $a_1x_1[n] + a_2x_2[n] \xrightarrow{H} a_1y_1[n] + a_2y_2[n]$

  where a_1 and a_2 are arbitrary constants.

Classification of discrete-time systems

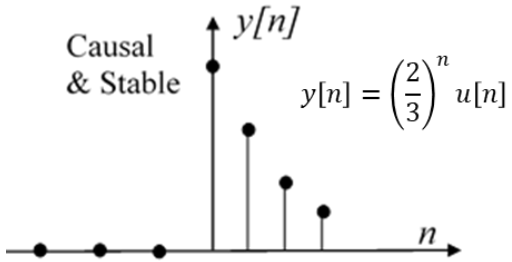
4. Causal systems

- A system is said to be causal if the output of the system at any time '*n*' depends only on present and past inputs, but does not depend on future inputs.
- If a system does not satisfy this definition, it is called *noncausal*. Such a system has an output that depends not only on present and past inputs but also on future inputs.

Examples:

$$y[n] = x[n] - x[n-1] \quad \rightarrow \text{Causal}$$
$$y[n] = ax[n] \quad \rightarrow \text{Causal}$$
$$y[n] = x[n] + 3x[n+4] \quad \rightarrow \text{Noncausal}$$
$$y[n] = x[-n] \quad \rightarrow \text{Noncausal}$$

- A discrete - time sequence is called **causal** if it has zero values for $n < 0$.



5. Stable systems

- A discrete signal *x*[*n*] is bounded if there exists a finite *M* such that $|x[n]| < M$ for all *n*.
- A discrete-time system in Bounded Input-Bounded Output (**BIBO**) stable if every bounded input sequence *x*[*n*] produced a bounded output sequence.

$$\text{If } x[n]_{max} \leq A, \text{ then } y[n]_{max} \leq B$$

Example: The following discrete-time system is at rest {i.e. *y*[-1]=0}. Check if the system is BIBO stable.

$$y[n] = ny[n-1] + x[n], \quad n > 0$$

Model answer:

If *x*[*n*]=*u*[*n*], then $|x[n]| \leq 1$. But for this bounded input, the output is unbounded (see below).

$$n = 0 \rightarrow y[0] = x[0] = 1$$
$$n = 1 \rightarrow y[1] = 1y[0] + x[1] = 2$$
$$n = 2 \rightarrow y[2] = 2y[1] + x[2] = 5$$
$$\vdots \qquad\qquad \vdots$$

y[0]=1 → y[1]=2 → y[2]=5 → increasing; Hence the system is unstable.

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Topic 2: Discrete-Time Systems (pp. 2-5)

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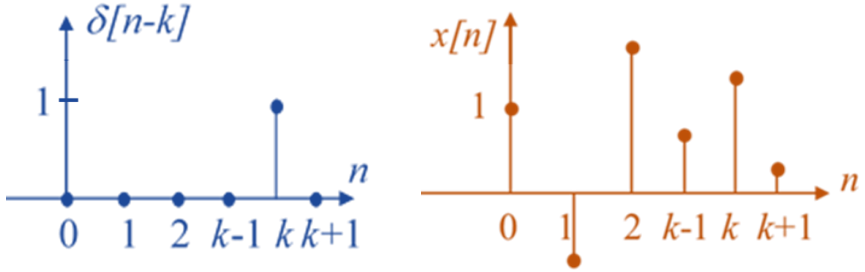
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Transformation of Discrete Time Signals

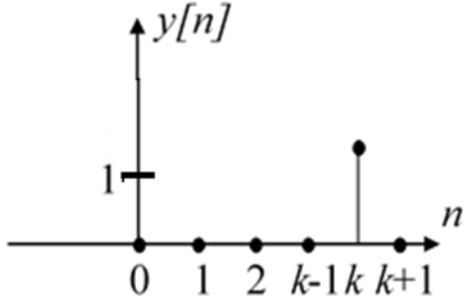
- A discrete-time signal, $x[n]$ may be shifted in time (delayed or advanced) by replacing the variables n with $n-k$ where $k > 0$ is an integer

  x[$n-k$] => $x[n]$ is delayed by k samples
  $x[n+k]$ => $x[n]$ is advanced by k samples

- For example, consider a shifted version of the unit impulse function and an arbitrary signal x[n] both shown below .

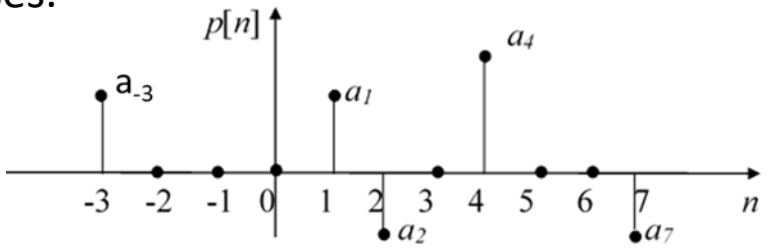


- If we multiply x[n] by $\delta[n-k]$, we obtain a signal that is zero everywhere, except at n = k.



$\therefore$ y[n] = x[n] · δ[n-k]
= x[k] · δ[n-k]

- An arbitrary sequence can then be expressed as a sum of scaled and delayed unit impulses.



$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_4\delta[n-4] + a_7\delta[n-7]$$

- More generally, the discrete-time sequence can be expressed as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

- For a real-time signal with a finite number of samples N:

$$x[n] = \sum_{k=0}^{N-1} x[k]\,\delta[n-k]$$

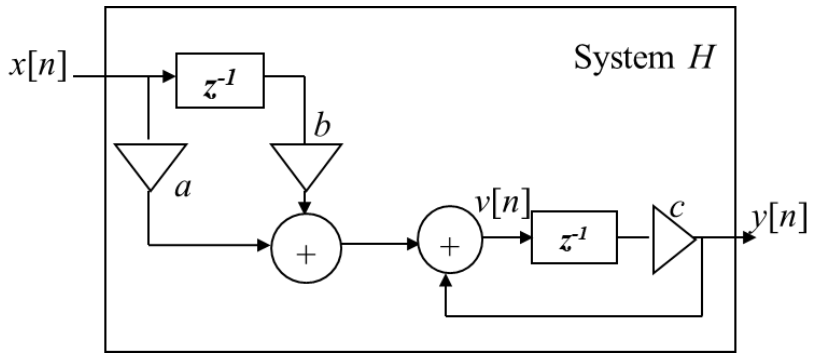
This equation is a special form of **convolution.**

- Generally, the **convolution** of two sequences $x[n]$ and $y[n]$ is defined as:

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]y[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]y[k]$$

Impulse Response of a LTI System

- For example, consider the discrete-time system, H, shown below.



- The difference equation for the system *H* is given by:

$$v[n] = ax[n] + bx[n-1] + y[n]$$

$$y[n] = cv[n-1]$$

- From the above difference equations, y[n] can be determined for a given input.

Let $x[n] = \delta[n]$ (Unit impulse)

You may assume v[n] = 0 for n $\leq$ 0; therefore y[n] is also initially zero for n $\leq$ 0. Substituting n = 0, 1 ,2 ,... in the above equation , we obtain,

$n = 0$

$v[0] = ax[0]+bx[-1]+y[0] = a\cdot 1+b\cdot 0+0 = a$

$\Rightarrow \mathbf{y[1]} = c\cdot v[0] = \mathbf{a.c}$

$n = 1$

$v[1] = a\,x[1] + b\,x[0] + y[1] = a\cdot 0 + b\cdot 1 + a\cdot c = b + a\,c$

$\Rightarrow \mathbf{y[2]} = c\,v[1] = \mathbf{c\,(b+ac)}$

$n = 2$

$v[2] = a\,x[2] + b\,x[1] + y[2] = 0 + 0 + c\,(b+ac)$

$\Rightarrow \mathbf{y[3]} = c\,v[2] = \mathbf{c^2\,(b+ac)}$

.....

$n = n - 1$

$v[n\text{-}1] = c^{\,n-2}\,(b+ac) = bc^{n-2} + ac^{n-1}$

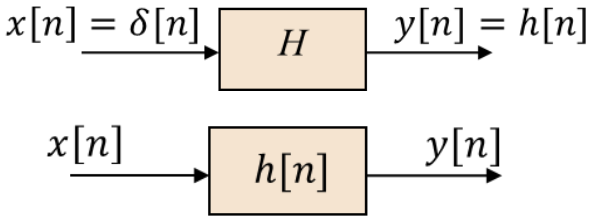
$\therefore \mathbf{y[n]} = c\,v[n\text{-}1] = \mathbf{bc^{n-1} + ac^{n}}$

$$\therefore y[n]\Big|_{x[n]=\delta[n]} = \boldsymbol{h[n]} = bc^{n-1} + ac^{n}$$

- The response $y[n] \overset{\Delta}{=} h[n]$ to an impulse excitation ($x[n] = \delta[n]$) is known as the impulse response and it is a very important characteristic of a discrete system.

Impulse Response of a LTI System

- If $x[n] = \delta[n]$, then $y[n] = h[n]$. The output tells us the system behaviour as the system, *H*, is being hit by all input frequencies. $h[n]$ completely characterizes the system, H.



- The response to an arbitrary input signal $x[n]$ is the **convolution** of $x[n]$ with the impulse response *h[n]* of the system as given by the equation below:

$$y[n] = x[n] * h[n]$$

Convolution

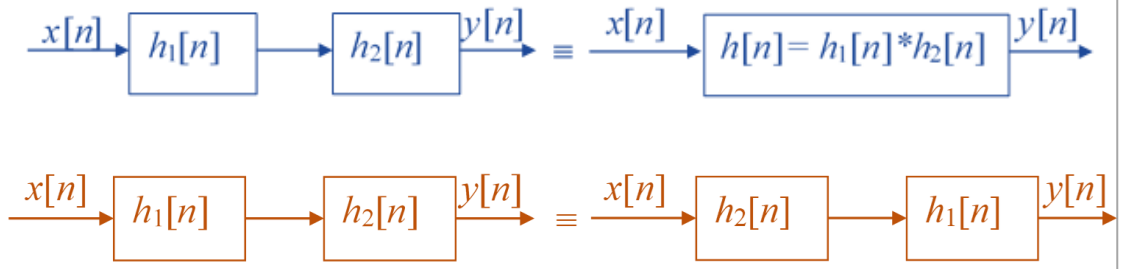
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

where $h[n]=H\{\delta[n]\}$

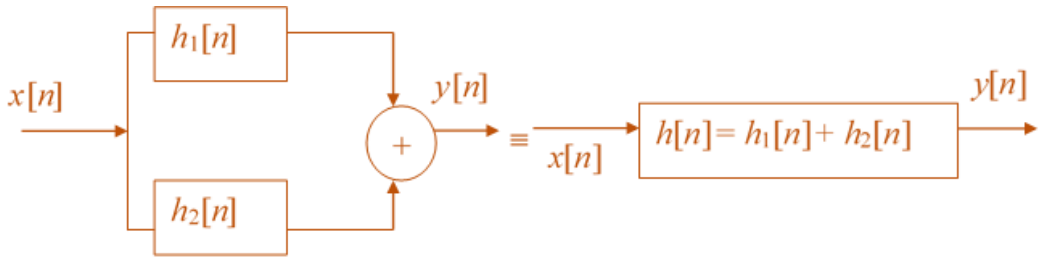
Associative Law

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$



Distributive Law

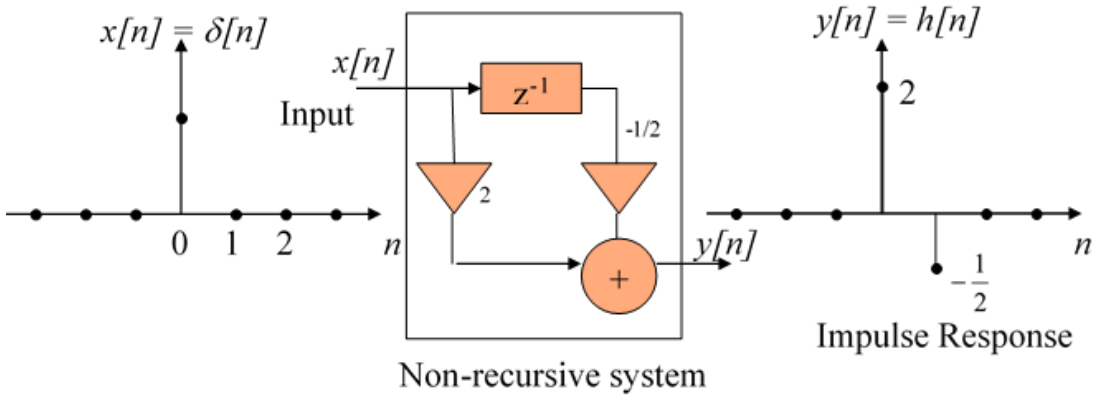
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



Impulse Response of a LTI System

Finite Impulse Response (FIR) System

- If the impulse response of a LTI system is of finite duration, the system is said to be Finite Impulse Response (FIR). It is also called **FIR filter.**



- Consider the system above; The difference equation is given by $y[n] = 2x[n] - 0.5\,x[n-1]$. Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

***n* = 0** $y[0] = h[0] = 2\,x[0] - 0.5\,x[-1] = 2$

***n* = 1** $y[1] = h[1] = 2\,x[1] - 0.5\,x[0] = -0.5$

***n* = 2** $y[2] = h[2] = 2x[2] - 0.5x[1] = 0$

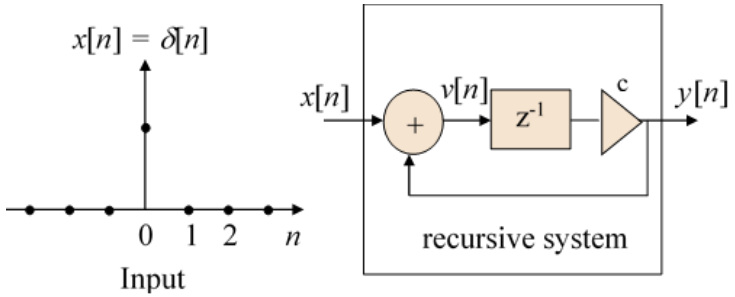
……

$n = n$ $y[n] = 0$

$$h[n] = \{2\,,\,0.5\}$$
$$h[n] = 0 \quad n \geq 2$$

Infinite Impulse Response (IIR) System

- If the impulse response of a linear time-invariant system is of infinite duration, the system is said to be an Infinite Impulse Response (IIR) system. It is also called **IIR filter.**



- Consider the system above; The difference equation is given by $v[n] = x[n] + y[n]$; $y[n] = c.v[n-1]$ ($c < 1$) Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

***n* = 0** $v[0] = x[0]+y[0] = 1$; ∴ $\boldsymbol{y[1]} = c \cdot v[0] = \boldsymbol{c}$

***n* = 1** $v[1] = x[1] + y[1] = c$; ∴ $\boldsymbol{y[2]} = c\,v[1] = \boldsymbol{c^2}$

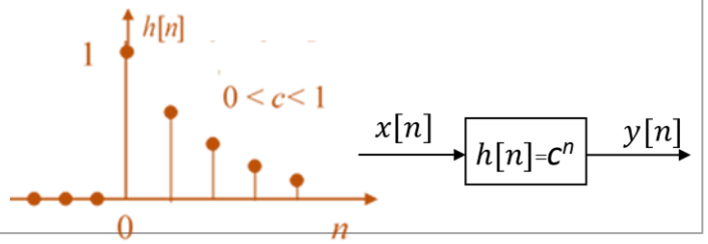
***n* = 2** $v[2] = x[2] + y[2] = c^2$; ∴ $\boldsymbol{y[3]} = c\,v[2] = \boldsymbol{c^3}$

……

***n* = *n* – 1** $v[n-1] = c^{n-1}$;

∴ $\boldsymbol{y[n]} = c\,v[n-1] = \boldsymbol{c^n}$

∴ $\boldsymbol{h[n]} = c^n$ for $n \geq 0$



Examples

Example 1:

- Find the impulse response $h[n]$ of the following fourth order non-recursive system.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + a_3x[n-3] + a_4x[n-4]$$

Answer: To find $h[n]$, we let $x[n] = \delta[n]$.

n=0 ➔ $h[0] = a_0\delta[0] + a_1\delta[-1] + a_2\delta[-2] + a_3\delta[-3] + a_4\delta[-4] = a_0$

n=1 ➔ $h[1] = a_0\delta[1] + a_1\delta[0] + a_2\delta[-1] + a_3\delta[-2] + a_4\delta[-3] = a_1$

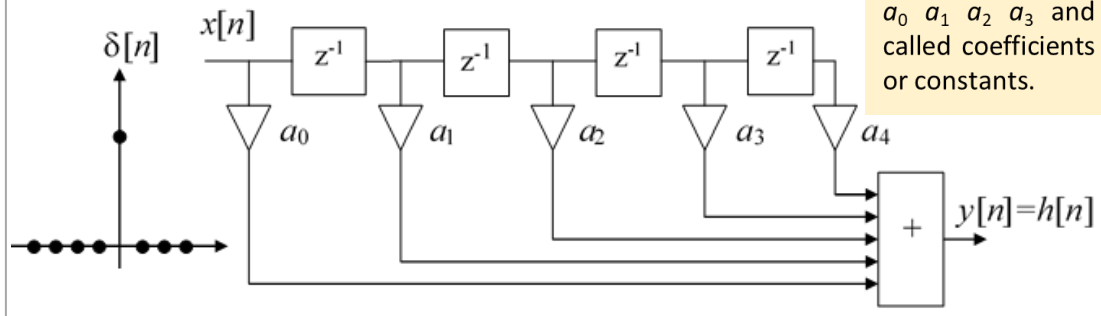
n=2 ➔ $h[2] = a_0\delta[2] + a_1\delta[1] + a_2\delta[0] + a_3\delta[-1] + a_4\delta[-2] = a_2$

n=3 ➔ $h[3] = a_0\delta[3] + a_1\delta[2] + a_2\delta[1] + a_3\delta[0] + a_4\delta[-1] = a_3$

n=4 ➔ $h[4] = 0 + 0 + 0 + 0 + a_4\delta[0] = a_4$

n=5 ➔ $h[5] = 0 + 0 + 0 + 0 + a_4\delta[1] = 0$

For n ≥ 5, h[n] = 0, since the nonzero value of δ[n] has moved out of the memory of this system.

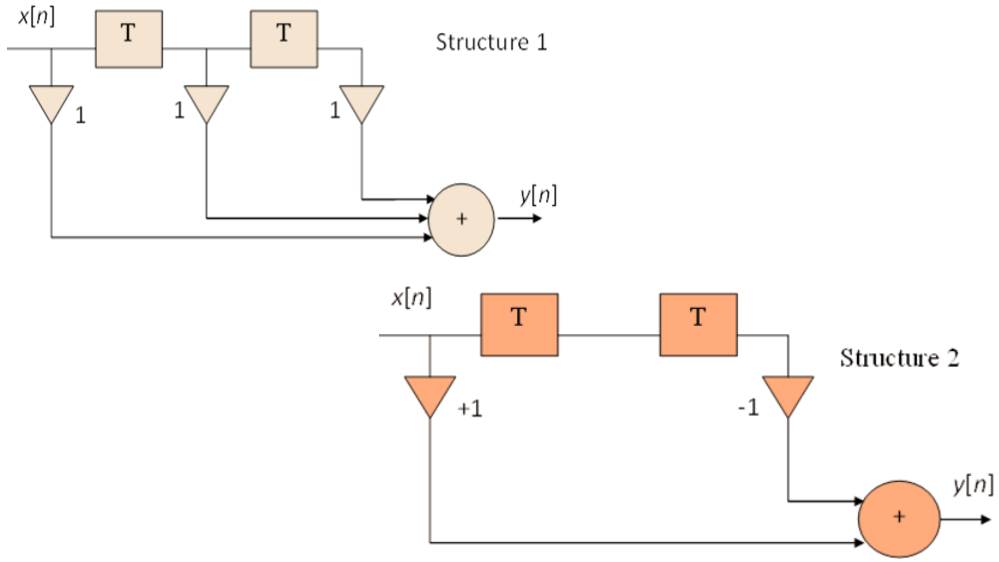


a_0 a_1 a_2 a_3 and a_4 are called coefficients (+ or -) or constants.

$h[n] = \{a_0 \;\; a_1 \;\; a_2 \;\; a_3 \;\; a_4\}$ and $h[n] = 0 \;\; n \geq 5$

Example 2:

- Two FIR systems (Structures) are shown below:
  (a) Write the difference equation;
  (b) Calculate the impulse response.



Structure 1:

$y[n] = x[n] + x[n-1] + x[n-2]$;

$h[n] = \{1\ 1\ 1\}$ and $h[n] = 0 \quad n \geq 3$

Structure 2:

$y[n] = x[n] - x[n-2]$

$h[n] = \{1\ \ 0\ \ -1\}$ and $h[n] = 0 \quad n \geq 3$

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

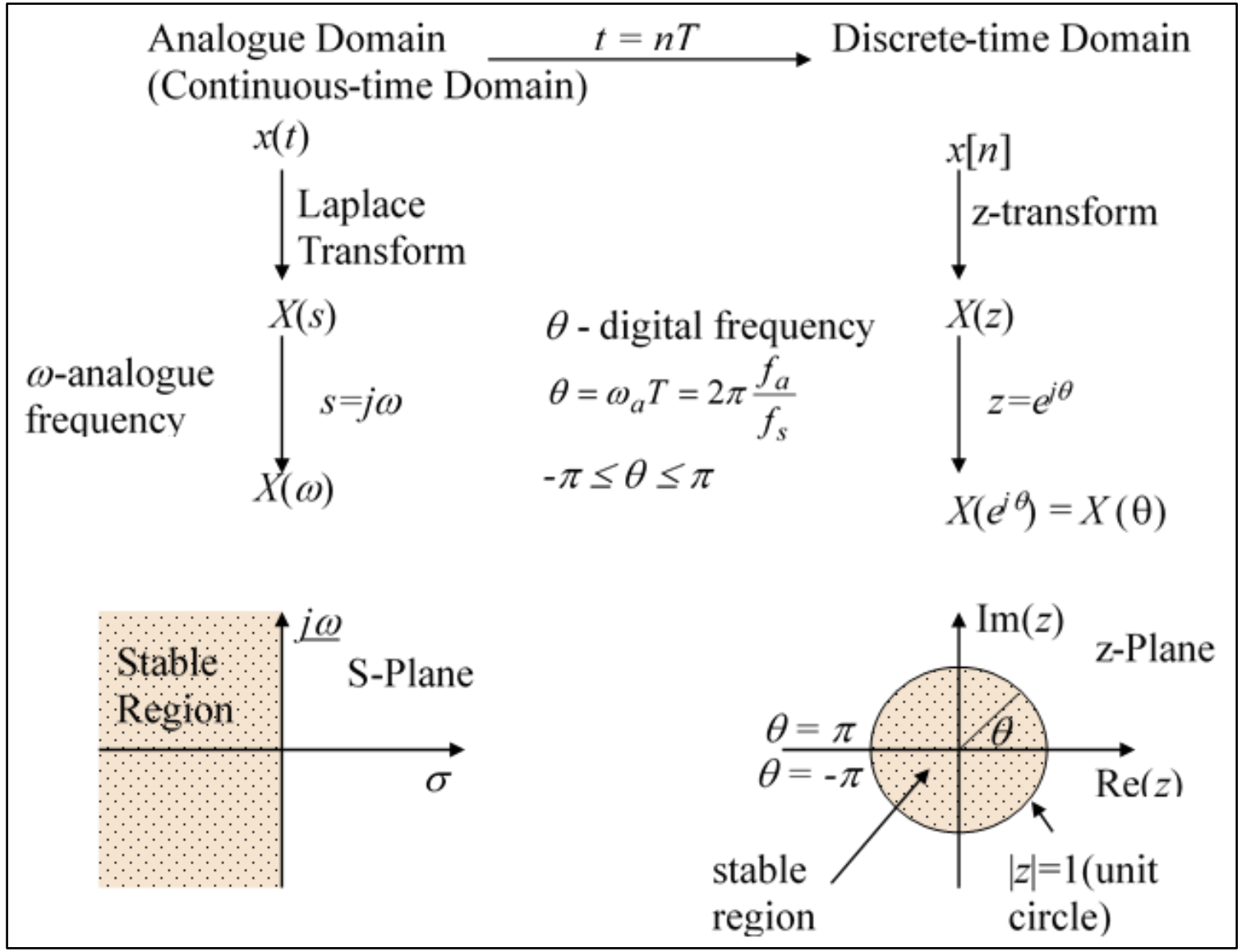
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Term 3, 2021

z-Transform

- The primary role of the Laplace transform in engineering is transient and stability analysis of causal LTI system described by differential equations.
- The primary roles of the z-transform are the study of system characteristics and derivation of computational structures for implementing discrete-time systems on computers.
- The z-transform is also used solve difference equations.
- We concentrate on the operational properties of the z-transform and its applications in system Analysis.



z-Transform

- The **two-sided z-transform** of a discrete-time signal is given by :

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$z = re^{j\theta}$, θ = digital frequency; where z is a complex variable

- In causal systems, $x[n]$ may be zero when $n < 0$. The **one-sided z-transform** of a discrete-time signal is given by :

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

- Clearly, the z-transform is a power series with an infinite number of terms and so may not converge for all values of z.

- The region where the z-transform converges is known as the **region of convergence (ROC)** and in this region the values of $X(z)$ are finite.

- **The step sequence:** $x[n] = \begin{cases} 1 & 0 \le n \le \infty \\ 0 & n < 0 \end{cases}$

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = 1 + z^{-1} + z^{-2} + \cdots = \frac{1}{1 - z^{-1}} = \frac{z}{z-1}$$

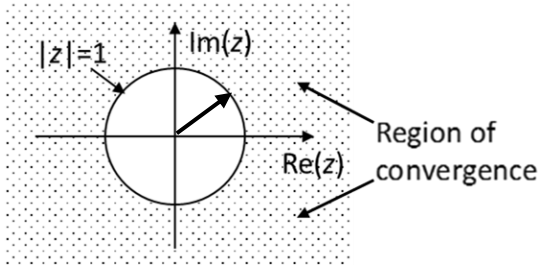
This is a geometric series with a common ratio of z^{-1}. The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$ and diverges if $|z| < 1$.

Let z =2 , $X(z) = 1 + 2^{-1} + 2^{-2} + \ldots = \frac{1}{1-2^{-1}} = 2$

Let $z = 0.5$, $X(z) = 1 + \left(\frac{1}{2}\right)^{-1} + \left(\frac{1}{2}\right)^{-2} + \cdots = 1 + 2 + 4 + \cdots$

- In this case, the z-transform is valid everywhere outside a circle of unit radius whose centre is at the origin (see below):

$|z|=1$ is a circle of unit radius referred to as the 'unit circle'



- The Region of Convergence (ROC) is seen to be bounded by the circle $|z|=1$, the radius of the **pole** of $X(z)\left\{= \frac{z}{z-1}\right\}$.

z-Transform

The delta sequence: δ[n]

$$Z\{\delta[n]\} = \sum_{n=0}^{\infty} \delta[n] z^{-n} = 1$$

The geometric sequence: $x[n] = a^n$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a} \qquad \left|\frac{a}{z}\right| < 1$$

or equivalently |z| > |a|

when $a = 1$, $x[n] = 1$ for $n \geq 0$ ie. $x[n] = u[n]$

$$X(z) = \frac{z}{z-1} \qquad \text{ROC: } |z|>1$$

The complex exponential sequence: $x[n] = e^{jn\theta}$

$$Z\{e^{j\theta n}\} = \sum_{n=0}^{\infty} e^{j\theta n} z^{-n} = \frac{1}{1-\frac{e^{j\theta}}{z}} = \frac{z}{z-e^{j\theta}}$$

$$\therefore Z\{\cos n\,\theta\} = \frac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$$

$$\therefore Z\{\sin n\,\theta\} = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Properties of z-Transform

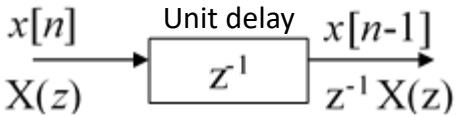
Linearity: $ax[n] + by[n] \overset{Z}{\longleftrightarrow} aX(z) + bY(z)$

Shifting property (Delay theorem):

$$x[n-k] \overset{Z}{\longleftrightarrow} z^{-k}X(z)$$

- A very important property of the z-transform is the delay theorem.
- Example: (1) $Z\{x[n-1]\} = z^{-1}X(z)$

  (2) $Z\{x[n-2]\} = z^{-2}X(z)$



Time reversal (even Symmetry): $x[-n] \overset{Z}{\longleftrightarrow} X\left(\frac{1}{z}\right) = X(z^{-1})$

Time reversal (odd Symmetry): $-x[-n] \overset{Z}{\longleftrightarrow} -X\left(\frac{1}{z}\right) = -X(z^{-1})$

Example: Find the z-transform of $x[n] = u[-n]$.

$$u[n] \overset{Z}{\longleftrightarrow} \frac{z}{z-1}; \quad u[-n] \overset{Z}{\longleftrightarrow} \frac{z^{-1}}{z^{-1}-1} = \frac{1}{1-z}$$

ROC: $|z| < 1$

z-Transform

Properties of z-Transform

Multiplication by exponential sequence:

$$a^n x[n] \overset{z}{\longleftrightarrow} X(a^{-1}z)$$

As a special case if $x[n]$ is multiplied by $e^{jn\theta}$

$$e^{j\theta n} x[n] \overset{z}{\longleftrightarrow} X(e^{-j\theta}z)$$

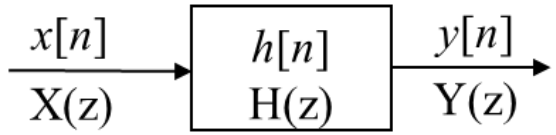
Differentiation in the z-domain:

$$nx[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}X(z)$$

Discreate convolution:

If $y[n] = x[n] * h[n]$

then $Y(z) = X(z) \cdot H(z)$



- **Convolution** operation in time domain transforms to a product in the z-domain.

Example 3:

- **Discrete convolution in z-domain:** Compute the convolution $y[n]$ of the discrete-time signals given by (assume that both sequences start at n=0]:

$x_1[n] = [1, -2, 1]$; $\{x_1[0] = 1, x_1[1] = -2, x_1[1] = -2\}$

$x_2[n] = 1$ for $0 \le n \le 5$,

$x_2[n] = 0$ elsewhere

Ans: $y[n] = x_1[n] * x_2[n] \Rightarrow Y(z) = X_1(z) \cdot X_2(z)$

$X_1(z) = 1 - 2z^{-1} + z^{-2}$ or $X_1(z) = (1 - z^{-1})^2$

$X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5}$ or $X_2(z) = \frac{1-z^{-6}}{1-z^{-1}}$

$Y(z) = X_1(z) \cdot X_2(z)$

$= 1 - z^{-1} - z^{-6} + z^{-7}$

↓ Inverse z-transform

$y[n] = [1, -1, 0, 0, 0, 0, -1, 1]$

Note:

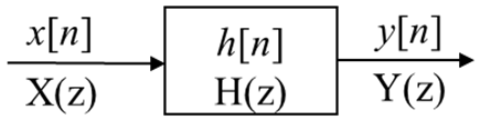
Length of $y[n]$ = Length of $x_1[n]$ + Length of $x_2[n]$ - 1

[8 samples = 3 samples + 6 samples - 1]

z-Transform

Example 4:

- Convolution of a Finite Sequence: The convolution of two finite-length sequences is also of finite length. Find the convolution of $h[n]$ = [2, 5, 0, 4] and $x[n]$ = [4, 1, 3];



If $y[n] = x[n] * h[n]$
then $Y(z) = X(z) \cdot H(z)$

Method1: Convolution using z-transform

$Y(z) = X(z) \cdot H(z)$

$X(z) = 4 + 1\,z^{-1} + 3z^{-2}$

$H(z) = 2 + 5z^{-1} + 4\,z^{-3}$

$Y(z) = 8 + 22z^{-1} + 11z^{-2} + 31z^{-3} + 4z^{-4} + 12z^{-5}$

↓ Inverse z-transform

$y[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

Method 2: Convolution in Discrete-time domain:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

- The flipped sequence is $h[-i]$ = [4, 0, 5, 2]. Line up flipped sequence below x[n] to begin overlap and shift it successively, summing the product sequence to obtain the discrete convolution

0 shift

```
x[n]:      4  1  3
   4  0  5  2        ← h[-i]
y[0] = Sum of products = 8
```

1 shift

```
x[n]:      4  1  3
      4  0  5  2     ← h[1-i]
y[1] = (5x4)+(2x1) = 22
```

2 shifts

```
x[n]:      4  1  3
         4  0  5  2  ← h[2-i]
y[2] = 0 + 5 + 6 = 11
```

3 shifts

```
x[n]:      4  1  3
            4  0  5  2 ← h[3-i]
y[3] = 16 + 0 + 15 = 31
```

4 shifts

```
x[n]:      4  1  3
               4  0  5  2 ← h[4-i]
y[4] = 4 + 0 = 4
```

5 shifts

```
x[n]:   4  1  3
                  4  0  5  2 ← h[5-i]
y[5] = 12
```

$\therefore y[n] = x[n] * h[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

z-Transform

Transfer Function

- A transfer function is defined as the ratio of the output *Y(z)* and the input *X(z)* or as the z-transform of the system impulse response *h[n]*.
- Given the following difference equation, find the transfer function $H(z) = \frac{Y(z)}{X(z)}$.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + b_1y[n-1] + b_2y[n-2]$$

Take the z-transform of both sides:

$$Y(z) = a_oX(z) + a_1z^{-1}\,\mathrm{X(z)} + a_2z^{-2}\,\mathrm{X(z)} + b_1z^{-1}\,Y(z) + b_2z^{-2}\,Y(z)$$

$$Y(z) - b_1\,z^{-1}\,Y(z) - b_2\,z^{-2}\,Y(z) = a_o\,X(z) + a_1\,z^{-1}\,\mathrm{X(z)} + a_2\,z^{-2}\,\mathrm{X(z)}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1\,z^{-1} + a_2\,z^{-2}}{1 - b_1\,z^{-1} - b_2\,z^{-2}}$$

- The transfer function is thus a ratio of polynomials in z.
- A LTI system may be described by its **transfer function**, its **impulse response**, its **difference equation** or its **poles** (roots of the denominator) and **zeros** (roots of the numerator)
- Given one form, it is possible to obtain any of the other forms.

Example 5:

- Find the difference-equation of the following transfer function: $H(z) = \frac{5z+2}{z^2+3z+2}$

Step1: First rewrite *H*(*z*) as a ratio of polynomials in z^{-1}

$$H(z) = \frac{Y(z)}{X(z)} = \frac{5z^{-1} + 2z^{-2}}{1 + 3z^{-1} + 2z^{-2}}$$

$$Y(z) + 3\,Y(z)z^{-1} + 2\,Y(z)z^{-2} = 5\,\mathrm{X(z)}\,z^{-1} + 2\mathrm{X(z)}\,z^{-2}$$

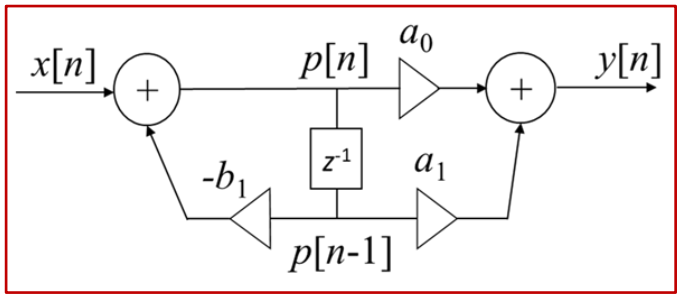
Step 2: Take inverse z-transform

$$y[n] + 3y[n-1] + 2y[n-2] = 5x[n-1] + 2x[n-2]$$

$$y[n] = 5x[n-1] + 2x[n-2] - 3y[n-1] - 2y[n-2]$$

Example 6:

Determine the Transfer Function H(z):



$$\frac{P(z)}{X(z)} = \frac{1}{1 + b_1z^{-1}}$$

$$\frac{Y(z)}{P(z)} = a_0 + a_1z^{-1}$$

$$\frac{Y(z)}{X(z)} = \frac{a_0 + a_1z^{-1}}{1 + b_1z^{-1}}$$

z-Transform

Basic z-Transform

Signal	Transform	ROC
$\delta[n]$	1	all z
$U[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\|z\| > \|\alpha\|$
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\|z\| > \|\alpha\|$
$\cos(n\theta)\, u[n]$	$\frac{1-z^{-1}\cos\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\|>1$
$\sin(n\theta)\, u[n]$	$\frac{z^{-1}\sin\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\| > 1$
$r^n\cos(n\theta)\, u[n]$	$\frac{1-z^{-1}r\cos\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\| > r$
$r^n\sin(n\theta)\, u[n]$	$\frac{1-z^{-1}r\sin\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\|>r$

z-Transform Properties

Signal	Transform
$x[n]$	$X(z)$
$ax[n] + by[n]$	$aX(z) + bY(z)$
$x[n-k]$	$z^{-k}X(z)$
$a^n x[n]$	$X\left(\frac{z}{a}\right)$
$x[-n]$	$X\left(\frac{1}{z}\right)$
$x[n]*y[n]$	$X(z)\cdot Y(z)$
$nx[n]$	$-z\frac{d}{dz}X(z)$

Note:

$$S_N = 1 + z^{-1} + z^{-2} + \ldots + z^{-(N-1)} = \frac{1-z^{-N}}{1-z^{-1}}$$

If $N \to \infty$ $S_\infty = \frac{1}{1-z^{-1}}$, $|z^{-1}| < 1$

Inverse z-Transform

Partial Fraction Method

- The inverse z-transform allows us to recover the discrete-time sequence.

$$x[n] = \mathbf{Z^{-1}}[X(z)]$$

where $X(z)$ is the z-transform of $x[n]$.

Example 7:

- Find $x[n]$ for the following:

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}}$$

Answer:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)} = z\left[\frac{A}{z-0.75} + \frac{B}{z+0.5}\right] = z\left[\frac{\left(\frac{4}{5}\right)}{z-0.75} + \frac{\left(-\frac{4}{5}\right)}{z+0.5}\right]$$

$$X(z) = \left[\frac{\left(\frac{4}{5}\right)}{1 - 0.75z^{-1}} - \frac{\left(\frac{4}{5}\right)}{1 + 0.5z^{-1}}\right]$$

Inverse z-transform

$$x[n] = \frac{4}{5}\left[(0.75)^n - (-0.5)^n\right] \text{ for } n > 0.$$

Example 8:

- Find h[n] for the following system and show that the system is stable if $|a|<1$:

$$H(z) = \frac{1}{1 + az^{-1}} - \frac{az^{-1}}{1 + az^{-1}}$$

Inverse z-transform

Answer:

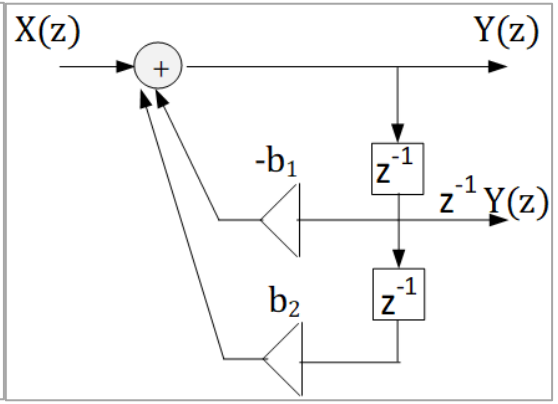
$$h[n] = \{(-a)^n u[n]\} - a\{(-a)^{n-1}\, u[n-1]\}$$

Since $|a|<1$, $(-a)^n$ and $(-a)^{n-1}$ are then decaying function. Therefore, h[n] is a decaying function. Hence the system stable.

Example 9:.

- Draw a structure for the following Transfer function:

$$H(z) = \frac{z^{-1}}{1 + b_1 z^{-1} - b_2 z^{-2}}$$



ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

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- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
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Term 3, 2021

Pole-Zero Description of Discrete Time Systems

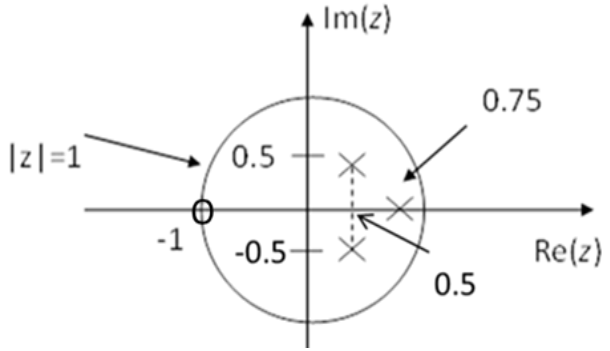
Poles and Zeros

- The **zeros** of a z-transform *H(z)* are the values of *z* for which *H(z)=0*. The **poles** of a z-transform are the values of *z* for which *H(z)=∞* . If *H(z)* is a rational function , then

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + \ldots + a_M z^{-M}}{1 + b_1 z^{-1} + \ldots\ldots + b_L z^{-L}}$$

$$= a_0\, z^{L-M}\, \frac{(z - z_1)(z - z_2)\ldots\ldots(z - z_M)}{(z - p_1)(z - p_2)\ldots\ldots(z - p_L)}$$

- The complex quantities (or may be real) z_1, z_2, z_3 …. are called zeros of *H(z)* and the complex quantities (or may be real) p_1, p_2, p_3 … are called the poles of *H(z)*.
- We thus see that *H(z)* is completely determined , except for the constant a_0, by the values of poles and zeros.
- The information contained in the z-transform can be conveniently displayed as a poles-zero diagram (see figure on the right)
- In the diagram, 'X' marks the position of a pole and 'O' denotes the position of a zero

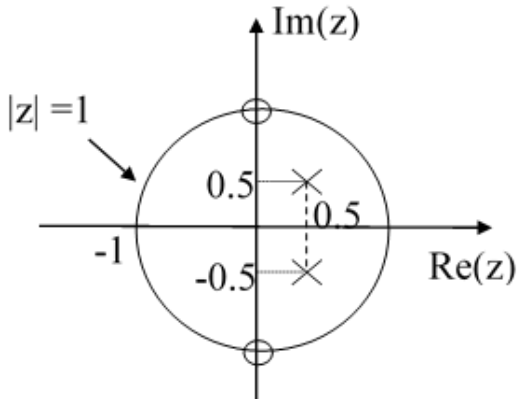


- The poles are located at $z = 0.5 \pm 0.5j$ and $z = 0.75$; a single zero is at $z = -1$.
- An important feature of the pole-zero diagram is the unit circle $|z|=1$.
- The pole-zero diagram provides an insight into the properties of a given discrete-time system.
- From the locations of the poles and zeros we can infer the frequency response of the system as well as its degree of stability.
- **For a stable system, all the poles must lie inside the unit circle. Zeros may lie inside, on, or outside the unit circle.**

Pole-Zero Description of Discrete Time Systems

Example 10:

- Determine the transfer function *H(z)* of a discrete-time system for the pole-zero diagram shown below:

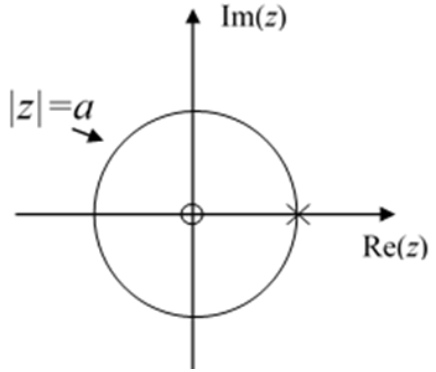


$$H(z) = \frac{K(z-j1)(z+j1)}{(z-0.5-0.5j)(z-0.5+0.5j)}$$

$$H(z) = \frac{K(1+z^{-2})}{1-z^{-1}+0.5z^{-2}}$$

Example 11:

- Determine the pole-zero plot for $H(z) = \frac{z}{z-a}$



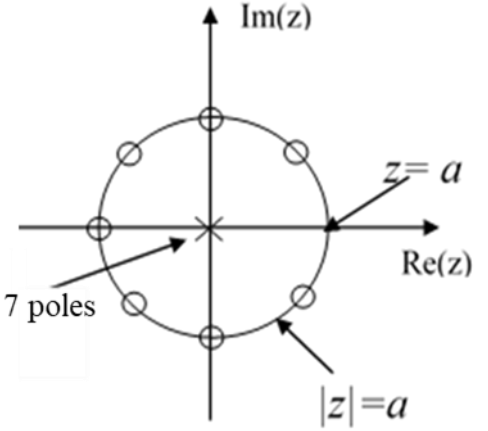
Example 12:

- Determine the pole-zero plot for

$$H(z) = \frac{z^M - a^M}{z^{M-1}(z-a)}$$

$$= \frac{(z-z_1)(z-z_2).....(z-(M-1))}{z^{M-1}}$$

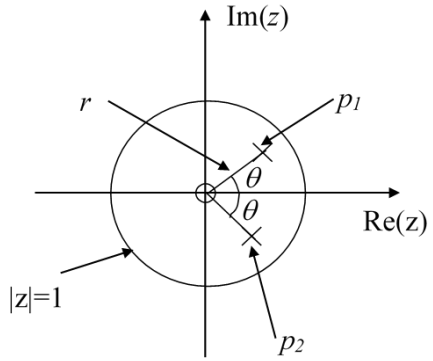
- The zero z = *a* cancels the pole at z = *a*. Thus, *H(z)* has *M*-1 zeros and *M*-1 poles as shown in the diagram below for *M* = 8.



Pole-Zero Description of Discrete Time Systems

Time domain behaviour of a system with a pair of complex conjugate poles

- Consider a system, *H(z)* with two complex conjugate poles in the z-plane:



$$p_1 = re^{j\theta}$$
$$p_2 = re^{-j\theta}$$
} Poles
$$z_1 = 0 \quad (zero)$$

- The transfer function corresponding to the pole-zero pattern:

$$H(z) = \frac{z}{(z-re^{j\theta})(z-re^{-j\theta})} = z\left[\frac{A}{z-re^{j\theta}} + \frac{B}{z-re^{-j\theta}}\right]$$

$$= z\left[\frac{\frac{1}{2jr\sin\theta}}{z-re^{j\theta}} - \frac{\frac{1}{2jr\sin\theta}}{z-re^{-j\theta}}\right] = \frac{1}{j(2r\sin\theta)}\left[\frac{1}{1-re^{j\theta}z^{-1}} - \frac{1}{1-re^{-j\theta}z^{-1}}\right]$$

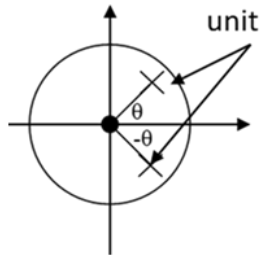
$$h(n) = \frac{1}{j2r\sin\theta}\left[(re^{j\theta})^n - (re^{-j\theta})^n\right] = \frac{r^n}{j2r\sin\theta}\left[e^{j\theta n} - e^{-j\theta n}\right]$$

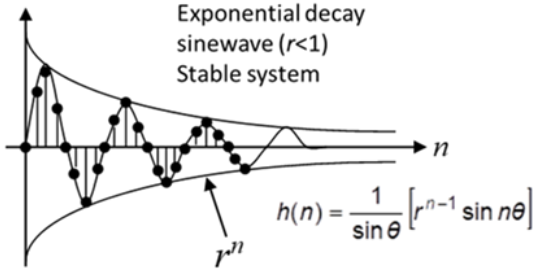
$$\boldsymbol{h(n) = \frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]}$$,← θ = frequency of oscillation

- This is the impulse response of the 2nd order system with complex poles. We note that the impulse response will decay away to zero provided r is less than one [$r < 1$].

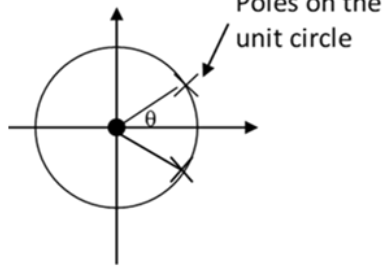
- Recall that *r* is also the distance from the origin in the z-plane to the poles p_1 or p_2, so that system will be stable if the poles in the z-plane lie inside the unit circle.

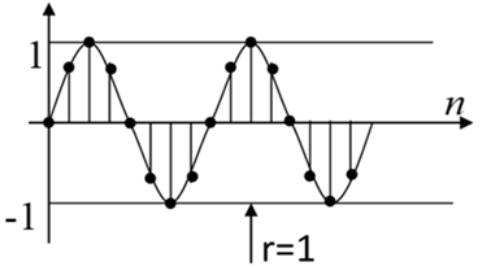
Case1:



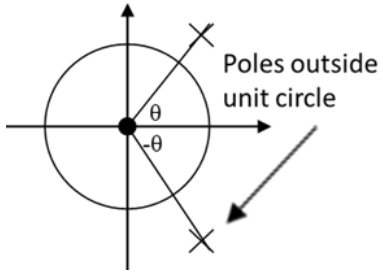


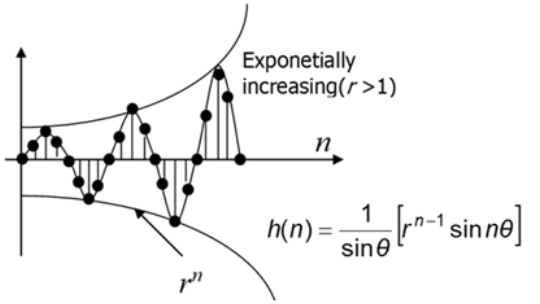
Case 2:





Case3:





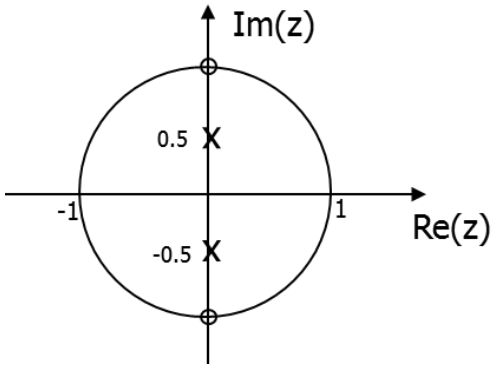
Pole-Zero Description of Discrete Time Systems

Stability Criteria:

- A system that is both **stable and causal** must have all its poles inside that unit circle within the z-plane.
- We cannot have a pole outside the unit circle, since the inverse transform of a pole located outside the circle will contribute either a right sided increasing exponential term, which **is not stable**, or a left-sided decaying exponential term that is not causal.

Example 13:

- Determine the transfer function H(z) of a discrete-time system with the pole-zero map given below:

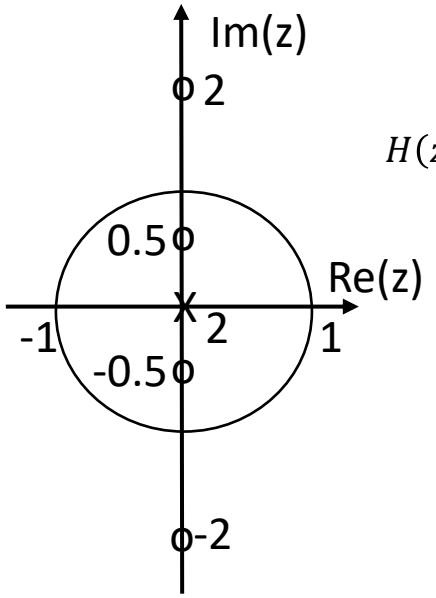


$$H(z) = K\frac{(z-j)(z+j)}{(z-0.5j)(z+0.5j)}$$

$$= K\frac{(z^2+1)}{(z^2+0.25)}$$

Example 14:

- What is the z-transform corresponding to the pole-zero pattern shown in the diagram below. Does it represent a symmetrical signal?



$$H(z) = K\frac{(z-j0.5)(z+j0.5)(z-j2)(z+j2)}{z^2}$$

$$H(z) = K\frac{(z^4+4.25z^2+1)}{z^2}$$

$$H(z) = K\,(z^2+4.25+z^{-2})$$

$$h[n] = K\,\{1, 0, 4.25, 0, 1\}$$

n = -2 n = 2

h[n] is even symmetry

- Checking for symmetry , we find that $H(z) = H(z^{-1})$ and thus h[n] s even symmetric.

ELEC3104: Digital Signal Processing

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Term 3, 2021

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

DTFT

- The Fourier Transform is a very important tool in many aspects of engineering.
- In the case of digital signal processing, it allows signals to be represented in terms of their constituent frequency components, i.e. as signals in the 'frequency domain'.
- In some cases this means signal processing operations can be executed on frequency domain rather than time domain signals, which can have a number of advantages.
- Before proceeding, it is worth noting that in signal processing what is referred to as 'the Fourier transform' may in fact be one of four possible types of Fourier transform, illustrated in the adjacent Table.
- The difference between each type of transform is whether the time domain signal is continuous – time or discrete-time and whether its frequency domain representation is continuous (i.e. a function of real numbers ω or θ) or discrete (i.e. a function of integers n or k)

Types of Fourier transforms for signal processing

		Frequency domain signal	
		Continuous	Discrete $X[k]$
Time domain signal	Continuous $x(t)$	Fourier transform $X(\omega)$	Fourier series X_n
	Discrete $x[n]$	Discrete Time Fourier Transform (DTFT) $X(\theta)$	Discrete Fourier Transform (DFT) $X[k]$

- Fourier Transform of an arbitrary discrete signal is also called Discrete Time Fourier Transform (DTFT).
- For discrete-time signals, the Fourier transform pair is defined by:

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

- where θ is the digital frequency or relative frequency ($-\pi \le \theta \le \pi$) and $\theta = \omega T$. $X(\theta)$ is periodic with period 2π. (ie. $X(\theta + 2\pi) = X(\theta)$). Note that although $x[n]$ is discrete, ***X(θ)* is a continuous function** of frequency θ (hence the integral in the inverse transform).
- Note: The Fourier transform of an analogue signal is not periodic.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

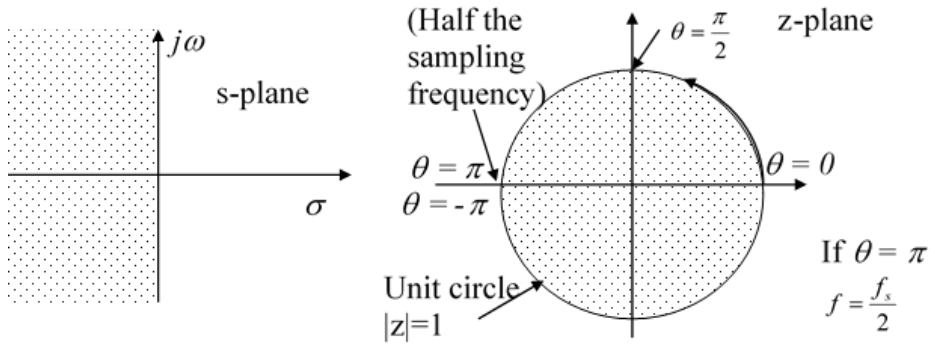
Relationship between the z-transform and Laplace transform

- If we let $z = e^{sT}$, then $z = e^{(\sigma + j\omega)T}$ (where T is the sampling period)

$$z = e^{\sigma T} \cdot e^{j\omega T} = e^{\sigma T} \cdot e^{j\theta} \qquad -\pi \le \theta \le \pi$$

$$\therefore |z| = e^{\sigma T} \text{ and } \arg(z) = \omega T = 2\pi \frac{f}{f_s} = \theta \quad (\theta = \text{digital frequency})$$

- As ω varies from -∞ to ∞ the s-plane is mapped to the z-plane as shown in figure below:



Mapping of the s-plane to the z-plane.

- The entire jω axis ($\sigma = 0$) in the s-plane is mapped onto the unit circle ($\sigma = 0$; $|z| = e^{\sigma T} = 1$)The left-hand s-plane is mapped to the inside of the unit circle and the right-hand s-plane maps to the outside of the unit circle.

- In terms of frequency response, the jω axis is the most important in the s-plane. In this case $\sigma = 0$ and the frequency points in the s-plane are related to points on the z-plane unit circle by :

$$z = e^{\sigma T} \cdot e^{j\omega T} = 1\, e^{j\theta} \qquad \boldsymbol{z = e^{j\theta}}$$

Frequency Response Estimation

- There are many instances when it is necessary to evaluate the frequency response of discrete-time systems.
- The frequency response of a system can be readily obtained from its z-transforms.
- For example, if we set $z = e^{j\theta}$, that is evaluate the z-transform around the unit circle, we obtain the Discrete Time Fourier Transform (DTFT) of the system.

DTFT: $H(\theta) = H(z)|_{z=e^{j\theta}}$ $-\pi \le \theta \le \pi$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 15:

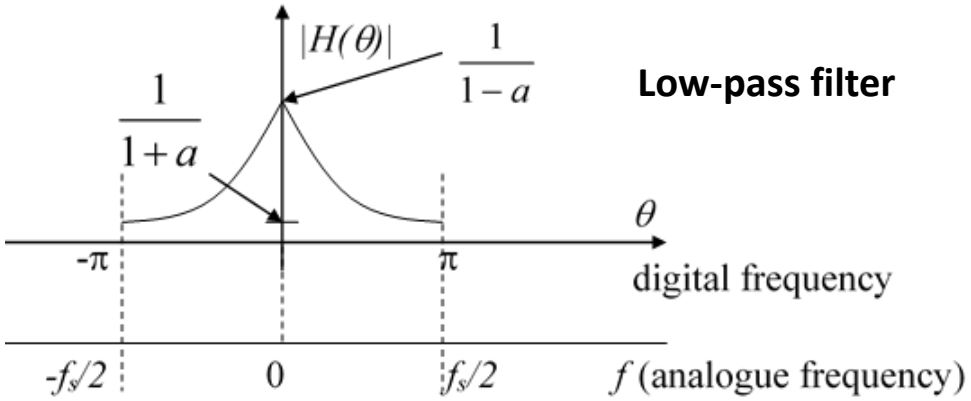
- Determine and sketch an approximate magnitude response for the following system:

$$H(z) = \frac{1}{1-az^{-1}} \qquad 0 < a < 1$$

- To find $H(\theta)$ (the frequency response) , substitute $z = e^{j\theta}$

$$H(\theta) = H(z)|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}} = \frac{1}{(1-a\cos\theta)+ja\sin\theta}$$

$$|H(\theta)| = \frac{1}{\sqrt{(1-a\cos\theta)^2+(a\sin\theta)^2}} = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$



Low-pass filter

- **DC gain or DC value : $|H(0)| = {}^{1}/_{1-a}$** {when θ=0}

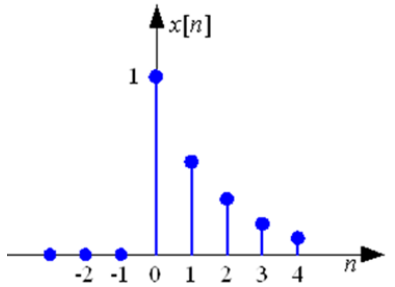
Example 16:

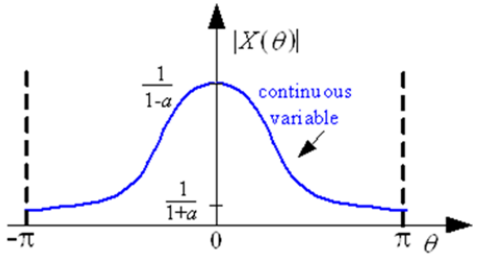
- Determine DTFT of the following sequence and sketch an approximate magnitude response.

$$x[n] = \begin{cases} 0 & n < 0 \\ a^n & n \geq 0, |a| < 1 \end{cases} \qquad X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

$$X(\theta) = \sum_{n=0}^{\infty} a^n e^{-jn\theta} = \sum_{n=0}^{\infty} (ae^{-j\theta})^n = \frac{1}{1-ae^{-j\theta}}$$

$$X(\theta) = \frac{1}{(1-a\cos\theta)+ja(\sin\theta)} \quad \therefore |X(\theta)| = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$





- *Note:* The above $X(\theta)$ can be obtained using the z-transform as well: $x[n] = a^n u[n] \Rightarrow X(z) = \frac{1}{1-az^{-1}}$

$$X(\theta) = X(z)\Big|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}}$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 17:

- Consider a system described by $y[n] = x[n] - x[n-1]$. Find the (a) impulse response; (b) transfer function $H(z) = \frac{Y(z)}{X(z)}$; (c) Discrete time Fourier Transform $H(\theta)$. Sketch the approximate magnitude response.

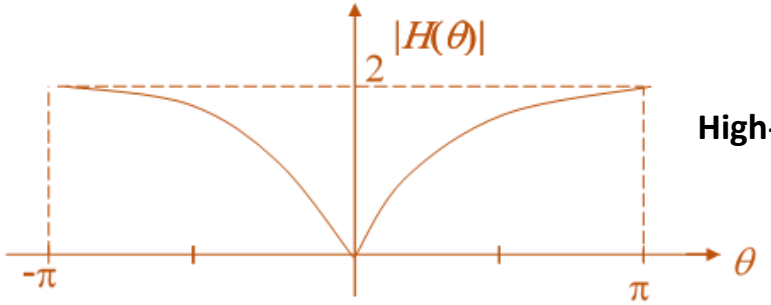
Answer:

$(a)\ h[n] = \{1, -1\}$;

$(b)\ Y(z) = X(z) - z^{-1}X(z);\ \ H(z) = 1 - z^{-1}$

$(c)\ H(\theta) = 1 - e^{-j\theta}$

$$|H(\theta)| = \sqrt{(1-\cos\theta)^2 + (-\sin\theta)^2} = \sqrt{2 - 2\cos\theta}$$



High-pass filter

- **DC gain : $|H(0)| = 0$** {when $\theta = 0$}

Properties of DTFT

$x[n] \overset{\text{DTFT}}{\rightarrow} X(\theta)$ $\quad x_1[n] \overset{\text{DTFT}}{\rightarrow} X_1(\theta)$ $\quad x_2[n] \overset{\text{DTFT}}{\rightarrow} X_2(\theta)$

- Linearity: $ax_1[n] + bx_2[n] \overset{\text{DTFT}}{\leftrightarrow} aX_1(\theta) + bX_2(\theta)$
- Time shift: $x[n-k] \overset{\text{DTFT}}{\leftrightarrow} e^{-j\theta k}X(\theta)$
- Frequency Shift: $x[n]e^{jn\theta_0} \overset{\text{DTFT}}{\leftrightarrow} X(\theta - \theta_0)$
- Frequency Differentiation: $nx[n] \overset{DTFT}{\leftrightarrow} j\frac{d}{d\theta}X(\theta)$
- Convolution: $x_1[n] * x_2[n] \overset{DTFT}{\leftrightarrow} X_1(\theta)X_2(\theta)$
- Multiplication: $x_1[n]x_2[n] \overset{DTFT}{\leftrightarrow} X_1(\theta) * X_2(\theta)$
- Parseval's Theorem:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\theta)|^2 d\theta$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 18:

- The frequency response of an ideal low pass filter, in the fundamental interval $-\pi \le \theta \le \pi$ is given by

$$H(\theta) = \begin{cases} \frac{1}{2} & 0 \le |\theta| \le \theta_C \\ 0 & \theta_C < |\theta| \le \pi \end{cases}$$

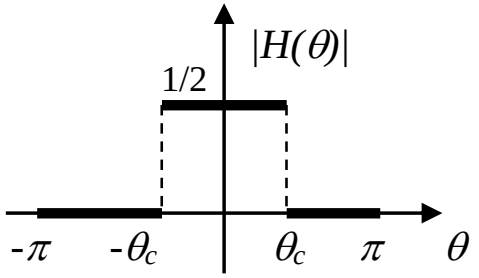
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\, d\theta$$

(a) Find the impulse response of the ideal low pass filter;

(b) Sketch the impulse response for $\theta_C = \frac{\pi}{4}$.

Answer:

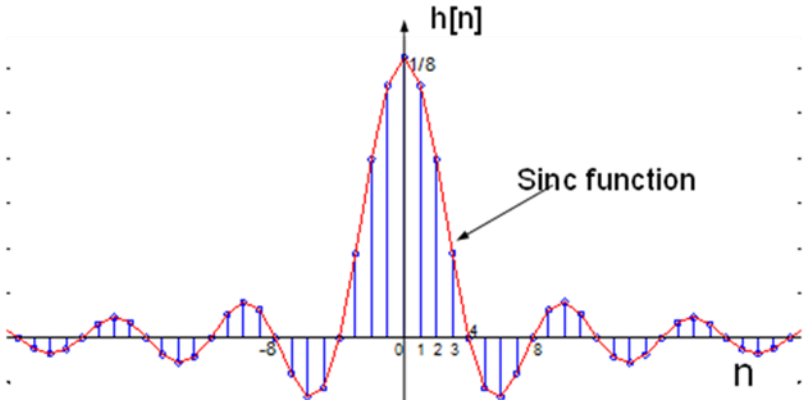
(a)



$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)e^{jn\theta}d\theta = \frac{1}{2\pi}\int_{-\theta_C}^{\theta_C}\frac{1}{2}e^{jn\theta}d\theta$$

$$= \frac{1}{4\pi}\frac{e^{j\theta_C n} - e^{-j\theta_C n}}{jn} = \frac{1}{2\pi}\frac{\sin\theta_c n}{n}$$

(b) $\theta_c = \frac{\pi}{4}$ $\quad h(n) = \frac{1}{2\pi}\frac{\sin\frac{n\pi}{4}}{n} = \frac{1}{8}\frac{\sin\frac{n\pi}{4}}{\frac{n\pi}{4}}$



Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Discrete Fourier Transform (DFT)

- The importance of the discrete Fourier transform (DFT) for practical applications is hard to overstate.
- If you've ever seen a real time spectrum of just about anything, chances are it was generated using a DFT.
- The DFT allows us to determine the frequency content of a signal, that is, to perform spectral analysis.
- The DFT plays a central role in the implementation of a variety of digital signal processing algorithms, as a result of the existence of the efficient algorithm for the Fast Fourier Transform (FFT).

$$x[n] \xleftrightarrow{DTFT} X(\theta) \quad \text{DTFT}$$

$$x[n] \xleftrightarrow{DFT} X[k] \quad \text{DFT}$$

{DFT mapping $x[n]$ to another sequence}

- The DFT denoted by $X[k]$ (complex valued sequence), is obtained by sampling the Discrete Time Fourier Transform $X(\theta)$ at a finite number of frequency points
- This sampling is conventionally performed at equally spaced points over the period extending over $-\pi \leq \theta \leq \pi$.

(1) Fourier transform of a discrete signal (DTFT) is

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}, \quad -\pi \leq \theta \leq \pi$$

(2) $X(\theta) = X(z)|_{z=e^{j\theta}}$ {Evaluate z-transform on the unit circle}

(3) Discrete Fourier Transform (DFT) or N-point DFT

DFT: $X[k] = \sum_{n=0}^{N-1} x[n]e^{-jn\left(\frac{2\pi k}{N}\right)}$

IDFT: $x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{jn\left(\frac{2\pi k}{N}\right)}$

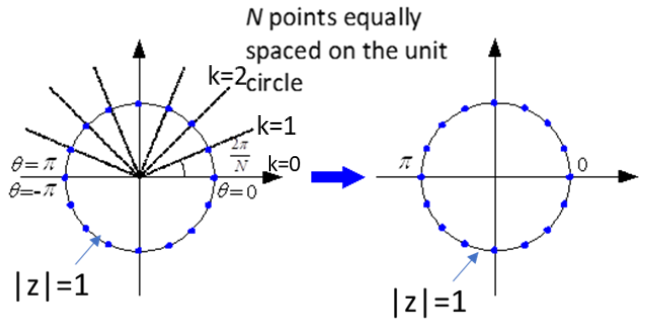
- i.e. sampling $X(\theta)$ at N equally spaced interval, k = 0, 1, 2, … N-1. Note that one of the most important properties of the DFT is implied periodicity.

Number of time samples = N
Number of frequency samples (k) = N

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Sampling the unit circle at N points equaly spaced in angle:

- The DFT corresponds to sampling the z-transform of *X(z)* at *N*-points equally spaced in angle around the unit circle ($\theta = \frac{2\pi k}{N}$; $(k = 0, 1, 2, \dots N-1)$

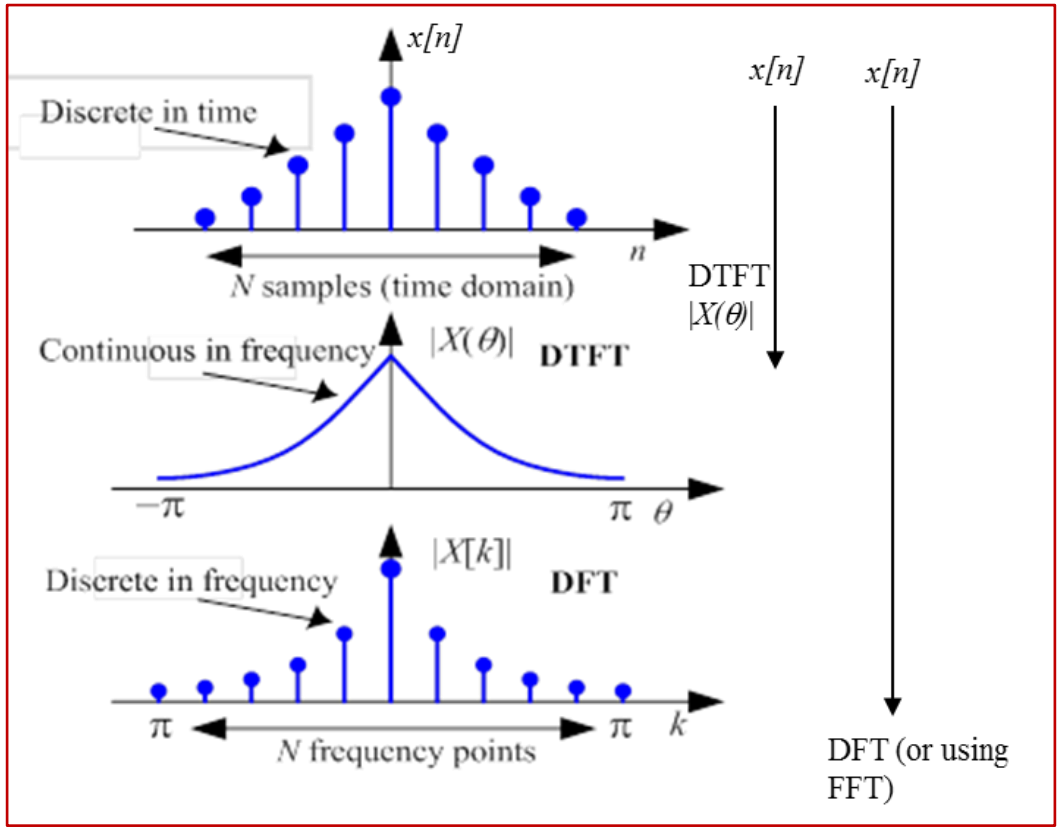


- The diagram on the right shows an example of a sequence $x[n]$ and its corresponding DTFT, X(θ), and the sampled version of DTFT known as DFT, X[k].

Approximating the DTFT by the DFT

- When the DFT of a block of *N* samples (i.e. a window of samples) is calculated, the assumption is that the original signal actually repeats itself periodically, with period *N*.
- Clearly, for all real signals this will not be true.
- Even for artificial signals such as a pure sinusoid this will only be true if *N* is a multiple of the period of the sinusoid.
- This "windowing" process introduces a slight distortion into the frequency representation of the signal being analysed.

If x[n] is a finite N-point signal with $n = 0, 1, 2 \dots . N-1$, the DFT is an exact match of its DTFT $X(\theta)$ at $\theta = \frac{2\pi k}{N}$ $(k = 0, 1, 2, \dots N-1)$ and the IDFT results in perfect recovery of x[n].



Number of time samples = *N*
Number of frequency samples (*k*) = *N*

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

DFT calculation:

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-jn\frac{2\pi k}{N}} \qquad k = 0,1,2,3,\ldots N$$

$$X[k] = \sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi}{N}nk\right) - j\sum_{n=0}^{N-1} x[n]\sin\left(\frac{2\pi}{N}nk\right)$$

$$X[k] = A[k] - jB[k]$$

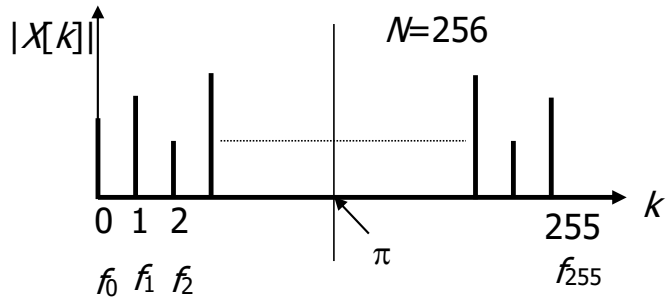
$$|X[k]| = \sqrt{A[k]^2 + B[k]^2} \qquad k = 0,1,2,3,\ldots N-1$$

$$\Delta\theta = \frac{2\pi}{N}$$

$$\Delta f = \frac{f_S}{N}$$

We have $f_k = k\left(\frac{f_s}{N}\right) = k\Delta f \quad (k = 0, 1, 2, \ldots, N-1)$

- N is the length of the sequence x[n] to be analysed using DFT and $|X[k]|$ is the magnitude of the sampled spectrum at index k.



The frequency resolution ($\Delta f = f_s/N$) can be made as small as desired by increasing the value of N.

Example 19:

- Let f_s = 8000 Hz, number of sample N = 1000
- Frequency resolution $\Delta f = \frac{f_S}{N} = \frac{8000}{1000} = 8Hz$
- $f_0 = 0$; $f_1 = 8$Hz, $f_2 = 16$ Hz, …. $f_{999} = 7992$Hz

$$X[k] = \sum_0^{999} x[n]e^{-jn\left(\frac{2\pi k}{1000}\right)}, \; k = 0, 1, 2, 3, 4..$$

Example 20:

- A speech signal is sampled at a rate of 20000 samples/sec. A sequence of length (N) 1024 samples is selected and the 1024-point DFT is computed.

(a) What is the time duration of segment of speech?

Duration = no of samples × sampling period.

= 1024 (1/20000) = 51.2 ms

(b) What is the frequency resolution (spacing in Hz) between the DFT values?

$$\text{Resolution} = \frac{f_S}{N} = \frac{20000}{1024} = 19.531Hz$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 21:

- Consider the finite length sequence: $x[n] = \delta[n] + 0.2\delta[n-2]$

  Find the *N*-point DFT of *x*[*n*] for *N* = 50.

Answer:

$$X(z) = 1 + 0.2\, z^{-2} \Rightarrow X(\theta) = 1 + 0.2\, e^{-j2\theta}$$

The *N*-point DFT is obtained by evaluating $X(\theta)$ at *N* points equally spaced around the unit circle: $\theta = \frac{2\pi k}{N}$

Therefore, the DFT is given by:

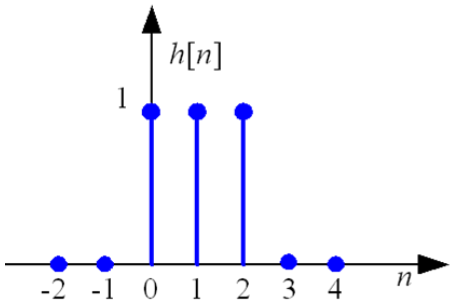
$$X[k] = 1 + 0.2e^{-j\left(\frac{2\pi k}{N}\right)2}$$

$$= 1 + 0.2e^{-j\left(\frac{2\pi k}{25}\right)} \qquad k = 0, 1, 2, \ldots. 49$$

Example 22:

- Find the DFT of the three-sample averager.

$$h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$



$$H[k] = \sum_{n=0}^{N-1} h[n]e^{-jn(\frac{2\pi}{N}k)} = \sum_{n=0}^{2} \frac{1}{3} e^{-jn(\frac{2\pi}{N}k)}$$

$$H[k] = e^{-j(\frac{2\pi}{N}k)} \frac{(1 + 2\,cos(\frac{2\pi k}{N}))}{3}$$

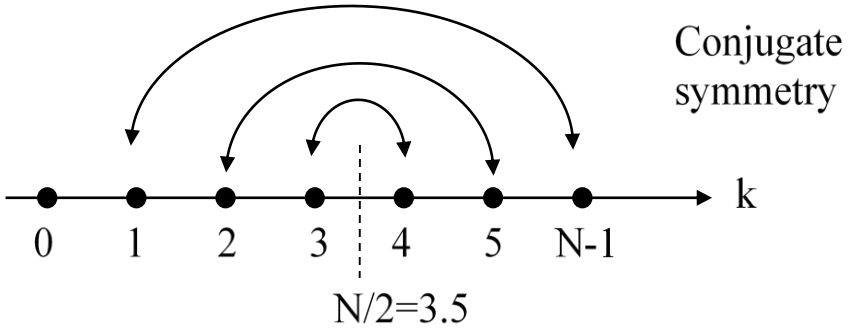
$$|H[k]| = \frac{1}{3}\left|[1 + 2\,cos(\frac{2\pi k}{N})]\right| \quad \text{where } k = 0,1,2$$

When k= 0, we obtain the DC gain or DC value. $|H[0]| = 1$

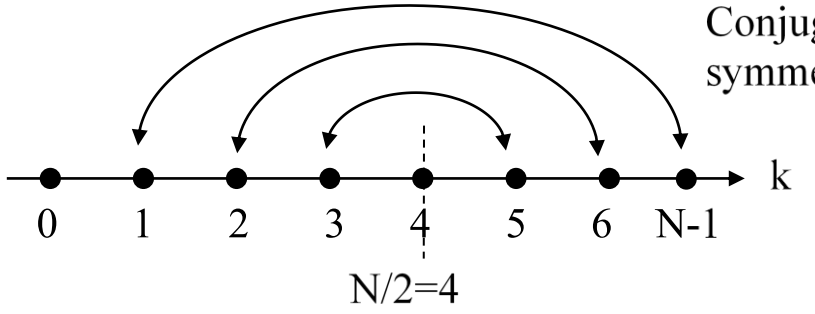
Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Conjugate Symmetry:

- The DFT of a real sequence possesses conjugate symmetry about the origin with $X[-k] = X^*[k]$
- Since the DFT is periodic, $X[-k] = X[N-k]$
- **N is odd** (N=7)



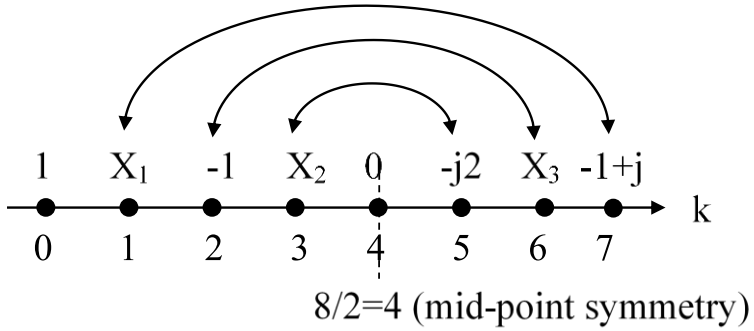
- **N is even** (N=8)



Example 23:

- The DFT of a real signal is $\{1, X_1, -1, X_2, 0, -j2, X_3, -1+j\}$

  Find X_1, X_2 and X_3.



$$X_1 = X^*[7] = -1 - j$$

$$X_2 = X^*[5] = j2$$

$$X_3 = X^*[2] = -1$$

- **Note:** 8 frequency points will result in 8 time samples when IDFT is applied.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Parseval's Theorem

- DFT is an energy-conserving transformation that allows to find the signal energy from the signal or its spectrum.
- Therefore, the sum of squares of the signal samples is related to the sum of squares of the magnitude of the DFT samples:

$$\sum_{n=0}^{N-1}|x[n]|^2=\frac{1}{N}\sum_{k=0}^{N-1}|X[k]|^2$$

Example 24:

- The DFT of a real signal is: $\{1, X_1, -1, X_2, -7, -j2, X_3, -1+j\}$. What is the signal energy? (Note: First value is at k=0 and the last value is at k = 7)

Ans: X_1, X_2 and X_3 can be found first using conjugate symmetry property

$$\text{Signal Energy}=\frac{1}{8}\sum|X[k]|^2$$

$$=\frac{1}{8}\left[1^2+(\sqrt{2})^2+(-1)^2+2^2+7^2+2^2+1^2+\left(\sqrt{2}\right)^2\right]$$

$$\therefore \text{Signal Energy}=8$$

Inverse Discrete Fourier Transform (IDFT)

- The inverse DFT equation comes directly from the Inverse Fourier Transform equation

$$x[n]=\frac{1}{N}\sum_{k=0}^{N-1}X(k)e^{jn\frac{2\pi}{N}k}\quad 0\le n\le N-1$$

However, the signal which it produces will be a periodic signal repeating every *N* samples

Example 25:

- If $X[k]$ = {4, -*j*2, 0, *j*2}, find its IDFT.

Answer:

N=4, $\therefore x[n]=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{2\pi kn}{4}}=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{\pi kn}{2}}$

- $n=0\quad x[0]=\frac{1}{4}[4-j2+0+j2]=1$
- $n=1\quad x[1]=\frac{1}{4}[4-j2e^{j\frac{\pi}{2}}+0+j2e^{j\frac{3\pi}{2}}]=2$
- $n=2\quad x[2]=\frac{1}{4}\left[4-j2e^{j\pi}+0+j2e^{j3\pi}\right]=1$
- $n=3\quad x[3]=\frac{1}{4}[4-j2e^{j\frac{3\pi}{2}}+0+j2e^{j\frac{9\pi}{2}}]=0$
- The IDFT is $x[n]$={1, 2, 1, 0}(Note: When $n=0, x[0]=1$)

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Padding with Zeros and Frequency Resolution

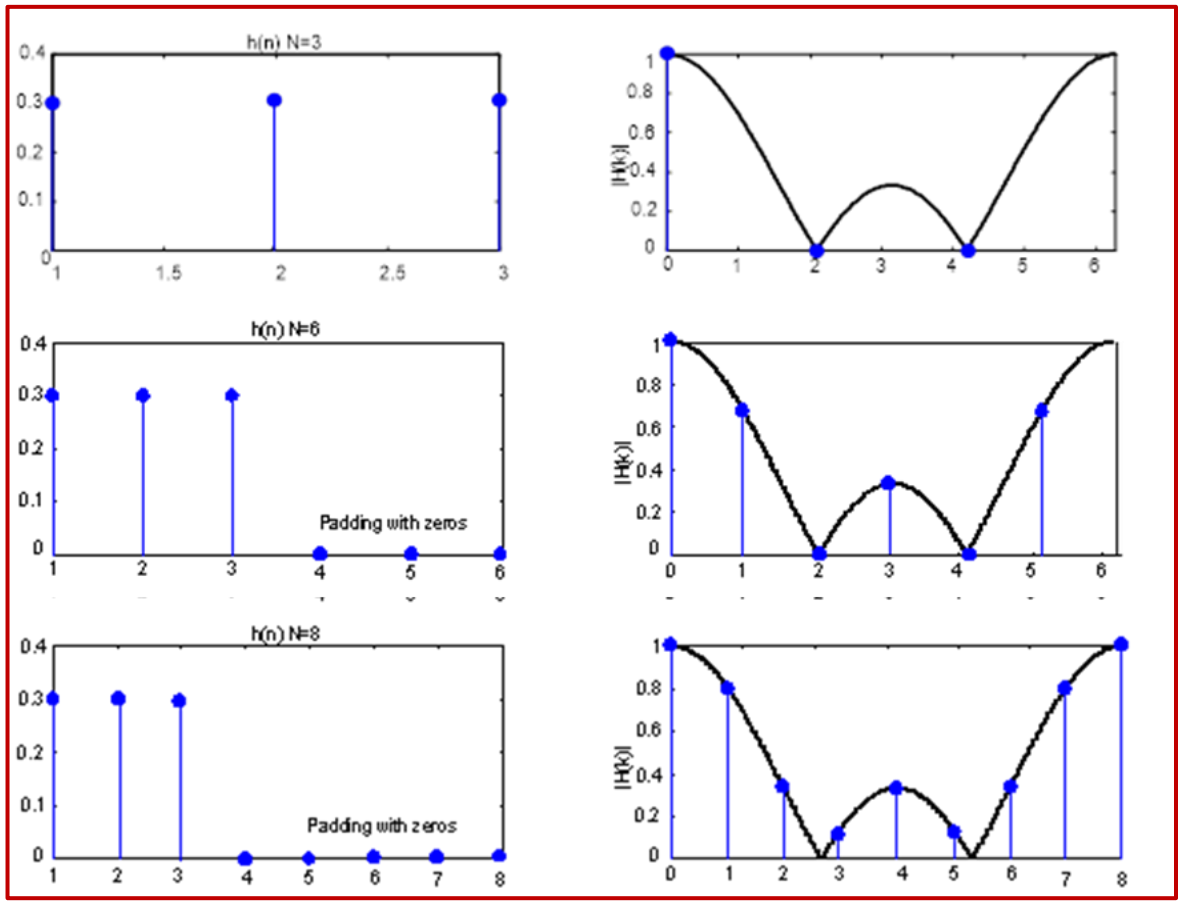
- DFT: $X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\left(\frac{2\pi k}{N}\right)}$
- The frequency resolution depends directly on the number of time samples *N*, i.e. the longer the sequence *x*[*n*], the more frequency points in the DFT domain (frequency domain).
- To obtain more points in the DFT sequence, we can always increase the duration of *x[n]* by adding additional zero-valued elements. This procedure is called padding with zeros.
- These zero-valued elements contribute nothing to the sum in the above equation, but act to decrease the frequency spacing ($2\pi/N$).
- The zero padding gives us a high-density spectrum and provided a better displayed version for plotting. But it does not give us a high resolution spectrum because no new information is added to the signal. Only additional zeros are added in the data.

Example 26:

- The three sample averager can be transformed using both the DTFT and the DFT. Because the DTFT is determined for all $n \in [-\infty,\infty]$, it is unaffected by the window length *N*.
- On the other hand, adding zeros to the three sample average (Example 22) gives a different view of the magnitude spectrum, depending on the length *N*.
- Lengths of *N* = 3, 6 and 8 are shown on the next page .
- Note that although the frequencies at which the DFT is evaluated are different in each case, the underlying spectrum (given by the DTFT) is exactly the same.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Three Sample Averager



Decreasing spectral spacing by zero padding to increase the number of samples. Padding with 3 zeros (N=6) and padding with 5 zeros (N=8)

Fast Fourier Transform (FFT)

- The Fast Fourier Transform (FFT) is simply a mathematical technique to accelerate the calculation of the DFT. It was developed by Cooley and Tukey (1965) requires *N* to be a power of 2.
- Typically, if the DFT is calculated for a block of 2*n* samples e.g. 512 or 1024 samples (*N*) it would make the calculation of the DFT quite demanding.
- The FFT simply uses repetition and redundancy in the calculation to speed it up.
- The FFT is simply a TECHNIQUE to calculate the DFT, NOT a different transform.

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
- Tutorial questions (pp. 43-47)

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

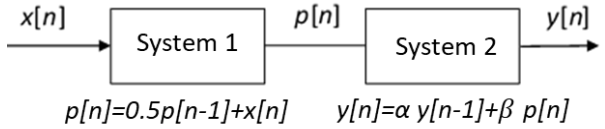
Topic 1: Tutorial Questions (Level 2)

SOLO – Level 2

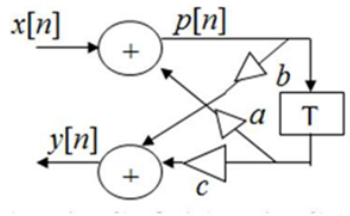
1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ of $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

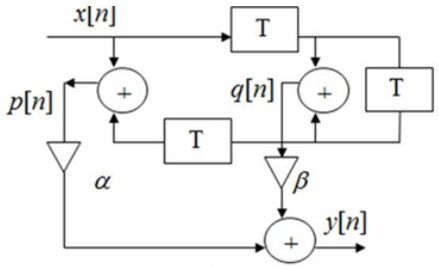
Determine α and β.



2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



(a)



(b)

3. Draw a system implementation for each of the following equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

b) $y[n] = x[n] - x[n-N]$

c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$

4. A difference equation for a particular filter is given by

$$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$$

Find the impulse response of the above filter.

SOLO – Level 2

5. Consider the system described by the difference equation

$$y[n] = ay[n-1] + bx[n]$$

Determine '*b*' in terms of '*a*' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

6. Let $x[n]$ = {1 4 0 2} and $h[n]$ = {1 2 1}. Find their convolution (Both sequences start at n=0).

7. Find the impulse response h[n] for the following system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

8. Consider the following system:

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

a) Sketch the pole-zero pattern. Is this system stable?
b) Determine the impulse response of the system.

9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ 0<a<1

10. A speech signal is sampled with a sampling period of 125ms. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

11. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

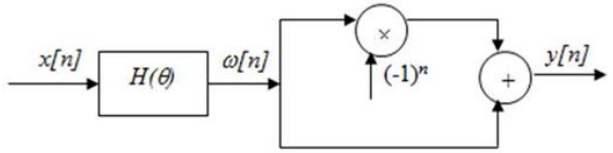
12. Find the 8-point DFT of $x[n]$ ={1, 1, 0, 0, 0, 0, 0, 0}, using conjugate symmetry property of $X[k]$.

Topic 1: Tutorial questions (Levels 2 & 3)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$



14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

SOLO – Level 3

1. If $y[n] = x[n] * h[n]$, then show that Y(z) = X(z) . H(z).

2. Find the z-transform, G(z), if $g[n] = \begin{cases} na^n & n \ge 0 \\ 0 & n < 0 \end{cases}$ where a is a constant

3. The frequency response of a band stop filter is given by

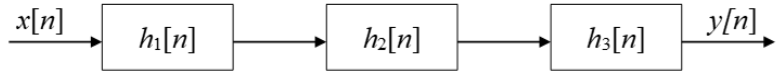
$$H_d(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{6} \\ 0 & \frac{\pi}{6} < |\theta| < \frac{\pi}{3} \\ 1 & \frac{\pi}{3} \le |\theta| \le \pi \end{cases}$$

Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n] = \delta[n] - \frac{\sin\left(\frac{\pi}{3}n\right)}{\pi n} + \frac{\sin\left(\frac{\pi}{6}n\right)}{\pi n}$$

SOLO – Level 3

4. Three systems with the following impulse responses are connected in cascade: $h_1[n] = \delta(n) - \delta(n-1)$; $h_2[n] = g[n]$; $h_3[n] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

5. Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

for all *n*, where *u*[*n*] is the unit step function and $a \neq b$, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

6. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\, y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

Topic 1: Tutorial questions (Level 4)

SOLO – Level 4

1. A system L is known to be linear. The responses of L to inputs $x_1[n]$ and $x_2[n]$ are given by $y_1[n]$ and $y_2[n]$ respectively (see Figure 1).

   (a) Determine whether the system L could be time-invariant.

   (b) If the input x[n] to the system is $\delta[n]$, what is the system response y[n]?

2. Consider the interconnection of LTI systems as shown in Figure 2:

   (a) Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.

   (b) Determine *h[n]* when

   $h_1[n] = \{\frac{1}{2}\ \underset{\uparrow}{\frac{1}{4}}\ \frac{1}{2}\}$; $h_2[n] = h_3[n] = (n+1)\,u[n]$; $h_4[n] = \delta[n-2]$

   (c) Determine the response of the system in part(b) if

   $x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

3. Find a difference equation to implement a system that has a unit impulse response:

   $$h[n] = (\frac{1}{4})^n \cos(n\frac{\pi}{3})\, u[n]$$

4. An Lth order moving average filter is a linear time invariant system that, for an input $x[n]$, produces the output $y[n] = \frac{1}{L+1}\sum_{k=0}^{L} x[n-k]$. Show that the frequency response of this system is $H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \frac{\sin[(L+1)\theta/2]}{\sin(\frac{\theta}{2})}$

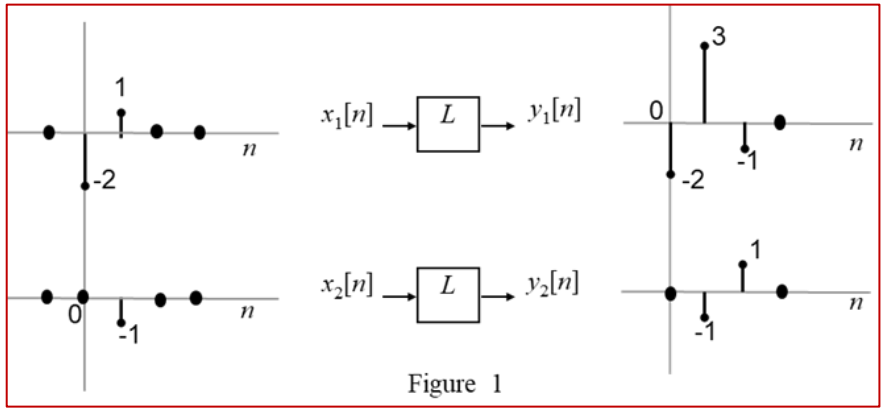


Figure 1

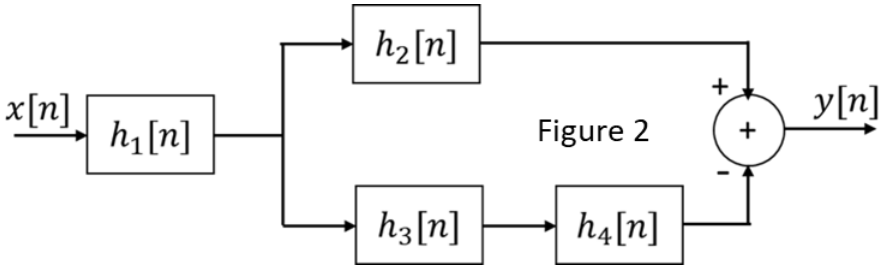


Figure 2

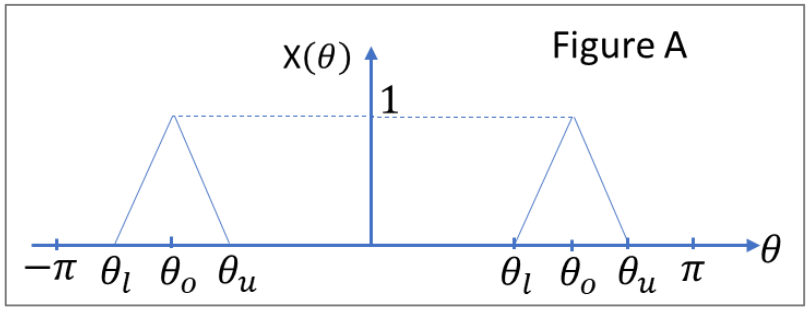
Topic 1: Tutorial questions (Level 5)

SOLO – Level 5

1. Consider a bandpass filter with a spectrum X(θ) as shown in Figure A. Show that x[n] can be formulated as:

$$x[n] = (k_p + k_q)\cos(n\theta_0) - k_p\cos(n\theta_u) - k_q\cos(n\theta_l)$$

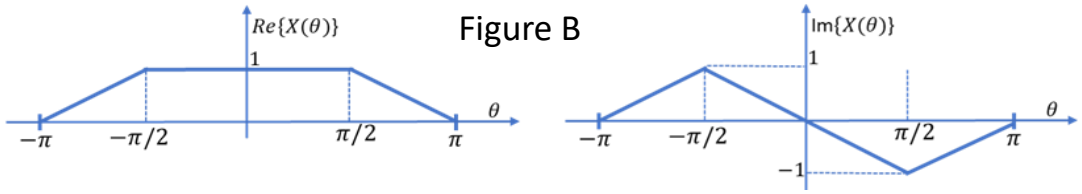
$$k_p = \frac{1}{\pi n^2(\theta_u - \theta_0)}; \qquad k_q = \frac{1}{\pi n^2(\theta_0 - \theta_l)}; \qquad \theta = 2\pi\frac{f}{f_s}$$



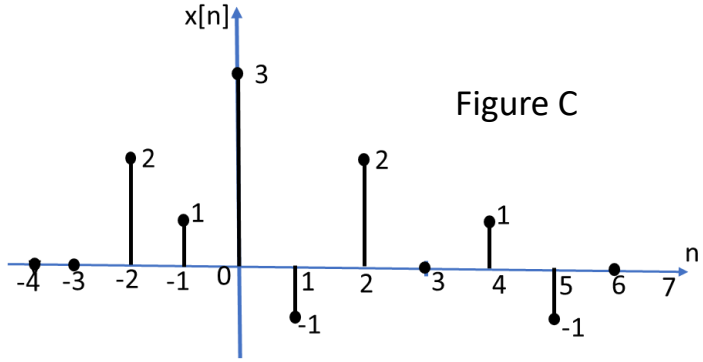
2. The centre of gravity of a sequence is defined by $c = \frac{\sum_{n=-\infty}^{\infty} n\,x[n]}{\sum_{n=-\infty}^{\infty} x[n]}$ and is used as a measure of the *time delay* of a sequence: Find (using the properties of DTFT)

   (a) an expression for *c* in terms of the DTFT of x[n].

   (b) the value of *c* for the sequence x[n] that has a DTFT as shown in Figure B.



3. For the sequence x[n] plotted in Figure C, evaluate the integral $\int_{-\pi}^{\pi}\left|\frac{d}{d\theta}X(\theta)\right|^2 d\theta$.



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Digital Filter Fundamentals:

Introduction

- A digital filter is a discrete-time system that is designed to remove some frequency components or modify some characteristic of a signal, but often the two terms (systems or filters)are used interchangeably.
- There are two types of digital filters:
  (a) FIR (Finite Impulse Response) filters or Non-recursive filters (no feedback path)
  (b) IIR (Infinite Impulse Response) filters or Recursive filters (at least one feedback path present)
- A linear time invariant (LTI) discrete-time system described by the following equation is commonly called a digital filter:

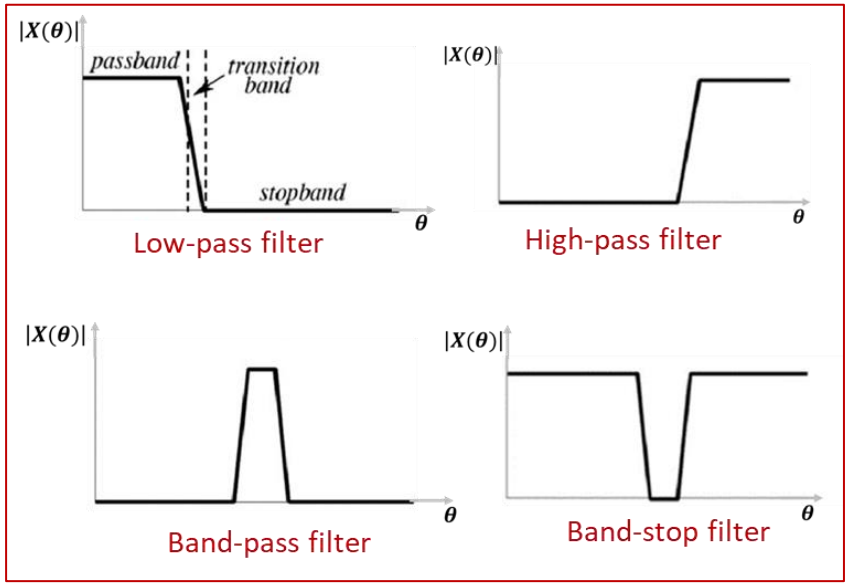
$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

Feed forward Feedback

where $x[n]$ is the input signal, $y[n]$ is the output signal. a_0, a_1 , a_2, ..., a_M; b_1, b_2, b_3, ..., b_L are constants (filter coefficients). These coefficients determine the characteristics of the filter.

- **When $b_k = 0$, the filter is an FIR filter ; when $b_k \neq 0$, the filter is an IIR filter.**





Four Common Digital Filters

ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

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Digital Filter Fundamentals: FIR filter

FIR filter

- Any filter that has an impulse response of finite duration is called Finite Impulse Response (FIR) filter.
- If $b_k = 0$, then the calculation of *y[n]* does not require the use of previously calculated samples of the output (see the equation below):

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

$$\therefore \quad y[n] \quad = \quad \sum_{k=0}^{M} a_k x[n-k]$$

$$= \quad a_0 x[n] + a_1 x[n-1] + \cdots + a_M x[n-k]$$

- This is recognised as a convolution sum:

$$y[n] = \sum_{k=0}^{M} h[k] x[n-k]$$

- Therefore, the impulse response is identical to the coefficients, that is,

$$h[n] = \begin{cases} a_n & 0 \le n \le M \\ 0 & otherwise \end{cases}$$

Example 1:

- The following is a (non-recursive) second order FIR filter:

$$y[n] \quad = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2]$$

$$H(z) = \frac{Y(z)}{X(z)} = a_0 + a_1 z^{-1} + a_2 z^{-2}$$

FIR filter properties

A property of the FIR filter is that it will always be stable:

- **Stability** requires that there should be no poles outside the unit circle. This condition is automatically satisfied since there are no poles at all outside the origin. (In fact, all poles are located at the origin.)
- Another property of FIR filter is that we can make filters with exactly **linear phase** characteristics.
- The ability to have an exactly linear phase response is one of the most important properties of a LTI system (filter).

Digital Filter Fundamentals: FIR filter – Phase Delay & Group Delay

Phase Delay or Group Delay

- When a signal passes through a filter, it is modified in amplitude and/or phase.
- The nature and extent of the modification of the signal is dependent on the amplitude and phase characteristics of the filter.
- The **phase delay** or **group delay** of the filter provides a useful measure of how the filter modified the phase characteristic of the signal.



Phase Delay

- If we consider a signal that consists of several frequency components (eg. speech waveform) the *phase delay* of the filter is the amount of time delay each frequency component of the signal suffers in going through the filter.

$$Phase_delay(T_p) = -\frac{\emptyset(\theta)}{\theta}$$

T_p = the negative of the phase angle divided by frequency

Group Delay

- The *group delay* on the other hand is the average time delay the composite signal suffers at each frequency as it passes from the input to the output of the filter.

$$Group_delay(T_g) = -\frac{d\emptyset(\theta)}{d\theta}$$

T_g = the negative of the derivative of the phase with respect to frequency

Digital Filter Fundamentals: Linear phase FIR filter

Linear Phase Filter

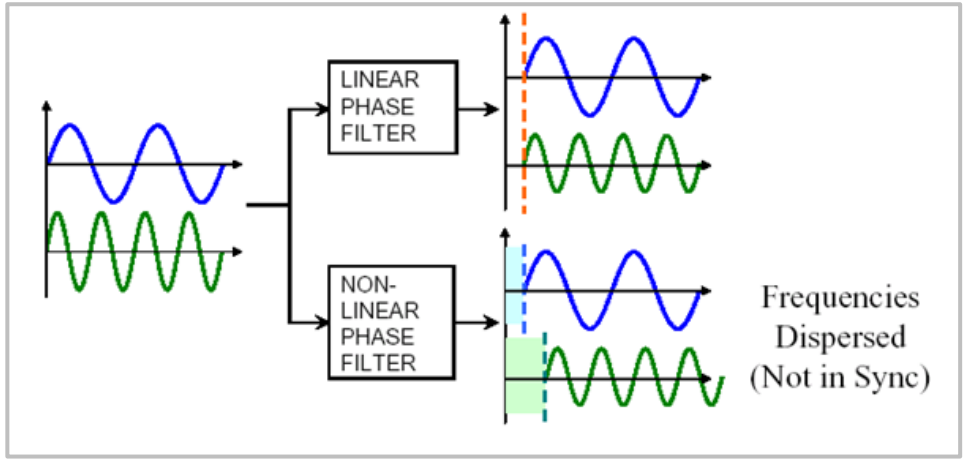
- A *constant group delay* means that signal components at different frequencies receive the same delay in the filter.
- A *linear phase filter* gives same time delay to all frequency components of the input signal. A filter with a nonlinear phase characteristic will cause a phase distortion in the signal that passes through it.
- This is because the frequency components in the signal will each be delayed by an amount not proportional to frequency, thereby altering their harmonic relationship. Such a distortion is undesirable in many applications, for example music, video etc.
- A filter is said to have a **linear phase response** if its phase response satisfies one of the following relationships:

$$\emptyset(\theta) = -a\theta$$

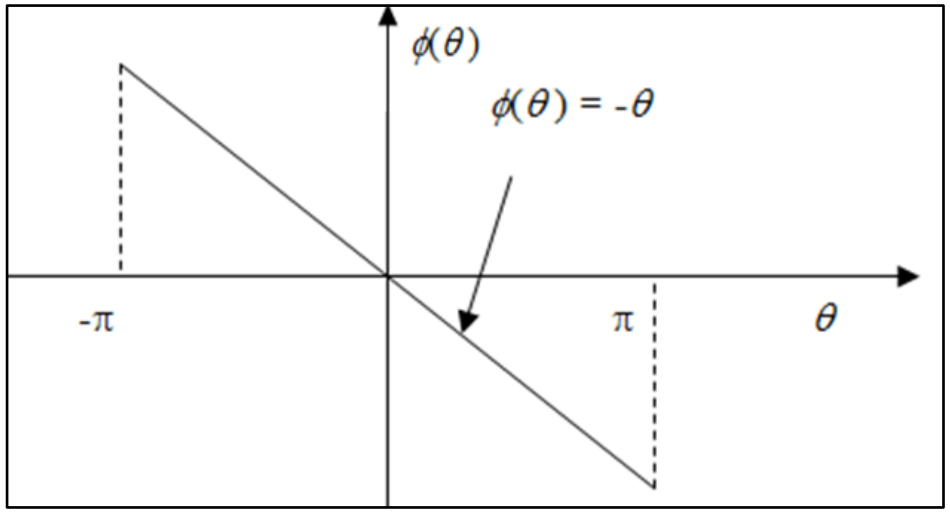
$$\emptyset(\theta) = b - a\theta$$

where '*a*' and '*b*' are constants.

Comparing a linear phase filter with a non-linear phase filter



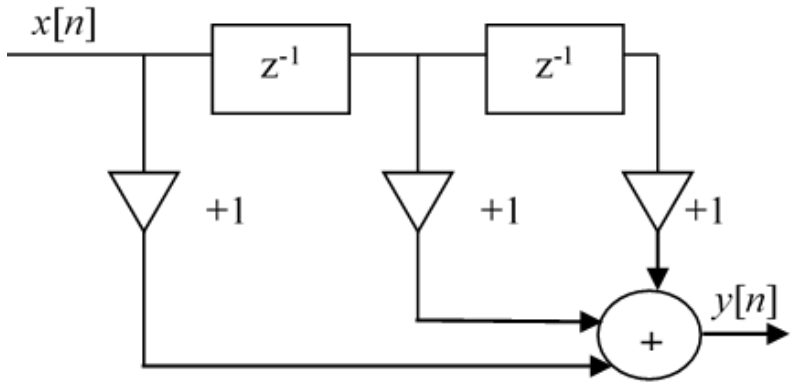
Phase Response of a linear Phase filter



Phase varies linearly with frequency

Digital Filter Fundamentals: Linear phase FIR filter

Example 2: Two filter structures are shown below. Show that both filters have linear phase.



$$y[n] = x[n] + x[n-1] + x[n-2]$$

$$H(z) = 1 + z^{-1} + z^{-2}$$

$$H(\theta) = 1 + e^{-j\theta} + e^{-j2\theta}$$

$$= e^{-j\theta}\left(e^{j\theta} + e^{-j\theta} + 1\right)$$

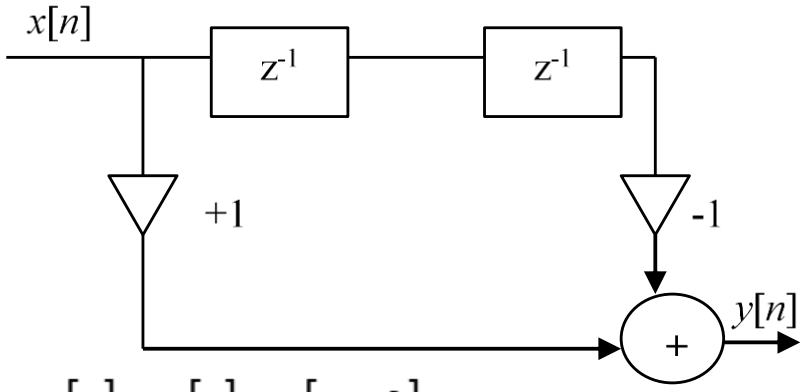
$$= e^{-j\theta}(1 + 2\cos\theta)$$

Let $B(\theta) = 1 + 2cos\theta$

Phase: $\phi(\theta) = -\theta \quad B(\theta) \geq 0$

$\phi(\theta) = -\theta \ \pm\pi \quad B(\theta) < 0$

- If $\emptyset(\theta) = -\theta\ (Linear\ Phase)$ the impulse response is always even symmetric about its midpoint.



$$y[n] = x[n] - x[n-2]$$

$$H(z) = 1 - z^{-2}$$

$$H(\theta) = 1 - e^{-j2\theta}$$

$$= 2je^{-j\theta}\left(\frac{e^{j\theta} - e^{-j\theta}}{2j}\right)$$

$$= 2e^{j\frac{\pi}{2}}e^{-j\theta}\sin\theta$$

$$= e^{j\left(\frac{\pi}{2}-\theta\right)}(2\sin\theta)$$

Let $B(\theta) = 2sin\theta$

Phase: $\phi(\theta) = \pi/2 - \theta \quad B(\theta) \geq 0$

$\phi(\theta) = (\pi/2 - \theta) \pm\pi \quad B(\theta) < 0$

- If $\emptyset(\theta) = \pi/2 - \theta$, the impulse response is always odd symmetric about its midpoint

The frequency response of linear phase filter may be written as

$$H(\theta) = A(\theta)\, e^{j\alpha\theta}$$

(for even symmetric h[n])

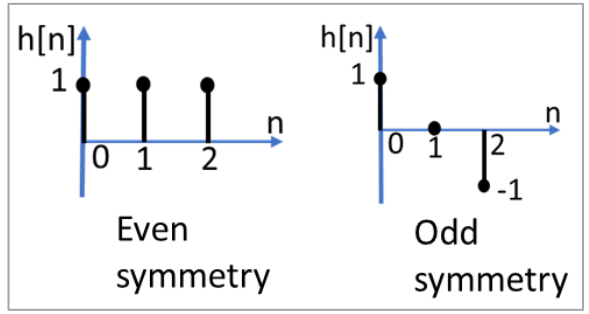
or $H(\theta) = je^{j\alpha\theta}A(\theta)$

$$= e^{j\left(\alpha\theta+\frac{\pi}{2}\right)}A(\theta)$$

(for odd symmetric h[n])

where $A(\theta)$ is real.

For a linear phase FIR filter its coefficients are symmetrical about its midpoint

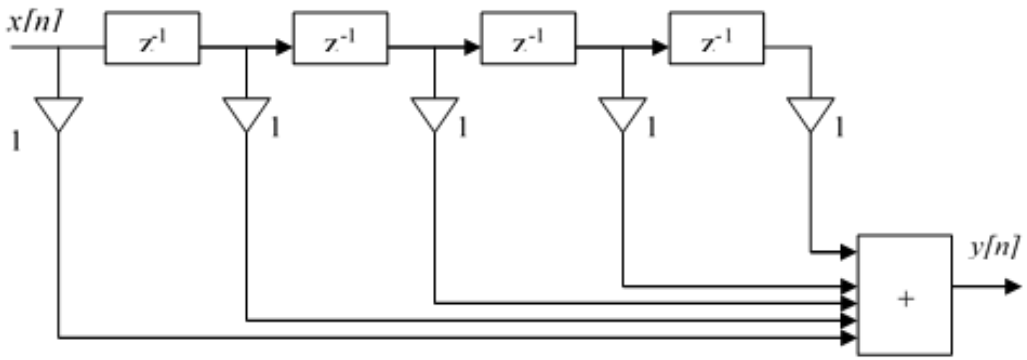


Digital Filter Fundamentals: Linear phase FIR filter

Example 3:

a) Show that the filter structure shown on the right has a linear phase characteristic equation given by: $\phi(\theta) = -2\theta$.

b) Show for the above liner phase filter , both the phase delay and the group delay are constant.



Answer:

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

Let N = number of coefficients of the liner phase filter

$$H(z) = \frac{1 - z^{-N}}{1 - z^{-1}} \Rightarrow H(\theta) = \frac{1 - e^{-j\theta N}}{1 - e^{-j\theta}}$$

$$H(\theta) = \frac{e^{-j\frac{\theta N}{2}}}{e^{-j\frac{\theta}{2}}} \frac{[e^{j\frac{\theta N}{2}} - e^{-j\frac{\theta N}{2}}]}{[e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}]} = e^{-j\frac{N-1}{2}\theta} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\frac{\theta}{2}}$$

$$N = 5 \quad \Rightarrow \quad H(\theta) = e^{-j2\theta}\left\{\frac{\sin\frac{5\theta}{2}}{\sin\frac{\theta}{2}}\right\} = B(\theta)$$

$\therefore \phi(\theta) = -2\theta \qquad B(\theta) \geq 0$

$\phi(\theta) = -2\theta \ \pm\pi \quad B(\theta) < 0$

$$Phase_delay(T_p) = -\frac{\varphi(\theta)}{\theta} = -\frac{-2\theta}{\theta}$$

$$= 2\ (constant)$$

$$Group_delay(T_g) = -\frac{d\varphi(\theta)}{d\theta} = -\frac{d[-2\theta)]}{d\theta}$$

$$= 2\ (constant)$$

Digital Filter Fundamentals: IIR filter

IIR filter

- Every recursive (Infinite Impulse Response) digital filter must contain at least one closed loop. Each closed loop contains at least one delay element.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

- For recursive digital filters $b_k \neq 0$. For example, $a_k = 0$ for $k \geq 1$ and & also $b_k = 0$ for $k \geq 1$.

$$\therefore \quad y[n] = a_0 x[n] - b_1 y[n-1]$$

$$H(z) = \frac{a_0}{1+b_1 z^{-1}} \qquad \text{[IIR filter]}$$

- A recursive filter is an infinite impulse response filter (IIR).

Second Order Filters

Transfer function	Description
$H_1(z) = a_0 + a_1 z^{-1} + a_2 z^{-2}$	all-zero second order FIR filter (excluding poles at origin)
$H_2(z) = \dfrac{1}{1+b_1 z^{-1} + b_2 z^{-2}}$	all-pole second order IIR filter (excluding zeros at origin)
$H_3(z) = \dfrac{1+a_1 z^{-1} + a_2 z^{-2}}{1+b_1 z^{-1} + b_2 z^{-2}}$	pole-zero second order IIR filter

Note: Poles and zeros can be real or imaginary

Digital Filter Fundamentals: IIR filter – DC gain and 3 dB bandwidth

Example 4: DC gain calculation

- Consider the 1st order IIR filter

$$H(z) = \frac{1}{1 - az^{-1}} \quad 0 < a < 1$$

The difference equation is:

$$y[n] = x[n] + a\,y[n-1].$$

- The DC gain of $H(z)$ can be obtained by substituting $\theta = 0$.

$$H(\theta)|_{\theta=0} = \frac{1}{1 - ae^{-j(0)}} = \frac{1}{1-a}$$

- If DC gain is undesirable, introduce a constant gain factor of (1-a), so that $H(z)$ becomes

$$H(z) = \frac{1-a}{1-az^{-1}} \quad \text{[DC gain =1]}$$

$$\therefore\ y[n] = (1-a)\,x[n] + a\,y[n-1]$$

Example 5: 3dB bandwidth calculation

- Consider a lowpass filter $y[n] = ay[n-1] + bx[n], 0 < a < 1$

(i) Determine '*b*' so that $|H(0)| = 1$.
(ii) Determine the 3dB bandwidth for the normalised filter in part (i).

Answer:

$$Y(z) = a\,Y(z)z^{-1} + b\,X(z) \Rightarrow H(z) = \frac{b}{1-az^{-1}} \Rightarrow H(\theta) = \frac{b}{1-ae^{-j\theta}}$$

we have $|H(0)| = 1;\ H(0) = \frac{b}{1-a} = 1 \Rightarrow b = 1 - a$

(ii) $H(\theta) = \frac{1-a}{1-ae^{-j\theta}} = \frac{1-a}{(1-a\cos\theta)+ja\sin\theta};\ |H(\theta)| = \frac{1-a}{\sqrt{1+a^2-2a\cos\theta}}$

Second Method: $|H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$ {$H^*(\theta)$ is the complex conjugate of $H(\theta)$}

$$\left\{|H(\theta)|^2 = \frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1-2a\cos\theta+a^2}\right\}$$

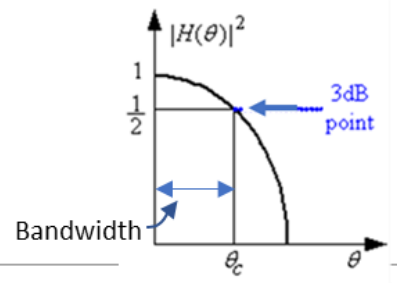
θ_c is the cut-off frequency at which:

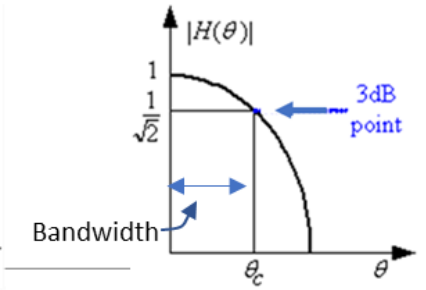
$$|H(\theta)|^2_{\theta=\theta_c} = \frac{1}{2}|H(0)|^2 \quad \textbf{or} \quad |H(\theta)|_{\theta=\theta_c} = \frac{1}{\sqrt{2}}|H(0)|$$

We have $|H(0)|^2 = 1$ *when* $\theta = 0$;

$$\frac{1}{2} = \frac{(1-a)^2}{1+a^2-2a\cos\theta_c}$$

$$\theta_c = \cos^{-1}\left[\frac{-a^2+4a-1}{2a}\right]$$





ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (pp. 3-10)
- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
- Minimum, Maximum, Mixed Phase Filters (PP. 24-28)
- All-Pass filters (pp. 29-31)
- Tutorial questions (pp. 32-38)

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Digital Filter Fundamentals: Digital filter Structures

Digital Filter Structures

- It is common for a digital filter to be implemented by combining simple digital filter structures (e.g. Direct forms I & II) to form a larger composite filter.
- There are two common ways of combining simple structures, either in **cascade** or in **parallel**.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] - \sum_{k=1}^{L} b_k y[n-k]$$

$$Y(z) = \sum_{k=0}^{M} a_k X(z) z^{-k} - \sum_{k=1}^{L} b_k Y(z) z^{-k}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} a_k z^{-k}}{1 + \sum_{k=1}^{L} b_k z^{-k}}$$

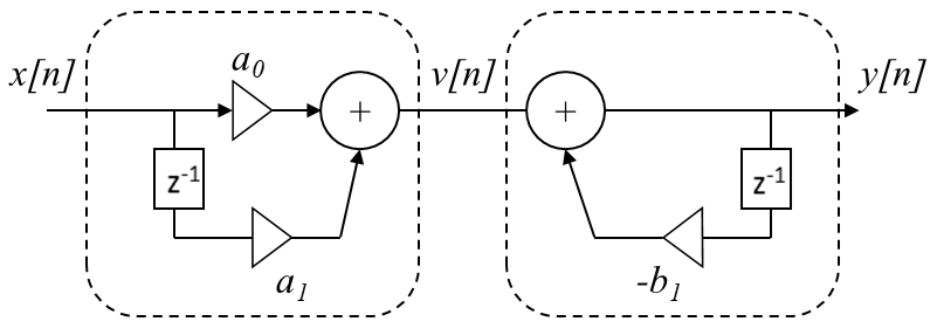
- **Direct form I:**

$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$

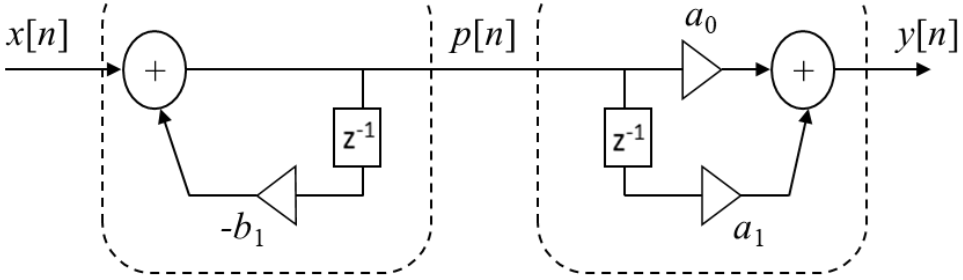
- **Direct form II:**

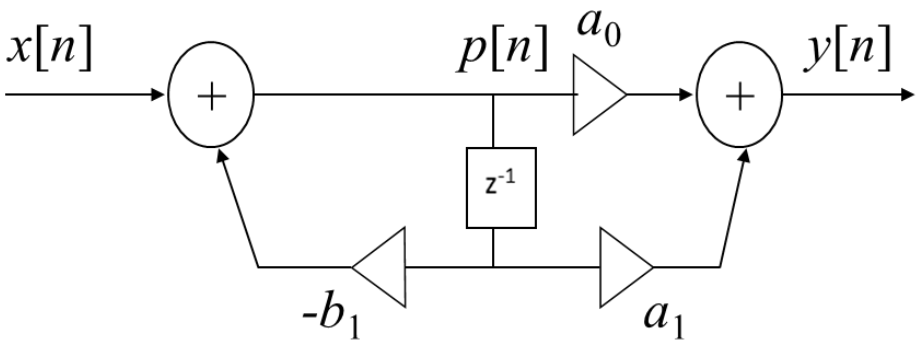
$$H(z) = \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

Direct form I



Direct form II





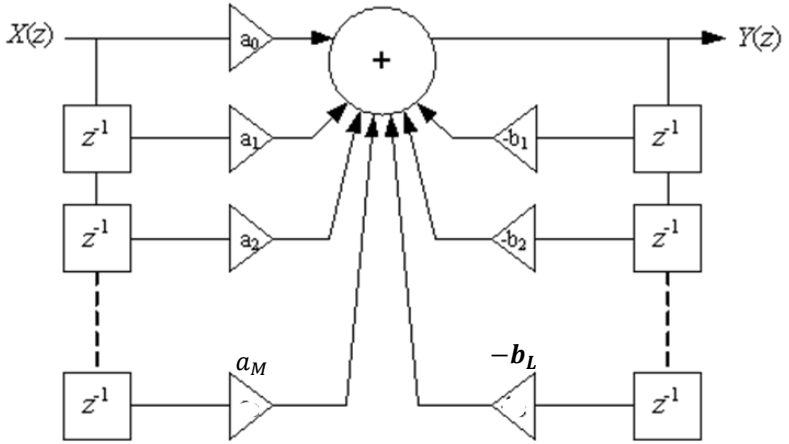
Canonic form

Digital Filter Fundamentals: Digital filter Structures

Direct Form I:

- **Direct form I:**

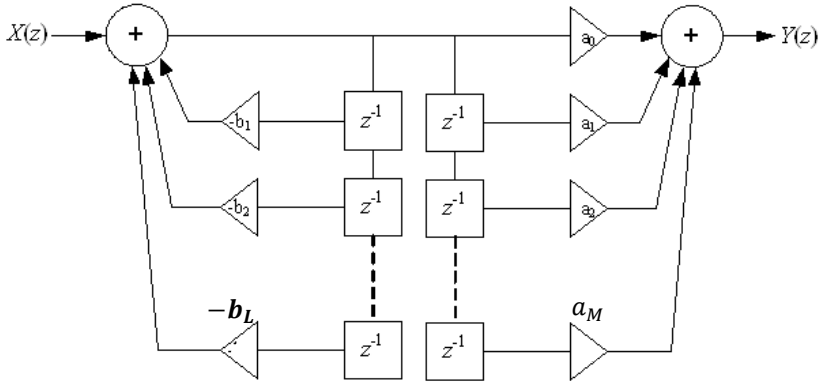
$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$



Direct Form II:

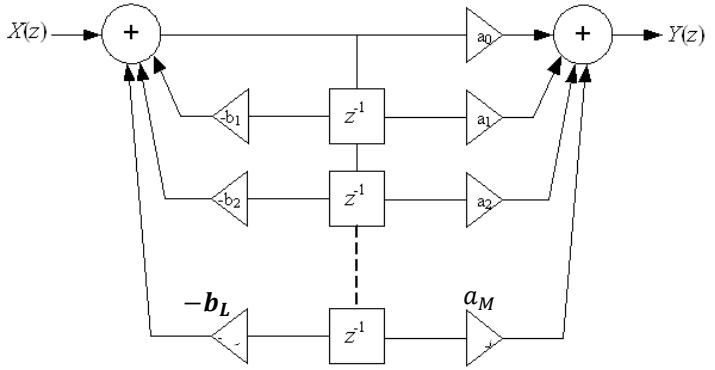
- **Direct form II:**

$$H(z) = \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

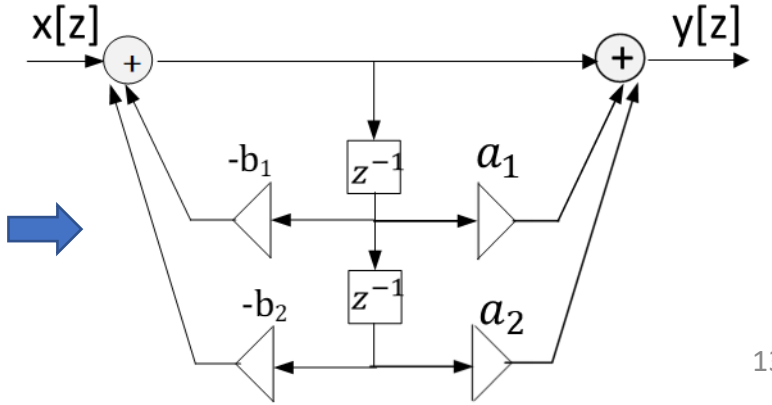


Canonic Form:

- In the case when *L* = *M*, we have canonic form realisation.



A discrete-time filter is said to be canonic if it contains the minimum numbers of delay elements necessary to realise the associated frequency response.



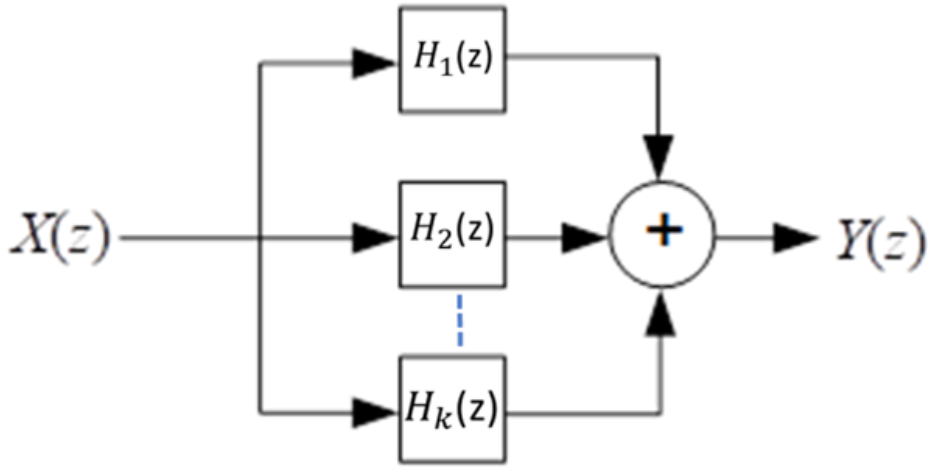
Digital Filter Fundamentals: IIR filter – Parallel Structure

Parallel Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$= \underbrace{\sum_{i=1}^{k} H_i(z) \quad = \quad H_1(z) + H_2(z) + H_3(z) + \ldots + H_k(z)}_{parallel_structure}$$

Note: Use partial fraction to obtain $H_i(z)$



Parallel Structure

Example 7:

- Draw a parallel realisation of a third order system $H(z)$ as given below: $H(z) = \frac{z^3 - z^2 - 5z - 3}{5z(z^2 - 4)}$

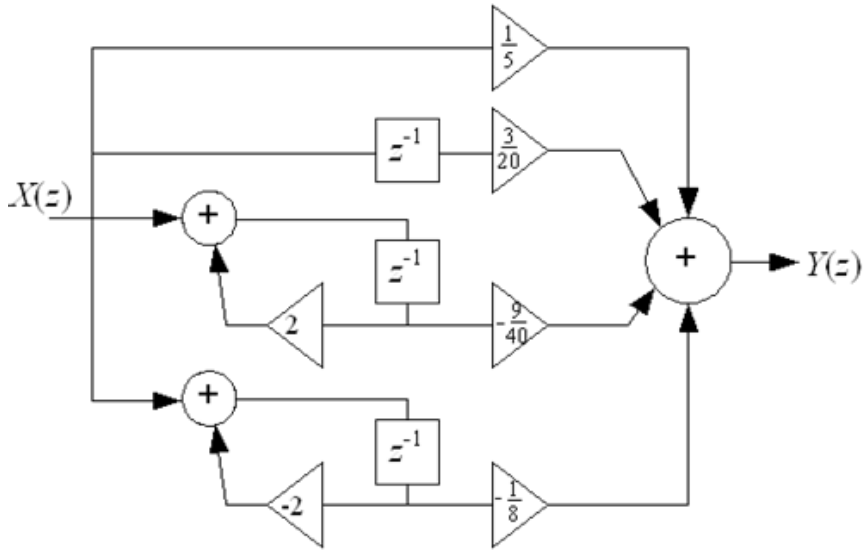
Answer:

By using a partial fraction decomposition, $H(z)$ becomes:

$$H(z) = A + \frac{B}{z} + \frac{C}{z-2} + \frac{D}{z+2}$$

$$H(z) = \frac{1}{5} + \frac{3}{20z} - \frac{9}{40(z-2)} - \frac{1}{8(z+2)}$$

$$H(z) = \frac{1}{5} + \frac{3}{20} z^{-1} - \frac{9z^{-1}}{40(1-2z^{-1})} - \frac{z^{-1}}{8(1+2z^{-1})}$$



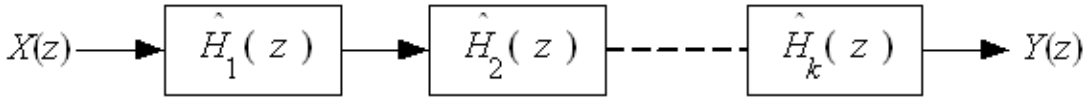
Digital Filter Fundamentals: IIR filter – Cascade Structure

Cascade Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$H(z) = \prod_{i=1}^{k} \hat{H}_i(z) = \underbrace{\hat{H}_1(z) \cdot \hat{H}_2(z) \cdot \hat{H}_3(z) \ldots \hat{H}_k(z)}_{cascade\ structure}$$

Note: H(z) now becomes the product of lower order transfer functions $\hat{H}_1(z)$, $\hat{H}_2(z)$ *etc* i.e. 1st or 2nd order sections. The cascade structure is the most popular form in digital filter implementation.



Cascade Structure

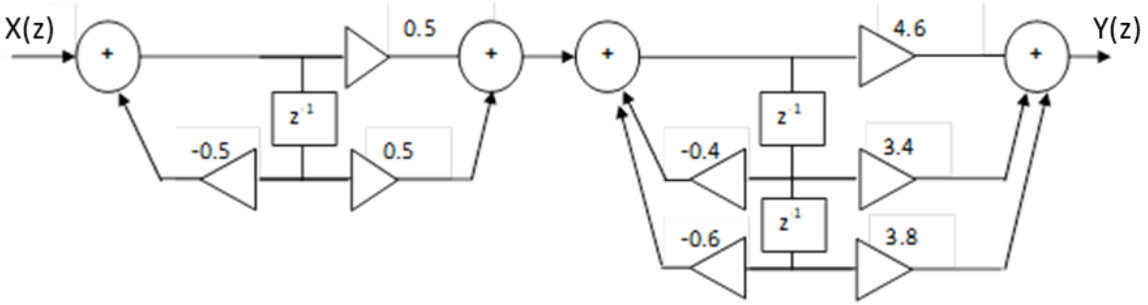
Example 8:

- A cascade realisation of a third-order filter is given by

$$H(z) = \frac{23 + 40z^{-1} + 36z^{-2} + 19z^{-3}}{10 + 9z^{-1} + 8z^{-2} + 3z^{-3}} = \frac{(1 + z^{-1})}{(2 + z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5 + 2z^{-1} + 3z^{-2}}$$

$$= \frac{(1 + z^{-1})}{2(1 + 0.5z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5(1 + 0.4z^{-1} + 0.6z^{-2})} = \left(\frac{(0.5 + 0.5z^{-1})}{1 + 0.5z^{-1}}\right)\left(\frac{4.6 + 3.4z^{-1} + 3.8z^{-2}}{1 + 0.4z^{-1} + 0.6z^{-2}}\right)$$

1st order 2nd order



Example 9:

- Draw a cascade realisation of the filter H(z):

$$H(z) = \frac{1}{(1 - az^{-1})^3}$$

Note: No parallel structure exists for H(z), because partial fraction expansion cannot be performed



ELEC3104: Digital Signal Processing

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Term 3, 2021

Digital Filter Fundamentals: Magnitude and Phase Responses

Phase Response

- We can show that the magnitude response is an **even** function of frequency.
- The phase response is an **odd** function of frequency.
- For example, let us consider the following linear-phase form of the transfer function:

$$H(\theta) = e^{-jk\theta} B(\theta) \quad k > 0$$

The real-valued function of $B(\theta)$ that can take positive and negative values.

$$H(\theta) = B(\theta)\cos(-k\theta) + B(\theta)j\sin(-k\theta)$$

$$H(\theta) = B(\theta)\cos(k\theta) - jB(\theta)\sin(k\theta)$$

Let the phase angle be ϕ:

$$\tan\phi = -\frac{B(\theta)\sin(k\theta)}{B(\theta)\cos(k\theta)} = -\tan(k\theta)$$

$\therefore \tan\phi = \tan(-k\theta) \Rightarrow \phi(\theta) = -k\theta$

- The phase function $\phi(\theta)$ includes linear phase term and also accommodates for the sign changes in $B(\theta)$.
- Since -1 can be expressed as $e^{\pm j\pi}$, phase jumps of $\pm\pi$ will occur at frequencies where $B(\theta)$ changes sign. Therefore,

If $B(\theta) > 0$, then $\phi(\theta) = -k\theta$.
If $B(\theta) < 0$, then $\phi(\theta) = -k\theta \pm \pi$..

Phase Jumps

- A jump of $\pm 2\pi$ occurs to maintain the phase function within the principal value range of [$-\pi$ and π]
- A jump of $\pm\pi$ occurs when $B(\theta)$ undergoes a change of sign
- The sign of the phase jump is chosen such that the resulting phase function is odd and, after the jump, lies in the range [$-\pi$ and π]

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 10: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & -1 \le n \le 1 \\ 0 & otherwise \end{cases}$$

Answer:

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n} = \sum_{n=-1}^{1} h(n)z^{-n} = \frac{1}{3}z^{-1} + \frac{1}{3}z^{0} + \frac{1}{3}z^{1}$$

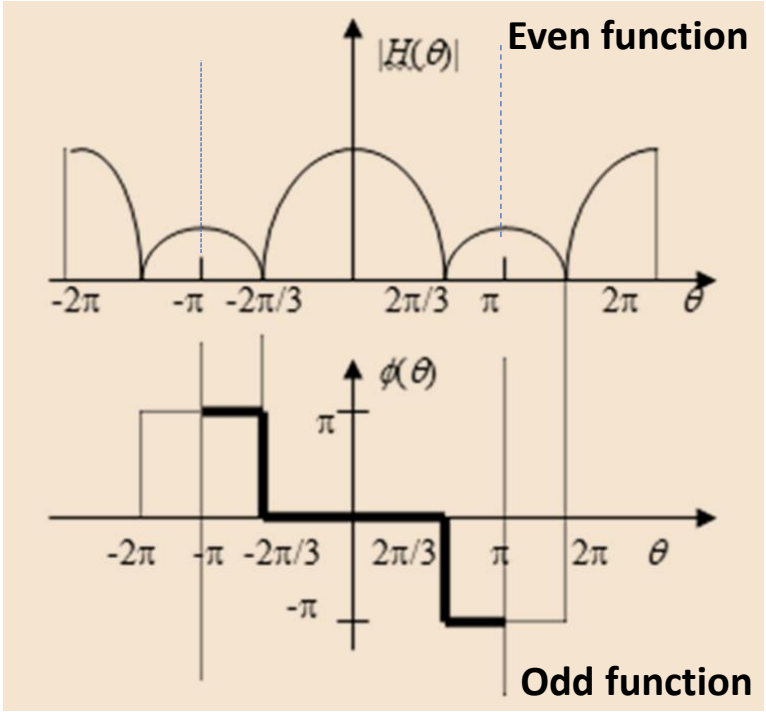
$$H(z) = \frac{1}{3}[z^{-1} + 1 + z^{1}]$$

$$H(\theta) = H(z)\Big|_{z=e^{j\theta}} = \frac{1}{3}\left[e^{-j\theta} + 1 + e^{j\theta}\right] = \left[1 + e^{j\theta} + e^{-j\theta}\right] \cdot \frac{1}{3}$$

$$H(\theta) = \frac{1}{3}[1 + 2\cos\theta] \Rightarrow |H(\theta)| = \left|\frac{1}{3}[1 + 2\cos\theta]\right|$$

$$\begin{cases} \emptyset(\theta) = 0 & H(\theta) \ge 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = 0 \pm \pi & H(\theta) < 0 \quad -\pi \le \theta \le -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta < \pi \end{cases}$$

- The appropriate sign of π must be chosen to make $\phi(\theta)$ an odd function of frequency.



Digital Filter Fundamentals: Magnitude and Phase Responses

Example 11: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Answer:

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2} \Rightarrow H(\theta) = \frac{1}{3} + \frac{1}{3}e^{-j\theta} + \frac{1}{3}e^{-j2\theta}$$

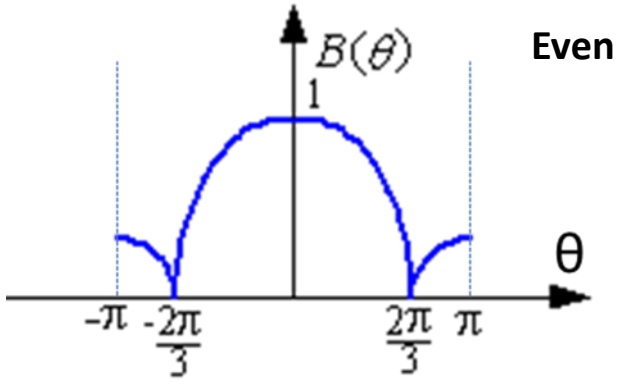
$$H(\theta) = \frac{1}{3}e^{-j\theta}\left[e^{j\theta} + 1 + e^{-j\theta}\right]$$

$$\therefore H(\theta) = e^{-j\theta}\underbrace{\frac{1}{3}[1 + 2\cos\theta]}_{B(\theta)}$$

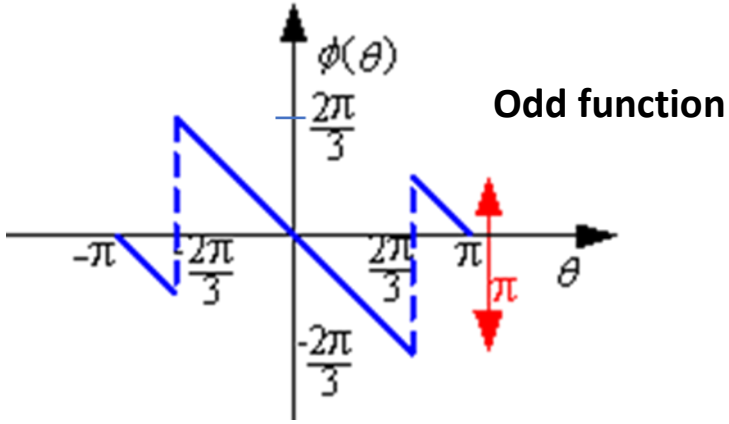
$$|H(\theta)| = |B(\theta)| = \frac{1}{3}[1 + 2\cos\theta] \quad \text{and} \quad \emptyset(\theta) = -\theta;$$

$$\begin{cases} \emptyset(\theta) = -\theta; & B(\theta) \ge 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = -\theta \pm \pi & B(\theta) < 0 \quad -\pi \le \theta \le -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta < \pi \end{cases}$$

- Phase is undefined at points $|H(\theta)| = 0$ or B(θ)=0.



Even function



Odd function

Phase Jumps: A jump of ± π occurs when $B(\theta)$ undergoes a change of sign.

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 12:

- Find the magnitude and phase response of the following:

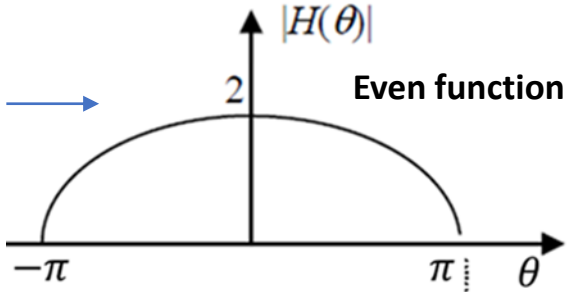
$h(0) = \frac{1}{2}, \quad h(1) = 1, \quad h(2) = \frac{1}{2}, \quad h(n) = 0$, otherwise.

Answer:

$$H(z) = \frac{1}{2}z^0 + 1z^{-1} + \frac{1}{2}z^{-2} = \frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}$$

$$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta} = e^{-j\theta}\left[\frac{1}{2}e^{j\theta} + 1 + \frac{1}{2}e^{-j\theta}\right]$$

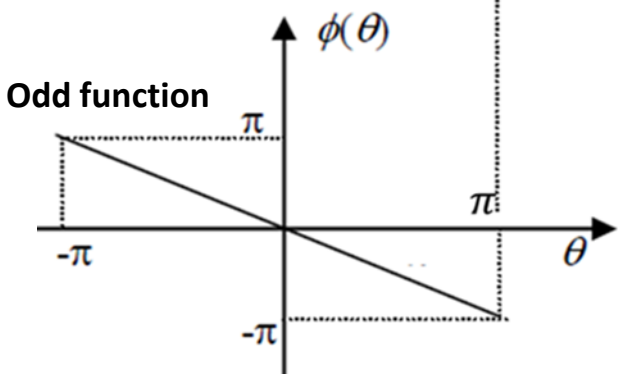
$$H(\theta) = e^{-j\theta}\underbrace{[1 + cos\,\theta]}_{B(\theta)};$$



B(θ)is never negative; therefore, there is no phase jumps of $\pm\pi$.

Phase Response:

$\emptyset(\theta) = -\theta$

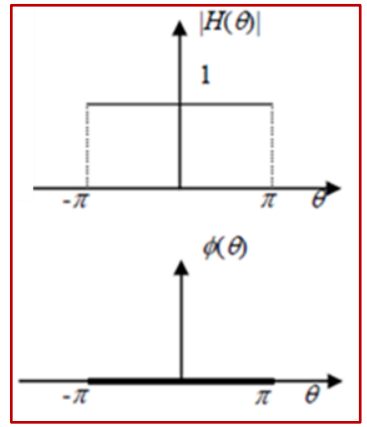


Example 13:

- **Case 1:**

$$\left.\begin{array}{ll}\delta(n) = 1 & n = 0 \\ \delta(n) = 0 & otherwise\end{array}\right\}$$

$$\therefore\ H(\theta) = 1\ ;$$
$$\emptyset(\theta) = 0$$



- **Case 2:** $h[n] = \delta[n-k];$

$$\therefore H[z] = 1\,z^{-k};$$

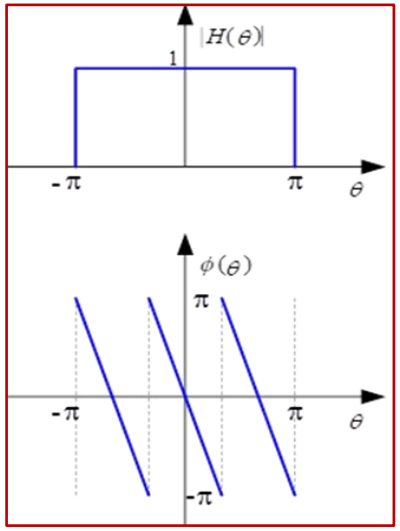
$$H(\theta) = 1\,e^{-jk\theta}$$

$$\mathrm{H}(\theta) = \mathrm{B}(\theta) = 1\ ;$$

$$\phi(\theta) = -\theta\,k;$$

Assume k =3 ; $\emptyset(\theta) - 3\theta$

When phase exceeds ±π range a jump of ±2π is needed to bring the phase back into ±π range.



Digital Filter Fundamentals: Magnitude and Phase Responses

Example 14: Magnitude and Phase Responses of a Low-pass IIR filter

- Determine and sketch the magnitude and phase response of 1st order recursive filter (IIR filter):

$$y[n] = x[n] + ay[n-1] \qquad 0 < a < 1$$

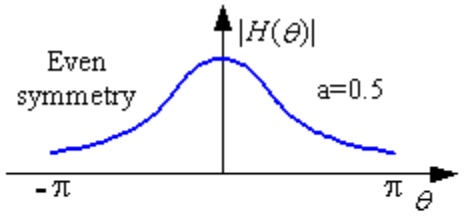
Answer:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1-az^{-1}}; \qquad H(\theta) = \frac{1}{1-ae^{-j\theta}}$$

$$H(\theta) = \frac{1}{1-ae^{-j\theta}} \cdot \frac{1-ae^{j\theta}}{1-ae^{j\theta}} = \frac{1-a\cos\theta}{1-2a\cos\theta+a^2} - \frac{ja\sin\theta}{1-2a\cos\theta+a^2}$$

$$\tan\emptyset = \frac{-\dfrac{a\sin\theta}{1-2a\cos\theta+a^2}}{\dfrac{1-a\cos\theta}{1-2a\cos\theta+a^2}} = \frac{-a\sin\theta}{1-a\cos\theta}$$

Phase: $\emptyset(\theta) = \tan^{-1}\left[\frac{-a\sin\theta}{1-a\cos\theta}\right]$;

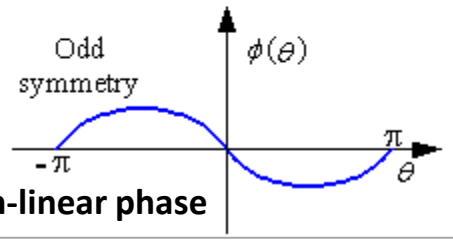


Magnitude:

$$|H(\theta)|^2 = H(\theta).H^*(\theta);$$

$$= \frac{1}{1-ae^{-j\theta}} \cdot \frac{1}{1-ae^{j\theta}}$$

$$|H(\theta)| = \frac{1}{1-2a\cos\theta+a^2};$$



Non-linear phase

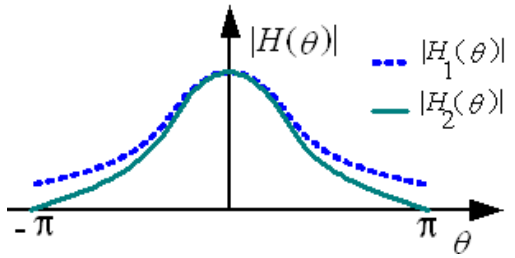
Example 15: Magnitude Responses of IIR filters

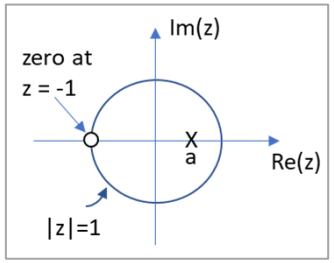
(a) $H_1(z) = \frac{k_0}{1-az^{-1}} \Rightarrow \frac{1-a}{1-az^{-1}} \qquad 0 < a < 1$

The gain k_0 can be selected as $1 - a$, so that the **low-pass filter** has unity gain at $\theta = 0$.

(b) $H_2(z) = k_0\frac{1+z^{-1}}{1-az^{-1}}$; In this case, $k_0 = \frac{1-a}{2}$ (for unity gain at $\theta = 0$).

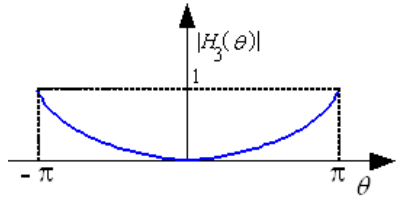
The addition The addition of a zero at z = -1 further attenuates the response of the filter at high frequencies (see magnitude response below)

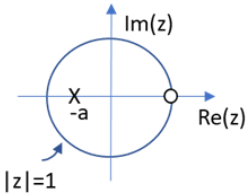




(c) We can obtain simple **high-pass filters** by reflecting (folding) the pole-zero locations of the lowpass filters about the imaginary axis in the z-plane

$$H_3(z) = \frac{1-a}{2} \cdot \frac{1-z^{-1}}{1+az^{-1}}$$





Digital Filter Fundamentals: Magnitude Responses

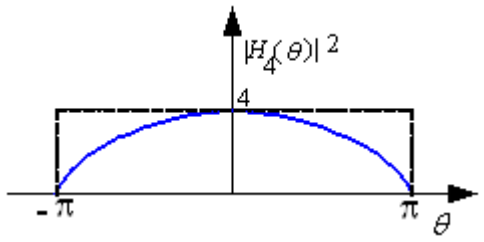
Example 16: FIR filters

- **Low-pass filter:**

$H_4(z) = 1 + z^{-1}$;

$y[n] = x[n] + x[n-1]$

$|H_4(\theta)|^2 = 2(1+\cos\theta)$

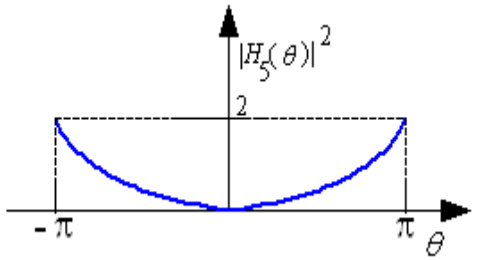


- **High-pass filter:**

$H_4(z) = 1 - z^{-1}$;

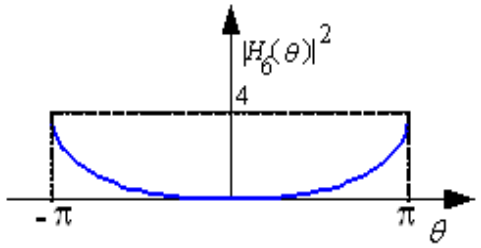
$y[n] = x[n] - x[n-1]$

$|H_5(\theta)|^2 = 2(1-\cos\theta)$



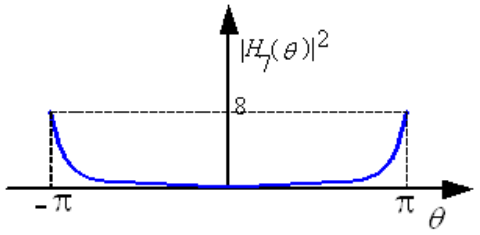
- **High-pass filter:**

$H_6(z) = (1 - z^{-1})^2$



- **High-pass filter:**

$H_7(z) = (1 - z^{-1})^3$

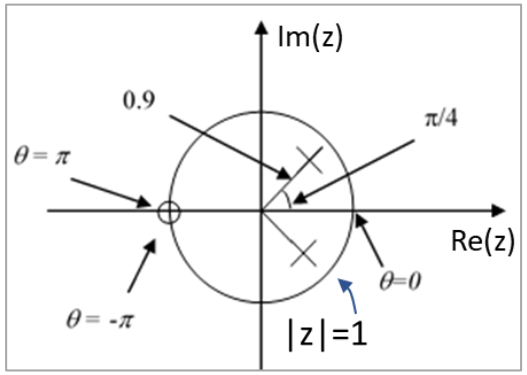


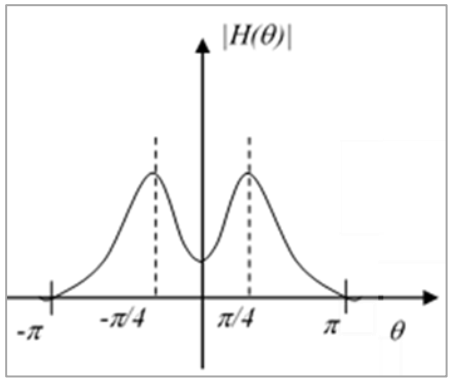
Example 17: IIR Resonant filters

- Sketch an approximate magnitude response for the system having the transfer function

$$H(z) = \frac{1 + z^{-1}}{(1 - 0.9e^{j\frac{\pi}{4}}z^{-1})(1 - 0.9e^{-j\frac{\pi}{4}}z^{-1})}$$

- The system has a zero at z = -1 & poles at $z = 0.9\, e^{-j\frac{\pi}{4}}$
- Magnitude response will be zero at $\theta = \pi$ and large at $\theta_0 = \pm\, \pi/4$ because the poles are close to the unit circle.





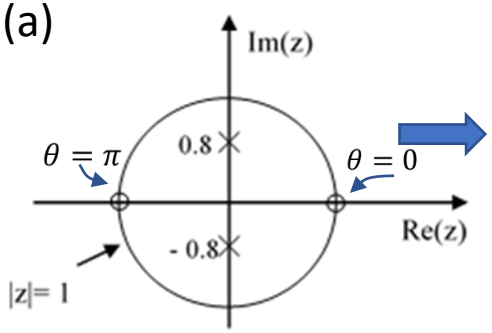
- A complex pole at θ_0 causes a peak in the magnitude response at that frequency.
- A zero on the unit circle will force the magnitude response $|H(\theta)|$ at that frequency to zero.

Digital Filter Fundamentals: Magnitude Responses

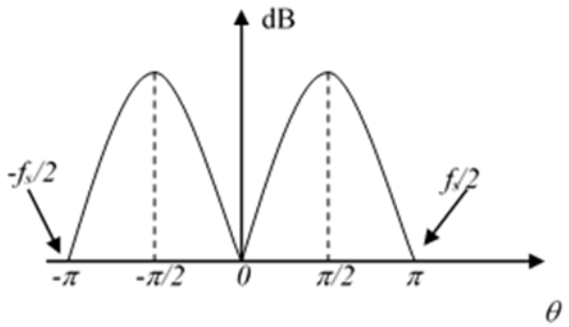
Example 18: IIR resonant filters

- Sketch an approximate magnitude response for each of the pole-zero maps given below:

(a)



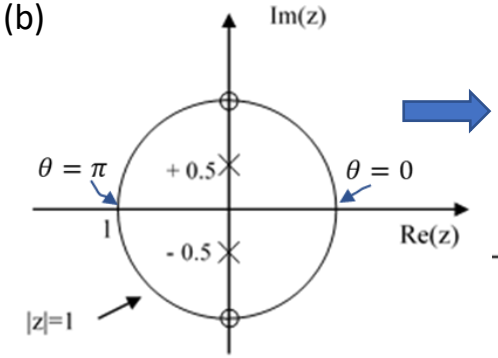
Pole-zero map



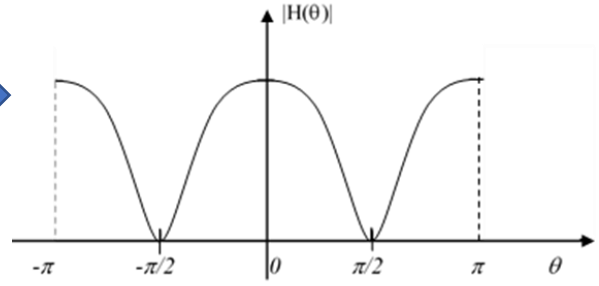
Magnitude response

A complex pole at $\theta_0 = \frac{\pi}{2}$ causes a peak in the magnitude response at θ_0 . Zeros on the unit circle at $\theta = 0$ and $\theta = \pi$ will force the magnitude response at those frequencies to zero.

(b)



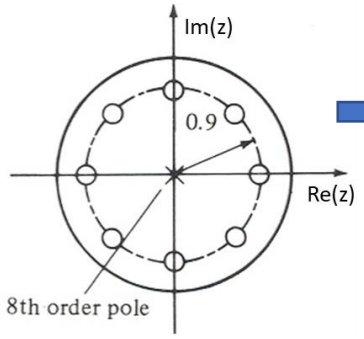
Pole-zero map

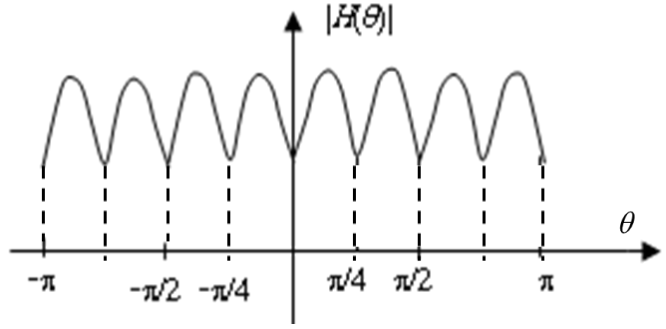


Magnitude response

Example 19: FIR resonant filters

- Sketch roughly the magnitude response corresponding to the pole-zero pattern given below:

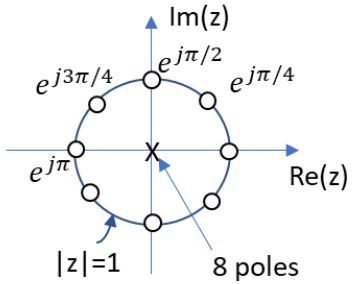


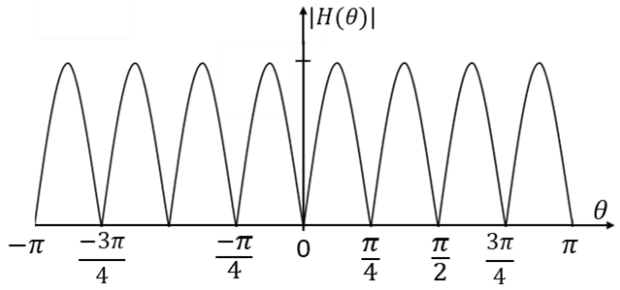


Magnitude response

Note: Zeroes near the unit circle cause a dip in the magnitude response.

- Sketch roughly the magnitude response corresponding to the pole-zero pattern (there *N =8* equally spaced zeros over the unit circle with the first zero at *z* = 0) given below:





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Topic 3: Digital Filter Fundamentals

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- Digital Filter Structures (pp. 11-15)
- Magnitude and Phase Responses (pp. 16-23)
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- All-Pass filters (pp. 29-31)
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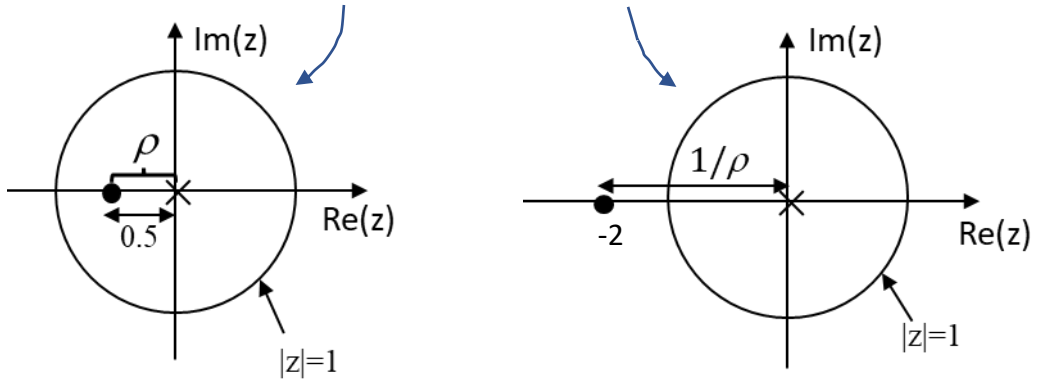
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Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Filters

FIR filter: Reciprocal roots

- The magnitude characteristics of two FIR filters $H_1(z)$ and $H_2(z)$ are **identical** , because the zeros of the FIR filters are reciprocals.
- If we **reflect** a zero with magnitude $|z| = \rho$, that is inside the unit circle into a zero with magnitude $|z| = 1/\rho$ outside the unit circle the magnitude characteristic of the system is unaltered, but the phase response changes.
- Let us consider two FIR filters of transfer functions :

$$H_1(z) = 1 + \frac{1}{2}z^{-1}; \qquad H_2(z) = \frac{1}{2} + z^{-1}$$

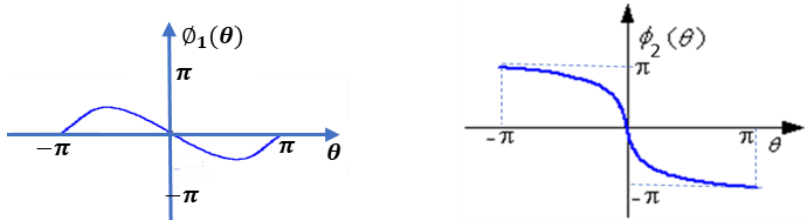


- $H_2(z)$ is the reverse of the system $H_1(z)$. This is due to the reciprocal relationship between the zeros of $H_1(z)$ and $H_2(z)$.

$$H_1(\theta) = 1 + (1/2)\, e^{-j\theta}; \qquad H_2(\theta) = 1/2 + e^{-j\theta}$$

$$|H_1(\theta)| = |H_2(\theta)| = \sqrt{5/4 + \cos\theta}$$

- Phase Response:



- We observe that the phase characters $\phi_1(\theta)$ begins at zero phase at frequency $\theta = 0$ and terminates at zero phase at the frequency $\omega = \pi$. Hence the net phase change. This filter is called **minimum phase** filter

$$\emptyset_1(\pi) - \emptyset_1(0) = 0$$

- On the other hand, the phase characteristic for the filter with the zero outside the unit circle undergoes a net phase change. This filter is called **maximum phase** filter.

$$\emptyset_2(\pi) - \emptyset_2(0) = -\pi$$

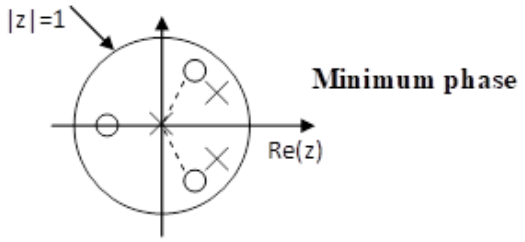
- If a filter with M zeros has some of its zeros inside the unit circle and the remaining outside the unit circle, it is called a **mixed-phase** system.

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Minimum, Maximum & Mixed Phase Filters

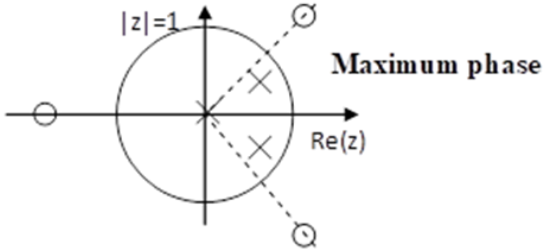
- A minimum-phase property of FIR filter carries over to IIR filter. Let us consider:

$$H(z) = \frac{B(z)}{A(z)}$$

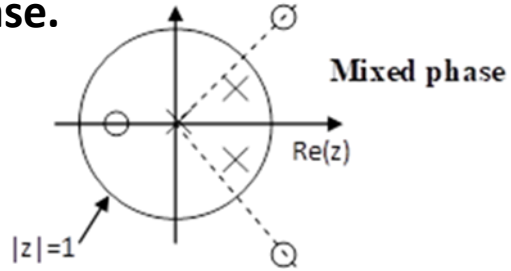


is called **minimum phase** if all its poles and zeros are inside the unit circle.

- If all the zeros lie outside the unit circle, the system is called **maximum phase**.



- If zeros lie both inside and outside the unit circle, the system is called **mixed-phase.**



Note:

- For a given magnitude response, the minimum-phase system is the causal system that has the smallest phase at every frequency (θ).

- Since the derivative of the phase characteristic of the filter (group delay) is a measure of the time delay that signal frequency components undergo in passing through the filter, a minimum phase characteristic implies a minimum delay function, while a maximum-phase characteristic implies that the delay characteristic is also maximum

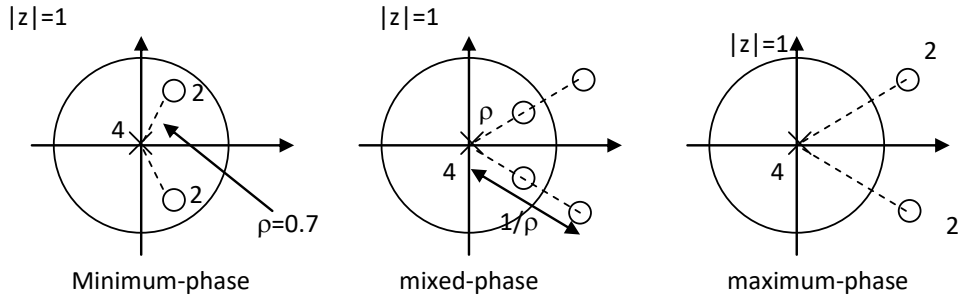
Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Example 20:

- Consider a fourth-order all-zero filter containing a double complex conjugate set of zeros located at

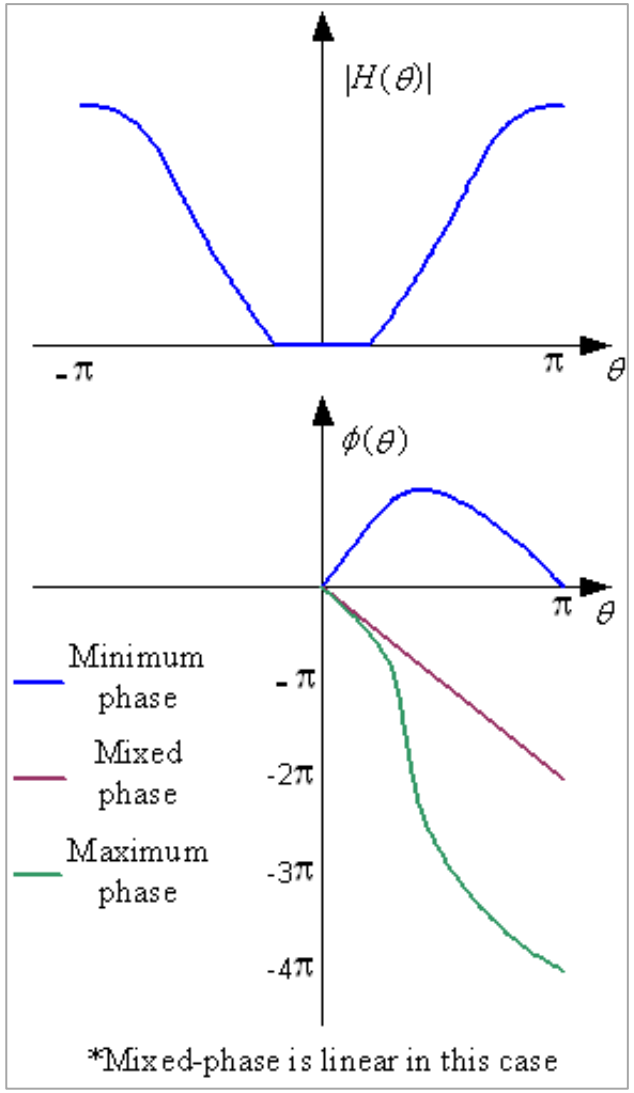
$$z = 0.7e^{\pm j\frac{\pi}{4}}$$

- The minimum-phase, mixed phase and maximum phase system pole-zero patterns having identical magnitude response are shown below:



- The magnitude response and the phase response of the three systems are shown on the right.

- The minimum-phase system seems to have the phase with the smallest deviation from zero at each frequency.



Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase System

Example 21:

- A third order FIR filter has a transfer function $G(z)$ given by:

$$G(z) = (6 - z^{-1} - 12z^{-2})(2 + 5z^{-1})$$

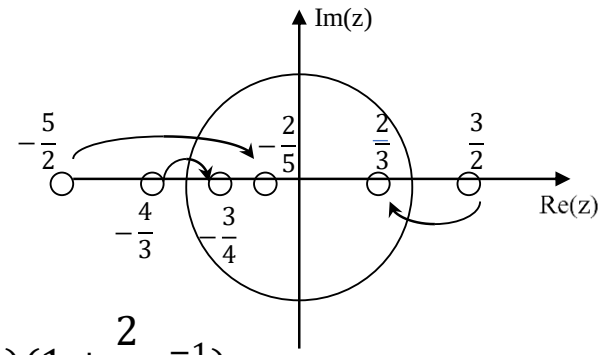
From $G(z)$, determine the transfer function of an FIR filter whose magnitude response is identical to that of $G(z)$ and has a minimum phase response.

Answer: First Factorise G(z):

$$G(z) = (2 - 3z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

$$G(z) = 12(1 - \frac{3}{2}z^{-1})(1 + \frac{4}{3}z^{-1})(1 + \frac{5}{2}z^{-1})$$

- All the zeros of G(z) are outside the unit circle; So, reflect all the zeros inside the unit circle to obtain the minimum-phase filter.

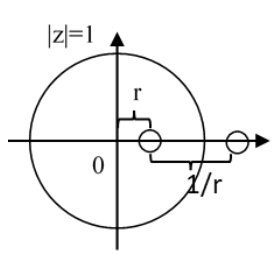


The minimum phase filter:

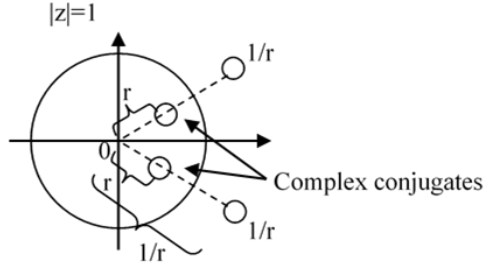
$$P(z) = 60(1 - \frac{2}{3}z^{-1})(1 + \frac{3}{4}z^{-1})(1 + \frac{2}{5}z^{-1})$$

Example 22: Reciprocal pairs of zeros for linear phase filters

- The zeros of a linear-phase filter must occur in reciprocal pairs and exhibit conjugate reciprocal symmetry as shown below.

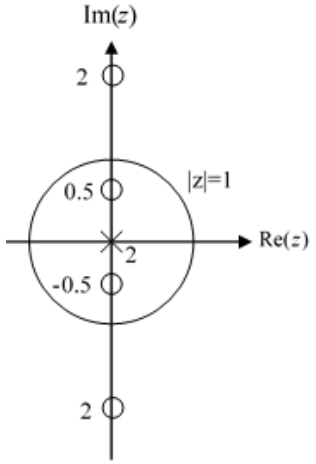


For a real zero



For a complex zero

- The pole-zero plot of a filter H(z) is given below. Is $H(z)$ a linear phase filter?



The four zeros at $z = j0.5$, $z = j2$, $z = -j0.5$ and $z = -j2$ are conjugate reciprocal symmetry. All poles are at $z = 0$.

Therefore, this is a linear phase filter.

$H(z) = z^2 + 4.25 + z^{-2}$

$\therefore h[n] = \{1, 0, 4.25, 0, 1\}$ <– even symmetry about n=0.

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Digital Filter Fundamentals: All-Pass filters

All-Pass Filters

- An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.
- The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k}$
- A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}}$$

where $a_0 = 1$ and all coefficients are real.

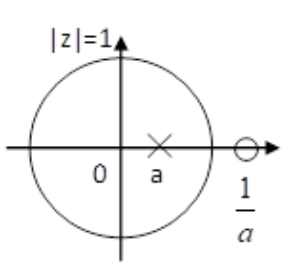
- If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

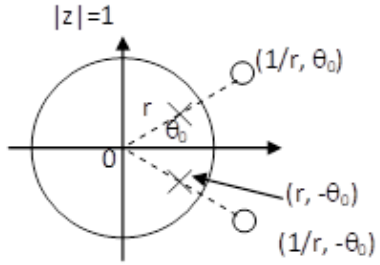
$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)}$$

- It can be shown that $|H(\theta)|^2 = H(z)\cdot H(z^{-1})|_{z=e^{j\theta}} = 1$ which is an all-pass filter.

- Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}.
- The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



All-pass



All pass filter

$$H(z) = \frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a| < 1 \text{ for stability}$$

- We can easily show that the magnitude response is constant.

$$|H(\theta)|^2 = |H(\theta)\cdot H^*(\theta) = H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$= \frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}} = \frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2} = 1/a^2$$

Digital Filter Fundamentals: All-Pass filters

Example 23:

- Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of a constant and an all-pass filter.

$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

Answer:

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{constant} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-passfilter}$$

Example 24: All-pass filter

- Consider the filter described by

$H(z) = \frac{a+cz^2}{c+az^2}$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

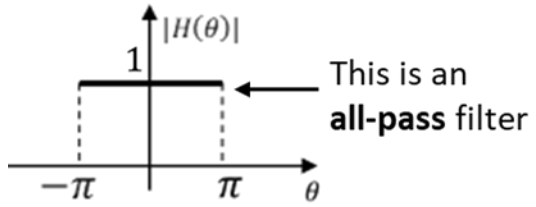
Answer:

$$H(z) = \frac{c+az^{-2}}{a+cz^{-2}}; \qquad H(\theta) = \frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}$$

$$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)$$

$$= \frac{c + ae^{-j2\theta}}{a + ce^{-j2\theta}} \cdot \frac{c + ae^{j2\theta}}{a + ce^{j2\theta}}$$

$$= \frac{c^2 + a^2 + ac[e^{j2\theta} + e^{-j2\theta}]}{a^2 + c^2 + ac[e^{j2\theta} + e^{-j2\theta}]} = 1$$



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Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

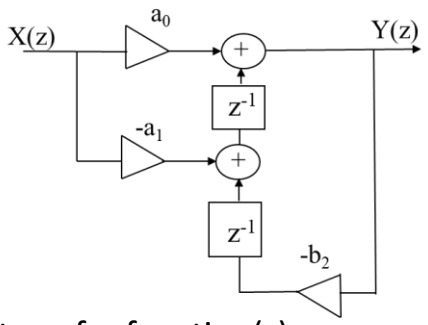
1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)

   (a) $1 + a_0 - a_1$

   (b) $\frac{a_0 - a_1}{1 + b_2}$

   (c) $\frac{a_0 - a_1}{1 - b_2}$

   (d) Non of the above



2. Tick which of the following transfer function(s) represent high pass filters, where $0 < \alpha < 1$.

   $(a)\ \ H(z) = 1 - \alpha\, z^{-1}$ $\quad (b)\ \ H(z) = \frac{1}{1 - \alpha\, z^{-1}}$

   $(c)\ \ H(z) = 1 + \alpha\, z^{-1}$ $\quad (d)\ \ H(z) = \frac{1}{1 + \alpha\, z^{-1}}$

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

   a) is a low-pass filter
   b) is a high-pass filter
   c) is an all-pass filter

4. A maximum phase filter has: (Tick the correct answer)

   a) All zeros and poles outside the unit circles
   b) All poles outside the unit circle
   c) All zeros outside the unit circle
   d) All poles and zeros inside the unit circle

SOLO – Level 1

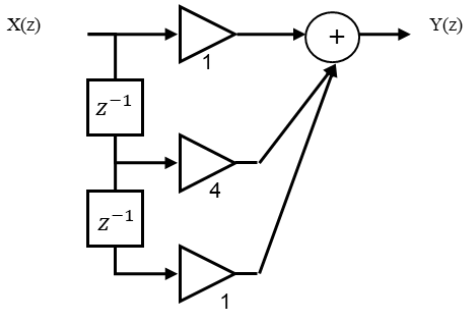
5. Which of the following is/are linear phase filters: (Tick the correct answer)

   (a) $H(z) = 0.25 + 0.5\, z^{-1} + 0.25\, z^{-2}$ (b) $H(z) = z^{-4}$

   (c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\, z^{-2} - z^{-3}$

6. What can be said of the following filter ? (Tick the correct answer)

   a) It has minimum phase and a finite impulse response.
   b) It has linear phase and an infinite impulse response.
   c) It has an infinite impulse response.
   d) It has linear phase and a finite impulse response



7. Which of the following system is linear, time invariant, causal and stable? (Tick the correct answer)

   a) y[n] = |x[n]|
   b) y[n] = x[n] + n x[n+1]
   c) y[n] = x[-n]
   d) $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

1. Compute the 3dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}}; \quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, \qquad 0 < a < 1$$

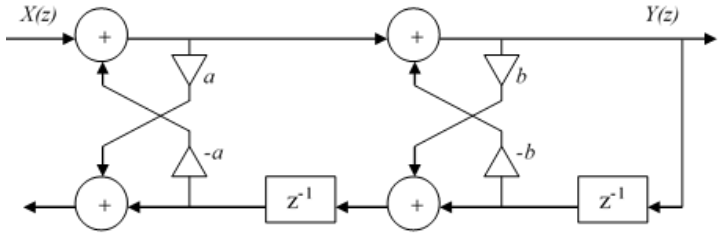
   Which filter has a smaller 3dB bandwidth?

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

   Draw the canonic realization of the filter *H(z)*.

3. Determine the frequency response $H(\theta)$ of the following filter:



4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1-b_1z^{-1})^2}$$

   Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

5. Implement a parallel structure for the following *H(z):*

$$H(z) = \frac{z^2-4z+3}{z(z+0.5)^2}$$

SOLO – Level 2

6. A notch filter is given by $H(z) = \frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta = 0, \frac{\pi}{2}, and\ \pi$. Sketch the magnitude response in the interval $-\pi \le \theta \le \pi$.

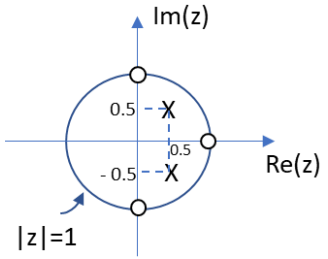
7. Determine and sketch the magnitude and phase response of the following filters

   (a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$

   (c) $y[n] = x[n-4]$

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

   (b) Using the pole-zero map , sketch an approximate magnitude response of the H(z) .



9. A Finite impulse filter filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$.Determine b_4 to b_6 such t hat this filter has linear phase characteristics.

10. Implement the following transfer function in cascade structure , in direct form II and in parallel structure. All coefficients are real.

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

1. Show that both digital filters given below have the same magnitude response:

   (a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

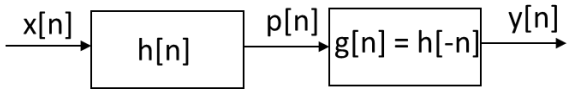
   where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

2. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2+z+0.5)}$$

   Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$. Is there any transfer function having the same magnitude response as *H(z)*?

3. Show that if *x*[n] is a real valued signal, the structure shown below has a zero-phase response; where *h*[n] is the impulse response of the filter H(z) given by; $H(z) = (\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2})$.



4. The difference equation of a digital filter is given by

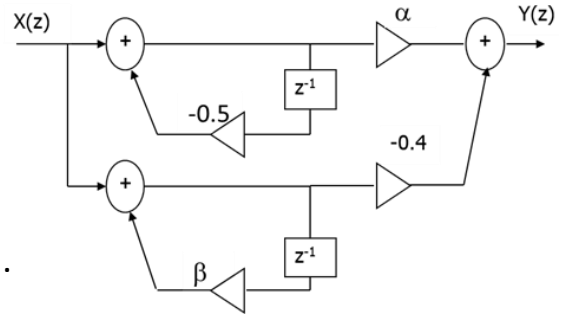
$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

   (a) Find the transfer function for the above filter.
   (b) Using the transfer function obtained in part (a) above, draw the pole zero map for the filter and sketch the corresponding magnitude response.

SOLO – Level 3

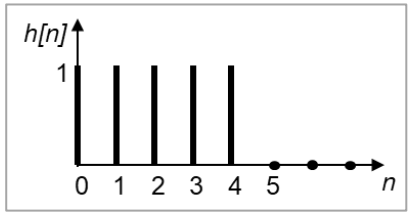
5. The following figure shows an IIR filter structure, and its transfer function given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$$



   Determine the values of the Multiplier Coefficients α and β.

6. A FIR filter has an impulse response as shown below:

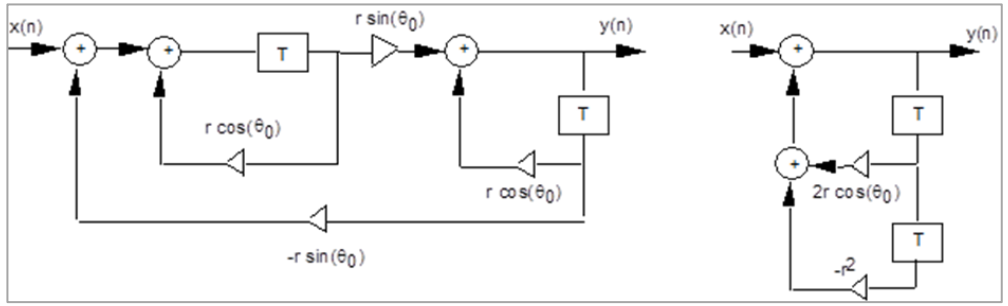


   (a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$

   (b) Draw an implementation of this filter as a cascade structure.

7. Show that the systems (i) and (ii) shown below are equivalent:



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

8. A third order FIR filter has a transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

From G(z), determine the transfer function of a FIR filter whose magnitude response is identical to that of G(z) and has

(a) a minimum phase response
(b) a maximum phase response

9. Analyse the digital structure shown on the right:

(a) Determine its transfer function $H(z)$ =(Y(z))/(X(z))

(b) What should be the value of the multiplier coefficient K so that the transfer function *H(z)* that you obtained in (a) above has a unity gain at $\theta = 0$?

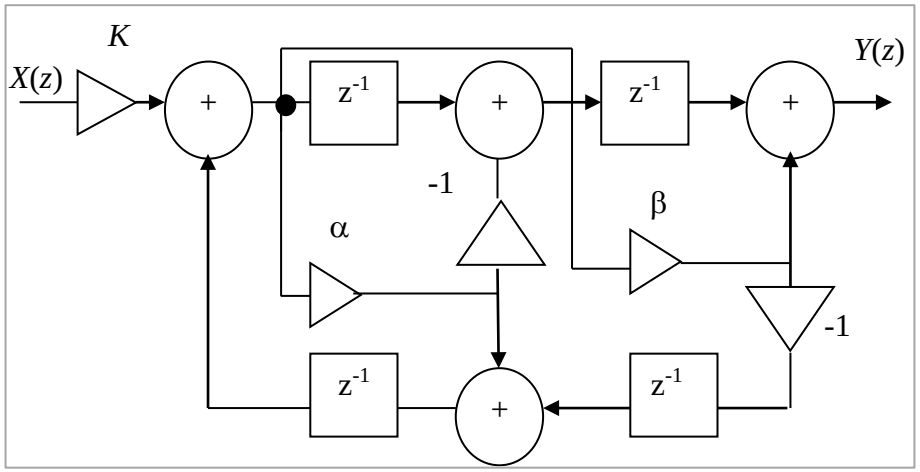
(c) What should be the value of the multiplier coefficient *K* so that *H*(*z*) has a unity gain at $\theta = \pi$?

(d) Is there a difference between these two values of *K* obtained in Question (b) and (c)? If not, why not?

10. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.
(c) Show that the above FIR filter has a constant group delay.



Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

1. The frequency response of an ideal differentiator is given by

$$H(\theta) = j\theta e^{-j\tau} \text{ - } \pi \leq \theta \leq \pi$$

This response is periodic with period 2π. The quantity τ is the delay of the system in samples.

   (a) Sketch the magnitude and phase responses of the system for $-\pi \leq \theta \leq \pi$.

   (b) Find the impulse response *h[n]* of the system as a sum of a sine and cosine functions.

2. Determine the magnitude and phase response of the multipath channel *y[n] = x[n] + x[n-M]*. At what frequencies $H(\theta) = 0$?

3. Let a causal linear time invariant discrete-time system be characterised by a real impulse response *h*[*n*] with a DTFT of $H(\theta)$. Consider the system shown in Figure A , where *x*[*n*] is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ $\{G(z) = {}^{Y(z)}/_{X(z)}\}$ in terms of $H(\theta)$ and show that it has a zero-phase response.

4. An analogue system may be implemented as shown in Figure B. Using the transformation $\frac{1}{s} \rightarrow k\frac{1+z^{-1}}{1-z^{-1}}$ a discrete time system may be obtained; where k is a constant.

   (a) Draw the digital structure of the system, using the minimum number of multipliers and delay units .

   (b) Write the difference equation that governs the behaviour of the system in the form $y[n] = 2\cos[\theta_0]\, y[n-1] - y[n-2]$ and find the value of $\cos\theta_0$.

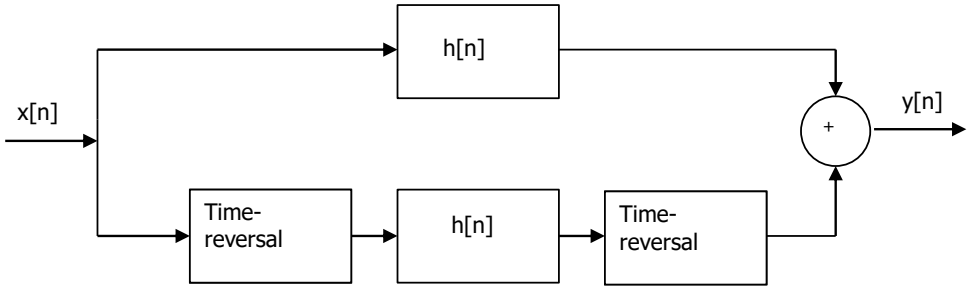


Figure A

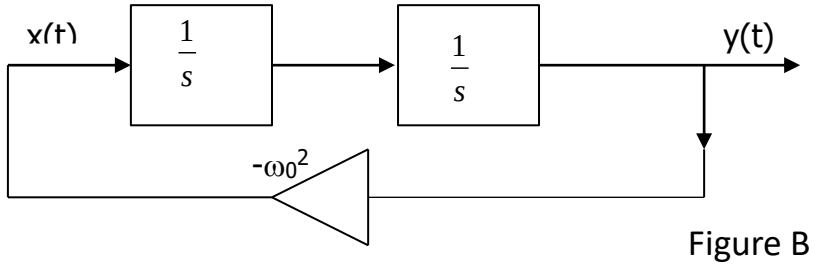


Figure B

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

1. (a) Find the group delay for each of the following system, where $|\alpha| < 1$

(i) $H_1(\theta) = 1 - \alpha e^{-j\theta}$ (ii) $H_2(\theta) = \frac{1}{1-\alpha e^{-j\theta}}$

(b) using the results obtained in (a) above, show that the group delay of an all-pass filter with a transfer function $H_0(z) = \frac{z^{-1} - \alpha}{1-\alpha z^{-1}}$ is $\frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos(\theta)}$

2. Find the minimum phase system that has a magnitude response given by

$$|H(\theta)|^2 = \frac{\frac{5}{4} - \cos\theta}{\frac{10}{9} - \frac{2}{3}\cos\theta}$$

3. A stable and causal system (H(z)) may always be factored into a product of a minimum phase system with an all-pass system: $H(z) = H_{min}(z) \cdot H_{ap}(z)$ **or** $H(z) = H_{min}(z) \cdot H_{lp}(z)$; where $H_{lp}(z)$ is a linear phase system.

$$H(z) = \frac{(1 - 0.7z^{-1})(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}$$

(a) Find a minimum phase system function $H_{min}(z)$ and an all-pass system $H_{ap}(z)$, for the above H(z).

(b) Find a minimum phase system function $H_{min}(z)$ and a linear phase system $H_{lp}(z)$, for the above H(z).

SOLO – Level 5

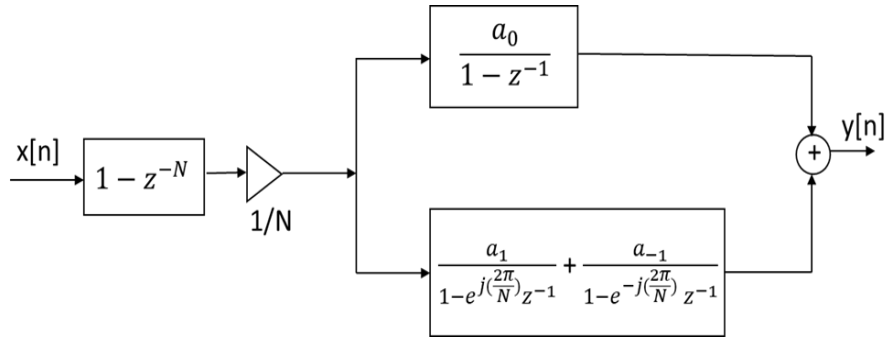
4. A filter structure with a transfer function $\{H(z) = \frac{Y(z)}{X(z)}\}$ is shown below. You may assume that $N = 3$, $a_0 = 3$ and $H(\theta)|_{\theta=2\pi/3} = 6+j3\sqrt{3}$.

Also $H\left(e^{j\frac{2\pi k}{N}}\right) = a_k$ and $H(e^{-j\frac{2\pi k}{N}}) = a_{-k}$

(a) Determine a_1 & a_{-1} such that this filter has a real valued impulse response *h[n];*

(b) Find H(z) and draw a filter structure using delays elements, multipliers and adders.

(c) Show that this filter structure can be realised as an FIR filter.



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- All-Pass filters (pp. 29-33) – two more slides have been added

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Term 3, 2021

Digital Filter Fundamentals: All-Pass filters

All-Pass Filters

- An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.
- The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k}$
- A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}}$$

where $a_0 = 1$ and all coefficients are real.

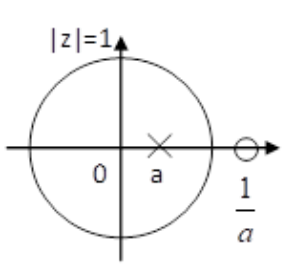
- If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

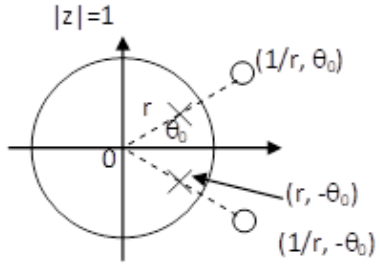
$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)}$$

- It can be shown that $|H(\theta)|^2 = H(z)\cdot H(z^{-1})|_{z=e^{j\theta}} = 1$ which is an all-pass filter.

- Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}.
- The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



All-pass



All pass filter

$$H(z) = \frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a| < 1 \text{ for stability}$$

- We can easily show that the magnitude response is constant.

$$|H(\theta)|^2 = H(\theta)\cdot H^*(\theta) = H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$= \frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}} = \frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2} = 1/a^2$$

Digital Filter Fundamentals: All-Pass filters

Example 23:

- Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of a constant and an all-pass filter.

$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

Answer:

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{constant} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-passfilter}$$

Example 24: All-pass filter

- Consider the filter described by

$H(z) = \frac{a+cz^2}{c+az^2}$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

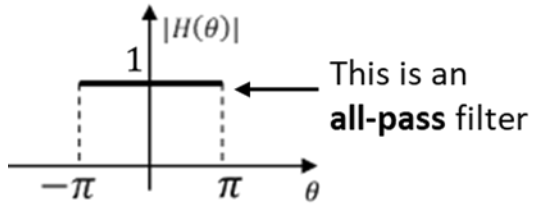
Answer:

$$H(z) = \frac{c+az^{-2}}{a+cz^{-2}}; \qquad H(\theta) = \frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}$$

$$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)$$

$$= \frac{c + ae^{-j2\theta}}{a + ce^{-j2\theta}} \cdot \frac{c + ae^{j2\theta}}{a + ce^{j2\theta}}$$

$$= \frac{c^2 + a^2 + ac[e^{j2\theta} + e^{-j2\theta}]}{a^2 + c^2 + ac[e^{j2\theta} + e^{-j2\theta}]} = 1$$



Digital Filter Fundamentals: All-Pass filters

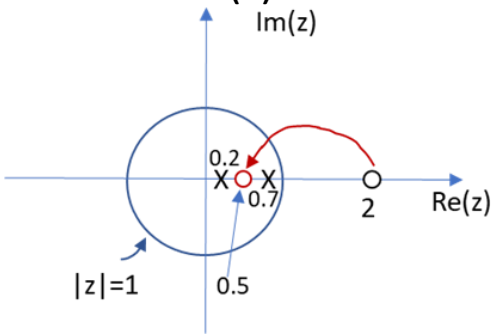
Example 25:

- A stable and causal non minimum phase filter may always be factored into a product of a minimum phase filter (with all poles and zeros inside the unit circle) with an all-pass filter.

$$H(z) = H_{min}(z).H_{allpass}(z)$$

- The procedure for performing this factorisation is as follows:
  - ❖ First, all of the zeros of H(z) that are outside the unit circle are reflected inside the unit circle to their conjugate reciprocal location. The resulting filter is minimum phase, $H_{min}(z)$.
  - ❖ Then, all-pass filter is selected so that it reflects the appropriate set of zeros of $H_{min}(z)$, back outside the unit circle.

- For the transfer function $H(z) = \frac{1-2z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})}$, find the factorization for H(z) of the form $H(z) = H_{min}(z).H_{allpass}(z)$.



- The minimum phase factor is: $H_{min}(z) = \frac{1-(\frac{1}{2})z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})}$
- Then , to reflect the zero at z=0.5 back outside the unit circle to z=2, we use the all-pass factor
- $H_{allpass}(z) = \underbrace{\frac{1-2z^{-1}}{1-(\frac{1}{2})z^{-1}}}_{All-passfilter}$

$$\therefore H(z) = \frac{1-(\frac{1}{2})z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})} \cdot \frac{1-2z^{-1}}{1-(\frac{1}{2})z^{-1}}$$

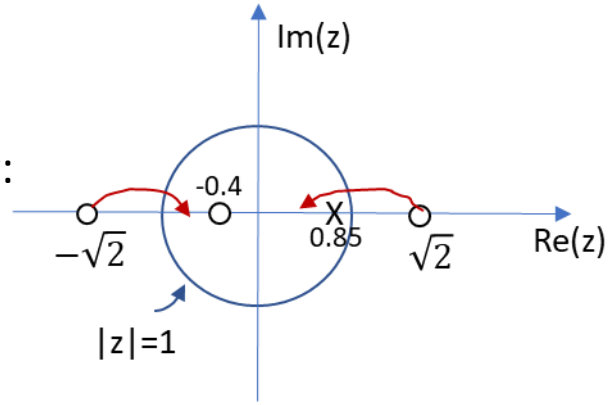
$$H(z) = H_{min}(z).H_{allpass}(z)$$

Digital Filter Fundamentals: All-Pass filters

Example 26:

- A causal LTI system has a system function: $H(z) = \frac{(1-2z^{-2})(1+0.4z^{-1})}{(1-0.85z^{-1})}$; Find a factorisation for H(z) of the form $H(z) = H_{min}(z).H_{allpass}(z)$, where $H_{min}(z)$ has minimum phase filter and $H_{allpass}(z)$ is a stable, all-pass, maximum phase filter .

The pole-zero map of H(z) is given by:



The system function H(z) has a nonminimum phase factor,$(1-2z^{-2})$, which may be written as the product of a minimum phase term and an all-pass factor as follows:

$$(1-2z^{-2}) = (1-\tfrac{1}{2}z^{-2}) \quad \frac{(1-2z^{-2})}{(1-\frac{1}{2}z^{-2})}$$

Therefore , H(z) may be written as the product of a minimum phase system with an all=pass system as follows:

$$H(z) = \underbrace{\frac{(1+0.4z^{-1})(1-\frac{1}{2}z^{-2})}{(1-0.85z^{-1})}}_{H_{min}(z)} \quad \underbrace{\frac{(1-2z^{-2})}{(1-\frac{1}{2}z^{-2})}}_{H_{allpass}(z)}$$

ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

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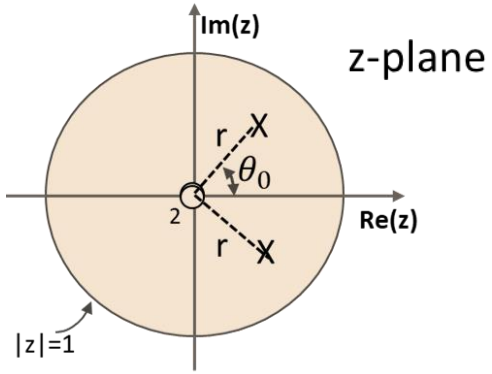
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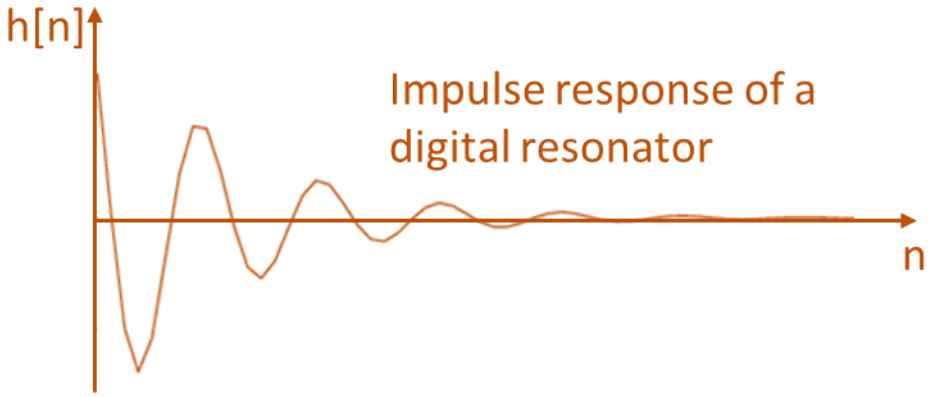
Digital Resonator Fundamentals

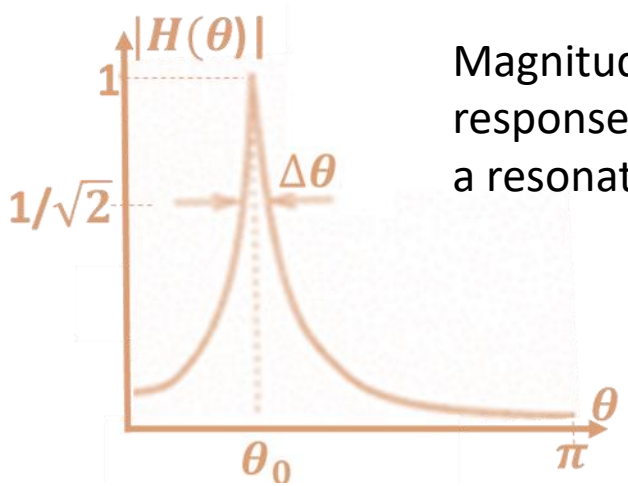
Introduction

- A digital resonator is normally a two-pole IIR filter having a complex conjugate pair of poles located inside the unit circle of the z-plane and a pair of zeros at z=0.
- The angle of the poles in polar co-ordinates determines the resonant frequency θ_0 of the resonator, while the distance (r) of the poles from the origin of the unit circle sets the bandwidth $\Delta\theta$.
- The conjugate poles closer to the unit circle ensure smaller bandwidth, but with a sharper peak at the resonant frequency θ_0 .
- The digital resonator is essentially a narrow-bandpass filter.









Magnitude response of a resonator

ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

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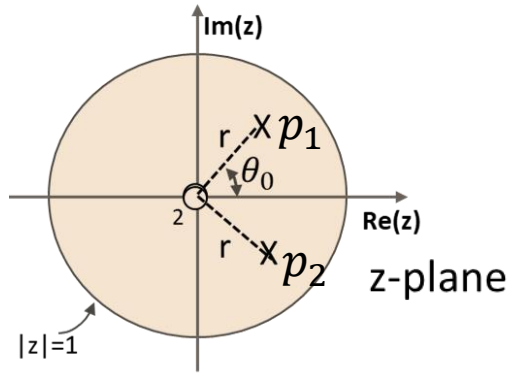
Digital Resonator Fundamentals: A second-order digital resonator

A second-order digital resonator

- One way to realise a second-order resonator with a peak at θ_0 is to place a pair of poles with the frequency of $\pm\theta_0$ ($p_1 = re^{j\theta_0}$ and $p_2 = re^{-j\theta_0}$) and a pair of zeros at z=0. (see diagram on the right)
- To ensure stability , we choose the pole radius r to be less than (but close to) unity.
- The transfer function of such a digital resonator with zeros at the origin is:

$$H(z) = \frac{z^2}{(z-p_1)(z-p_2)} = \frac{z^2}{(z-re^{j\theta_0})(z-re^{-j\theta_0})}$$

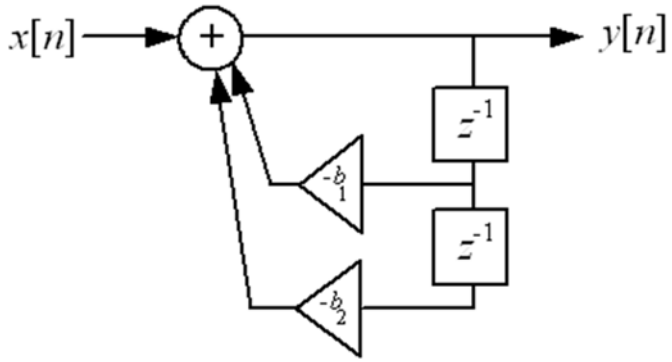
$$H(z) = \frac{z^2}{z^2 - r(e^{j\theta_0}+e^{-j\theta_0})z + r^2} = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2} \quad \text{..... (1)}$$



$$p_1 = re^{j\theta_0} = r\cos\theta_0 + jr\sin\theta_0$$
$$p_2 = re^{-j\theta_0} = r\cos\theta_0 - jr\sin\theta_0$$

From the second order realisation (see diagram on the right), we have

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \quad \text{.... (2)}$$



A second-order structure

Equations (1) and (2) are equivalent. By equating the coefficients, we obtain

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2; \quad \therefore \cos\theta_0 = -\frac{b_1}{2\sqrt{b_2}} \qquad \theta_0 = \frac{2\pi f_0}{f_s}$$

Digital Resonator Fundamentals

Magnitude Spectrum

- The digital resonator's magnitude squared function $|H(\theta)|^2$ can be obtained from H(z) as follows: $H(z) = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2}$

$$|H(\theta)|^2 \left|\frac{e^{j2\theta}}{e^{j2\theta} - 2r\cos\theta_0 e^{j\theta} + r^2}\right|^2 = \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\ [1-2r\cos(\theta+\theta_0)+r^2]}$$

- From the magnitude spectrum shown on the right, it can be seen, the peak value *A* occurs at $\theta = \theta_0$ (or very close to) the resonant frequency (θ_0) and is given by

$$A^2 = |H(\theta)|^2 = \frac{1}{[(1-r)^2]\ [1-2r\cos(2\theta_0)+r^2]}$$

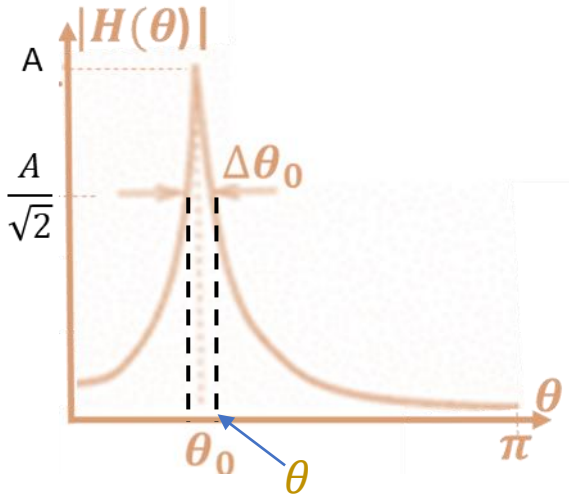
- The half-power bandwidth $\Delta\theta$ is found by locating the frequencies at which $|H(\theta)|^2 = 0.5\ |H(\theta_0)|^2 = 0.5\ A^2$.

$$\therefore \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\ [1-2r\cos(\theta+\theta_0)+r^2]} = \frac{0.5}{[(1-r)^2]\ [1-2r\cos(2\theta_0)+r^2]}$$

- Since the half-power points are very close to θ_0, we have $\theta + \theta_0 \approx 2\theta_0$ and $\theta - \theta_0 \approx \frac{1}{2}\Delta\theta$ and the above relationship simplifies to

$$1 - 2r\cos(\tfrac{1}{2}\Delta\theta) + r^2 = 2(1-r)^2$$

- This relationship between the pole radius r and the bandwidth $\Delta\theta$ can be simplified when the poles are very close to the unit circle (with r > 0.9 or so). The values of *r* and the peak magnitude, A, are then reasonably approximated by:



The peak gain can be normalised to unity by dividing the transfer function *H(z)* by A.

$$r \approx 1 - 0.5\,\Delta\theta$$

$$\Delta\theta = 2\pi\frac{\Delta f}{f_s};\ r \approx 1 - \pi\frac{\Delta f}{f_s}$$

$$A \approx \frac{1}{(1-r^2)\sin(\theta_0)}$$

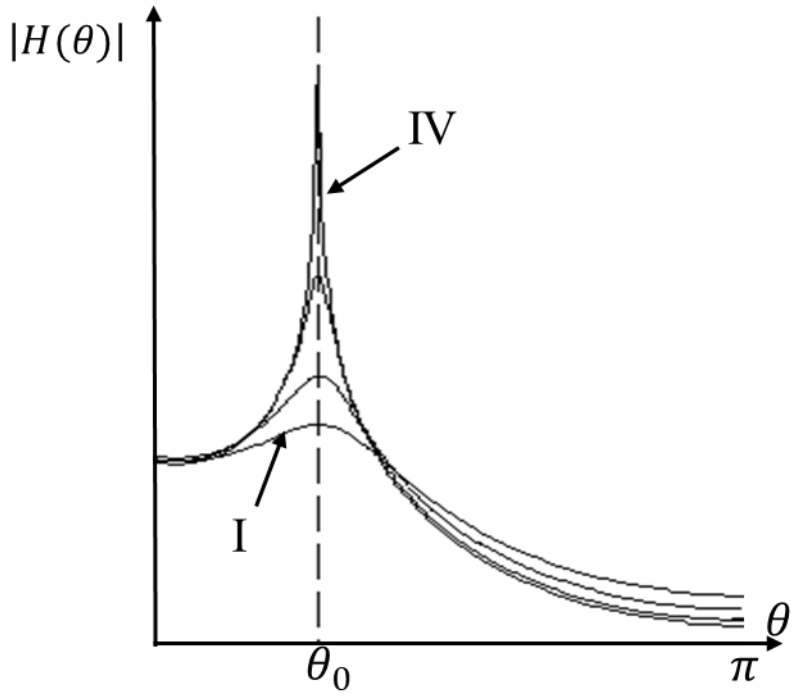
Digital Resonator Fundamentals

Magnitude Spectrum

- From the resonator equation H(z), we can derive the magnitude spectrum as discussed before:

$$H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}; \quad b_2 = r^2; \quad b_1 = -2r\,cos(\theta_0); \quad \theta_0 = cos^{-1}\left[\frac{-b_1}{2\sqrt{b_2}}\right]$$

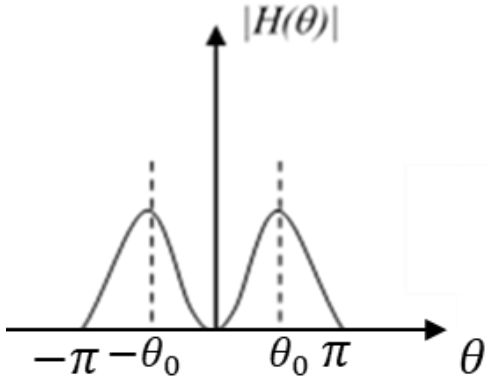
- The magnitude spectrum is shown below for various values of b_1 and b_2.



	b_1	b_2
I	-0.94	0.5
II	-1.16	0.7
III	-1.34	0.9
IV	-1.41	0.99

To null the magnitude response at low and high frequencies (θ=0 and θ= π) , the two zeros in H(z) may be located at z = 1 and z = -1 and this leads to the modified transfer function $H_1(z)$ and its magnitude response is shown below:

$$H_1(z) = \frac{1 - z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$



Magnitude response

Digital Resonator Fundamentals

Example 1 [Reference 2]:

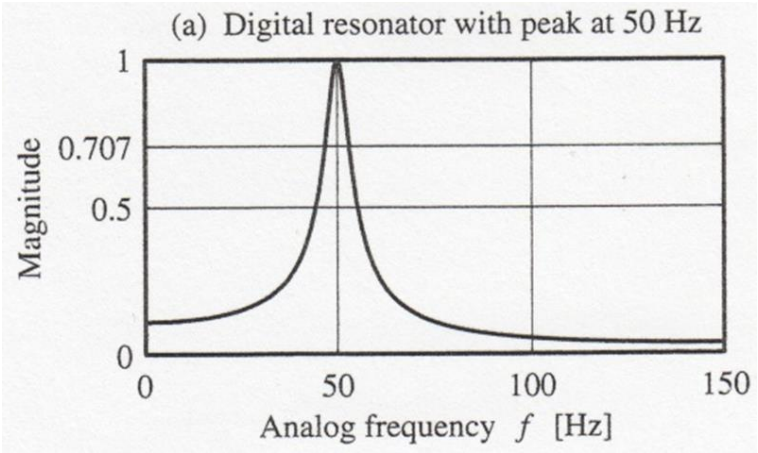
- Design a digital resonator with a peak gain of unity ay 50Hz and a 3-dB bandwidth of 6 Hz , assuming the sampling frequency of 300Hz.
- The resonant frequency $\theta_0 = \frac{2\pi(50)}{300} = \frac{\pi}{3}$;
- The 3-dB bandwidth $\Delta\theta = \frac{2\pi(6)}{300} = 0.04\pi$.
- The pole radius (r) = 1- 0.5 $\Delta\theta$ = 1-0.02 π = 0.9372
- The transfer function of the digital resonator H(z) is:

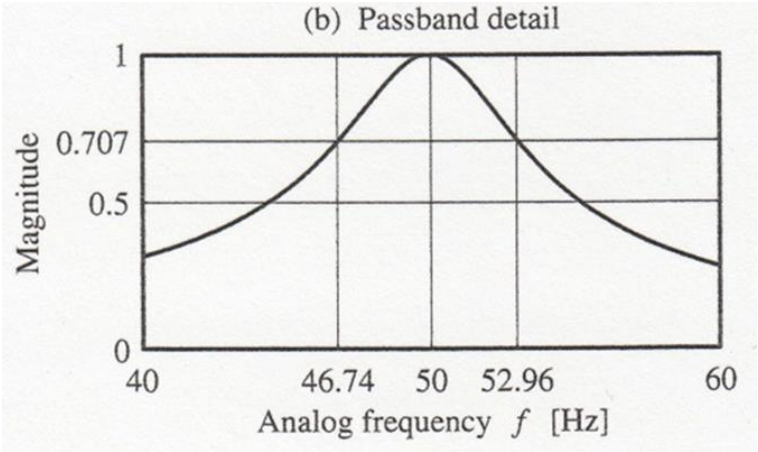
$$H(z) = \frac{k_0 z^2}{z^2 - 2r\cos\theta_0 z + r^2} = \frac{k_0 z^2}{z^2 - 0.9372z + 0.8783}$$

- For a peak gain of unity, we choose $k_0 = \frac{1}{A} = (1 - r^2)\sin(\theta_0) = 0.1054$

$$\therefore H(z) = \frac{0.1054 z^2}{z^2 - 0.9372z + 0.8783} = \frac{0.1054}{1 - 0.9372z^{-1} + 0.8783z^{-2}}$$

- The magnitude spectrum and the passband details are shown on the right
- The passband details shows that the 3 dB frequencies are located at 46.74 Hz and 52.96 Hz and the bandwidth is = (52.96) – (46.74) ≈ 6Hz





Digital Resonator Fundamentals : Notch filters

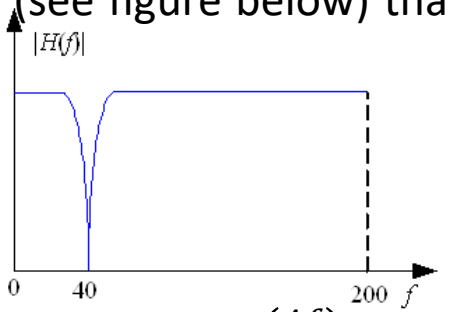
Notch filters

- When a zero is placed at a given point on the unit circle, the frequency response will be zero at the corresponding point.
- A pole on the other hand produces a peak at the corresponding frequency point.
- Poles that are close to the unit circle give rise large peaks, whereas zeros close to or on the unit circle produces troughs or minima.
- Thus, by strategically placing poles and zeros on the z-plane, we can obtain simple low pass or other frequency selective filters (**notch filters**).

Example 2:

Obtain, by the pole-zero placement method, the transfer function of a sample digital notch filter (see figure below) that meets the following specifications:

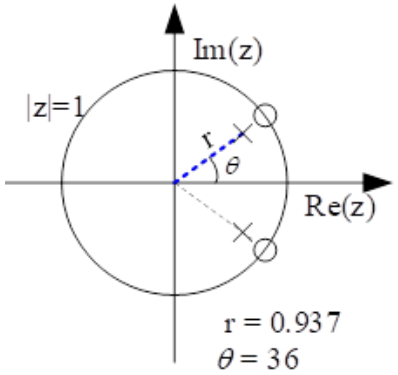
- Notch Frequency: 40Hz
- 3db width of the Notch: ± 4Hz
- Sampling frequency: 400 Hz



The radius, r of the poles is determined by: $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi$

- To reject the component at 40 Hz, place a pair of complex zeros at the points on the unit circle corresponding to ± 40 Hz. i.e. at an angles: $360^{\bullet} \times \frac{40}{400} = \pm 36^{\bullet} = \pm 0.2\pi$
- To achieve a sharp notch filter and improved magnitude response either side of the notch frequency, a pair of complex conjugate poles are placed at a radius $r < 1$, at the same frequency as the zeroes, as shown below.
- $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi = 1 - \left(\frac{8}{400}\right)\pi = 0.937$

$$H(z) = \frac{\left(z - e^{-j0.2\pi}\right)\left(z - e^{j0.2\pi}\right)}{\left(z - 0.937e^{-j0.2\pi}\right)\left(z - 0.937e^{j0.2\pi}\right)}$$



$$H(z) = \frac{1 - 1.6180z^{-1} + z^{-2}}{1 - 1.5161z^{-1} + 0.878z^{-2}}$$

ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Digital Resonator Fundamentals : Stability of a Second-Order Filter

Stability

- Consider a two-pole resonant filter given by:

$$H(z) = \frac{z^2}{z^2 + b_1 z + b_2} = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

where b_1 & b_2 are the coefficients.

- This system has two zeros at the origin and poles at

$$p_1, p_2 = -\frac{b_1}{2} \pm \frac{\sqrt{{b_1}^2 - 4b_2}}{2}$$

- The filter is stable if the poles lie inside the unit circle i.e.

$|p_1| < 1$ & $|p_2| < 1$

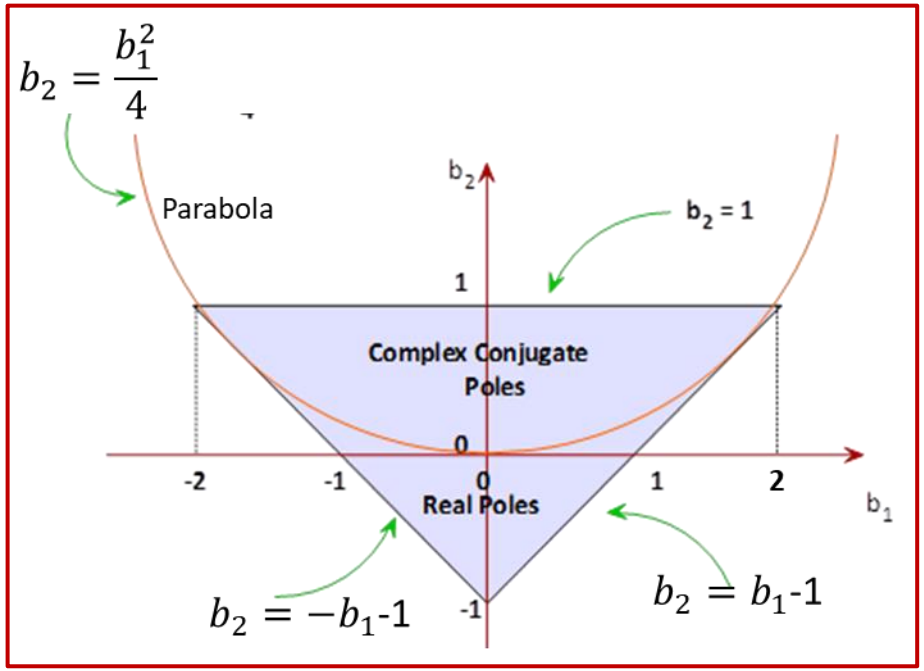
- For stability $b_2 < 1$. If $b_2 = 1$ then the system is an oscillator (**Marginally stable**)
- Assume that the poles are complex:

$${b_1}^2 - 4b_2 < 0 \quad \Rightarrow \quad {b_1}^2 < 4b_2 \quad \text{and} \quad b_1 < \pm 2\sqrt{b_2} \qquad b_2 > 0$$

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2$$

- If ${b_1}^2 - 4b_2 \geq 0$ then we get real roots.
- The stability conditions define a region in the coefficient plane (b_1, b_2) which is in the form of a triangle (see diagram on the right)

The system is only stable if and only if the point (b_1, b_2) lies inside the **stability triangle**.



Digital Resonator Fundamentals : Stability of a Second-Order Filter

Stability Triangle

- If the two poles (p_1 *and* p_2) are real, then they must have a value between -1 and 1 for the system to be stable.

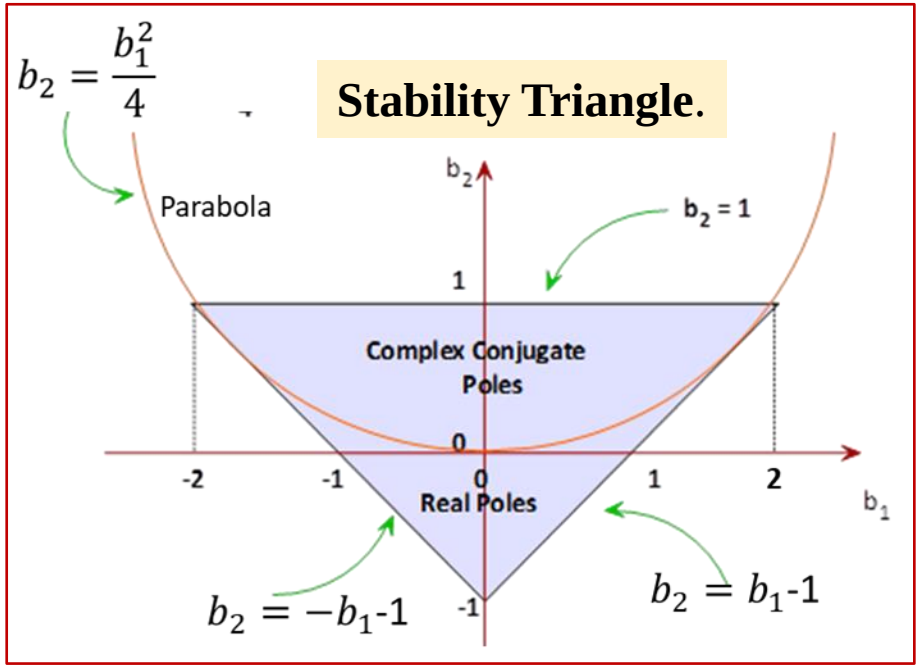
$$-1 < \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2} < 1 \quad \Rightarrow \quad -2 + b_1 < \pm\sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$\therefore -2 + b_1 < -\sqrt{b_1^2 - 4b_2} \quad \text{and} \quad \sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$(-2 + b_1)^2 > b_1^2 - 4b_2 \quad \text{and} \quad b_1^2 - 4b_2 < (2 + b_1)^2$$

$$b_1 - b_2 - 1 < 0 \quad \text{and} \quad b_1 + b_2 + 1 > 0$$

- The region below the parabola ($b_1^2 > 4b_2$) corresponds to real and distinct poles.
- The points on the parabola ($b_1^2 = 4b_2$) result in real and equal (double) poles.
- The points above the parabola correspond to complex-conjugate poles.



For a stable 2nd order system, the coefficients must satisfy the following inequality:

$$-2 \leq b_1 \leq 2$$
$$-1 \leq b_1 \leq 1$$

ELEC3104: Digital Signal Processing

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Digital Resonator Fundamentals : Digital Oscillators

Digital Oscillators

- A digital sinusoidal oscillator may be viewed as a limiting form of a two-pole resonator for which the complex-conjugate poles lie on the unit circle (see the pole-zero plot on the right)
- The impulse response, h[n], of the second-order filter with complex conjugate **poles on the unit circle** is a sinusoid and the system is called a digital sinusoid oscillator.

$$h[n] = A\ r^n \cos[n\theta] = A\ cos[n\theta]$$ [poles on the unit circle, r=1]

- From the z-transforms table, the transfer function of a causal oscillator that generates a pure sine wave $h[n] = A\ cos[n\theta]$ is given by ,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-(cos\ \theta)\ z^{-1}}{1-(2\ cos\ \theta)\ z^{-1}+z^{-2}}$$

- Taking inverse z-transform on both sides, we obtain

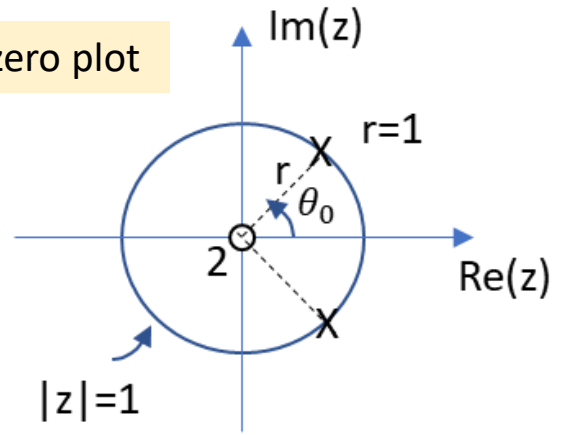
$$y[n] - 2\ cos\ \theta\ \ y[n-1] + y[n-2] = x[n] - cos\ \theta\ \ x[n-1]$$

- For an oscillator, the input x[n]=0 and x[n-1] =0; So, the equation of the digital oscillator becomes:

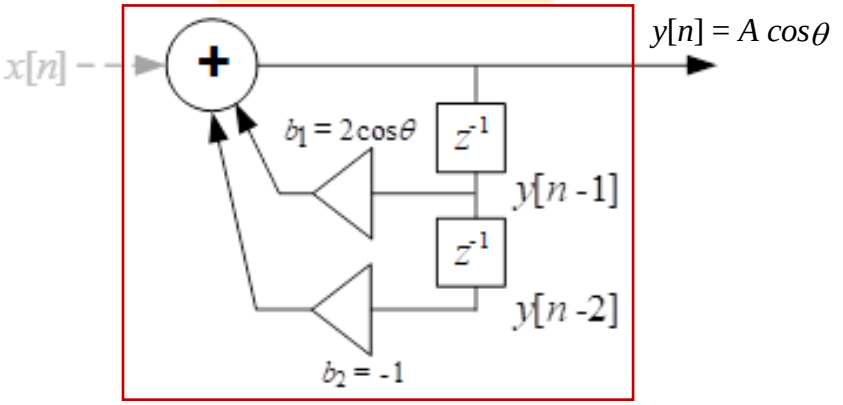
$$y[n] = 2\ cos\ \theta\ y[n-1] - y[n-2]$$

and its structure is shown below:

Pole-zero plot



Digital Oscillator



Digital Resonator Fundamentals : Digital Oscillators

Initial conditions

Case 1: Assume the input x[n] = 0:

- To obtain $y[n] = A\cos(n\theta)$, use the following initial conditions

$$y[0] = A\cos(0.\theta) = A \quad [for\ n=0]; \quad y[-1] = A\cos(-1.\theta) = A\cos\theta \quad [for\ n= -1]$$

- The frequency of the oscillator can be tuned by changing the coefficient b_1 (b_2 is a constant). The frequency θ of the oscillator is,

$$cos\,\theta = \frac{-b_1}{2\sqrt{b_2}} = -\frac{b_1}{2} \quad \text{[For an oscillator } b_2\text{=1]}$$

Case 2: Assume y[0]= 0 and y[-1]=0:

- In order to begin the sinusoidal oscillation, apply an impulse $\delta[n]$ at the input (see Figure A) when n= 0; Thereafter, the oscillation is self-sustaining because the system has no damping (i.e. r =1)

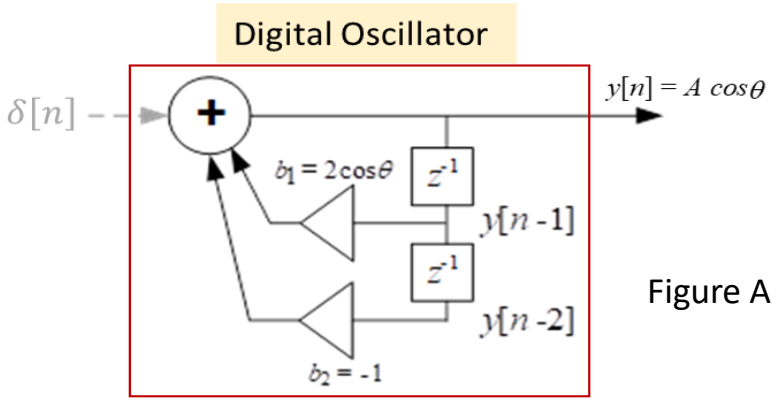


Figure A

Digital Oscillator Applications

- The touch tone phone (DTMF phone: Dual tone multi frequency) makes use of digital oscillators to generate audible tones by pressing buttons on a keypad as shown in Figure B.
- Pressing a button produces a two-tone signal containing, a high and a low frequency tone
- Each button is associated with a unique pair of low-frequency and high-frequency tones. The tones can be generated by using a parallel combination of two programmable digital oscillators

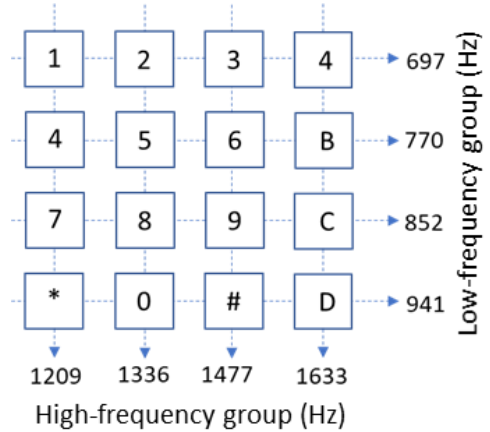
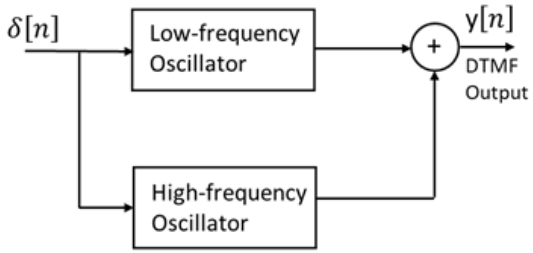


Figure B



Digital Resonator Fundamentals : Digital Oscillators

Example 3:

- A digital sinusoidal oscillator is shown in Figure C :

(a) Assuming θ_0 is the resonant frequency of the digital oscillator, find the values of b_1 and b_2 for sustaining the oscillation.

Answer:

$$H(z) = \frac{K}{1+b_1 z^{-1}+b_2 z^{-2}} \quad \text{----(1)}$$

$$H(z) = \frac{K\ z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} = \frac{Kz^2}{z^2 - 2r\cos\theta_0\, z + r^2} \quad \text{---(2)}$$

By equating (1) and (2), we obtain: $b_1 = -2\, r\cos\theta_0;\ \ b_2 = r^2$

However, for oscillation (marginally stable system) , $b_2 = 1$, $\therefore\ b_1 = -2\cos\theta_0$

(b)

(i) Write the difference equation corresponding to Figure C. You may assume that $x[n] = (A\sin\theta_0)\delta[n]$, and $y(-1) = y(-2) = 0$.

$$y[n] = -b_1 y[n-1] - y[n-2] + x[n]$$

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

(ii) Show, by analysing the difference equation, that the application of an impulse at n = 0 serves the purpose of beginning the sinusoidal oscillation and prove that the oscillation is self-sustaining thereafter.

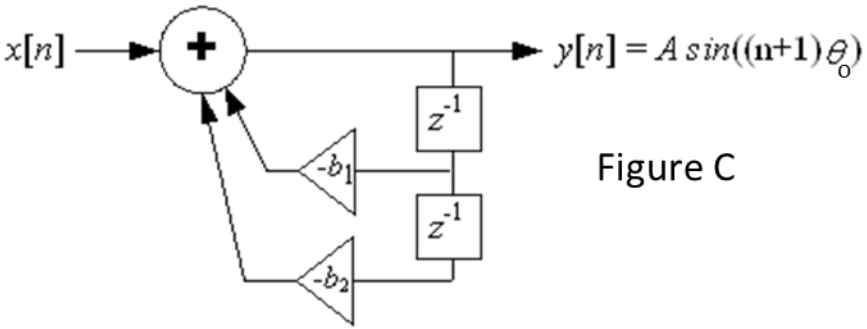


Figure C

Digital Resonator Fundamentals : Digital Oscillators

Example 3:

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

Let n = 0

$y[0] = 2\cos\theta_0\, y[-1] - y[-2] + A\sin\theta_0\, \delta[0]$ {$y[-1] = y[-2] = 0$; $\delta[0]=1$}

$\boldsymbol{y[0] = A\sin\theta_0}$

Let n = 1

$y[1] = 2\cos\theta_0\, y[0] - y[-1] + A\sin\theta_0\, \delta[1]$ {$y[-1] = 0$; $\delta[1]=0$}

$\boldsymbol{y[1] = 2\cos\theta_0\, A\sin\theta_0 = A\sin 2\theta_0}$

Let n = 2

$y[2] = 2\cos\theta_0\, y[1] - y[0] + A\sin\theta_0\, \delta[2]$ {$\delta[2] = 0$}

$= 2\cos\theta_0\, A\sin 2\theta_0 - A\sin\theta_0 = 2A\cos\theta_0\,[2\sin\theta_0\cos\theta_0] - A\sin\theta_0$

$= A\sin\theta_0\,[4\cos^2\theta_0 - 1] = A[3\sin\theta_0 - 4\sin^3\theta_0]$

$\boldsymbol{y[2] = A\sin 3\theta_0}$ and so forth.

$\therefore\ \boldsymbol{y[n] = A\sin(n+1)\theta_0}$ {The oscillation is self sustaining}

(c) By setting the input to zero and under certain initial conditions, sinusoidal oscillation can be obtained using the structure shown in Figure C. Find these initial conditions.

$$y[n] = 2\cos\theta\, y[n-1] - y[n-2] + x[n] \quad \{x[n] = 0 \text{ for an oscillator}\}$$

$$y[n] = 2\cos\theta\, y[n-1] - y[n-2] \ \boldsymbol{and}\ \boldsymbol{y[n] = A\sin(n+1)\theta_0}$$

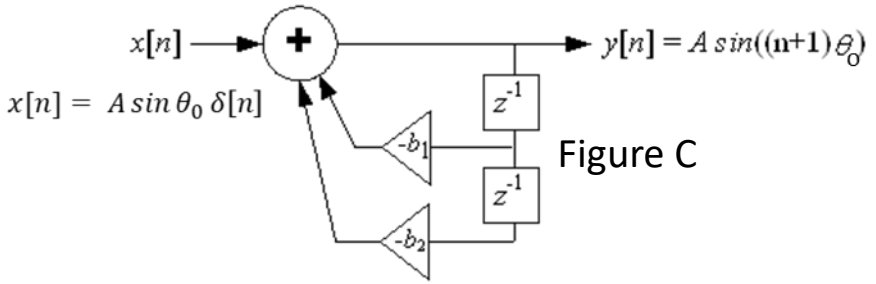


Figure C

Substitute n =-1 in $y[n] = A\sin(n+1)\theta_0$;
$\therefore$ y[-1] =0 - First initial conditions.

Substitute n = -2 in $y[n] = A\sin(n+1)\theta_0$,
$\therefore$ y[-2] =$-A\sin\theta_0$ – Second initial conditions.

Digital Resonator Fundamentals : Digital Oscillators

Sine and Cosine Oscillators

- Sinusoidal oscillators are normally used for generating carrier frequencies in modulation schemes .
- Some modulation schemes require both the sine and cosine oscillators. These signals can be generated simultaneously from the so-called coupled-form oscillator.
- Consider the trigonometric equation for $\cos(n\theta + \theta)$:

$$\cos(n\theta + \theta) = \cos(n\theta)\cos(\theta) - \sin(n\theta)\sin(\theta)$$

Let $\boldsymbol{y[n]} = \boldsymbol{cos}[\boldsymbol{n}\theta]$

$$\therefore y[n+1] = \cos(\theta)\, y[n] - \sin(\theta)\, x[n]$$

Replace *n* by *n*-1:

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{------- (1)}$$

- Similarly consider trigonometric equation for $\sin(n\theta + \theta)$

$$\sin(n\theta + \theta) = \sin(n\theta)\cos(\theta) + \cos(n\theta)\sin(\theta)$$

Let $\boldsymbol{x[n]} = \boldsymbol{sin}[\boldsymbol{n}\theta]$

$$\therefore x[n+1] = \sin(\theta)\, y[n] + x[n]\cos(\theta)$$

Replace *n* by *n*-1:

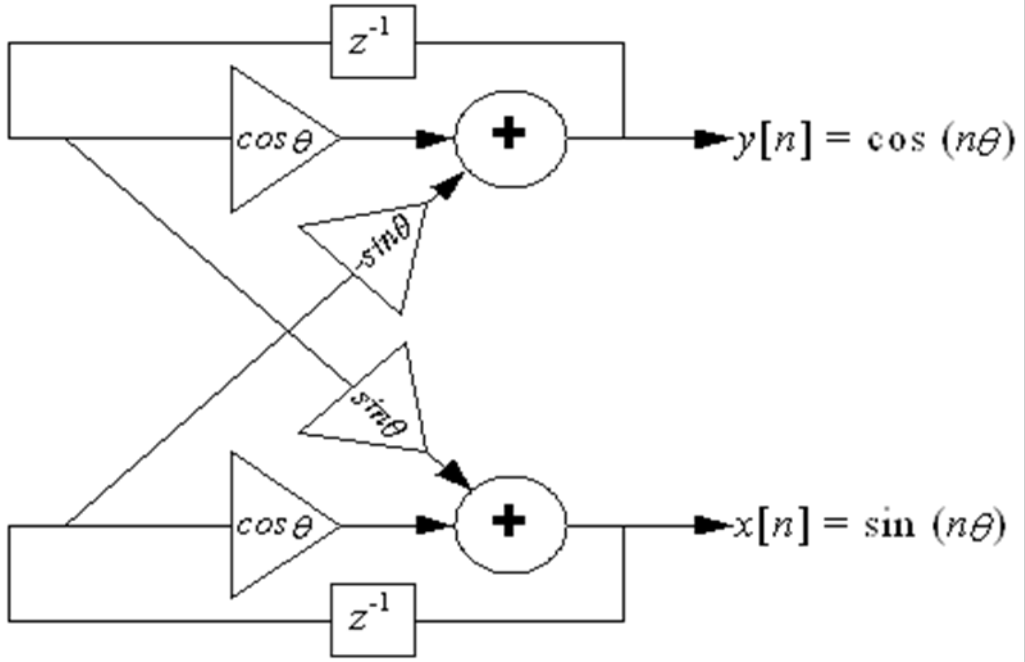
$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{-------- (2)}$$

- Thus, we obtain the two coupled difference equations:

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{--- (1)}$$

$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{--- (2)}$$

- Using equations (1) and (2), the structure for the realisation of the coupled-form oscillator is shown below:



Digital Resonator Fundamentals : Digital Oscillators

Example 4:

- An oscillator is given by the following coupled difference equations expressed in matrix form.

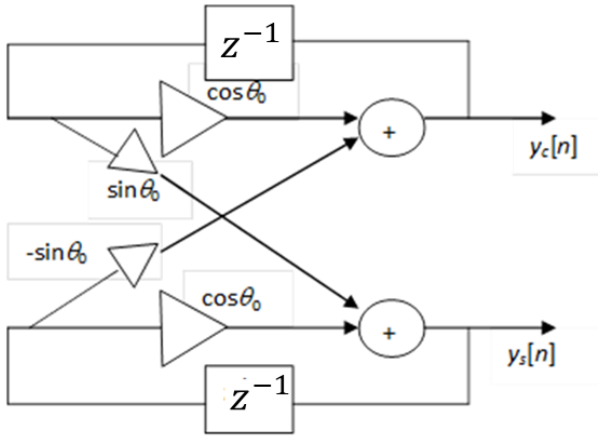
$$\begin{bmatrix} y_c[n] \\ y_s[n] \end{bmatrix} = \begin{bmatrix} cos\,\theta_0 & -sin\,\theta_0 \\ sin\,\theta_0 & cos\,\theta_0 \end{bmatrix} \begin{bmatrix} y_c[n-1] \\ y_s[n-1] \end{bmatrix}$$

(a) Draw the structure for the realisation of this oscillator, where θ_0 is the oscillation frequency.

(b) If the initial conditions $y_c[-1] = A cos\theta_0$ and $y_s[-1] = -A sin\theta_0$, obtain the outputs $y_c[n]$ and $y_s[n]$ using the above difference equations.

Answer:

$$y_c[n] = cos\,\theta_0\, y_c[n-1] - sin\,\theta_0\, y_s[n-1]$$
$$y_s[n] = sin\,\theta_0\, y_c[n-1] + cos\,\theta_0\, y_s[n-1]$$



(b)

$\boldsymbol{Let\ n = 0}$

$y_s[0] = \sin\theta_0\,(A\cos\theta_0) + \cos\theta_0\,(-A sin\theta_0) = 0$

$y_c[0] = \cos\theta_0\,(A cos\theta_0) - \sin\theta_0\,(-A sin\theta_0) = \boldsymbol{A}$

$\boldsymbol{Let\ n = 1}$

$y_c[1] = \cos\theta_0 . A - \sin\theta_0 .0 = \boldsymbol{A cos[1\ \theta_0]}$

$y_s[1] = A\sin\theta_0 + 0 = A\sin\theta_0$

Let $\boldsymbol{n = 2}$

$y_c[2] = \cos\theta_0\, yc[1] - \sin\theta_0\, ys[1]$

$= \cos\theta_0\, A\cos\theta_0 - \sin\theta_0\, A\sin\theta_0$

$= \boldsymbol{A}\cos[\boldsymbol{2}\ \theta_0]$

Let $\boldsymbol{n = n}$

$y_c[n] = \boldsymbol{A}\cos[\boldsymbol{n}\ \theta_0]$

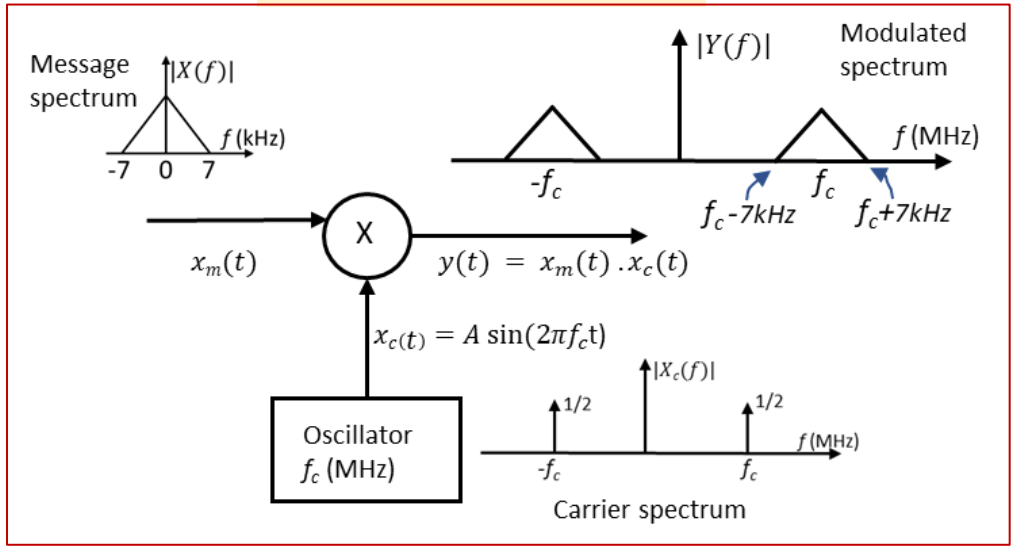
$Similarly \quad y_s[n] = A\sin[n\theta_0]$

Digital Resonator Fundamentals : Digital Oscillators

Note:

- Communication of information often requires transmission of message through space by converting electrical signals to electromagnetic waves using antennas.
- Efficient radiation requires antenna dimensions comparable to the wavelength of the signal .
- The wavelength λ of a radiated signal equals $\frac{c}{f}$, where c is the speed of light (at which electromagnetic waves travel) and f is the signal frequency.
- Transmission of low frequency messages, such as speech and music, would require huge antenna sizes (in km).
- A practical approach is to shift low-frequency messages to a higher frequency using *modulation* and to transmit the modulated signal instead.
- In turn, this requires a means of recovering the original message at the receiving end using a process called *demodulation*.
- Such modulation Scheme (known as amplitude modulation) is shown on the right, where the carrier frequency is generated by an oscillator.

Modulation Scheme



- The amplitude modulation scheme typically modulate the amplitude of a high frequency carrier by a message signal.
- In the above scheme (Double-sideband suppressed carrier), the carrier $x_c(t)$ is multiplied by the message $x_m(t)$.
- The message spectrum is centered about $\pm f_c$ as shown above.
- At the receiver, the message $x_m(t)$ can be recovered using demodulation

ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (pp. 3-8)
- Stability of a Second - Order Filter (pp. 9-11)
- Digital Oscillators (pp. 12-19)
- Tutorial questions (pp. 20-25)

Prof. Eliathamby Ambikairajah, School of EE&T

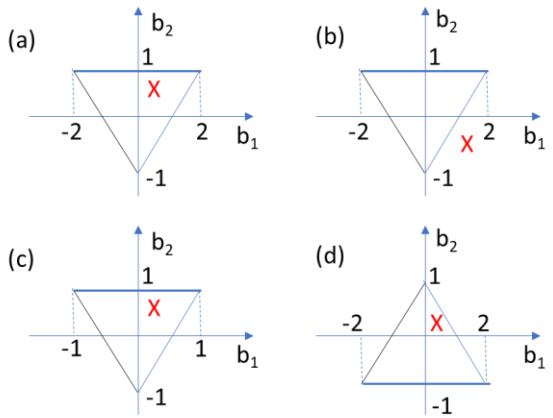
Term 3, 2021

Topic 4: Tutorial Questions (SOLO -Level 1)

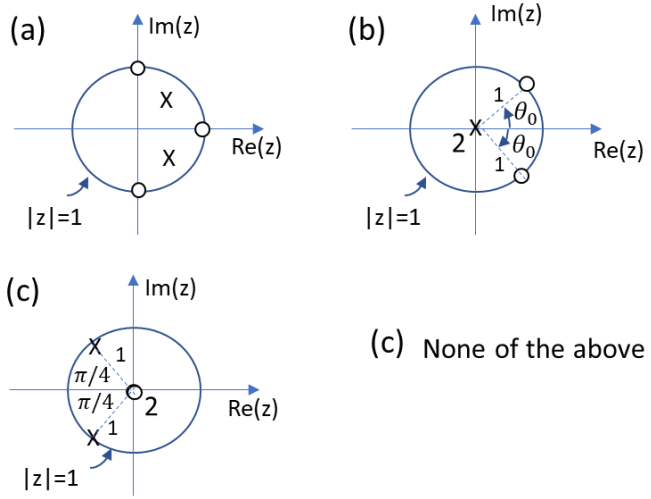
SOLO – Level 1

1. A second-order all-pole filter is described by $H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}$

   Which of the following diagram(s) is used to conclude that the system is stable ? (Tick the correct answer)



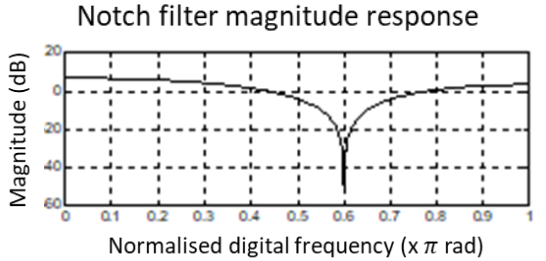
2. Which of the following pole-zero plot represents a digital oscillator? (Tick the correct answer)

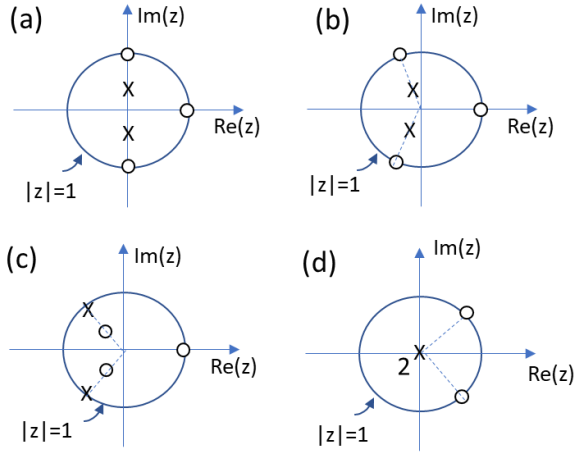


(c) None of the above

SOLO – Level 1

3. Which of the following pole-zero plot is most likely to produce the notch filter response shown below ?(Tick the correct answer)





4. An instrumentation and recording system require that the power-line frequency of 50Hz and harmonics to be eliminated. What type of filter would you choose? (tick the correct answer)

   (a) All-pass filter
   (b) Band-pass filter
   (c) Notch filter
   (d) High-pass filter
   (e) Low- pass filter
   (f) An oscillator
   (g) None of the above

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

1. A digital oscillator has a unit impulse response given by:

$$h[n] = \sin([n+1]\theta_0)\, u[n];$$

where θ_0 is the oscillator frequency;

   (a) Find the transfer function of this oscillator
   (b) Draw the structure for this transfer function using the transfer function obtained in part (a) above
   (c) By setting the input in part (b) to zero and under certain initial conditions, sinusoidal oscillation can be obtained. Find these initial conditions.

2. Using the following difference equation, you are required to implement (i.e. draw a structure) a digital oscillator that generates y[n]=sin(θ_0n) for n ≥0. Choose appropriate initial conditions. Note that: $x[n]$ is the input; $y[n]$ is the output and r is a constant. You are required to choose an appropriate value for r in order to sustain oscillation.

$$y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$$

3. Design a notch filter with notch frequency 60 Hz, stopband 5 Hz (bandwidth) and sampling frequency 300Hz.

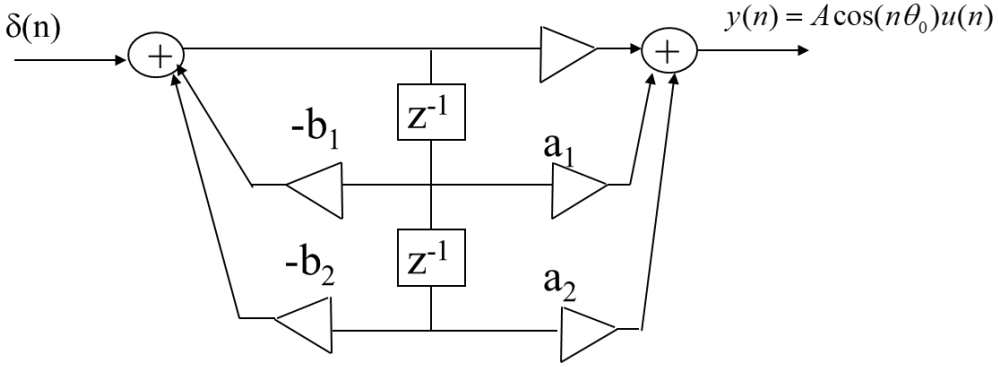
4. Design a bandpass filter with centre frequency of 100Hz, passband 10Hz, stopband edges at 50 Hz and 150 Hz , and sampling frequency 400Hz.
   Note: If the designed filter becomes non causal, then use double poles at each pole locations to get a causal filter.

5. An IIR oscillator which is initialised by an impulse is shown below. The desired impulse response is :

$$y[n] = A\cos[n\theta_0]u[n];$$

where θ_0 is the oscillator frequency;

Assuming θ_0 is the oscillator frequency of the digital oscillator, and by writing appropriate equations, find the values of b_1,b_2, a_0, a_1 and a_2 for sustaining the oscillation.



6. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.

Show that the gain b_0 is given by the following, when $|H(\theta_0)| = 1$,

$$b_0 = (1-r)\sqrt{1 + r^2 - 2r\cos(2\theta_0)}$$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

1. For the structure shown in Figure 1, write down the appropriate difference equations and hence state the function of this structure.

2. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0(1 - z^{-1})(1 + z^{-1})}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency. Determine the gain b_0 for the digital resonator so that $|H(\theta_0)| = 1$.

3. Using the z-transform table, (a) determine the difference equations for the digital oscillators that have impulse responses $h[n] = A\,sin[n\theta_0]u[n]$ and $h[n] = A\,cos[n\theta_0]u[n]$ respectively; (b) Determine the structure for the coupled-form oscillator by combining the structure for the digital oscillator obtained in part (a) above.

4. Design an FIR filter that will reject a very strong 60-Hz sinusoidal interference contaminating a 200Hz useful sinusoidal signal. Determine the gain of the filter so that the useful signal does not change amplitude. The filter operates at a sampling frequency f_s = 500 samples/sec.

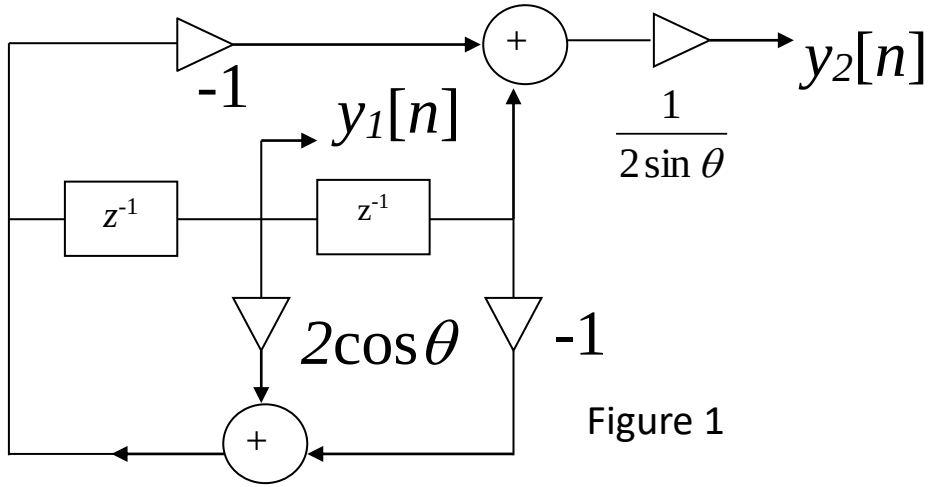


Figure 1

5. The transfer function of a digital resonator is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

The impulse response of this resonator can be found by partial fraction expansion . Show that the impulse response h[n] is given by:

$$h[n] = \frac{b_0\, r^n}{\sin(\theta_0)} \sin[(n+1)\theta_0]\, u[n]$$

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

1. For the system shown in figure 2,

   (a) Express $y_1[n]$ in terms of $y_1[n-1]$, $y_2[n-1]$, and $x[n]$; do the same for $y_2[n]$.

   (b) You may assume $A = \cos(\theta_0)$; $B = \sin(\theta_0)$; $y_1[-1] = \cos(-\theta_0)$; $y_2(-1) = \sin(-\theta_0)$. If $x[n] = 0$, show that : $y_1[n] = \cos(n\theta_0)$ and $y_2[n] = \sin(n\theta_0)$

   (c) Obtain the system transfer function

   $H_1(z) = \frac{Y_1(z)}{X(z)}$ and $H_2(z) = \frac{Y_2(z)}{X(z)}$ where A and B arbitrary values.

   (d) If $A = B = \frac{\sqrt{2}}{2}$, sketch the pole-zero pattern for $H_1(z)$ and $H_2(z)$

   (e) If $A = B = \frac{\sqrt{2}}{2}$ and $x[n] = \delta[n]$, find the impulse response $h_1[n]$.

2. Consider a filter with transfer function

$$H(z) = \frac{b_0\left(1 - e^{j\theta_0}z^{-1}\right)\left(1 - e^{-j\theta_0}z^{-1}\right)}{\left(1 - re^{j\theta_0}z^{-1}\right)\left(1 - re^{-j\theta_0}z^{-1}\right)}$$

   (a) Sketch the pole-zero pattern.

   (b) Show that for $r \approx 1$, , the system is a notch filter and provide a rough sketch of its magnitude response if $\theta_0 = 60^o$.

   (c) For $\theta_0 = 60^o$, choose b_0 so that the maximum value $|H(\theta)| = 1$.

   (d) Determine the approximate 3-dB bandwidth of this filter.

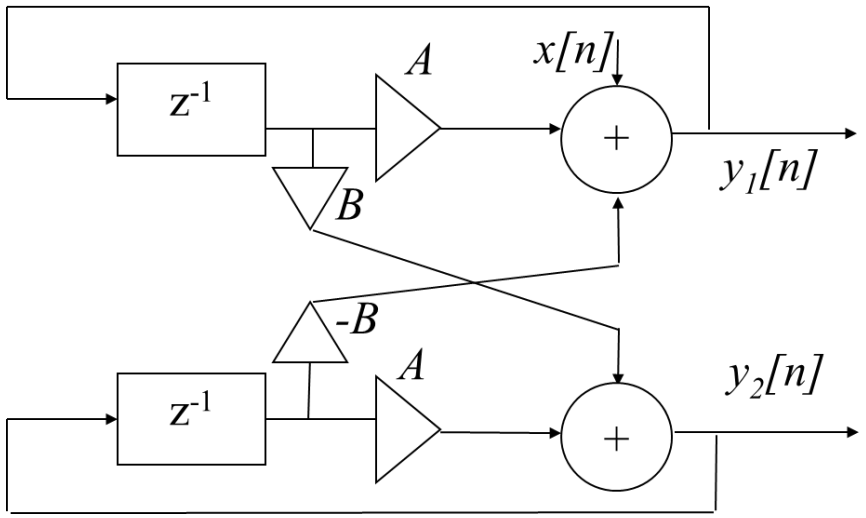


Figure 2

3. By calculating the magnitude response of the following filter with a transfer function

$$H(z) = 1 + z^{-1} + z^{-2} + \ldots\ldots + z^{-8}$$

   determine the frequencies of the analogue sinusoids that cannot pass through the filter. You may assume the sampling frequency $f_s = 1$ kHz.

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

1. Design a digital oscillator with adjustable phase, $\emptyset$, and produces an output

$$y[n] = \cos(n\theta_0 + \emptyset)\, u[n]$$

2. A digital notch filter is required to remove an undesirable 60 Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is fs = 500 samples/sec.

   (a) Design a second-order FIR notch filter and

   (b) A second-order pole-zero notch filter for this purpose

   In both cases choose the gain b_0 so that $|H(\theta)| = 1\ for\ \theta = 0$.

3. A comb filter may be viewed as a notch filter in which the spectral nulls occur periodically across the frequency band. Comb filters find applications in a wide range of practical systems, for example, in the rejection of power-line harmonics. A comb filter can be described using a moving average FIR filter with the difference equation

$$y[n] = \frac{1}{M+1}\sum_{k=0}^{M} x[n-k]$$

   (a) Show that the frequency response of the above filter is given by $H(\theta) = \frac{e^{-j\theta M/2}}{M+1}\frac{\sin\theta(\frac{M+1}{2})}{\sin(\frac{\theta}{2})}$

   (b) Sketch the magnitude characteristic of the comb filter (for M= 10) that you obtained in part (a) above, but clearly illustrating the existence of the periodically spaced zeros at $\theta_k = \frac{2\pi k}{M+1}\ for\ k = 1,2,\ldots.M$.

Week 7

Week 7: Introduction to Analogue Filters 2
7.1 Introduction..... 2
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7.3 Lowpass to High pass transformation 10
7.4 Low-pass to Band-pass Transformations 12
7.5 Low-pass to Band-stop Transformations..... 13

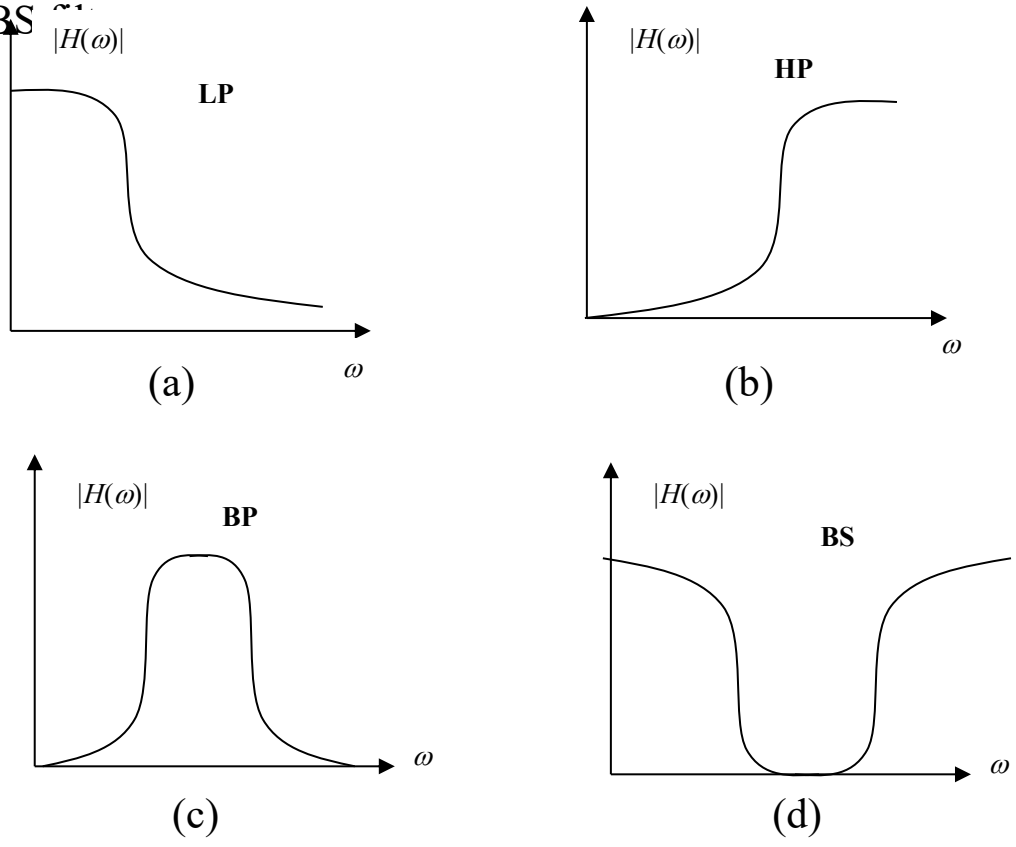
7.1 Introduction

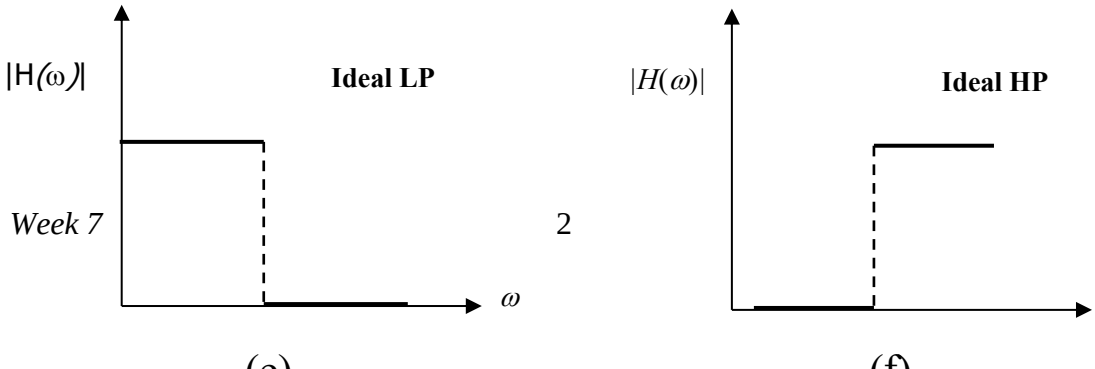
A very important approach to the design of digital filters is to apply a transformation to an existing analogue filter.

The commonly required types of frequency filter fall into four main groups:

- Low-pass (LP)
- High-pass (HP)
- Band-pass (BP)
- Band-stop (BS)

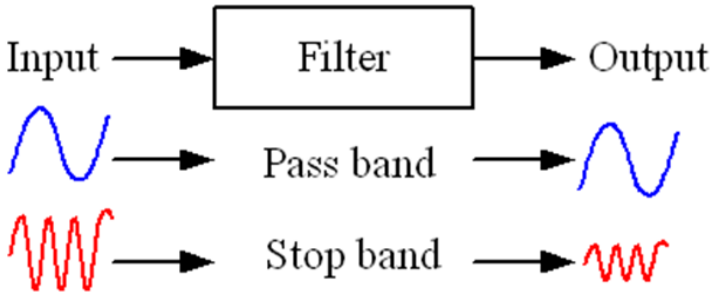
The frequency responses of these filters are shown in Figure **Figure** Error! No text of specified style in document.**.1** (a to d). Also shown are the frequency responses for the **ideal** LP, HP, BP and BS filters.





The analogue filter design procedure normally begins with a specification of the frequency response for the filter describing how the filter reacts to sinusoid inputs.

If an input sinusoid is not attenuated or attenuated less than a specific tolerance as it goes through the system, it is said to be in a **pass band** of the filter.



If it is attenuated more than a specified value it is said to be stopped and within the **stop band** of the filter.

Input sinusoids with neither a little nor a large amount of attenuation are said to be in the **transition band**. A typical frequency response is shown in **Figure** Error! No text of specified style in document.**.2**, showing the pass band, transition band and stop band.

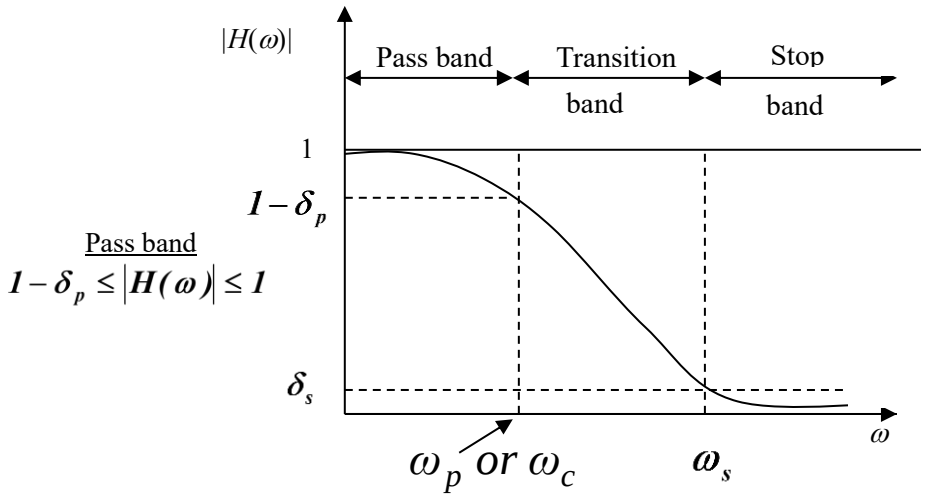


***Figure** Error! No text of specified style in document.**.2**: Low-pass filter frequency response*

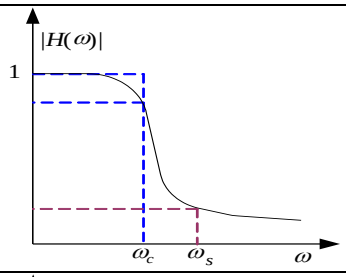
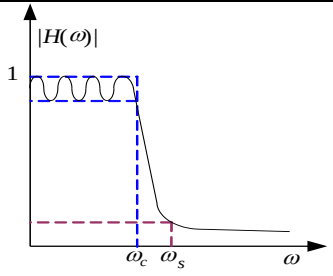
The filter with this type of frequency response is called a low-pass filter as it passes all frequencies less than a certain value ω_p (or ω_c), called the cut-off frequency and attenuates or stops all frequencies past ω_s, the stop band frequency.

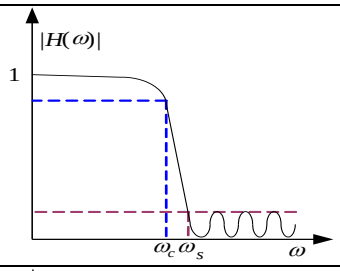
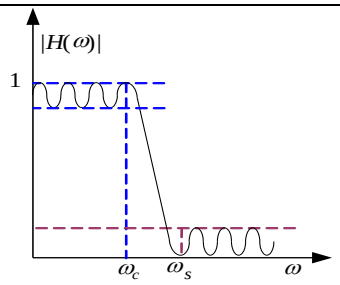
The analogue low pass filters are commonly used as **anti-aliasing** filters that are applied to continuous-time signals before A/D conversion and also used as reconstruction filters.

Note: The technique used to design analogue filters is to specify a prototype low-pass filter function which is normalised to provide a Cut-off frequency (ω_c) at 1 *rad/sec* and then apply transformations to achieve the actual desired cut-off frequencies and filter type.

Therefore, prime consideration will be given to low-pass filter design using the following:

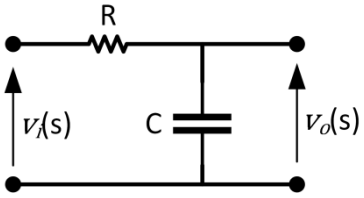
- Butterworth low-pass filter
- Chebyshev I low-pass filter
- Chebyshev II low-pass filter
- Elliptic low-pass filter
- Bessel low-pass filter

Name	**Property**	
Butterworth	Maximally flat in passband Stopband attenuation relatively poor	
Chebyshev I	Good match with ideal filter characteristic Maximal ripple in passband	

Chebyshev II	Good match with ideal filter characteristic Maximal ripple in stopband	
Elliptic	Minimal transition width Ripple in both passband and stopband	
Bessel	Maximally constant group delay $\{\tau_g = -\frac{d\phi(\theta)}{\theta}\}$	$\phi(\theta) = -a\theta$ *or* $\phi(\theta) = b - a\theta$

7.2 Passive RC filters

- Passive 1st order low-pass filter



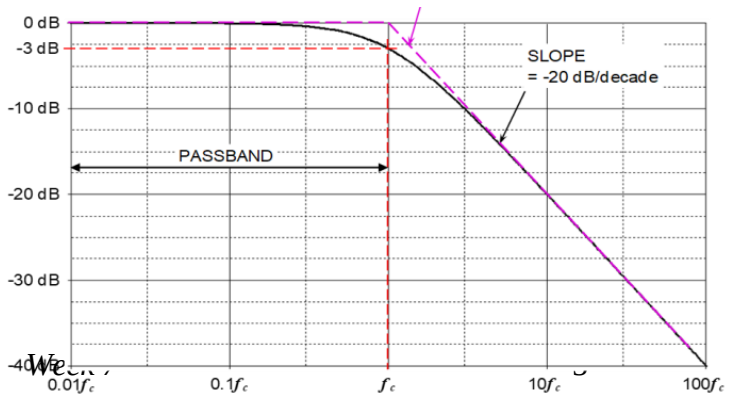
$$H(s) = \frac{v_o(s)}{v_i(s)} = \frac{\omega_c}{s + \omega_c} \quad \text{where } \omega_c = \frac{1}{RC}$$

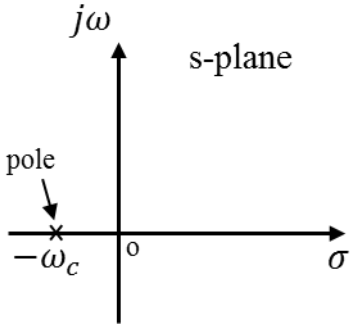
$$H(j\omega) = \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

$$|H(j\omega)|_{dB} = 20\ log\left[\frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}\right]$$

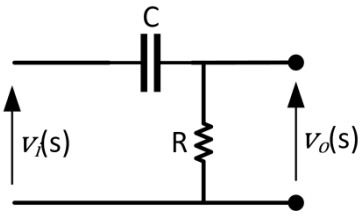
Note

- s - Complex frequency
- $s = j\omega$ sinusoid
- $s = j\omega + \sigma$ Exponential sinusoid
- $s = 0$ dc

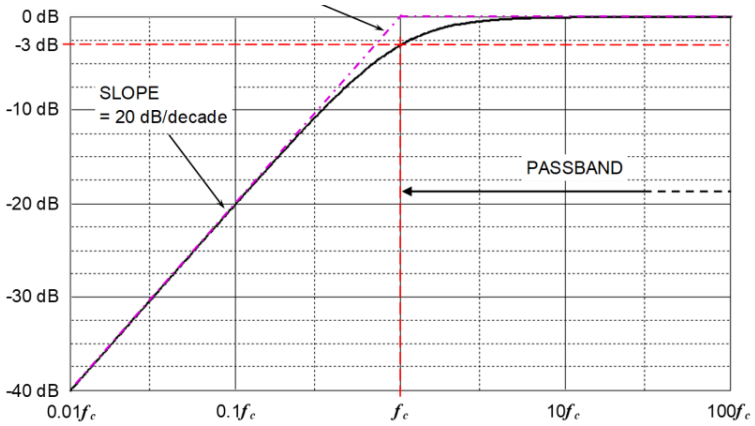


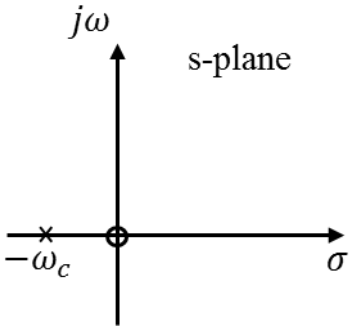


- Passive 1st order high-pass filter



$$H(s) = \frac{v_o(s)}{v_i(s)} = \frac{s}{s + \omega_c} \quad \text{where } \omega_c = \frac{1}{RC}$$





Note

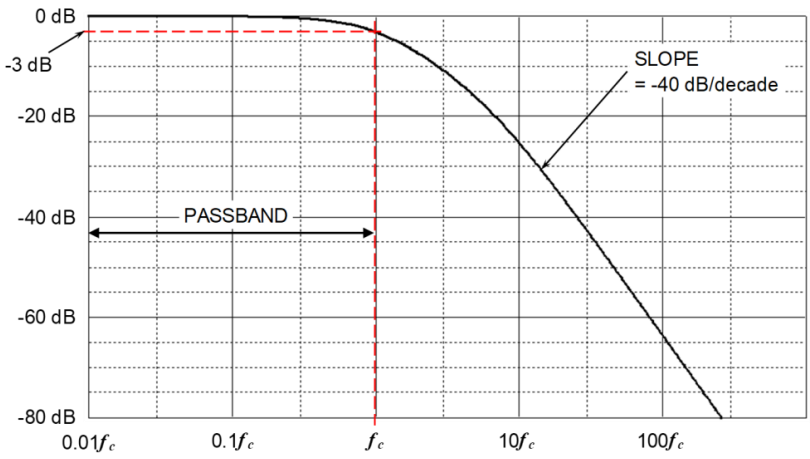
1st order LPF	1st order HPF
$H(s) = \frac{\omega_c}{s + \omega_c}$	$H(s) = \frac{s}{s + \omega_c}$

- Passive 2nd order low-pass filter

$$H(s) = \frac{{\omega_o}^2}{s^2 + b_o s + {\omega_o}^2} \qquad b_o = \frac{\omega_o}{Q}$$

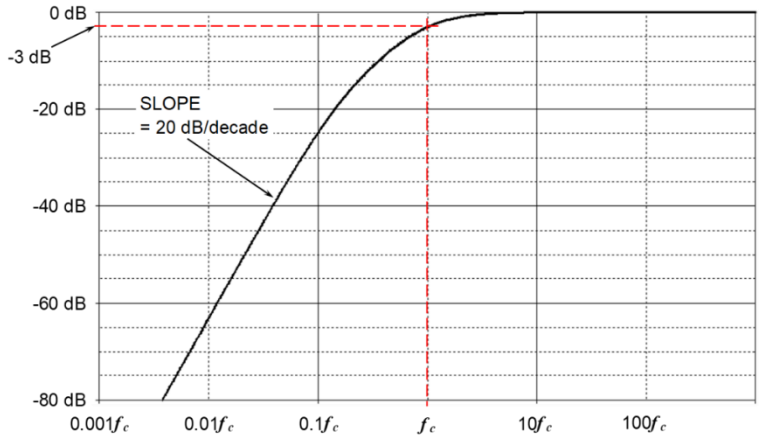
ω_o ← Undamped resonant frequency

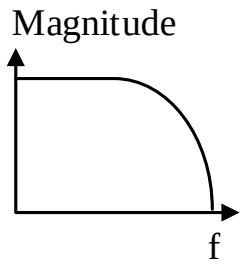
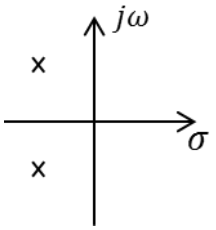
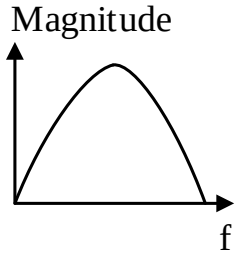
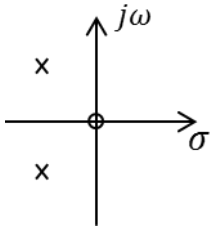
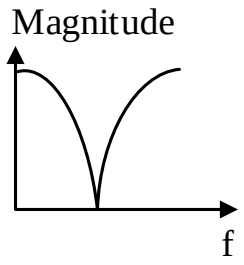
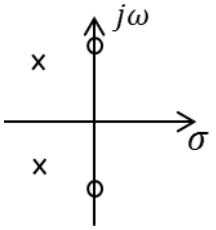
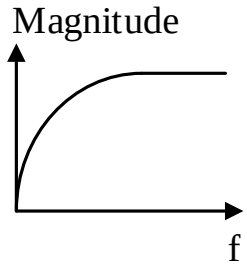
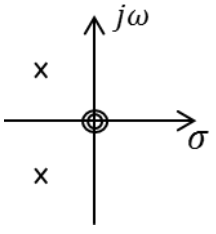
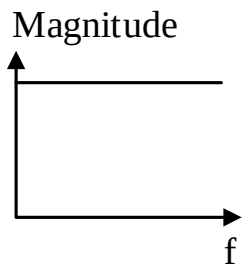
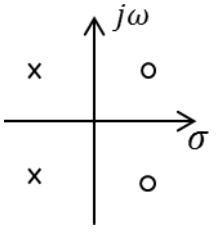
Q ← Q - factor
Q - Quality



- Passive 2nd order high-pass filter

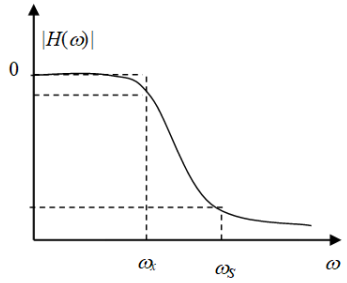
$$H(s) = \frac{s^2}{s^2 + b_o s + {\omega_o}^2}$$



Filter Type	Magnitude	Pole Location	Transfer function
Low-pass			$\frac{{\omega_o}^2}{s^2 + \frac{\omega_o}{Q}s + {\omega_o}^2}$
Band-pass			$\frac{\frac{\omega_o}{Q}s}{s^2 + \frac{\omega_o}{Q}s + {\omega_o}^2}$
Notch (Band-reject)			$\frac{s^2 + {\omega_z}^2}{s^2 + \frac{\omega_o}{Q}s + {\omega_o}^2}$
High-pass			$\frac{s^2}{s^2 + \frac{\omega_o}{Q}s + {\omega_o}^2}$
All-pass			$\frac{s^2 - \frac{\omega_o}{Q}s + {\omega_o}^2}{s^2 + \frac{\omega_o}{Q}s + {\omega_o}^2}$

Note: Lowpass Butterworth Filter

The magnitude response of a Butterworth filter is shown below:



- Lowpass Butterworth filter is a monotonically decreasing function of ω (i.e. throughout pass band and stop band).

The transfer function $H(s)$ is given by:

$$H(s) = \frac{1}{s+1} \quad 1^{st} \text{ order}$$

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad 2^{nd} \text{ order}$$

$$H(s) = \frac{1}{(s+1)(s^2+s+1)} \quad 3^{rd} \text{ order}$$

7.3 Lowpass to High pass transformation

To transform analogue low-pass filter $H(s)$ with unity cut-off frequency to low-pass filter $H(s)$ with cut-off frequency ω_c, we substitute

$$s \rightarrow \frac{s}{\omega_c}$$

Example: First order Butterworth prototype filter is given by

$$H(s) = \frac{1}{s+1} \qquad \longleftarrow \text{ Normalised}$$

To transform to new cut-off frequency $\omega_c = 5$, we replace s with $\frac{s}{\omega_c}$

$$H(s) = \frac{1}{\frac{s}{\omega_c}+1} = \frac{\omega_c}{s+\omega_c} = \frac{5}{s+5}$$

Note: To transform an analogue low-pass filter $H(s)$ with unit cut-off frequency to high-pass filter $H(s)$ with cut-off frequency ω_C, we substitute

$$s \rightarrow \frac{\omega_C}{s}$$

Example: First order Butterworth prototype filter is given by

$$H(s) = \frac{1}{s+1} \quad \longleftarrow \quad \text{Normalised}$$

To transfer to a high-pass filter with cut-off frequency $\omega_c = 2$

We replace $s \to \frac{\omega_c}{s}$; $\qquad H(s) = \frac{1}{\frac{\omega_c}{s}+1} = \frac{s}{s+2}$

Note: *High-pass filter contains zeros as well as poles*

Example:

$$H(s) = \frac{1}{(s+1)(s^2+s+1)} \quad \longleftarrow \quad 3^{rd} \text{ order Butterworth filter.}$$

Determine the transfer function of the corresponding high-pass filter with cut off frequency $\omega_c = 1$ (normalised)

$$s \to \frac{\omega_c}{s} \quad \text{i.e.} \quad s \to \frac{1}{s}$$

normalised

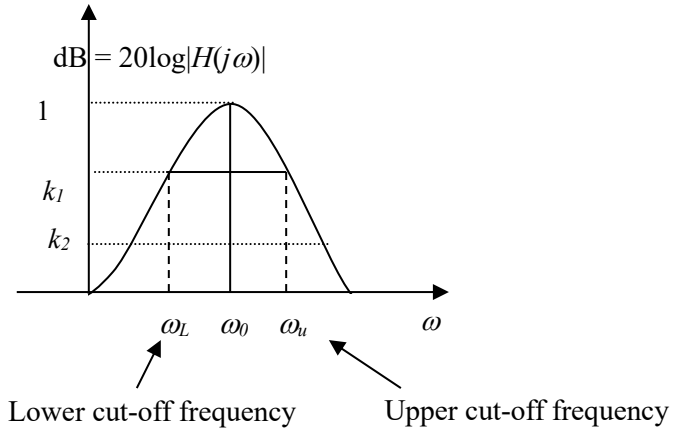
$$H(s) = \frac{1}{\left(\frac{1}{s}+1\right)\left(\left(\frac{1}{s}\right)^2 + \frac{1}{s}+1\right)} = \frac{s^3}{(s+1)\left(s^2+s+1\right)}$$

7.4 Low-pass to Band-pass Transformations

By definition, a band pass filter rejects both low and high frequency components and passes a certain band of frequencies some where between them. Thus the frequency response $H(j\omega)$, of a band-pass filter has the following properties.

1. $|H(j\omega)| = 0$ at both $\omega = 0$ & $\omega = \infty$

2. $|H(j\omega)| = 1$ for a frequency band centered on ω_0, where ω_0 is the mid frequency of the filter



$$B = \omega_u - \omega_L$$

Bandwidth of the band-pass filter

Method 1: To convert unity cut off low-pass filter $H(s)$ into a Band-Pass filter $H(s)$ with lower cut-off frequency ω_L and the upper cut-off frequency ω_u, we replace

$$s \rightarrow \frac{s^2 + \omega_L \omega_u}{s(\omega_u - \omega_L)}$$

Method 2: A lowpass to bandpass transformation can be performed by

$$s \rightarrow \frac{s^2 + {\omega_o}^2}{Bs}$$

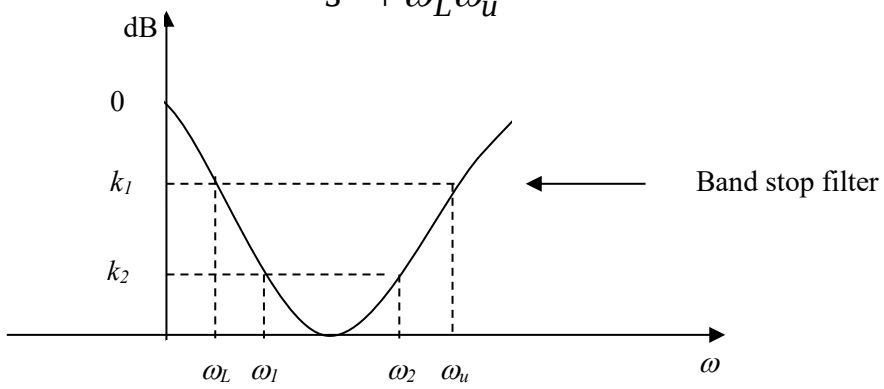
B - Bandwidth of the band-pass filter $(\omega_u - \omega_L)$

ω_0 - centre frequency.

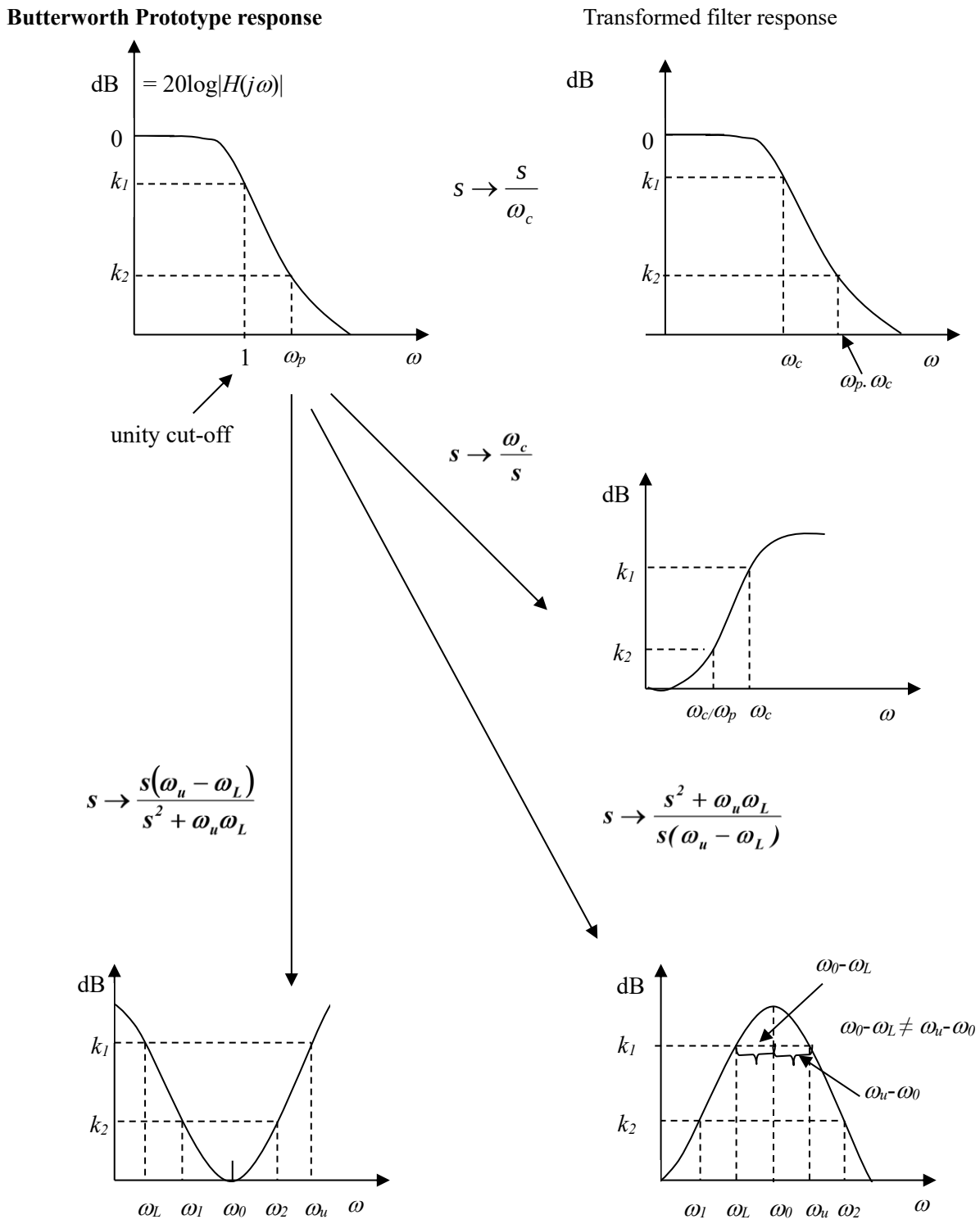
7.5 Low-pass to Band-stop Transformations

Similarly to convert unity cut-off lowpass filter $H(s)$ into a bandstop filter $H(s)$ with lower cutoff frequency ω_L and upper cut-off frequency ω_u, we replace:

$$s \rightarrow \frac{s(\omega_u - \omega_L)}{s^2 + \omega_L \omega_u}$$



Summary: Analogue to Analogue Transformation



Topic 5

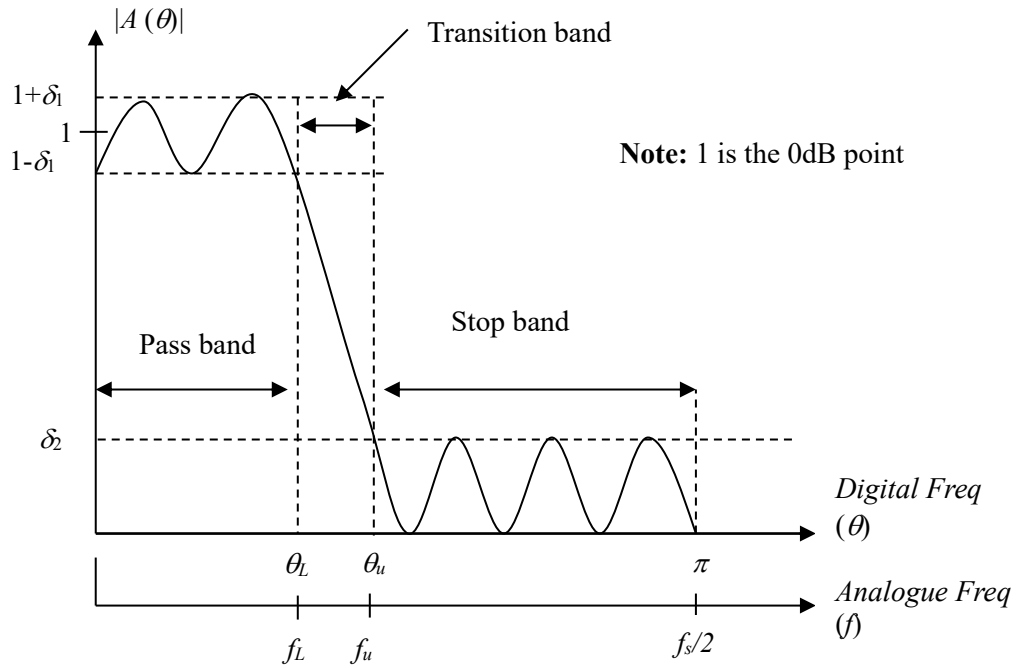
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Introduction

The design of a digital filter is the task of determining a transfer function which is a rational function of z^{-1} (e.g. $\frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$) in the case of a recursive filter (IIR) or a polynomial in z^{-1}

($a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}$) in the case of a non-recursive filter.

Performance specifications: A typical amplitude characteristic of a low-pass filter is shown below.



The goal of the design is to determine a transfer function H(z) so that its amplitude characteristic |H(θ)| satisfies the conditions.

$$1 - \delta_1 \le |H(\theta)| \le 1 + \delta_1 \quad \text{for } 0 \le \theta \leqslant \theta_L$$

$$|H(\theta)| \le \delta_2 \quad \text{for } \theta_u \le \theta \le \pi$$

Choosing between FIR and IIR filters

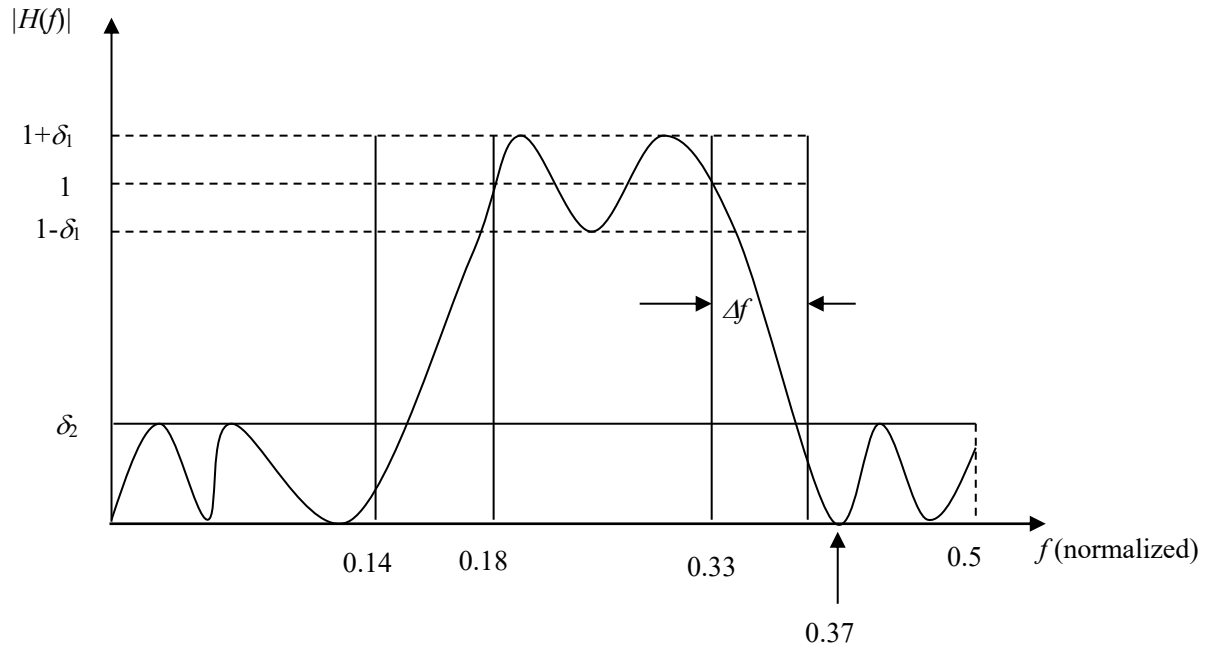
FIR Filters	IIR Filters
Difference equation $y[n] = b_0 x[n] + b_1 x[n-1] + ... + b_M x[n-M]$	Difference equation $y[n] = b_0 x[n] + b_1 x[n-1] + ... + b_M x[n-M]$ $- a_1 y[n-1] - ... - a_N y[n-N]$
Transfer function $H(z) = b_0 + b_1 z^{-1} + ... + b_M z^{-M}$	Transfer function $H(z) = \frac{b_0 + b_1 z^{-1} + ... + b_M z^{-M}}{1 + a_1 z^{-1} + ... + a_N z^{-N}}$
System function contain only zeros	Contain poles and zeros (normally)
Non-recursive or recursive structures are both possible; the best known is the non-recursive (transversal) structure.	Only recursive structure is possible; the most widely used form is the cascade connection of first-order and second order sections.
FIR Filters can have an exactly linear phase response. The implication of this is that no phase distortion is introduced into the signal by the filter.	The phase responses of IIR filters are nonlinear, especially at the band edges.
The effects of using a limited number of bits to implement filters such as round off noise and quantization errors are much less severe in FIR than in IIR.	Because of quantization of the filter coefficients, a pole can in principle move from a position inside the unit circle to a position outside the unit circle and hence cause instability.
FIR requires more coefficients for sharp cut-off filters than IIR. Thus for a given amplitude response specification, more processing time and storage will be required for FIR implementation.	IIR requires fewer coefficients for sharp cut off filters than FIR.
Complexity is proportional to the length of the impulse response.	No direct relation between the complexity and the length of the impulse response (which is infinite by definition) Filters with high selectivity can be realized with relatively low complexity.
FIR filters have no analogue counterpart. FIR design procedures are normally iterative procedures. Design equations do not exist.	Analogue filters can be readily transformed into equivalent IIR digital filters meeting similar specifications. IIR filters can be designed using design formulae.

Example: An FIR filter is to be designed to meet the following frequency response specifications.

- Pass band 0.18-0.33 (normalized)
- Transition band 0.04 (normalized)
- Stop-band deviation 0.001
- passband deviation 0.05

(i) Sketch the tolerance scheme for the filter.
(ii) Express the filter band edge frequencies in the standard unit of kHz, assuming a sampling frequency of 10 kHz and the stop band and pass band deviation in dBs.

The tolerance diagram for the filter is shown below:



f_s=10kHz , therefore

Pass band: 1.8-3.3 kHz
Stop band: 0-1.4 kHz and 3.7-5kHz
Stop band attenuation: $-20\log_{10}(0.001) = -60$dB
Pass band ripple: $20\log_{10}(1+0.05) = 0.42$ dB

Example: The following transfer functions represent two different filters meeting identical amplitude–frequency response specifications.

$$H_1(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} \quad \longleftarrow \quad \text{Filter 1}$$

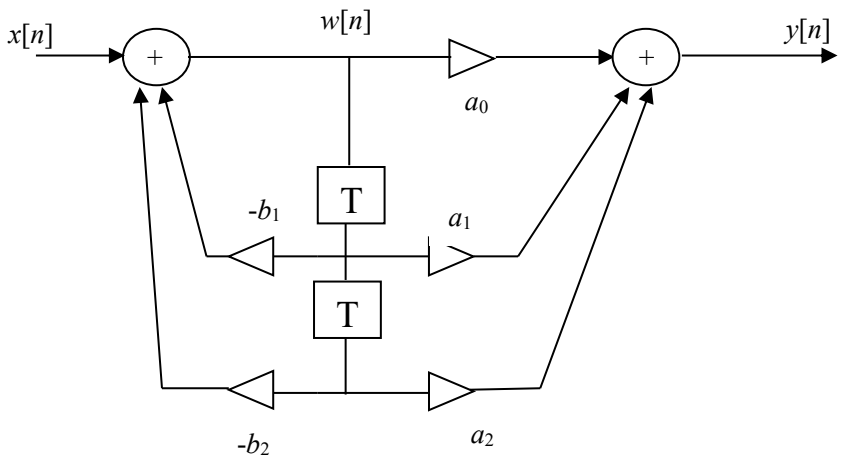
where $a_0 = 0.4981819$; $a_1 = 0.9274777$;
$a_2 = 0.4981819$; $b_1 = -0.6744878$; $b_2 = -0.3633482$;

$$H(z) = \sum_{k=0}^{11} h[k] z^{-k} \quad \longleftarrow \quad \text{Filter 2}$$

where

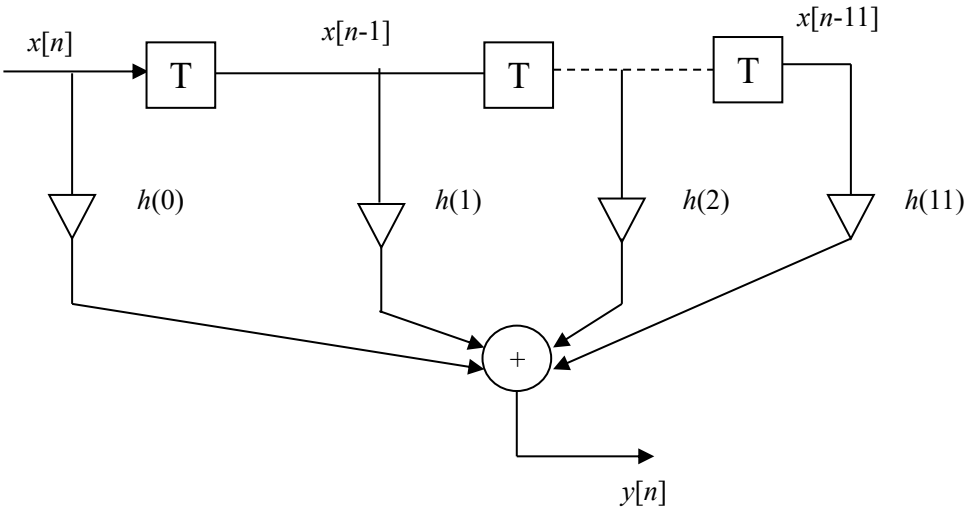
$$h[0] = 0.54603280 \times 10^{2} = h[11]$$
$$h[1] = -0.45068750 \times 10^{-1} = h)[10]$$
$$h[2] = 0.69169420 \times 10^{-1} = h[9]$$
$$h[3] = -0.55384370 \times 10^{-1} = h[8]$$
$$h[4] = -0.63428410 \times 10^{-1} = h[7]$$
$$h[5] = 0.57892400 \times 10^{0} = h[6]$$

Determine and comment on the computational and storage requirements.



Filter 1

$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$
$$y(n) = a_o w(n) + a_1 w(n-1) + a_2 w(n-2)$$



Filter 2

Computation and storage requirement of FIR and IIR

	Filter 1 (IIR)	**Filter 2 (FIR)**
Number of Multiplications	5	12
Number of Additions	4	11
Storage locations for coefficients and data	8	24

It is evident that the IIR filter is more economical in both computations and storage requirements than the FIR filter.

Design Techniques

The method used to calculate the filter coefficients (h_k for FIR, a_k and b_k for IIR) depends on whether the filter is IIR or FIR type.

There are several methods of calculating filter coefficients of which the following are the most widely used.

FIR digital filters	IIR digital filters
• window	• Impulse invariant transformation
• Frequency sampling	• Bilinear transformation
• Optimisation method (e.g Remez Algorithm)	• Pole-zero placement method
$\left\{\frac{Y(z)}{X(z)} = h_0 + h_1 z^{-1} + h_2 z^{-2} + \ldots\ldots + h_{N-1} z^{-(N-1)}\right\}$	$\left\{\frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + \ldots\ldots + a_n z^{-(N-1)}}{1 + b_1 z^{-1} + \ldots\ldots\ldots + b_L z^{-L}}\right\}$

We choose the method that best suits our particular applications.

In most cases, if the FIR properties are vital then a good candidate is the optimization method, where as, if IIR properties are desirable, then the bilinear method will in most cases suffice.

IIR Filter Design

In transforming an analogue filter to digital filter, we must obtain either $H(z)$ or $h[n]$ from the analogue filter design. In such transformations, we generally require that the essential properties of the analogue frequency response be preserved in the frequency response of the resulting digital filter. This implies that we want the imaginary axis of the s-plane to map into the unit circle of z-plane.

A second condition is that a stable analogue filter should be transformed to a stable digital filter. That is if the analogue

system has two poles only in the left half s-plane, then the digital filter must have poles inside the unit circle.

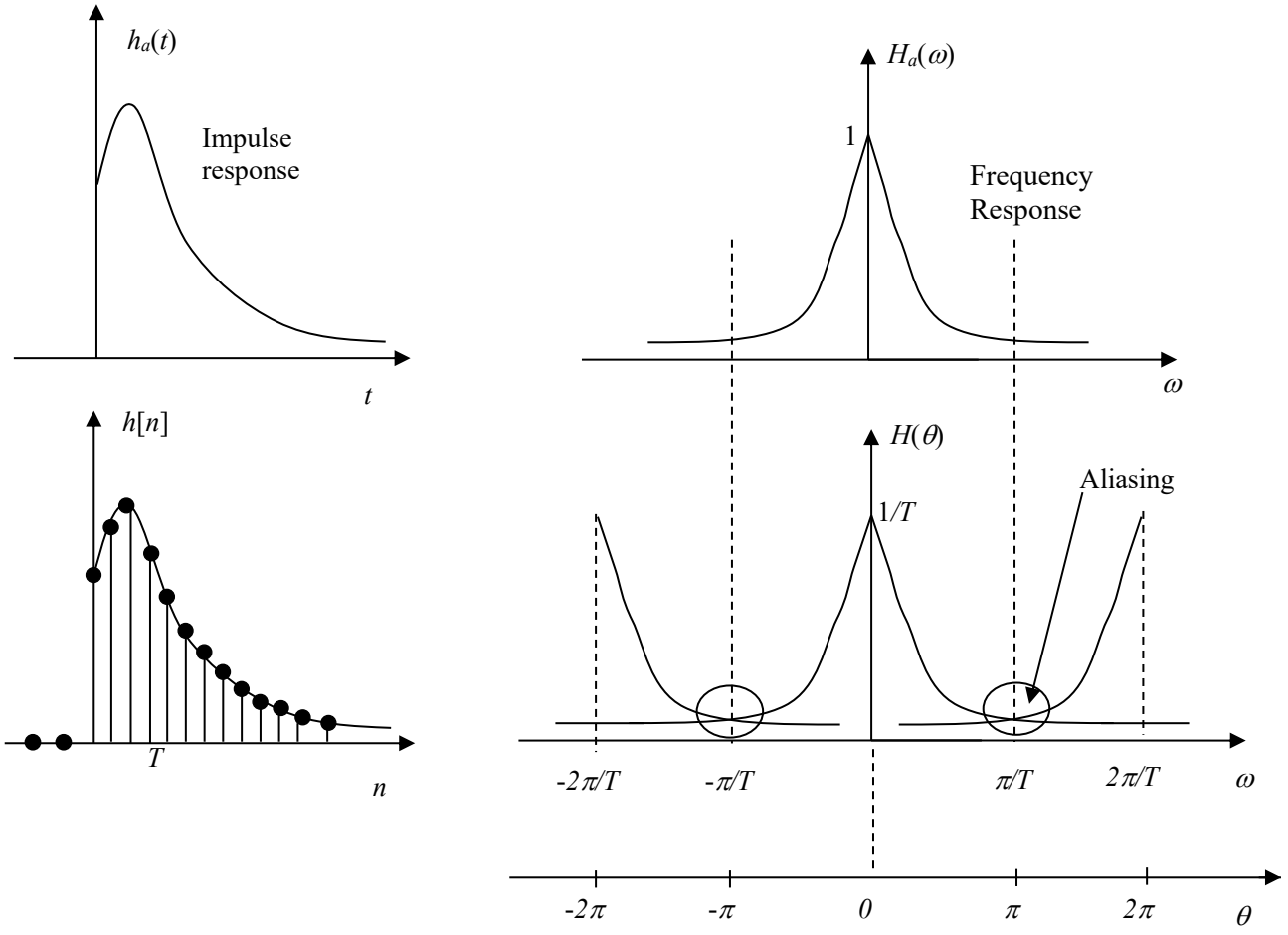
Impulse invariant method

In this method we start from an analogue filter of impulse response $h_a(t)$ and the system function $H_a(s)$.

The objective of our design is to realize an IIR filter with an impulse response $h[n]$ which satisfies :

$h[n] = h_a(nT)$ where T-sampling frequency

The characteristic property preserved by this transformation is that the impulse response of the resulting digital filter is a sampled version of the impulse response of the analogue filter.



We see that with this method there are problems to a greater or lesser extent depending on the choice of T.

The sampling frequency affects the frequency response of the impulse invariant discrete filter. A sufficient high sampling frequency is necessary for the frequency response to be close to that of the equivalent analogue filter. Thus due to aliasing, the frequency response of the digital filter will not be identical to that of the analogue filter. So how do we find the filter coefficients of the IIR filter in this design method?

To obtain the mapping let,

$$H_a(s) = \frac{1}{s+b}, \quad b > 0$$

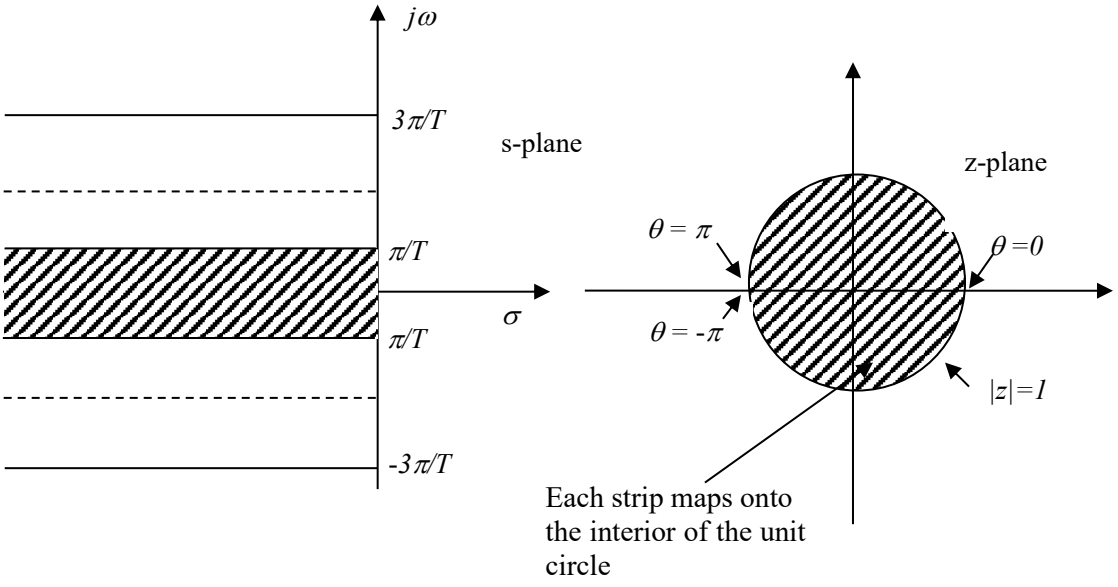
Inverse Laplace transform, $h_a(t) = e^{-bt}$

Sampled sequence, $h_a(nT) = e^{-bnT}$

Usually written as $h[n] = \begin{cases} e^{-bnT} & n \geq 0 \\ 0 & n < 0 \end{cases}$

z-transform, $H(z) = \dfrac{1}{1 - e^{-bT} z^{-1}}$

It is seen that $H(z)$ is obtained from $H_a(s)$ by using the mapping relationship, $\dfrac{1}{s+b} \rightarrow \dfrac{1}{1-e^{-bT}z^{-1}}$, $b > 0$, T-sampling period



In this kind of mapping, the perimeter is the imaginary axis.
Note: Mapping does not exist for the zeros.

Example:

$$H(s)=\frac{2}{(s+1)(s+3)}$$

Using partial fractions this can be written as

$$H(s)=\frac{1}{s+1}-\frac{1}{s+3}$$

$$\downarrow \quad \frac{1}{s+b} \rightarrow \frac{1}{1-e^{-bT}z^{-1}}$$

$$H(z)=\frac{1}{1-e^{-T}z^{-1}}-\frac{1}{1-e^{-3T}z^{-1}}$$

$$=\frac{\left(e^{-T}-e^{-3T}\right)z^{-1}}{1-(e^{-T}-e^{-3T})z^{-1}+e^{-4T}z^{-2}}$$

Example:

Use the impulse invariant method to design digital filter from an analogue prototype that has a system function.

$$H(s)=\frac{s+a}{(s+a)^2+b^2}$$

To design a filter using the impulse invariant method, expand *H(s)* in a partial fraction form:

$$H(s)=\frac{s+a}{\left[s+\left(a+jb\right)\right]\left[s+\left(a-jb\right)\right]}=\frac{A}{s+(a+jb)}+\frac{B}{s+(a-jb)}$$

$$=\frac{1/2}{s+(a+jb)}+\frac{1/2}{s+(a-jb)}$$

$$=\frac{1}{2}\left[\frac{1}{s+(a+jb)}\right]+\frac{1}{2}\left[\frac{1}{s+(a-jb)}\right]$$

Substituting $\frac{1}{s+c} \rightarrow \frac{1}{1-e^{-cT}z^{-1}}$ (Impulse invariant transform)

$$H(z) = \frac{1}{2}\left[\frac{1}{1-e^{-(a+jb)T}z^{-1}}\right] + \frac{1}{2}\left[\frac{1}{1-e^{-(a-jb)T}z^{-1}}\right]$$
$$= \frac{\frac{1}{2}\left[1-e^{-(a-jb)T}z^{-1}\right] + \frac{1}{2}\left[1-e^{-(a+jb)T}z^{-1}\right]}{(1-e^{-(a+jb)T}z^{-1})(1-e^{-(a-jb)T}z^{-1})}$$
$$= \frac{1-e^{-aT}\cos(bT)z^{-1}}{1-2e^{-aT}\cos(bT)z^{-1}+e^{-2aT}z^{-2}}$$

Hence,

$$H(z) = \frac{1+a_1z^{-1}}{1+b_1z^{-1}+b_2z^{-2}}$$

where

$$a_1 = e^{-aT}\cos(bT), \quad b_1 = -2e^{-aT}\cos(bT) \quad \text{and} \quad b_2 = e^{-2aT}$$

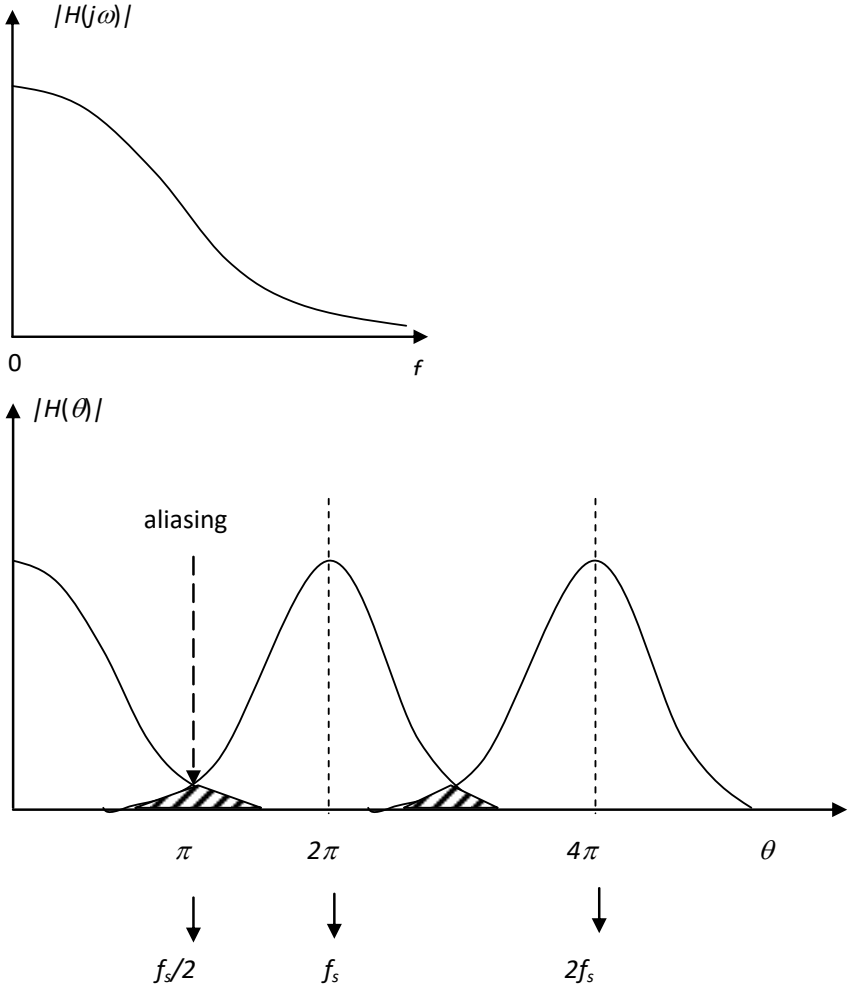
$$\frac{s+a}{(s+a)^2+b^2} \rightarrow \frac{1-e^{-aT}\cos(bT)z^{-1}}{1-2e^{-aT}\cos(bT)z^{-1}+e^{-2aT}z^{-2}}$$

Note that the zero at $s = -a$ is mapped to a zero at $z=e^{-aT}\cos(bT)$.

Thus, the location of the zero in the discrete time filter depends on the position of the poles as well as the zero in the analogue filter.

the digital filter obtained using impulse invariant transformation.

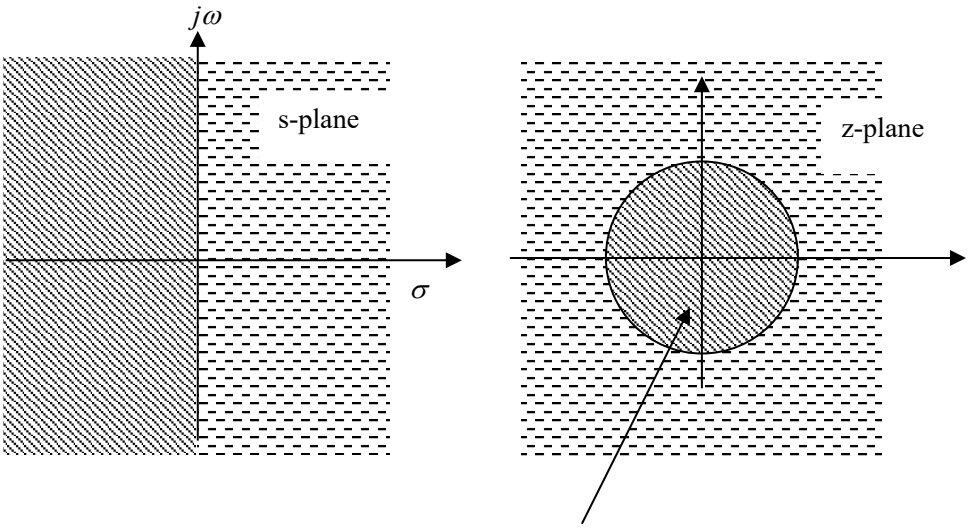
Note: A sufficient high sampling frequency is necessary for the frequency response to be closer to that of the equivalent analogue filter (see below)



Low degree of aliasing can be achieved by making the sampling frequency high.

Bilinear Transformation

The bilinear transformation yields stable digital filters from stable analogue filters (the impulse invariant technique may not). Also the bilinear transformation avoids the problem of aliasing encountered with the use of the impulse invariant transformation, because it maps the entire imaginary axis in the s-plane on to the unit circle in the z-plane.



$$z = e^{sT} = \frac{e^{\frac{sT}{2}}}{e^{\frac{-sT}{2}}} = \frac{1+\frac{sT}{2}+\left(\frac{sT}{2}\right)^2\cdot\frac{1}{2}+\ldots}{1-\frac{sT}{2}+\left(\frac{sT}{2}\right)^2\cdot\frac{1}{2}+\ldots}$$

$$z \approx \frac{1+\frac{sT}{2}}{1-\frac{sT}{2}}$$

(drop out higher order terms)

$$\therefore s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$

The price paid for the avoidance of aliasing is an introduction of distortion in the frequency axis. Consequently, the design of digital filters using the bilinear transformation is only useful when the distortion can be compensated.

Example: We have a system function $H_a(s)$ such that

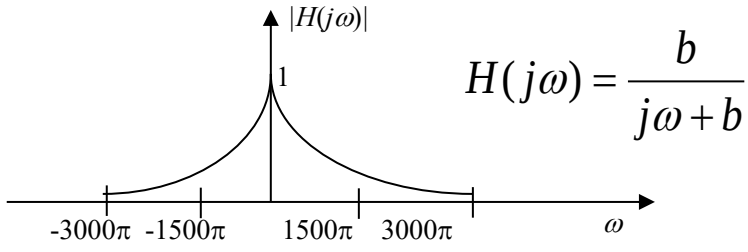
$$H_a(s) = \frac{b}{s+b}$$

Applying bilinear transformation,

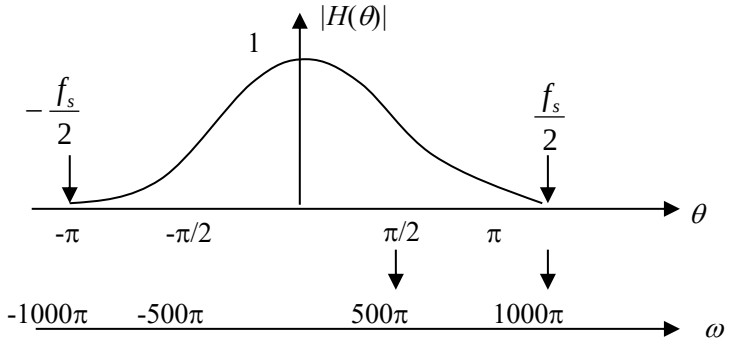
$$H(z) = \frac{b}{\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} + b} = \frac{bT(1+z^{-1})}{bT+2+(bT-2)z^{-1}}$$

Let $b = 1000$ & $T = \frac{1}{1000}$

The modulus of the frequency response $H(\omega)$ and $H(\theta)$ are shown below:



$$H(j\omega) = \frac{b}{j\omega + b}$$



There is a very important property of the bilinear transformation that can be seen in the above example. The entire frequency range ($-\infty \le \omega_a \le \infty$) of the continuous system maps into the fundamental interval ($-\pi \le \theta \le \pi$) of the discrete system, where $\omega = 0$ corresponds to $\theta = 0$, $\omega = \infty$ to $\theta = \pi$ and $\omega = -\infty$ to $\theta = -\pi$.

To demonstrate that this mapping has the property that the imaginary axis in the s-plane maps onto the unit circle, let $z = e^{j\theta}$ and s = $j\omega$. Then since $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$,

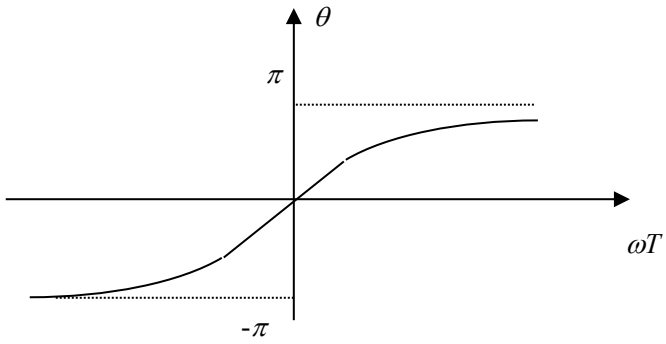
$$j\omega = \frac{2}{T}\frac{1-e^{-j\theta}}{1+e^{-j\theta}}$$

$$j\omega = \frac{2}{T}\, j\tan\left(\frac{\theta}{2}\right)$$

Hence,

$$\omega = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) \text{ or } \theta = 2\tan^{-1}\left(\frac{\omega T}{2}\right)$$

We see that a nonlinear relation exists between ω and θ. This effect is called 'Warping' and is shown below.



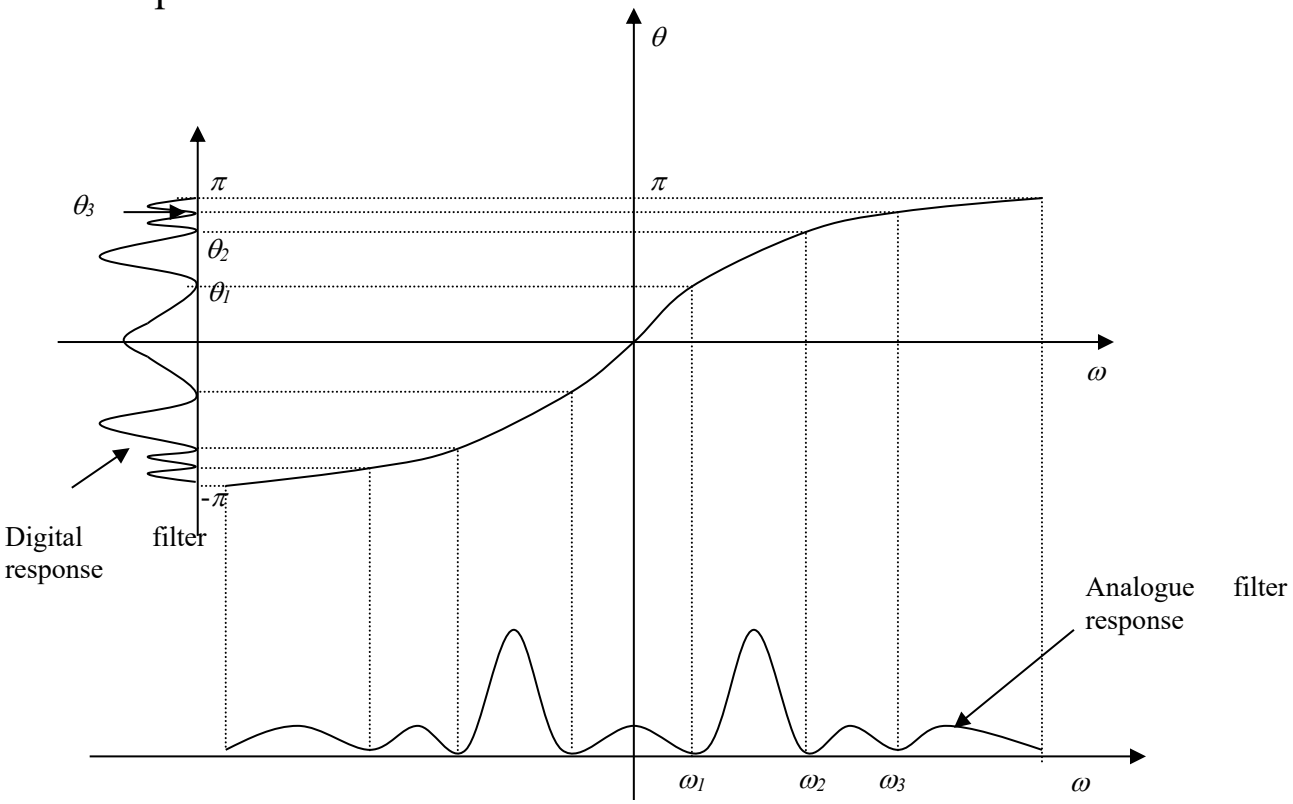
At low frequencies $\tan^{-1}\left(\frac{\omega T}{2}\right) = \frac{\omega T}{2}$.

$$\therefore \theta = 2 \times \frac{\omega T}{2} = \omega T$$

The great advantage of warping is that no aliasing of the frequency characteristic can occur in the transformation of an analogue filter to a discrete filter, which we encountered in the impulse-invariant method.

We must however check carefully just how the various characteristic frequencies of the continuous characteristic frequencies of the discrete filter.

We can illustrate this with the aid of a diagram (below) for a band-pass filter.



The effect of "warping" in the conversion of $|H(\omega)| \rightarrow |H(\theta)|$ is seen from the above diagram.

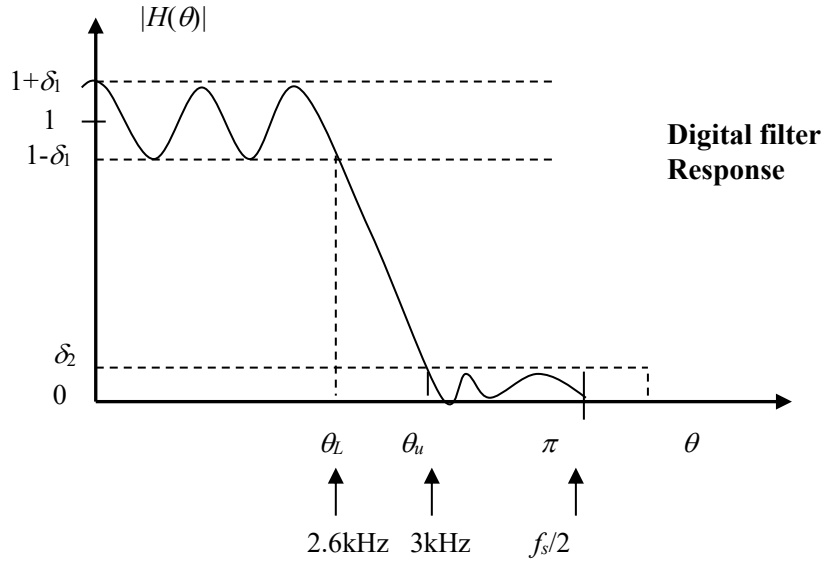
In designing a digital filter by this method we must first **pre-wrap** the given filter specifications to find the continuous filter to which we are going to apply the bilinear transformation.

Example: The specification of a desired digital low pass filter is shown below.

Sampling frequency: $f_s = 8\text{kHz}$ ($T = 1/f_s = 125\mu\text{s}$)

A pass band up to $f_L = 2.6$ kHz ($\theta_L = \omega_L T = 2\pi \frac{f_L}{f_s} = 0.65\pi$)

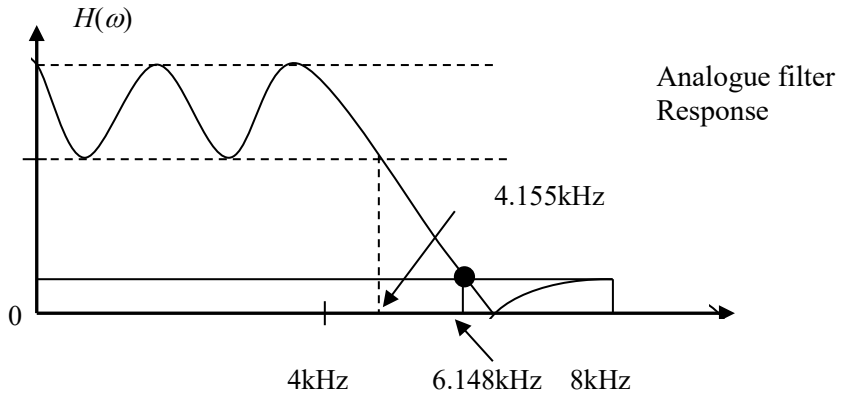
and a stop band, f_u, above 3kHz ($\theta_u = \omega_L T = 2\pi \frac{f_u}{f_s} = 0.75\pi$).



We must start from an analogue filter with

$$\omega_L = 2\pi f_l = \frac{2}{T}\tan\frac{\theta_L}{2} = 2\pi(4155) \Rightarrow f_L = 4.155\text{kHz}$$

$$\omega_u = 2\pi f_h = \frac{2}{T}\tan(\frac{\theta_u}{2}) = 2\pi(6148) \Rightarrow f_u = 6.148\text{kHz}$$



Example: Determine, using bilinear transformation method, the transfer function and difference equation for the digital equivalent of the RC filter. The normalized transfer function for the RC filter is

$$H(s) = \frac{1}{s+1}$$

Assume a sampling frequency of 150Hz and a cut-off frequency of 30Hz.

Solution: $\theta_c = \omega_c T = 2\pi f_c \cdot \frac{1}{f_s}$ before prewarping

$$\theta_c = 2\pi(30) \cdot \frac{1}{150} = 0.4\pi$$

The analogue frequency after prewarping is

$$\omega_c' = \frac{2}{T}\tan\left(\frac{0.4\pi}{2}\right) = 217.95 \text{ rad/sec}$$

Pre-warped frequency

f_c' is always > f_c (un-warped freq= 30 Hz) and after warping, $f_c' = 34.68$Hz.

The de-normalized analogue filter transfer function (this is achieved by replacing s with $\frac{s}{\omega_c}$) is obtained from $H(s)$ as

$$H_1(s) = H(s)\Big|_{s=\frac{s}{\omega_c'}} = \frac{\omega_c'}{s+\omega_s'} = \frac{\left(\frac{2}{T}\right)0.7265}{s+\frac{2}{T}\times 0.7265}$$

$$H(z) = H_1(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}\cdot} = \frac{0.7265(1+z)}{(1+7265)z+0.7265-1} = \frac{0.4208(1+z^{-1})}{1-0.1584z^{-1}}$$

Example:

It is required to design a digital filter to approximate the following analogue transfer function

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

using the Bi-linear transformation method obtain the transfer function, *H(s)* of the digital filter assuming a 3dB cut-off frequency of 150Hz and a sampling frequency of 1.28kHz.

Solution:

$$f_c = 150\text{Hz} \qquad f_s = 1280\text{Hz} \quad T = \frac{1}{f_s}$$

$$\downarrow$$

$$\theta_c = 2\pi\left(\frac{f_c}{f_s}\right) = 2\pi\left(\frac{150}{1280}\right) = \frac{15}{64}\pi$$

The analogue frequency after pre-warping

$$\omega_C^{'} = \frac{2}{T}\tan\left(\frac{\theta_c}{2}\right) = \frac{2}{T} \times 1280 \tan\left(\frac{\frac{15\pi}{64}}{2}\right) = 987.5009 = \frac{2}{T} \times 0.3859\, rad/\sec$$

$$f_c^{'} = \frac{987.5009}{2\pi} = 157.1656 Hz$$

$f_c^{'}$ is always greater than f_c.

Prewarped analogue filter is given by,

$$H_1(s) = \frac{\left(\frac{2}{T}\right)^2 (0.3857)^2}{s^2 + \frac{2}{T}(0.3857)\sqrt{2}s + \left(\frac{2}{T}\right)^2 (0.3857)^2}$$

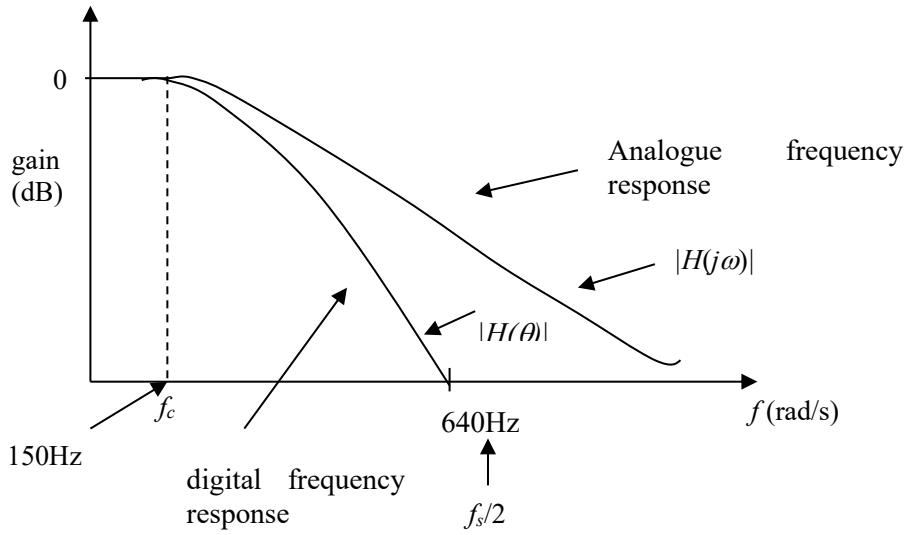
$$H(z) = H_1(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}} = \frac{\left(\frac{2}{T}\right)^2 (0.3857)^2}{\left(\frac{2}{T}\frac{z-1}{z+1}\right)^2 + \frac{2}{T} \times 0.3857 \times \sqrt{2}\left(\frac{2}{T}\frac{z-1}{z+1}\right) + \left(\frac{2}{T}\right)^2 (0.3857)^2}$$

$$H(z) = \frac{0.0878\left(1 + 2z^{-1} + z^{-2}\right)}{1 - 1.0048z^{-1} + 0.3561z^{-2}}$$

$$\frac{\left(\omega_c{}'\right)^2}{s^2 + \sqrt{2}\omega_c' s + \left(\omega_{c'}\right)^2} \xrightarrow{\textit{Bilinear transformation}} \frac{0.0878(1 - 2z^{-1} + z^{-2})}{1 - 1.0048z^{-1} + 0.3561z^{-2}}$$

All pole

Poles & zeros



Note:

- Same cut-off frequency
- increased roll off and attenuation in stop band.

Example: Convert the simple low pass filter $H(s)=\frac{1}{s+1}$ into an equivalent high pass discrete filter. Assume f_s = 150Hz, analogue cut-off f_c = 30Hz.

$$\theta_c = 2\pi\left(\frac{30}{150}\right) = 0.4\pi$$

$$\omega_c^{'} = \frac{2}{T}\tan(\frac{0.4\pi}{2}) = \frac{2}{T}\times 0.7265$$

LPF to HPF transformation

$$H^{'}(s) = H(s)\Big|_{s\to\frac{\omega_c^{'}}{s}} = \frac{1}{\frac{\omega_c^{'}}{s}+1} = \frac{s}{s+\omega_c^{'}}$$

$$H(z) = H^{'}(s)\Big|_{z=\frac{2}{T}\left(\frac{z-1}{z+1}\right)} = \frac{\frac{2}{T}\left(\frac{z-1}{z+1}\right)}{\frac{2}{T}\left(\frac{z-1}{z+1}\right)+\frac{2}{T}0.7265} = \frac{z-1}{z-1+0.7265(z+1)}$$

$$H(z) = 0.5792\frac{1-z^{-1}}{1+0.1584z^{-1}}$$

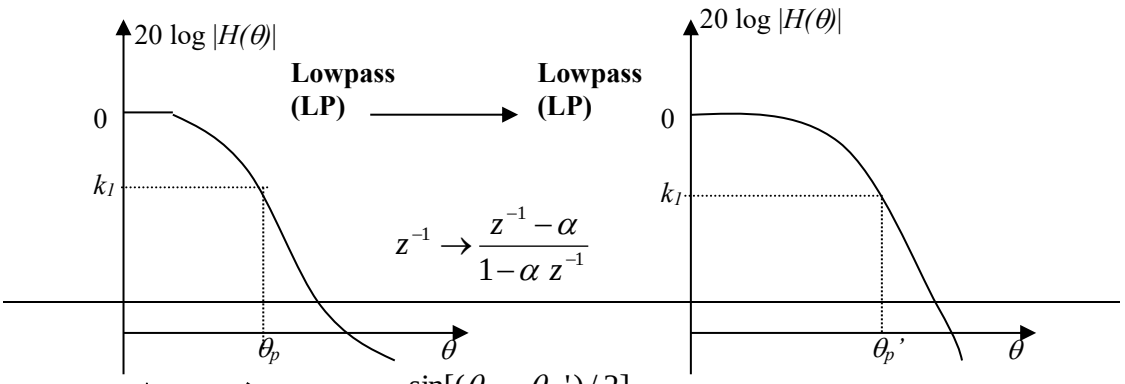
Digital-To-Digital Transformations

We have seen in the lecture notes that one method for the design of analogue filters relied on applying a transformation to an analogue low-pass filter with a unit bandwidth. It was shown that we could obtain low-pass, high-pass, band-pass and band-stop filters by selecting the appropriate transformation.

Similarly, a set of transformations can be formed that take a low-pass digital filter and turn it into high-pass, band-pass, and band-stop or another low-pass digital filter.

The transformations are given below:

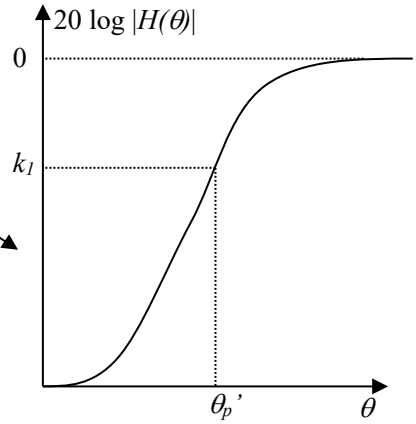
Digital to Digital Transformations



$$z^{-1} \rightarrow \frac{z^{-1}-\alpha}{1-\alpha\, z^{-1}}$$

$$\alpha = \frac{\sin[(\theta_p - \theta_p')/2]}{\sin[(\theta_p + \theta_p')/2]}$$

Lowpass to Highpass

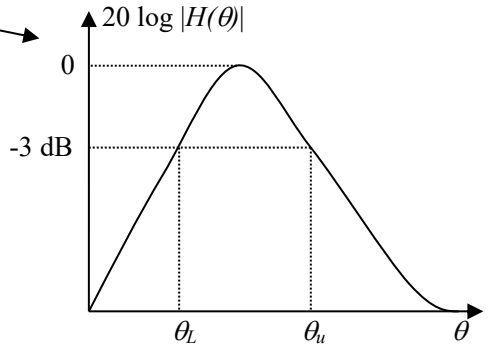


$$z^{-1} \rightarrow \frac{z^{-1}+\alpha}{1+\alpha\, z^{-1}}$$

Simplest transformation is to change the signs of poles and zeros of the LP → LP transformation

Lowpass to Bandpass

$$\alpha = \frac{\cos[(\theta_p + \theta_p')/2]}{\cos[(\theta_p - \theta_p')/2]}$$



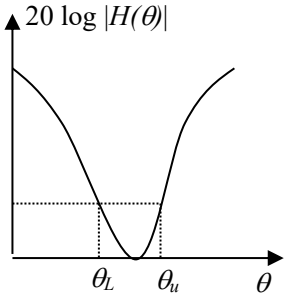
$$z^{-1} \rightarrow -\frac{z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + 1}$$

$$\alpha = \frac{\cos[(\theta_u + \theta_L)/2]}{\cos[(\theta_u - \theta_L)/2]}$$

$$k = \cot((\theta_u - \theta_L)/2) \cdot \tan\left(\frac{\theta_p}{2}\right)$$

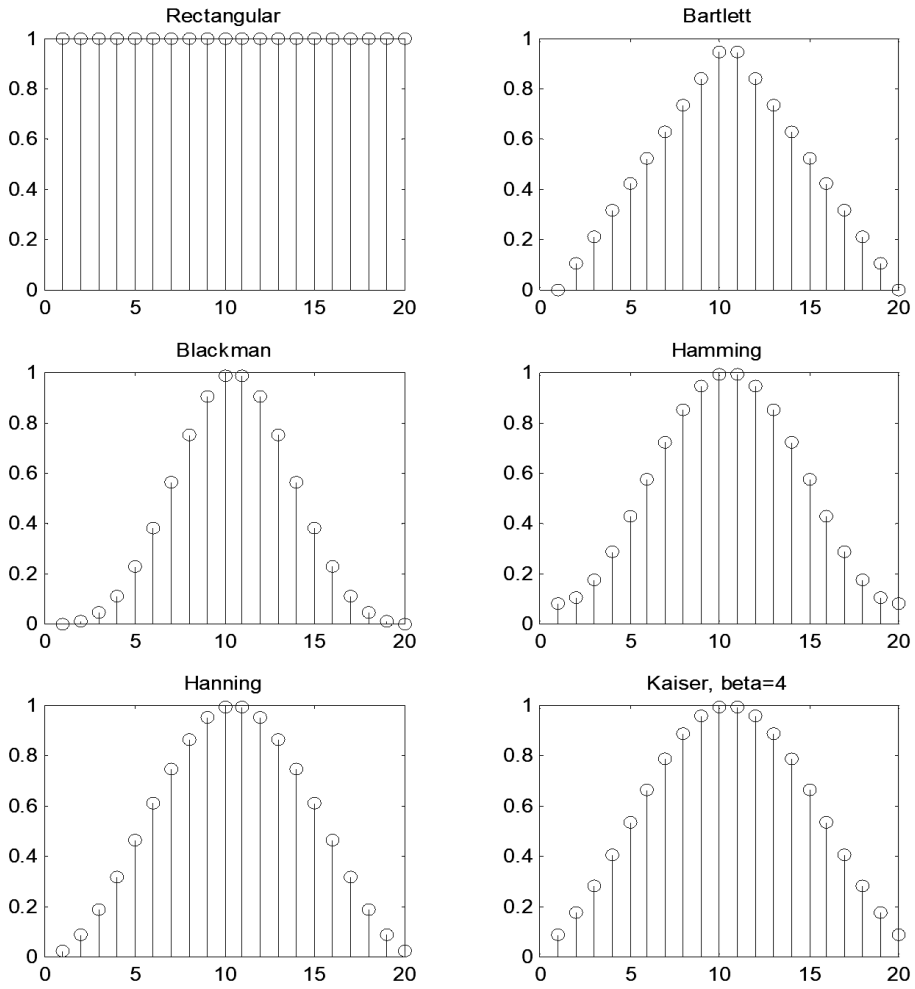
Lowpass to Bandstop

$$z^{-1} = \frac{z^{-2} - \frac{2\alpha}{k+1} z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha}{k+1} z^{-1} + 1}$$

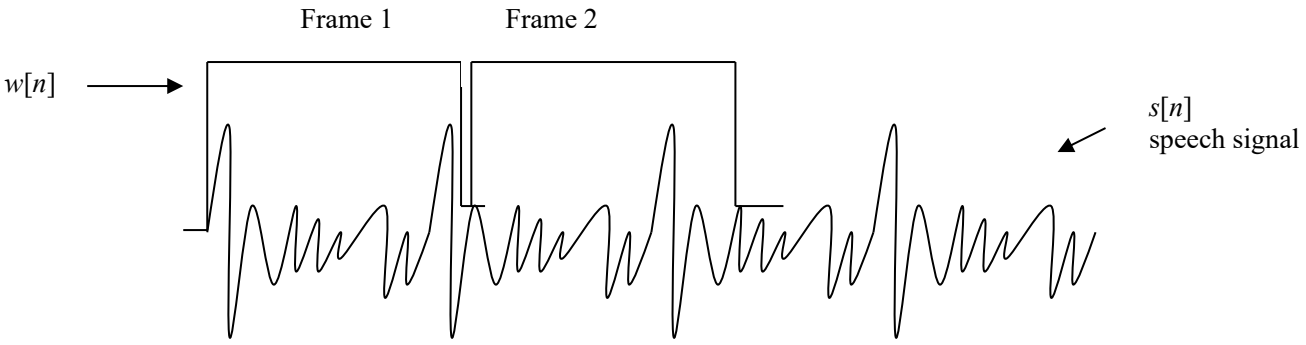


Window Functions

Window functions are used to truncate a signal or impulse response to produce a signal or impulse response of finite duration. Some of the most commonly used window functions are:



To analyse a **truncation** process we model it as a multiplication of the desired sequence by finite duration window sequence denote by $w[n]$. Truncation of a sequence $s[n]$ is equivalent to placing a rectangular time window around $s[n]$.



Vocal tract shape changes every 15ms. When using a sampling frequency of 8 kHz (T = 125μs) with 100 samples in each frame (12.5ms),

$$y[n] = s[n] \cdot w[n]$$

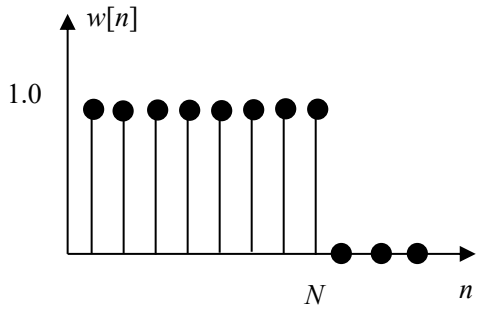
(Multiplication becomes convolution in the frequency domain)

In frequency domain,

$$Y(\theta) = S(\theta) * W(\theta)$$

Thus when window is applied, the frequency domain convolution causes distortion in the spectrum $Y(\theta)$. It can be shown that the rectangular window creates ripples in the spectrum $Y(\theta)$. To reduce the distortion, use Hamming or Kaiser Window.

Rectangular Window

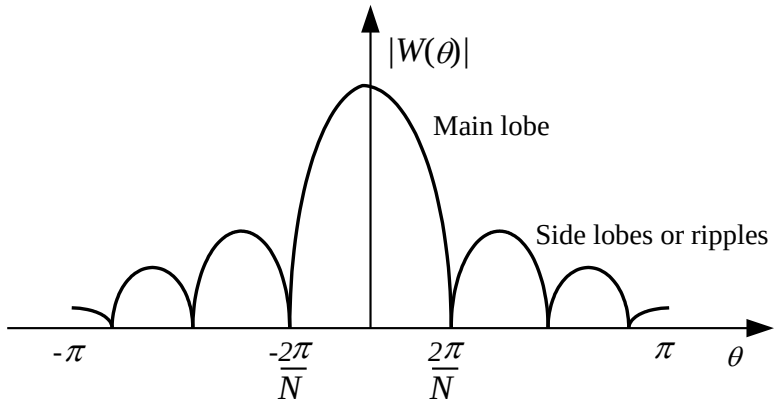


$$w[n]=\begin{cases}1 & 0\le n\le N-1\\ 0 & \text{otherwise}\end{cases}$$

$$W(z)=\sum_{n=0}^{N-1}w(n)z^{-n}=1+z^{-1}+z^{-2}+.....+z^{-(N-1)}=\frac{1-z^{-N}}{1-z^{-1}}$$

$$W(\theta)=W(z)\Big|_{z=e^{j\theta}}=\frac{1-e^{-jN\theta}}{1-e^{-j\theta}}$$

$W(\theta)=e^{-j\frac{N-1}{2}\theta}\sin\frac{\frac{N\theta}{2}}{\frac{\theta}{2}}$	$\phi(\theta)=-\left(\frac{N-1}{2}\right)\theta$	$\lvert W(\theta)\rvert=\left\lvert\frac{\sin\frac{N\theta}{2}}{\sin\frac{\theta}{2}}\right\rvert$



If N increases the width of the main lobe decreases but the peak amplitude of the side lobes grows in a manner such that the area under each lobe is constant while the width of each lobe decreases with N.

Design Methods for FIR filters

The most essential feature of FIR filters is, by definition, the finite length of the impulse response. Another important point is that it can be seem directly from the impulse response of an FIR filter whether we have a linear phase characteristic or not.

A filter is said to have a linear phase response of its response satisfies one of the following relationships.

$$\phi(\theta) = -a\theta \qquad (1)$$

$$\phi(\theta) = b - a\theta \qquad (2)$$

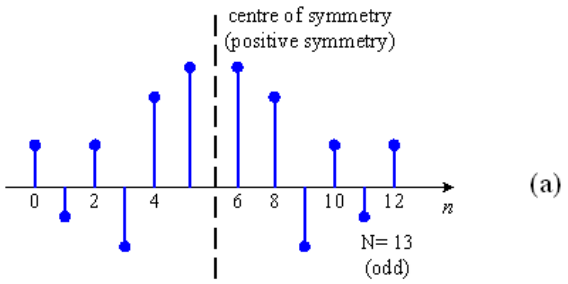
where a and b are constants.

It can be shown that for condition (1) above to be satisfied the impulse response of the filter must have positive symmetry.

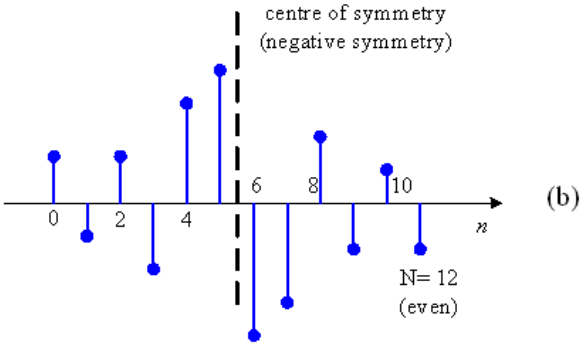
$$h[n] = h[N-n-1], \quad a = \frac{N-1}{2}$$

where N denotes the filter length. When the condition given in (2) is satisfied, the impulse response of the filter has negative symmetry.

$$h[n] = -h[N-n-1], \quad a = \frac{N-1}{2} \text{ and } b = \frac{\pi}{2}$$



(a)



(b)

Design of FIR filters using Windows.

The easiest way to obtain an FIR filter is to simply truncate the impulse response of an IIR filter. If $h_d[n]$ represents the impulse response of a desired IIR filter, then an FIR filter with impulse response $h[n]$ can be obtained as follows:

$$h[n] = \begin{cases} h_d[n] & N_1 \le n \le N_2 \\ 0 & \text{otherwise} \end{cases}$$

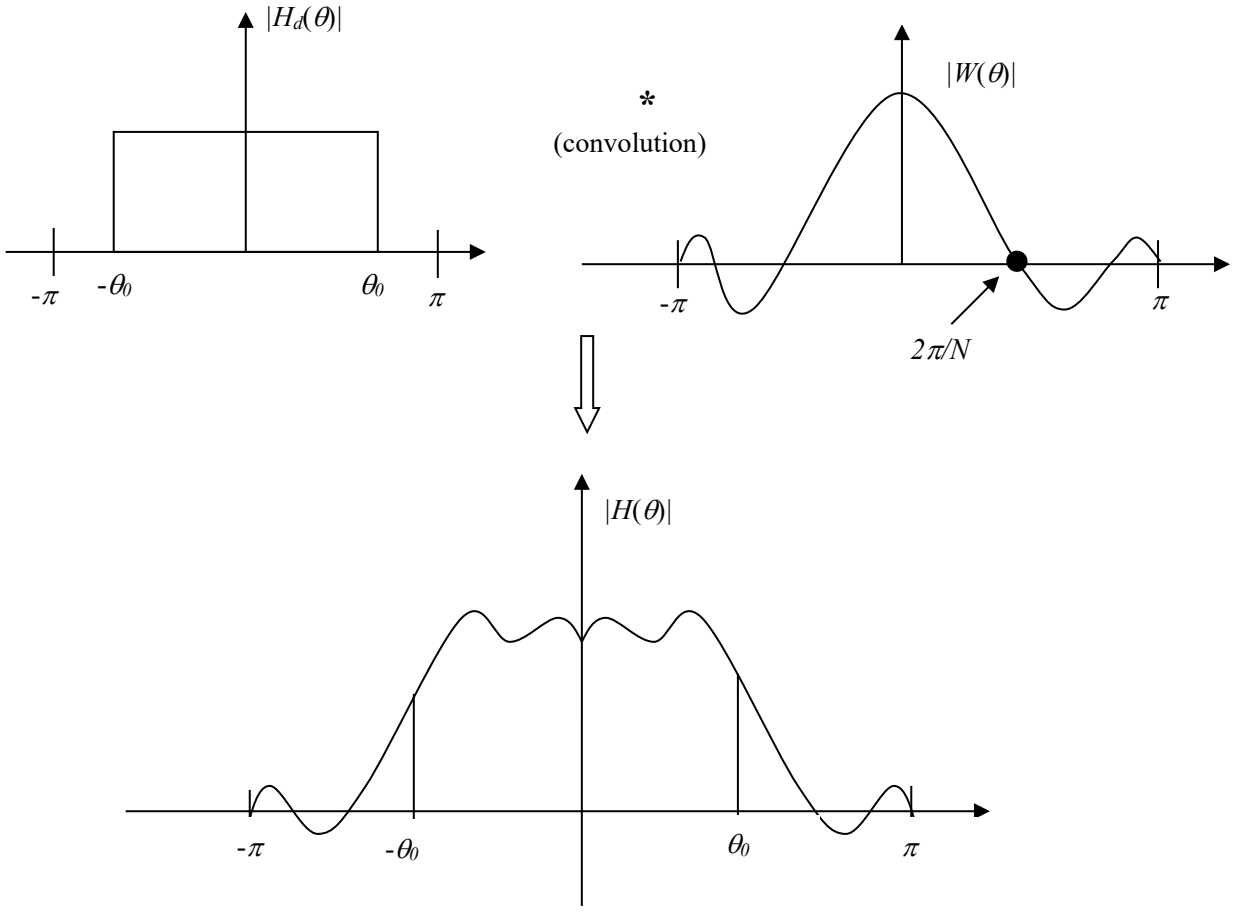
In general $h[n]$ can be thought of as being formed by the product $h_d[n]$ and a window function $w[n]$ as follows:

$$h[n] = h_d[n] \cdot w[n]$$

let it be, for example a rectangular window

$$H(\theta) = H_d(\theta) * W(\theta)$$

let it be an ideal low pass filter with cut off frequency θ_0



Therefore it is seen that the convolution produces a smeared version of the ideal low pass frequency response $H_d(\theta)$.

In general, the wider the main lobe of $W(\theta)$, the more spreading, where as the narrower the main lobe (larger N) the closer $|H(\theta)|$ comes to $|H_d(\theta)|$.

In general we are left with a trade off of making N large enough so that smearing is minimized, yet small enough to allow reasonable implementation.

Design Procedure

An ideal low pass filter with linear phase of slope $-\beta$ and cut-off ω_c can be characterized in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\theta\beta} & |w| \le \omega_c \\ 0 & w_c < |\omega| \le \pi \end{cases}$$

The corresponding impulse response $h_d[n]$ can be obtained by taking the inverse Fourier transform of $\boldsymbol{H_d(\theta)}$ and easily shown to be

$$h_d[n] = \frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}$$

A causal FIR filter with impulse response $h[n]$ can be obtained by multiplying $h_d[n]$ by a window beginning at the origin and ending at N-1 as follows :

$$h[n] = \frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}\omega[n]$$

For $h[n]$ to be a linear phase, β must be selected so that the resulting $h[n]$ is symmetric.

As $\frac{\sin[\omega_c(n-\beta)]}{\pi(n-\beta)}$ is symmetric about $n=\beta$ and the window symmetric is about $n=\frac{N-1}{2}$

$$\therefore \quad \beta = \frac{N-1}{2} \quad \longleftarrow \text{Symmetric about } \beta$$

Example:

(a) Determine the impulse response $h_d[n]$ of the lowpass filter whose frequency response is given by

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{3} \\ 0 & \frac{\pi}{3} < |\theta| \le \pi \end{cases}$$

(b) To obtain a finite impulse response from $h_d[n]$ a rectangular window of length $N = 9$ is used. Compute the coefficients of the FIR filter with a linear phase characteristic ands with this finite impulse response.

Solution:

(a)

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta) e^{j\theta n} d\theta$$

$$= \frac{1}{2\pi}\left[\frac{e^{j\theta n}}{jn}\right]_{\frac{-\pi}{3}}^{\frac{\pi}{3}} = \frac{1}{\pi}\left(\frac{\sin n\frac{\pi}{3}}{n}\right)$$

$$h_d[n] = \begin{cases} 1/3 & n = 0 \\ \frac{\sin(n\pi/3)}{n\pi} & n \ne 0 \end{cases}$$

(b) For a linear phase filter, $h_d[n]$ is symmetric about $n = 0$ and the window is symmetric about $n = \frac{(N-1)}{2} = \frac{9-1}{2} = 4$.

$$h[n] = h_d[n] \cdot \omega[n]$$

$$h[0] = 0.333$$

$$h[1] = \frac{\sqrt{3}}{2\pi}$$

$$h[2] = \frac{\sqrt{3}}{4\pi}$$

$$h[3] = 0$$

$$h[4] = -\frac{\sqrt{3}}{8\pi}$$

The coefficients are

	$\frac{-\sqrt{3}}{8\pi}$	0	$\frac{\sqrt{3}}{4\pi}$	$\frac{\sqrt{3}}{2\pi}$	0.333	$\frac{\sqrt{3}}{2\pi}$	$\frac{\sqrt{3}}{4\pi}$	0	$\frac{-\sqrt{3}}{8\pi}$
n	0	1	2	3	4	5	6	7	8

Symmetry about $n = 4$

Topic 6

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Multirate Digital Signal Processing

The increasing need in modern digital systems to process data at more than one sampling rate has led the development of a new sub-area in DSP known as multirate signal processing. The two primary operations in multirate proceeding are

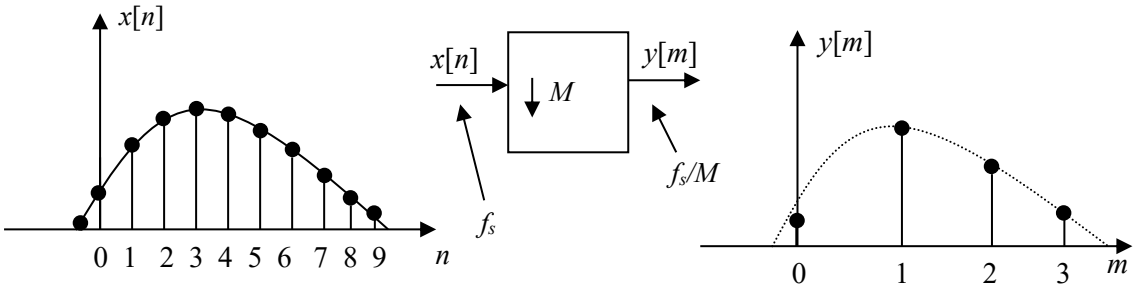
- **Decimation**
  (We can decrease the sampling rate f_s of a given signal $x[n]$: This is sometimes called **down sampling**).

- **Interpolation**
  (We can increase the sampling rate f_s of a given signal $x[n]$: This is sometimes called up **sampling**).

The process of decimation and interpolation are the fundamental operations in multirate signal processing.

Decimation

A sampling rate decreaser is shown below. We should confine our attention to a decrease by an integer factor M. (e.g. $M = 3$)



A discrete-time signal, x[n], is down-sampled by a factor of M=3, yielding another discrete-time signal, y[n].

Example: $x[n] = \{1, 2, 4, 3, 5, -6, -8, 2, -3, 2\}$

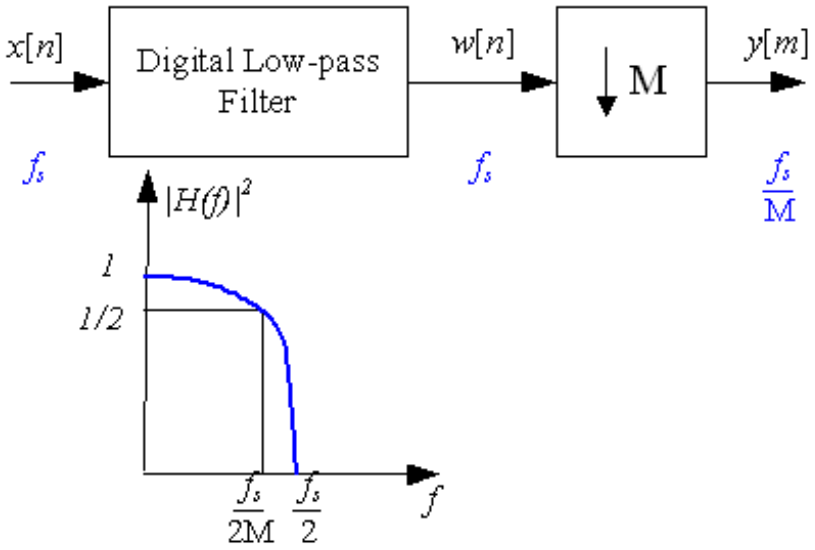
$\downarrow$ Down sample by 2

$y[m] = \{1, 4, 5, -8, -3\}$

The output signal $y[m]$ is obtained by taking every M^{th} sample of the input signal. If $M = 3$, we should just take every third sample of $x[n]$ to form the desired signal $y[m]$.

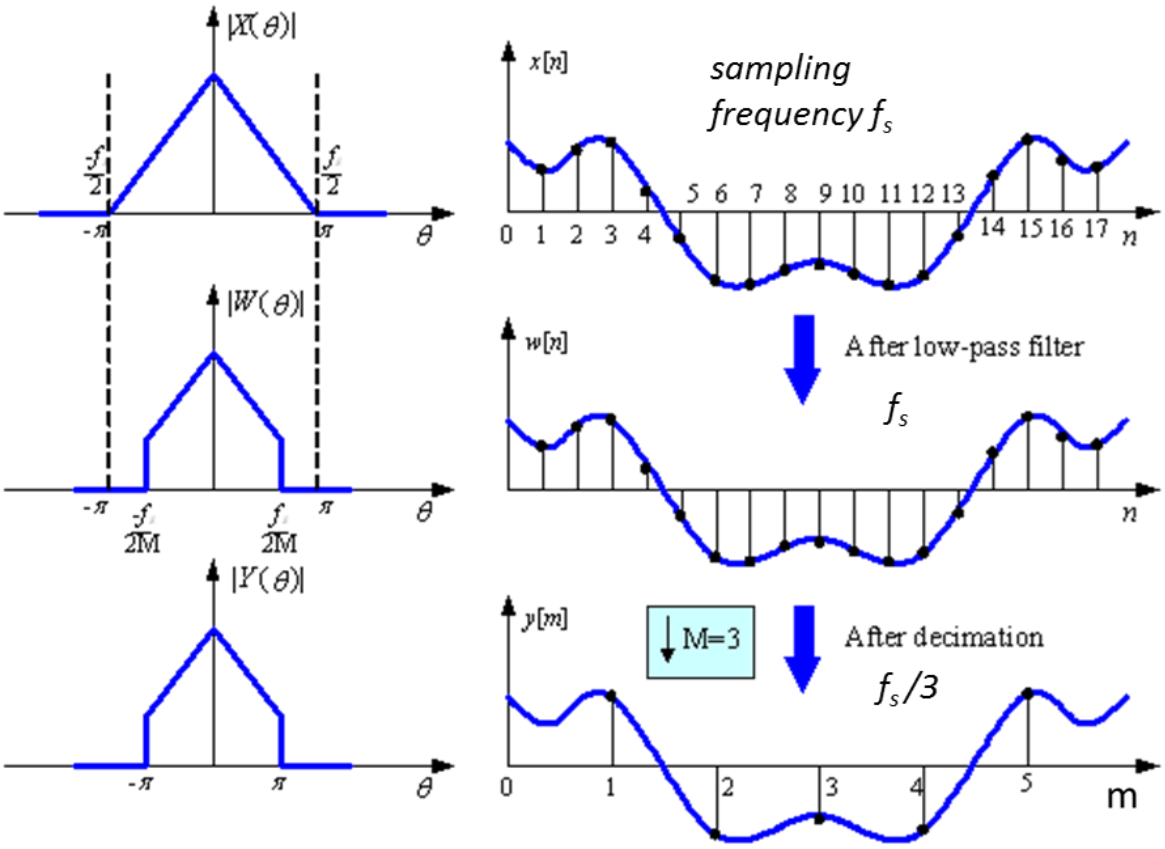
Obviously, it only makes sense to reduce the sampling rate if the information constant of the signal we wish to preserve is band limited to $\frac{f_s}{6}$;

Figure below shows a block representation of a times M decimator. We see that the signal $x[n]$ is first passed through a low pass filter that attenuates the band from $\frac{f_s/2}{M} = \frac{f_s}{2M}$ to $\frac{f_s}{2}$ to prevent aliasing

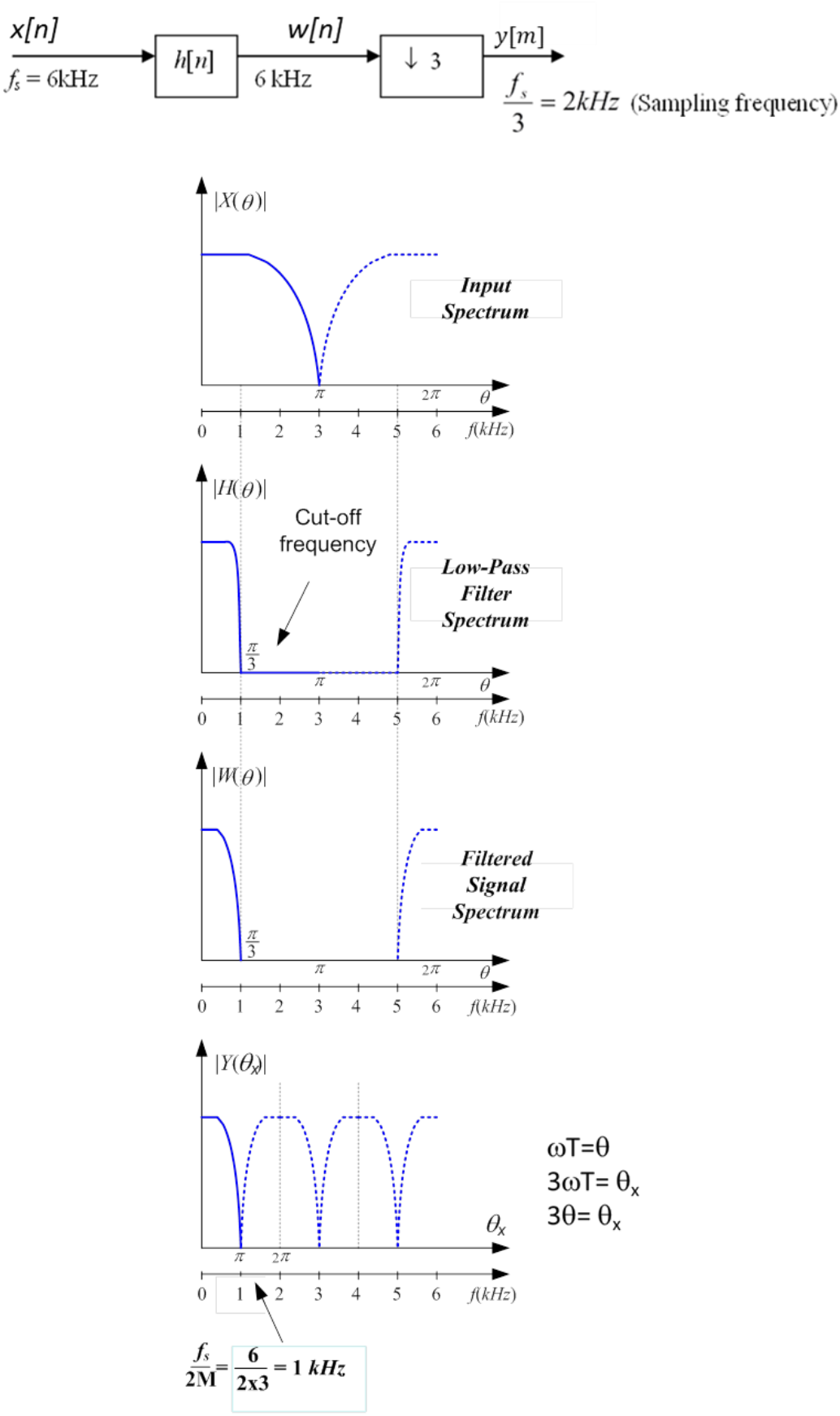


Prior to down-sampling, a discrete-time signal must first go through a low-pass filter to remove its high-frequency components.
This prevents aliasing.

The spectral description of the decimation process is shown below



Spectral description of decimation process



Spectral interpretation of decimation of a signal from 6kHz to 2kHz

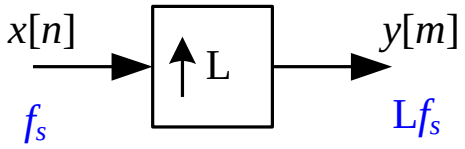
Note: Such a processing requirement may arise, for example when a speech signal is over sampled at f_s = 32 kHz. Since we are interested only in a band of 0-4 kHz, sampling rate can be reduced to 8 kHz, so the first step in the decimation process has to be the digital filtering of the signal $x[n]$ is band limited to $f_s/8$. Should we use IIR (recursive) or FIR (non-recursive) for the low pass filtering required?

Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we only have to compute every N^{th} output, since previous outputs are required to compute the Mth output. Thus no saving realized. On the other hand, *using an FIR filter*, in this case implies that we can do our computations at the rate $\frac{f_s}{M}$. Thus, using an FIR filter in the decimation process will lead to a significantly lower computation rate.

Another advantage of using an FIR filter is the fact that we can easily design linear phase filters and this is desirable in many applications.

Interpolation

The process of interpolation involves a sampling rate increase (see diagram below)



A general block diagram representation of up-sampling process.

Example:

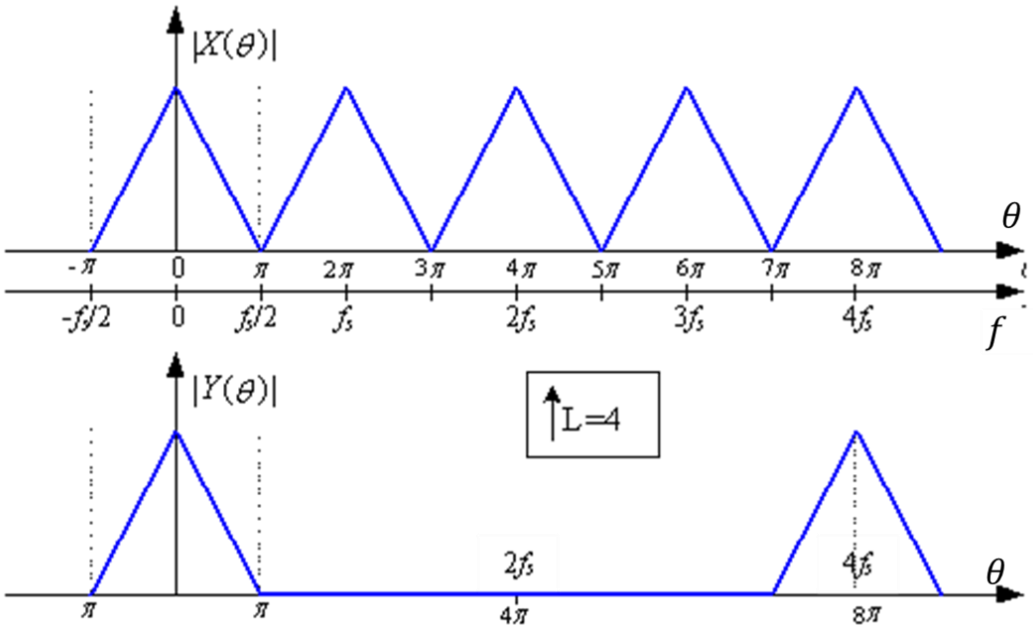
$x[n]=\{1,2,4,3,-5,6,-7,2,4,3\}$

$\uparrow 2$

$y[m]=\{1,0,2,0,4,0,3,0,-5,0,6,0,-7,0,2,0,4,0,3,0\}$

The sequence $x[n]$ was derived by sampling $x(t)$ at a sampling rate of f_s and we want to obtain a sequence $y[m]$ that approximates as closely as possible the sequence that would have been obtained had we sampled $x(t)$ at the rate Lf_s.

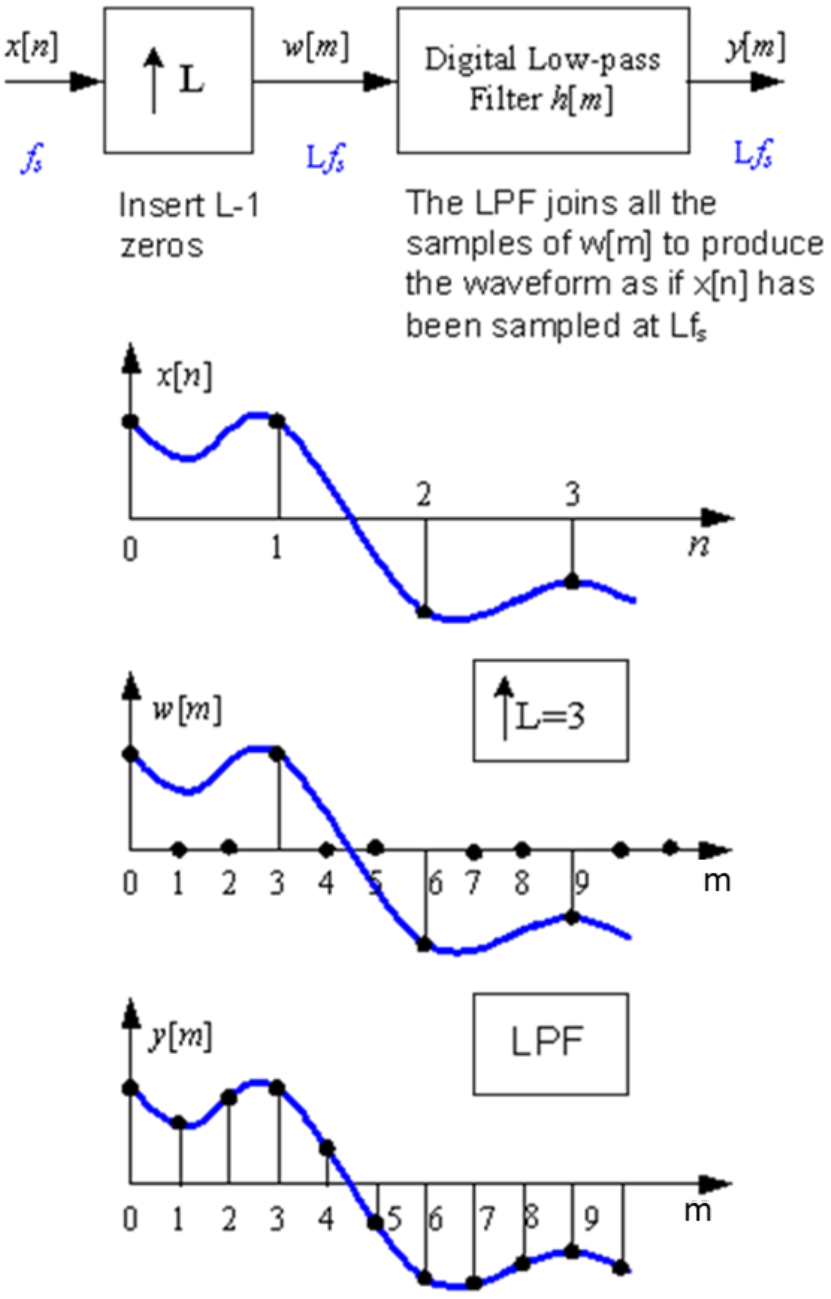
This involves inserting between any two samples $x[n]$ and $x[n\text{-}1]$ an additional L-1 samples.



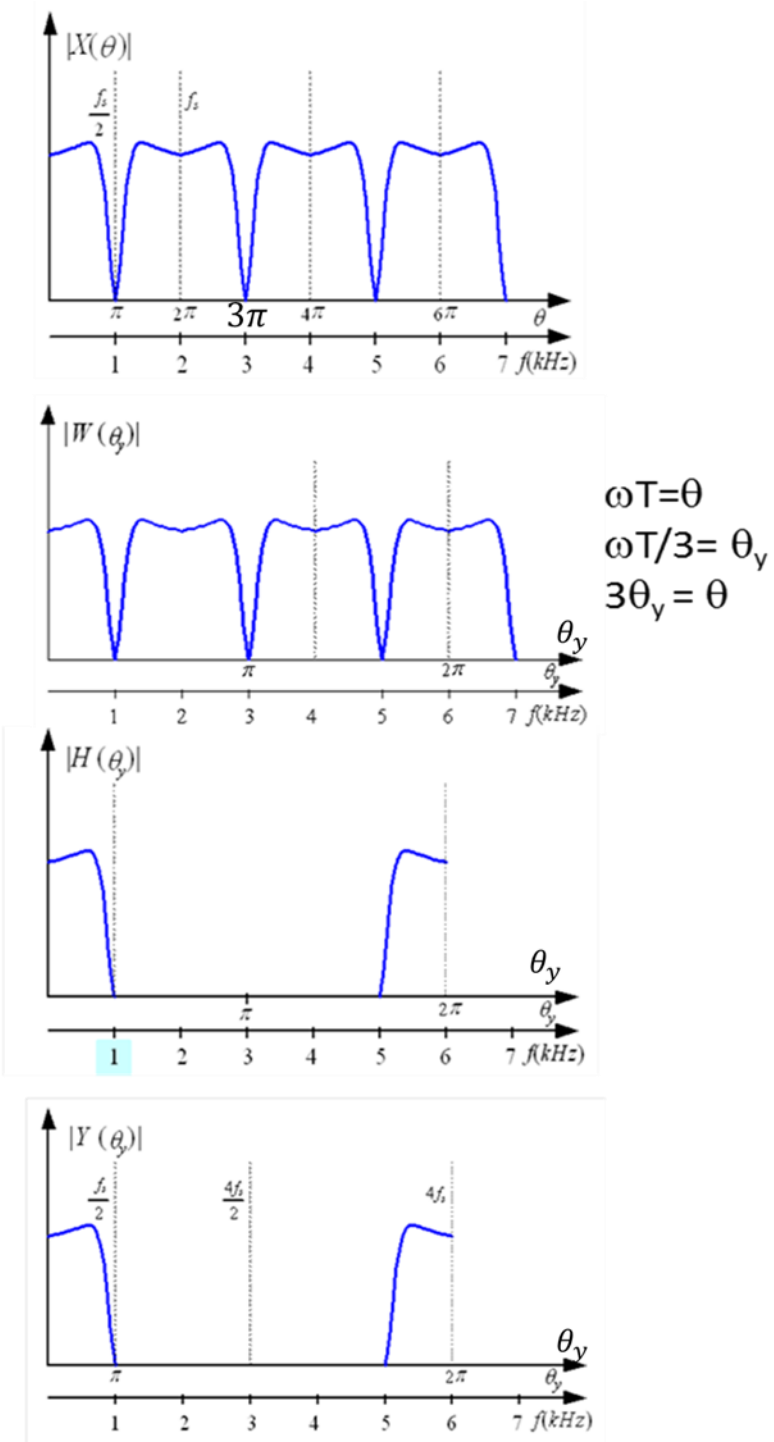
Spectral description of the interpolation process. Sampling frequency of y(n) = 4f_s .

We observe from above Figure that to go from $X(\theta)$ to $Y(\theta)$, we have to pass $x[n]$ through a lowpass digital filter designated at the Lf_s sampling rate that attenuates sufficiently any frequency components above $\frac{f_s}{2}$.

The frequency domain interpretation of the process is given below.



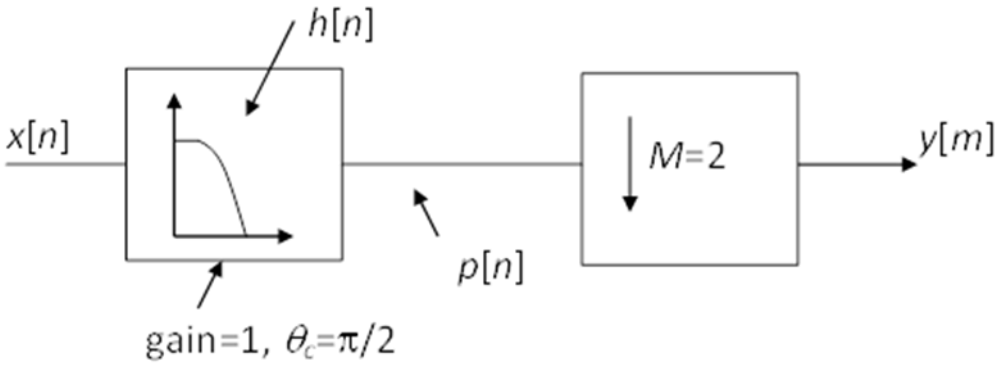
A general description of the interpolation process. A discrete-time signal is up-sampled by inserting zeros and then filtered by a low-pass filter.
Note that for each sample of x[n], three output samples y[m] are obtained.



Spectral interpretation of interpolation of a signal from 2kHz to 6 kHz.

Note: Obviously the same reasoning that led us to believe that FIR filters are preferable in the decimation holds here also.

Example: Decimation, $x[n] = \{2, 6, 4, 2, 6, 8, 4, 2, 4, 4\}$

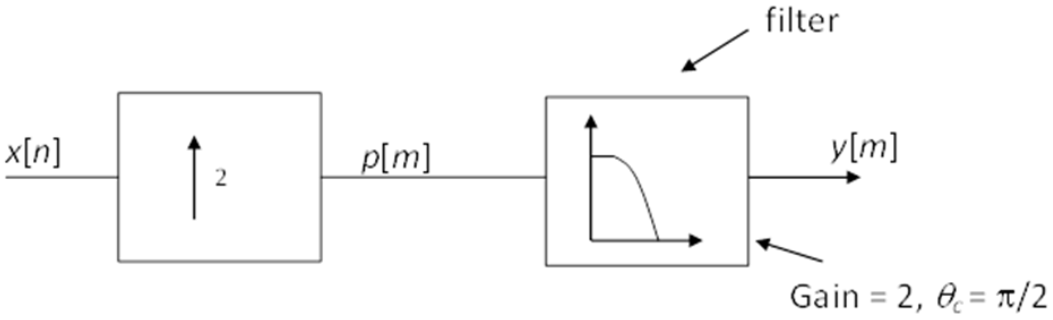


$$h[n] = \left\{\frac{1}{2}, \frac{1}{2}\right\}$$

$$p[n] = x[n] * h[n] = \{1,4,5,3,4,7,6,3,3,4,2\}$$

When p[n] is downsampled by 2, we obtain $y[n] = \{1,5,4,6,3,2\}$

Linear Interpolation



$x[n] = \{1,3,5,3,7\}$

$p[n] = \{1,0,3,0,5,0,3,0,7,0\}$ (insert zeros)

$$h[n] = \left\{\frac{1}{2}, 1, \frac{1}{2}\right\}$$

$y[n] = p[n] * h[n] = \{0.5,1,2,3,4,5,4,3,5,7,3.5\}$

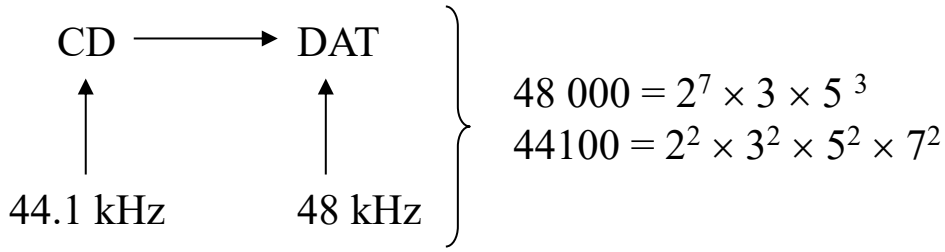
Sampling rate conversion by Non-integer factors

In some applications, the need often arises to change the sampling rate by a non-integer factor. An example is in digital audio applications where it may be necessary to transfer data from one storage system to another, where both systems employ different rates.

An example is transferring data from the compact disk system at a rate of 44.1 kHz to a digital audio Tape (DAT) at 48 kHz. This can be achieved by increasing the data rate of the CD by a factor of $\frac{48}{44.1}$, a non-integer.

In practice, such a non-integer factor is represented by a rational number that is a ratio of two integers, say L and M. The sampling frequency change is then achieved by first interpolating the data by L and then decimating by M.

Note: It is necessary that the interpolation process precede decimation otherwise the decimation process would remove some of the desired frequency components.



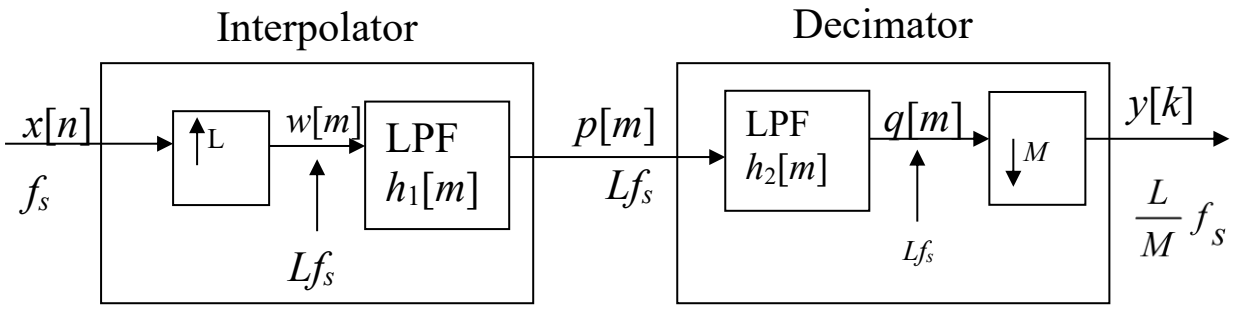
To change the sampling rate we require:

$$\frac{L}{M} = \frac{2^7 \times 3 \times 5^3}{2^2 \times 3^2 \times 5^2 \times 7^2} = \frac{160}{147}$$

Therefore if we up-sample by $L = 160$ and then down-sample by $M = 147$, we achieve the desired sampling rate conversion.

The diagram below shows that the sampling frequency change is

achieved by first interpolating the data by L and then decimating by M.

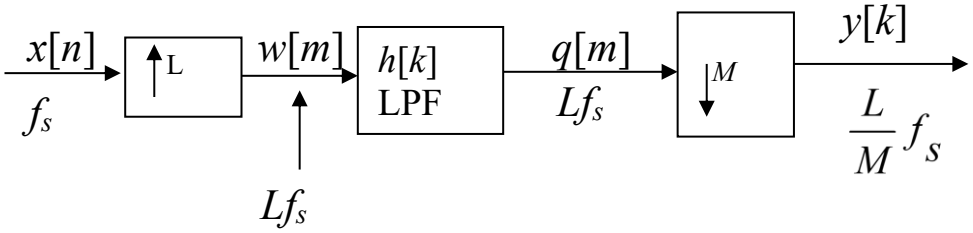


A discrete-time signal is interpolated and then decimated in order to achieve a non-integer conversion of sampling rate

The two LPFs, $h_1[m] \& h_2[m]$ can be combined into a single filter since they are in cascade and have a common sampling frequency.

If $M > L$, the resulting operation is a decimation process by a non-integer.

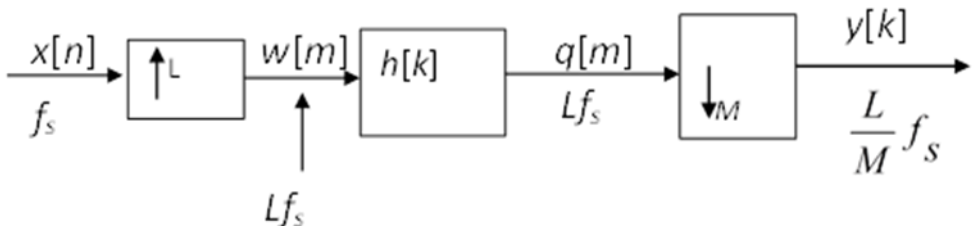
If $M < L$, it is interpolation



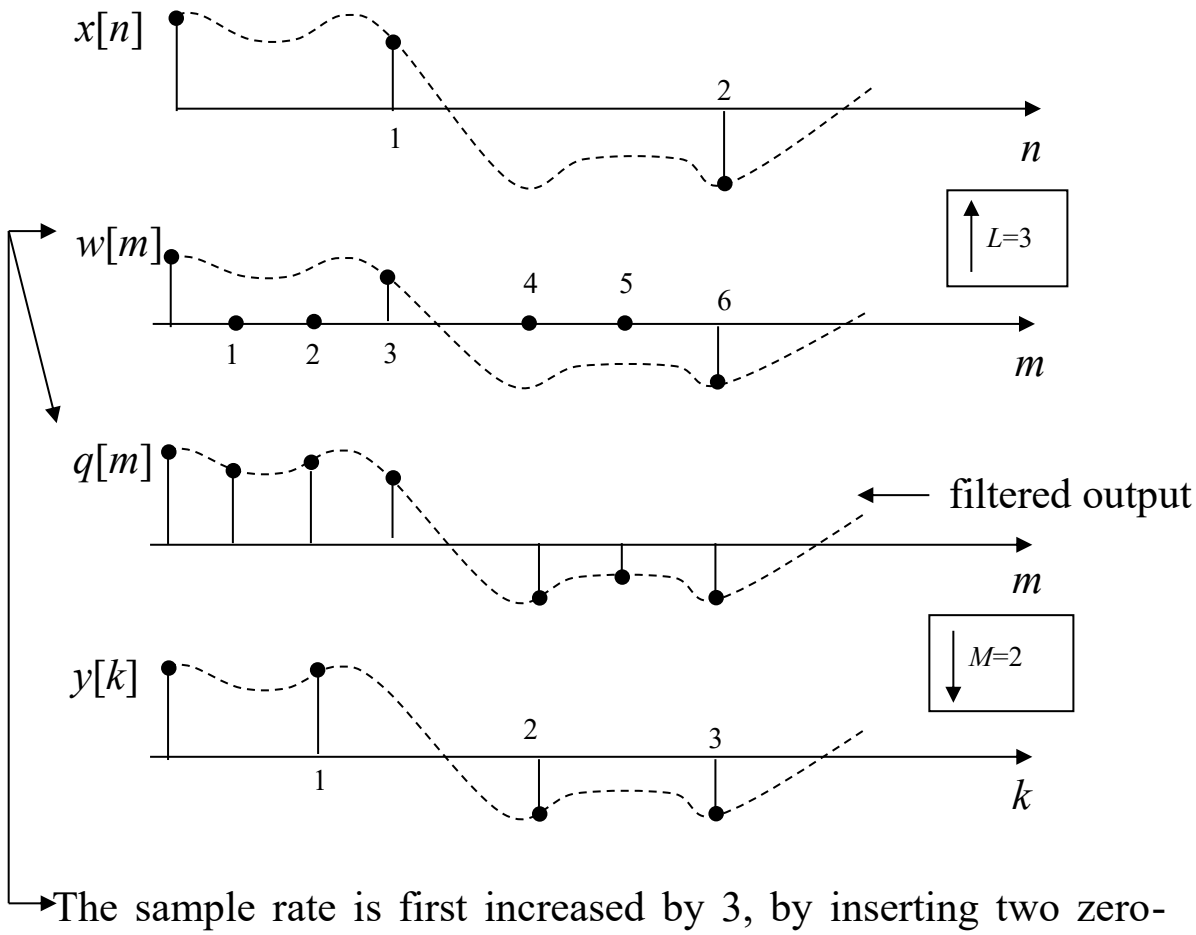
A block diagram representation of sampling rate conversion with combined low-pass filter

The lowpass filter that we require is the one that has a cut-off frequency, $\theta_c = \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right)$

Illustration of Interpolation by a factor 3/2



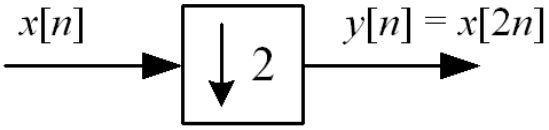
$$\frac{L}{M} = \frac{3}{2}$$



The sample rate is first increased by 3, by inserting two zero-value samples for each sample of $x[n]$ and low-pass filtered to yield $q[m]$.

The filtered data is then reduced by a factor of 2 by retaining only one sample for every two samples of $q[m]$.

Decimation by 2

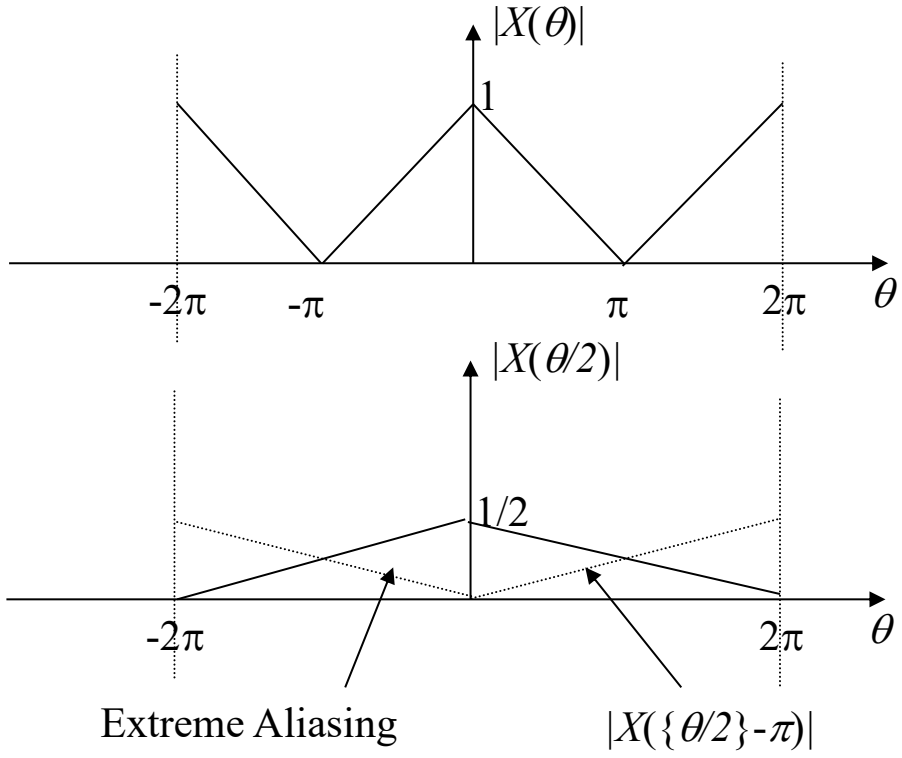


$$Y(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}}) \right]$$ Aliasing term

$$Y(\theta) = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(-e^{\frac{j\theta}{2}}) \right] = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(e^{j(\frac{\theta}{2}-\pi)}) \right]$$

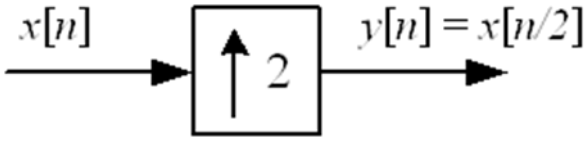
$$Y(\theta) = \frac{1}{2}\left[X(\frac{\theta}{2}) + X(\frac{\theta}{2}-\pi) \right]$$ Aliasing term

Stretch $X(\theta)$ by a factor 2 to obtain $X(\theta/2)$



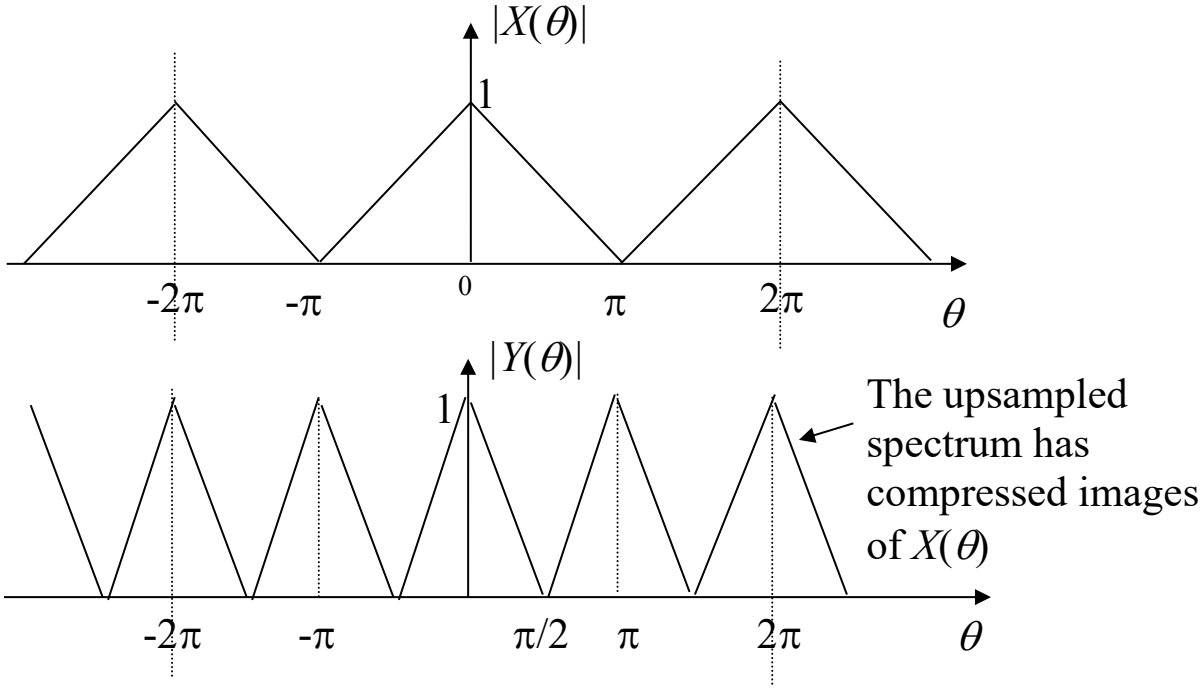
Note: The spectrum is stretched by down sampling.

Interpolation by 2



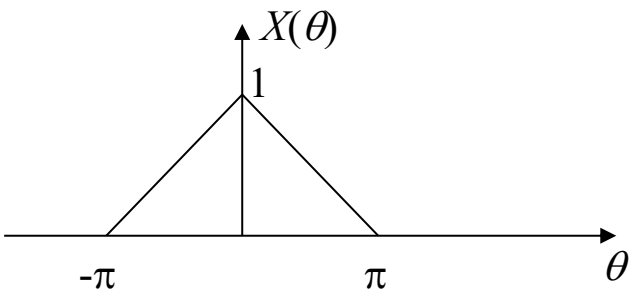
$$y[n] = x\left[\frac{n}{2}\right]; \quad Y(z) = X(z^2)$$

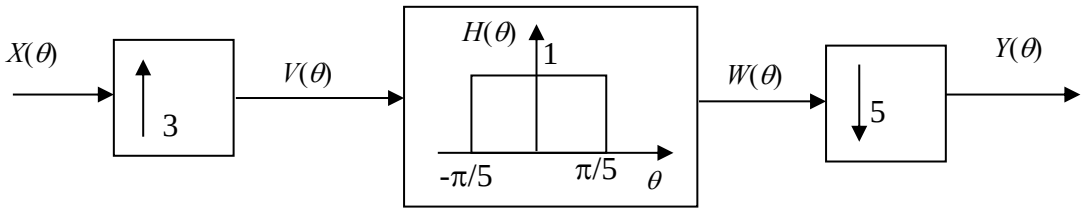
$$Y(\theta) = \left[X(e^{j2\theta}) \right] = \left[X(2\theta) \right]$$

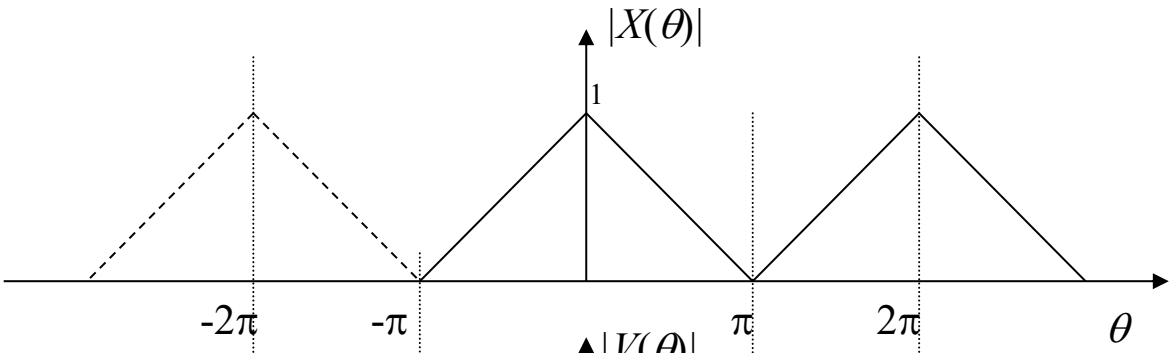


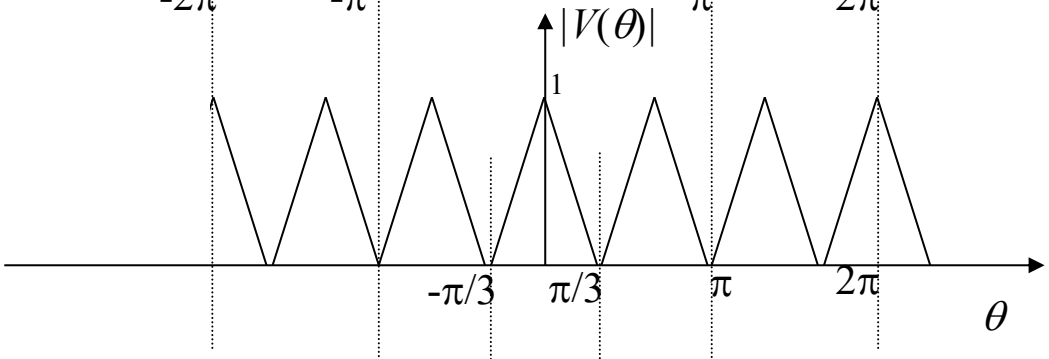
Note: The spectrum is compressed by upsampling.

<u>Example:</u> An input signal $x[n]$ with the spectrum $X(\theta)$ is applied to the system shown below. Sketch, one above another, $|X(\theta)|$, $|V(\theta)|$, $|W(\theta)|$, $|Y(\theta)|$ against θ.

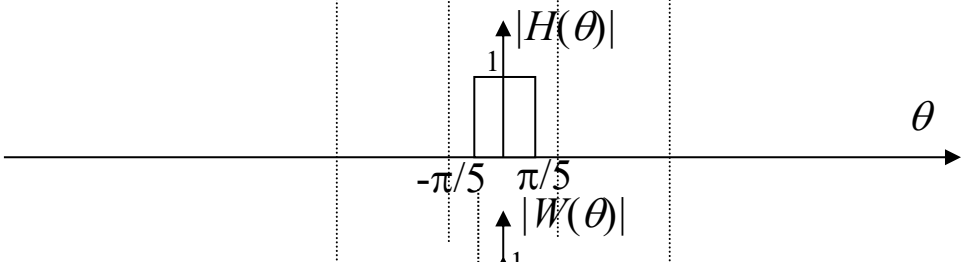


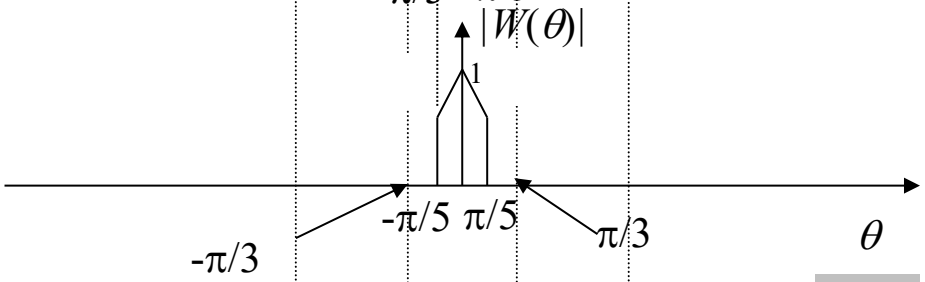

X(θ)
3
V(θ)
H(θ)
1
-π/5
π/5
θ
W(θ)
5
Y(θ)


|X(θ)|
1
-2π
-π
π
2π
θ

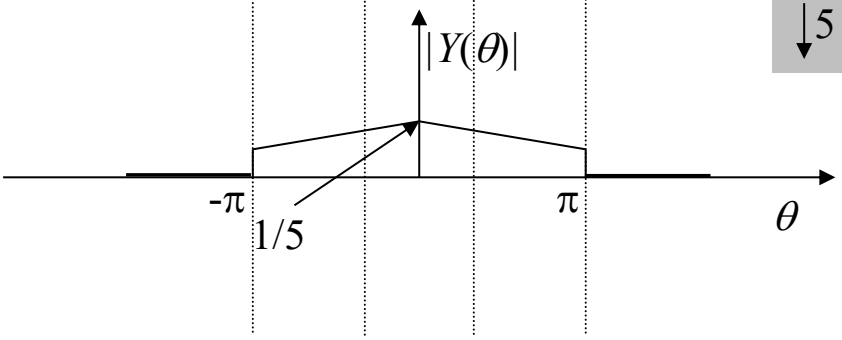

|V(θ)|
1
-π/3
π/3
π
2π
θ

↑3

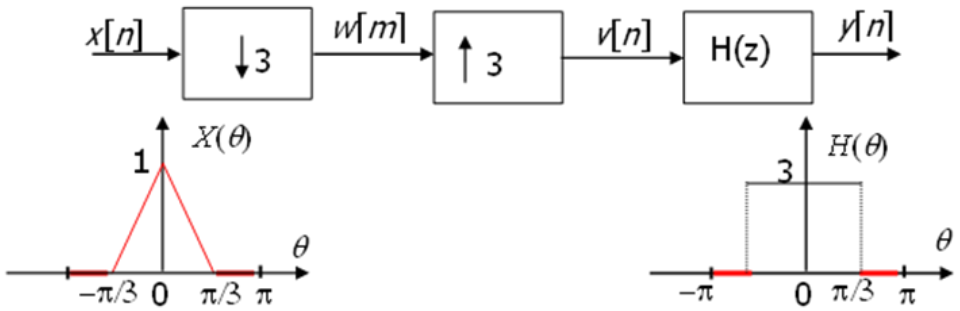

|H(θ)|
1
θ
-π/5
π/5


|W(θ)|
1
-π/5
π/5
π/3
θ
-π/3

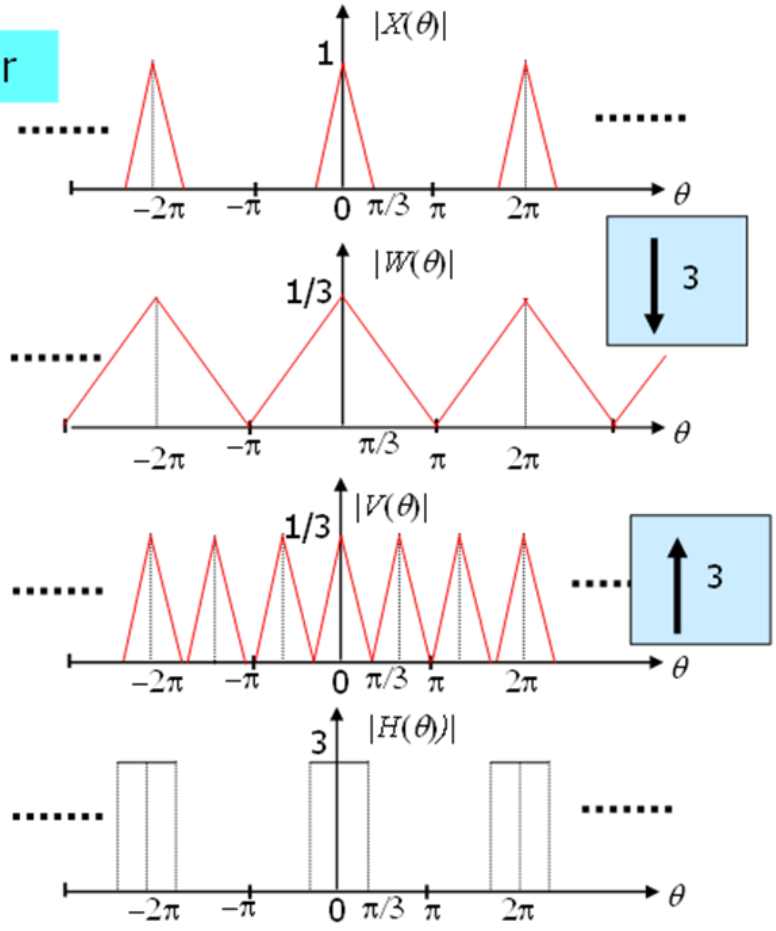
↓5

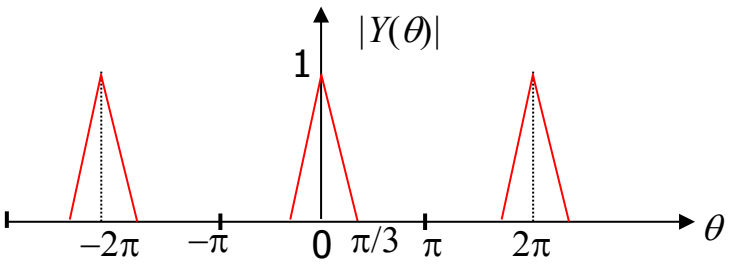

|Y(θ)|
-π
1/5
π
θ

Example: An input signal $x[n]$ with spectrum $X(\theta)$ is shown below. The input signal is applied to the system shown below. Sketch $|X(\theta)|$, $|W(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.



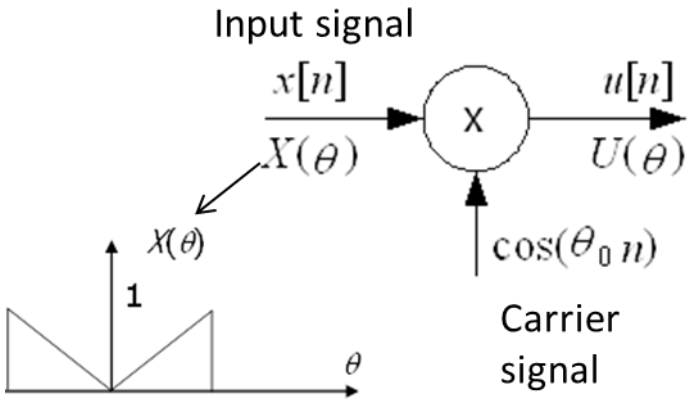
Answer





Modulation

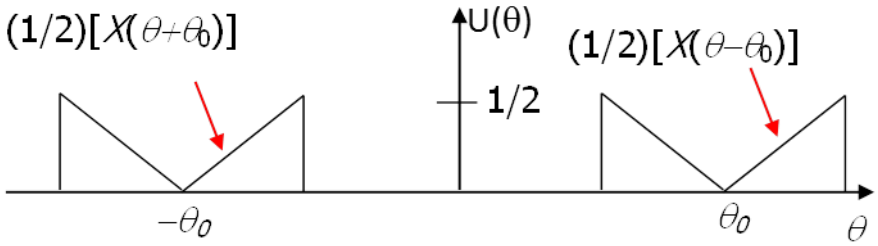
In the time domain, modulation is the process of multiplying an input signal $x[n]$ with a sinusoidal signal known as the carrier, illustrated in the diagram below.



According to the modulation property of the discrete-time Fourier transform:

$$x[n]\cos\theta_0 n \leftrightarrow \tfrac{1}{2}X(\theta+\theta_0)+\tfrac{1}{2}X(\theta-\theta_0),$$

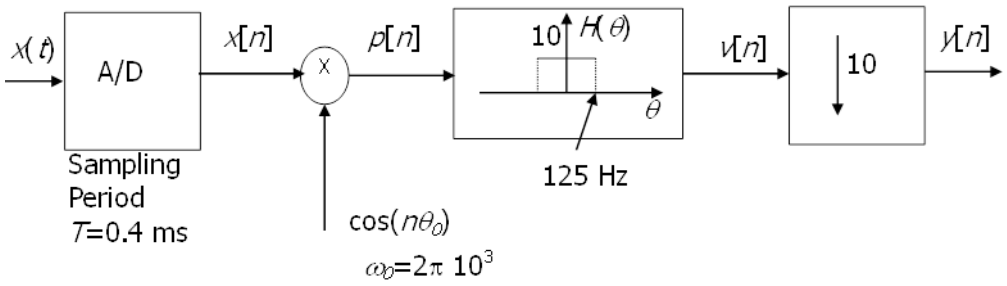
In the frequency domain, the modulated signal comprises two shifted versions of the original signal, translated by the carrier frequency.



Time Domain	Frequency Domain
$x(n)\cos(\theta_0 n)$	$\frac{1}{2}[X(\theta-\theta_0)+X(\theta+\theta_0)]$
$x(n)\sin(\theta_0 n)$	$\frac{1}{2j}[X(\theta-\theta_0)-X(\theta+\theta_0)]$
$x(n)e^{-jn\theta_0}$	$X(\theta+\theta_0)$

Example: $x(t)$ is the input signal for the system shown below. The analogue signal $x(t)$ has the spectrum $X(f)$ given by:

$$X(f)=\begin{cases}1 & 0.875kHz \le f \le 1.125kHz \\ 1 & -1.125kHz \le f \le -0.875kHz \\ 0 & \text{elsewhere}\end{cases}$$



$H(\theta)$ is an ideal lowpass filter (gain=10) with cut-off frequency f_c=125 Hz. Sketch, one above another, $|X(\theta)|$, $|P(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.

$$\text{Note}: \quad P(\theta)=\frac{1}{2}[X(\theta+\theta_0)+X(\theta-\theta_0)]$$

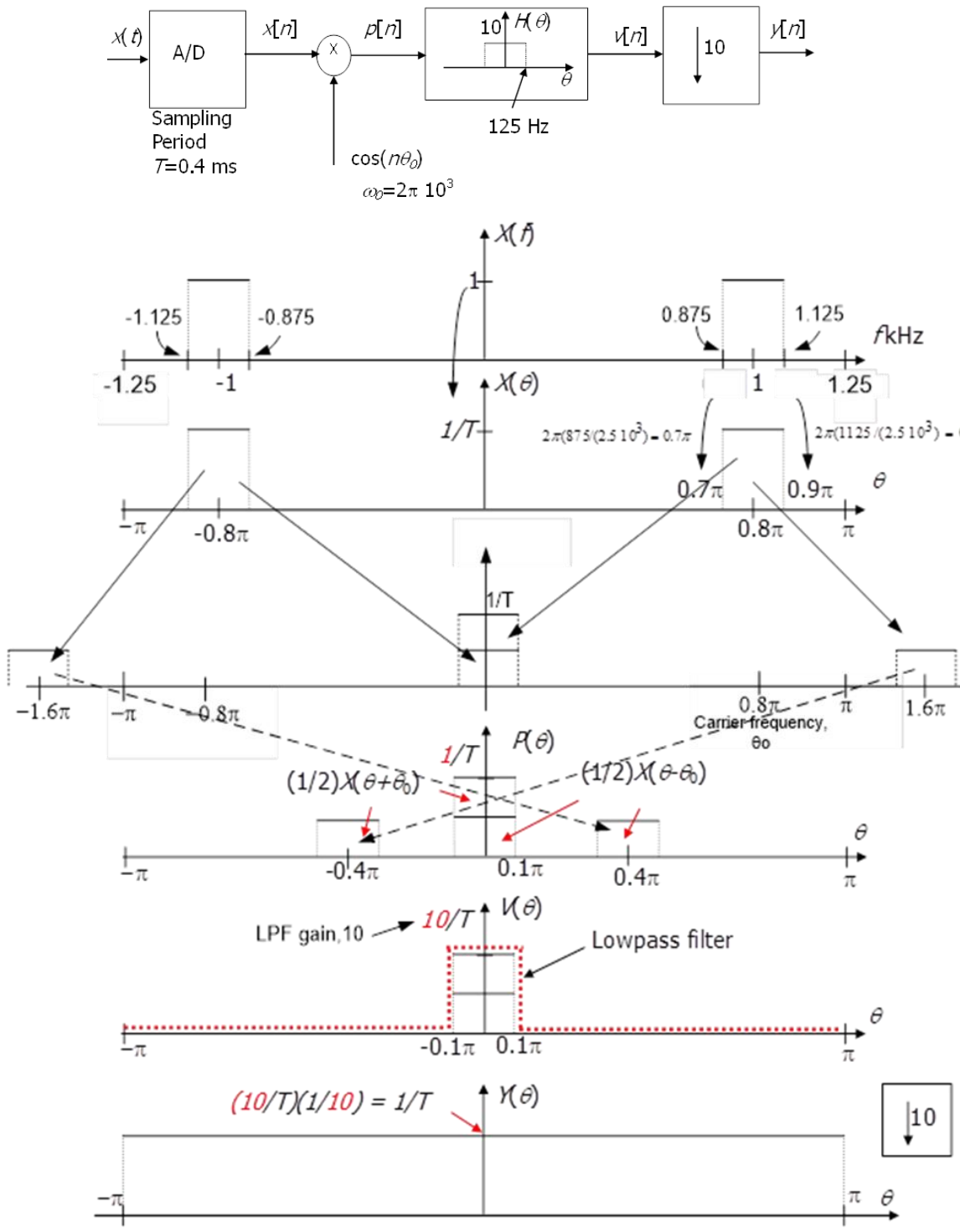
Solution:

Sampling Period, T = 0.4ms

$$f_S=\frac{1}{T}=2.5kHz;$$

Carrier Frequency: $\theta_0=2\pi 10^3(0.4\ 10^{-3})=0.8\pi;$

LPF Cutoff Frequency: $\theta_C=2\pi(125/(2.5\quad 10^3)=0.1\pi$


x(t)
A/D
x[n]
p[n]
10
H(θ)
θ
125 Hz
v[n]
10
y[n]
Sampling
Period
T=0.4 ms
cos(nθ0)
ω0=2π 10^3
X(f)
1
-1.125
-0.875
0.875
1.125
fkHz
-1.25
-1
1
1.25
X(θ)
1/T
2π(875/(2.5 10^3) = 0.7π
2π(1125/(2.5 10^3) = 0.9π
0.7π
0.9π
θ
−π
-0.8π
0.8π
π
1/T
−1.6π
−π
-0.8π
0.8π
π
1.6π
Carrier frequency, θo
P(θ)
1/T
(1/2)X(θ+θ0)
(1/2)X(θ-θ0)
θ
−π
-0.4π
0.1π
0.4π
π
V(θ)
LPF gain,10
10/T
Lowpass filter
θ
−π
-0.1π
0.1π
π
(10/T)(1/10) = 1/T
Y(θ)
10
−π
π
θ

Example

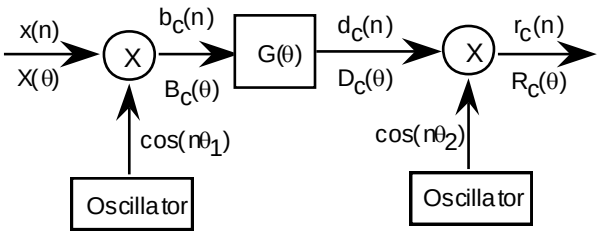
a. A signal has been sampled at 44 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 8 kHz.
   (i) Draw a block diagram to show how you would achieve this.
   (ii) Indicate the type of digital filter which you would choose for part (i) above (give reasons).

b. The sampling period T of the input signal shown in the figure below is 125 μs. The relative frequency is θ = ωT. The first oscillator generates a carrier with a relative frequency $\theta_1 = \omega_1 T$, where $\omega_1 = 2\pi.2.10^3\ rad/\sec$. The second oscillator generates a carrier with a relative frequency $\theta_2 = (\omega_C + \omega_1)T$, where $\omega_C = 2\pi.10^3\ rad/\sec$. The low-pass filter has the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

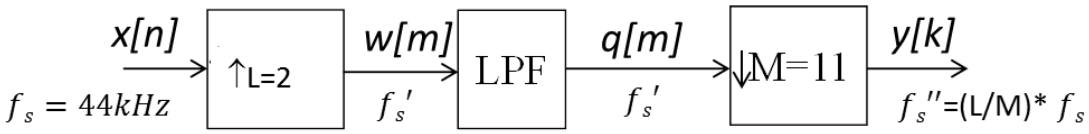
If the input spectrum is $X(\theta) = 1 + \cos(\theta) \quad -\pi \le \theta \le \pi$,

   (i) Write appropriate equations for $B_C(\theta)$, $D_C(\theta)$, and $R_C(\theta)$
   (ii) Sketch |X(θ)|, $|B_C(\theta)|$, $|D_C(\theta)|$, and $|R_C(\theta)|$ against θ,



Solution

(a)
(i)



$$\frac{L}{M} = \frac{8000}{44000} = \frac{2}{11}$$

(ii)
Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.

(b)

$$\theta_1 = 2\pi \times 2 \times 10^3 \times \frac{1}{8 \times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8 \times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta - \theta_1) + X(\theta + \theta_1)\right] = \frac{1}{2}\left[1 + \cos\left(\theta - \frac{\pi}{2}\right) + 1 + \cos\left(\theta + \frac{\pi}{2}\right)\right] = \frac{1}{2} \times 2 = 1$$

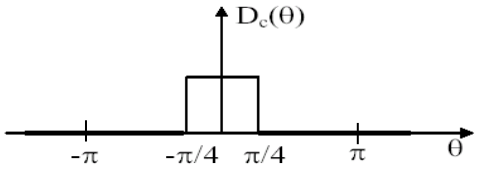
$$D_C(\theta) = B_C(\theta) \cdot G(\theta) = 1, \quad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta)=\frac{1}{2}\left[D_C(\theta-\theta_2)+D_C(\theta+\theta_2)\right]=\frac{1}{2}\left[D_C\left(\theta-\tfrac{3}{4}\pi\right)+D_C\left(\theta+\tfrac{3}{4}\pi\right)\right]$$

$$R_C(\theta)=\begin{cases}\frac{1}{2} & -\frac{\pi}{4}\le\theta-\frac{3\pi}{4}\le\frac{\pi}{4}\Rightarrow\frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\frac{\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\Rightarrow-\pi\le\theta\le-\frac{\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta),$ and $R_C(\theta)$.

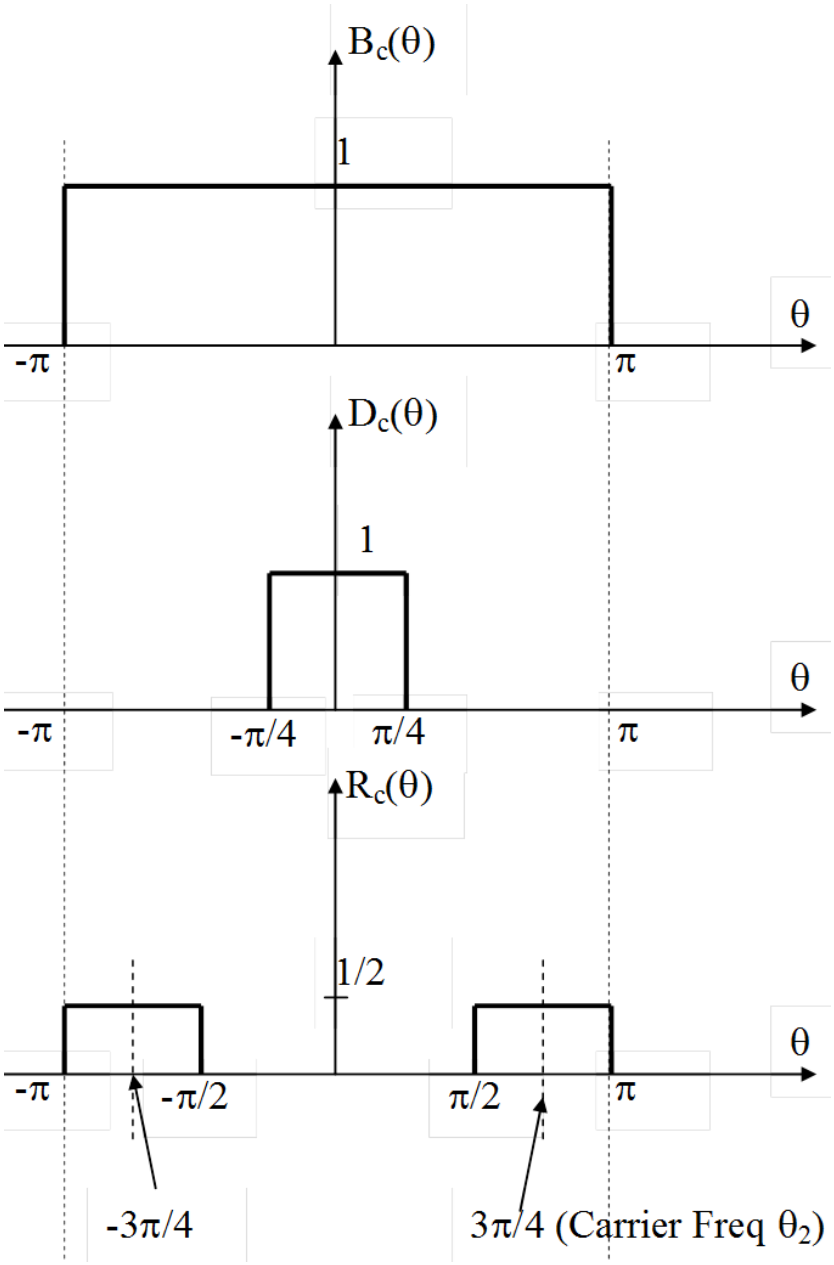
$\mathrm{B_c}(\theta)=1, \mathrm{D_c}(\theta)=1$



$$\mathrm{R_c}(\theta)=\frac{1}{2}\left\{\mathrm{D_c}(\theta-\theta_1)+\mathrm{D_c}(\theta+\theta_1)\right\}$$

$$=\frac{1}{2}\left\{\mathrm{D_C}(\theta-\frac{3\pi}{4})+\mathrm{D_c}(\theta+\frac{3\pi}{4})\right\}$$

$$R_c(\theta)=\begin{cases}\frac{1}{2} & -\frac{\pi}{4}\le(\theta-\frac{3\pi}{4})\le\frac{\pi}{4}\\ \frac{1}{2} & -\frac{\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\\ 0 & \text{elsewhere}\end{cases}=\begin{cases}\frac{1}{2} & \frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\pi\le\theta\le-\frac{\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$



End of Topic 6